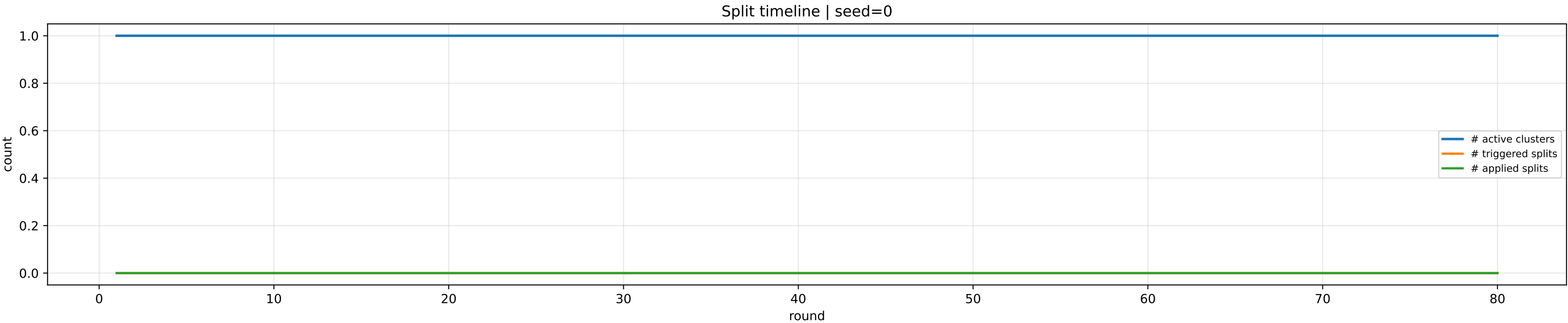


Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002



Seed summary | seed=0

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0			0	0:5

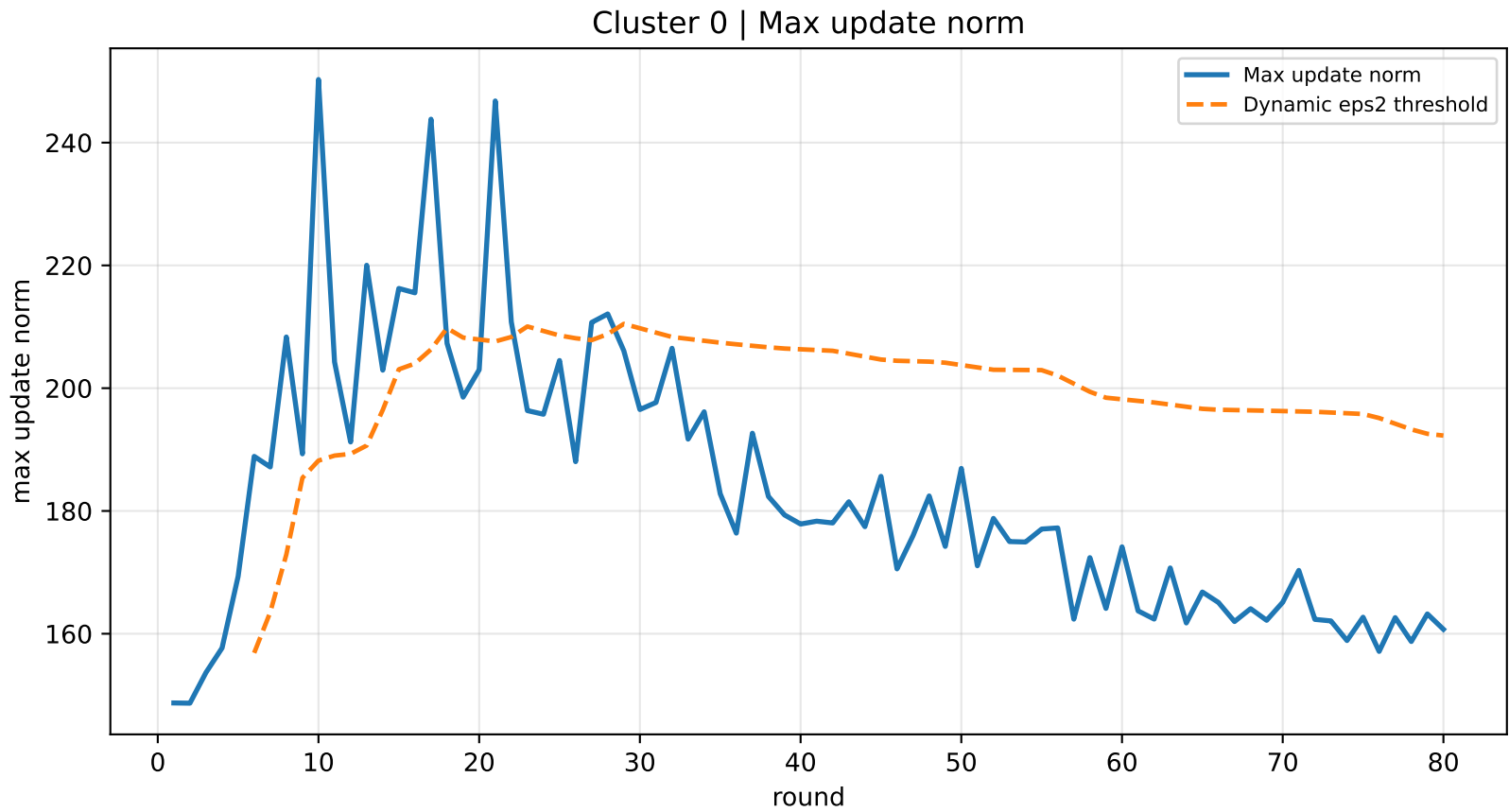
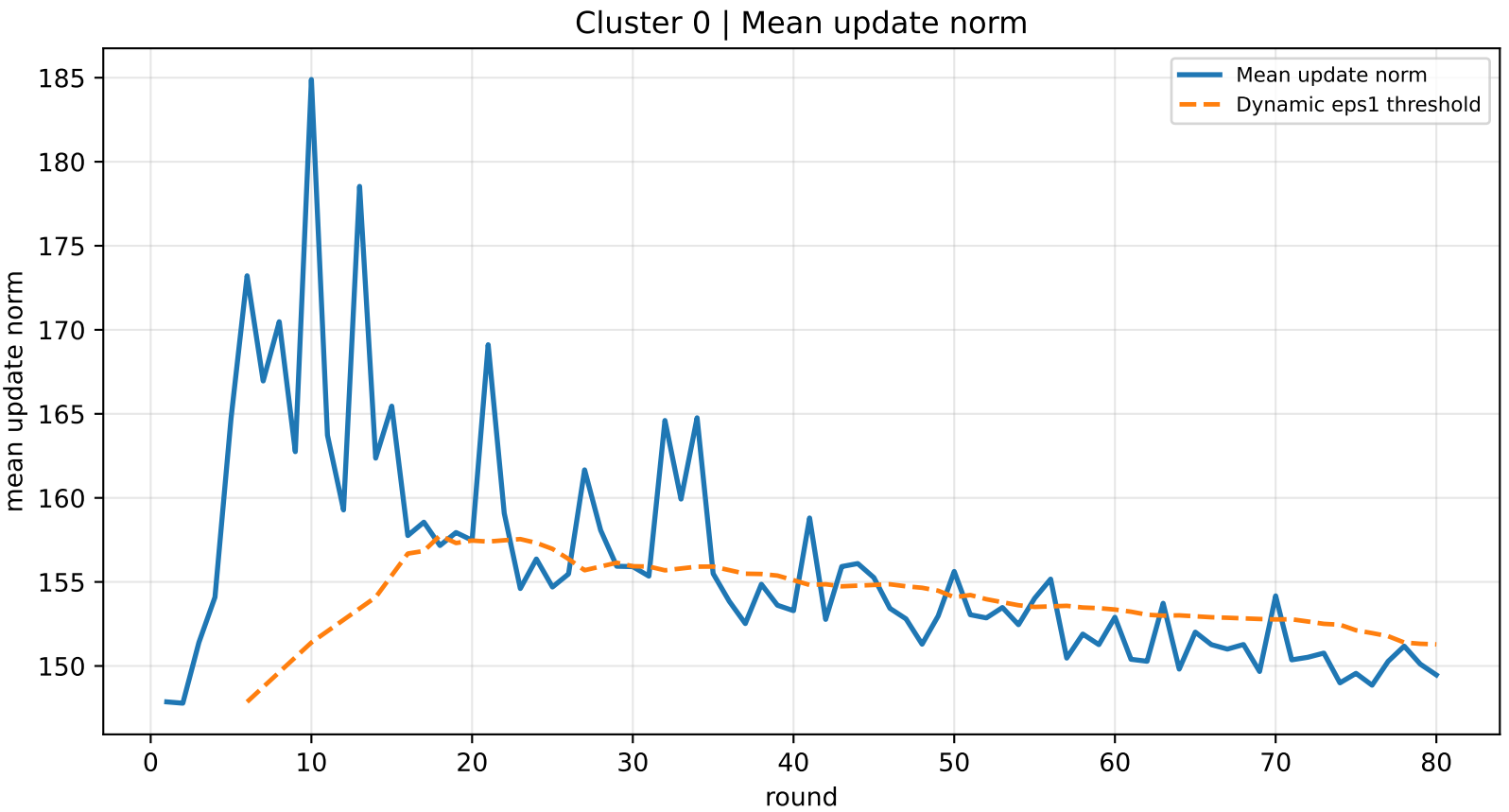
Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002

Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
0	5	1000000, 1000001, 1000002, 1000003, 1000004	0, 1, 2, 3, 4	masked_cycle2:1 , masked_cycle3:1 , masked_cycle4:1 , masked_cycle5:1 , masked_cycle6:1

Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5
clients | seed=0 | norm / threshold panels (0-0)
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002



Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

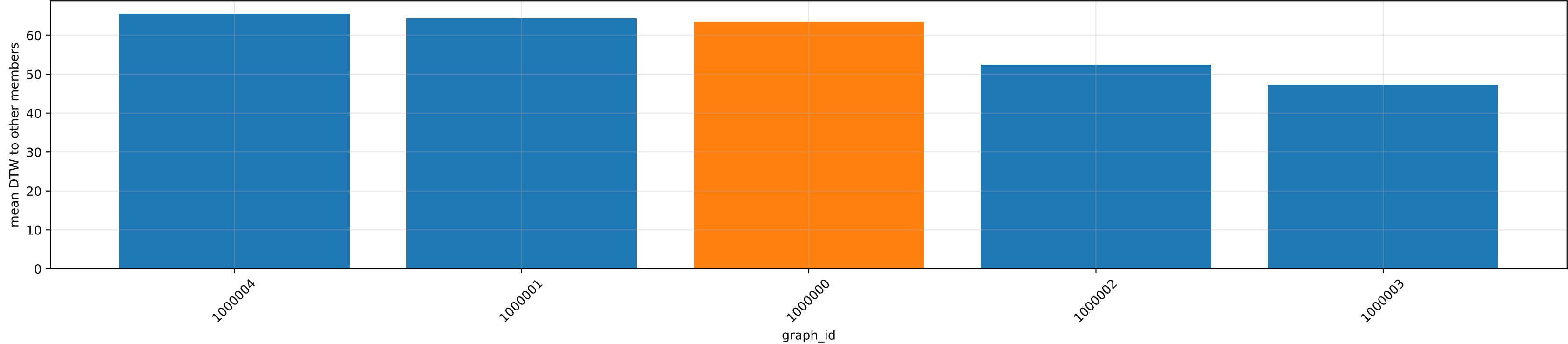
First applied split summary | seed=1

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
1	22	0	4 vs 1	1000000	1000004	no

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
63.362	55.380	nan

Per-member mean DTW



Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

First applied split pairwise DTW metadata | seed=1

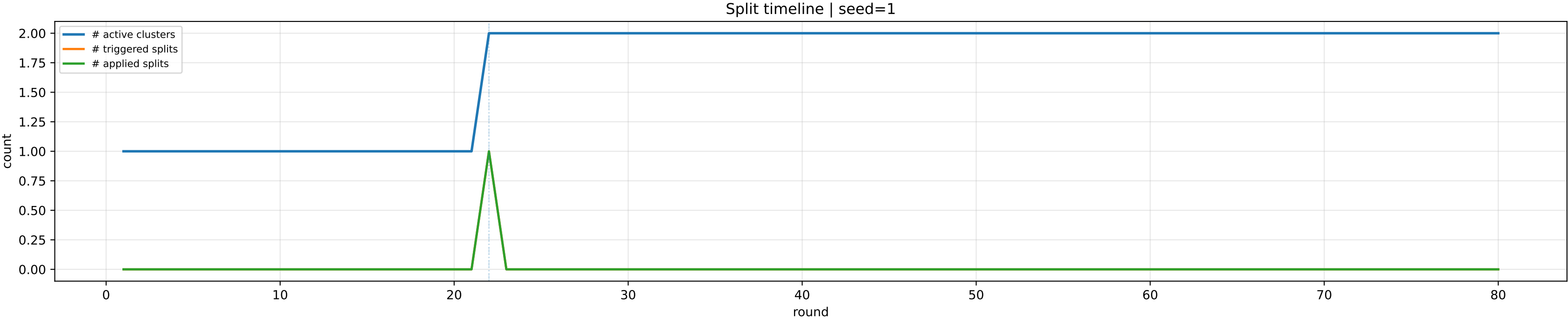
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
1	22	0	5	1000000.0	0.064	63.362	1

Pairwise DTW table | first applied split | seed=1
Global max DTW = 87.7 at (1000001, 1000004)

graph_id	1000000	1000001	1000002	1000003	1000004
1000000	0.0	68.0	61.2	46.4	77.9
1000001	68.0	0.0	45.0	57.0	87.7
1000002	61.2	45.0	0.0	46.1	57.2
1000003	46.4	57.0	46.1	0.0	39.4
1000004	77.9	87.7	57.2	39.4	0.0

Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

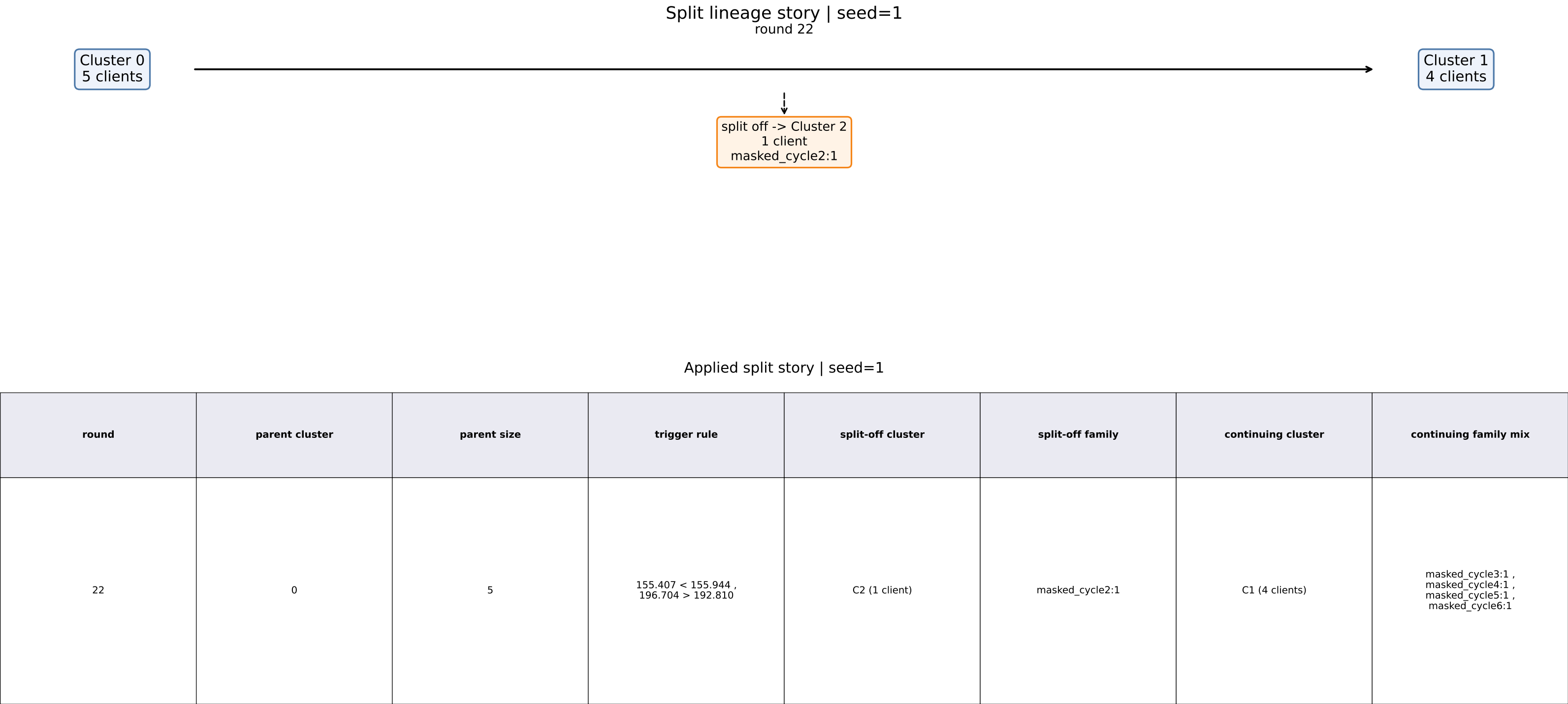


Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0	22	22	1, 2	1:4, 2:1

Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002



Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002

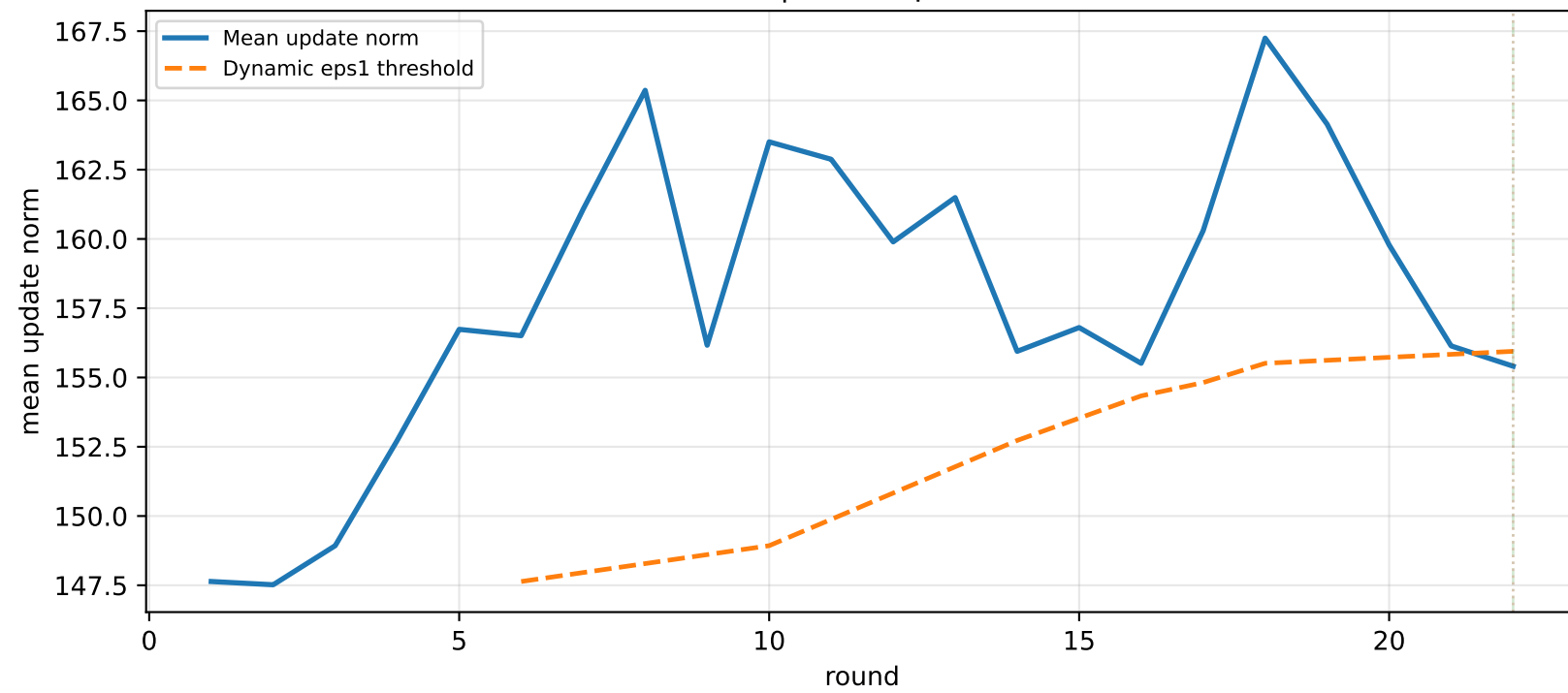
Final active clusters | seed=1

cluster_id	cluster_size	members	member_indices	family_mix
1	4	1000001, 1000002, 1000003, 1000004	1, 2, 3, 4	masked_cycle3:1 , masked_cycle4:1 , masked_cycle5:1 , masked_cycle6:1
2	1	1000000	0	masked_cycle2:1

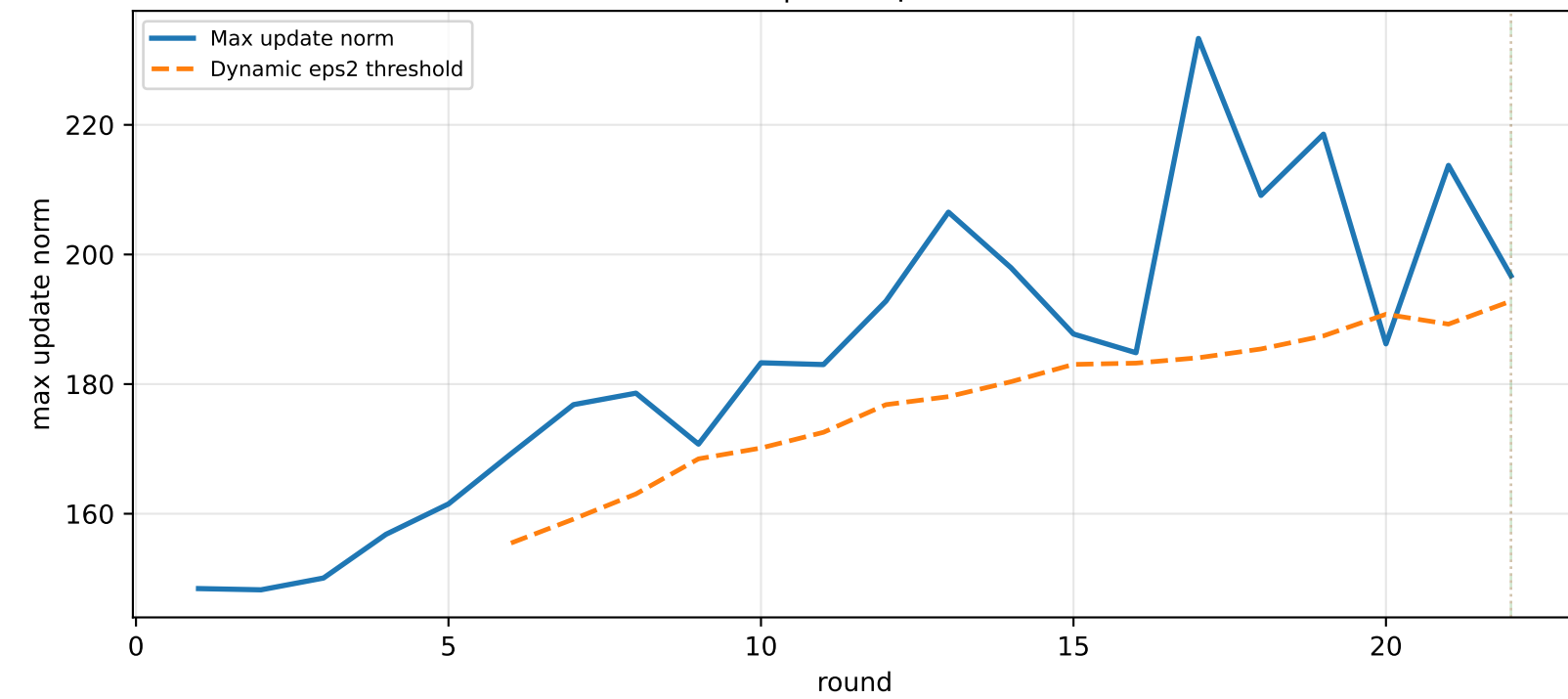
Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | norm / threshold panels (0-1)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002

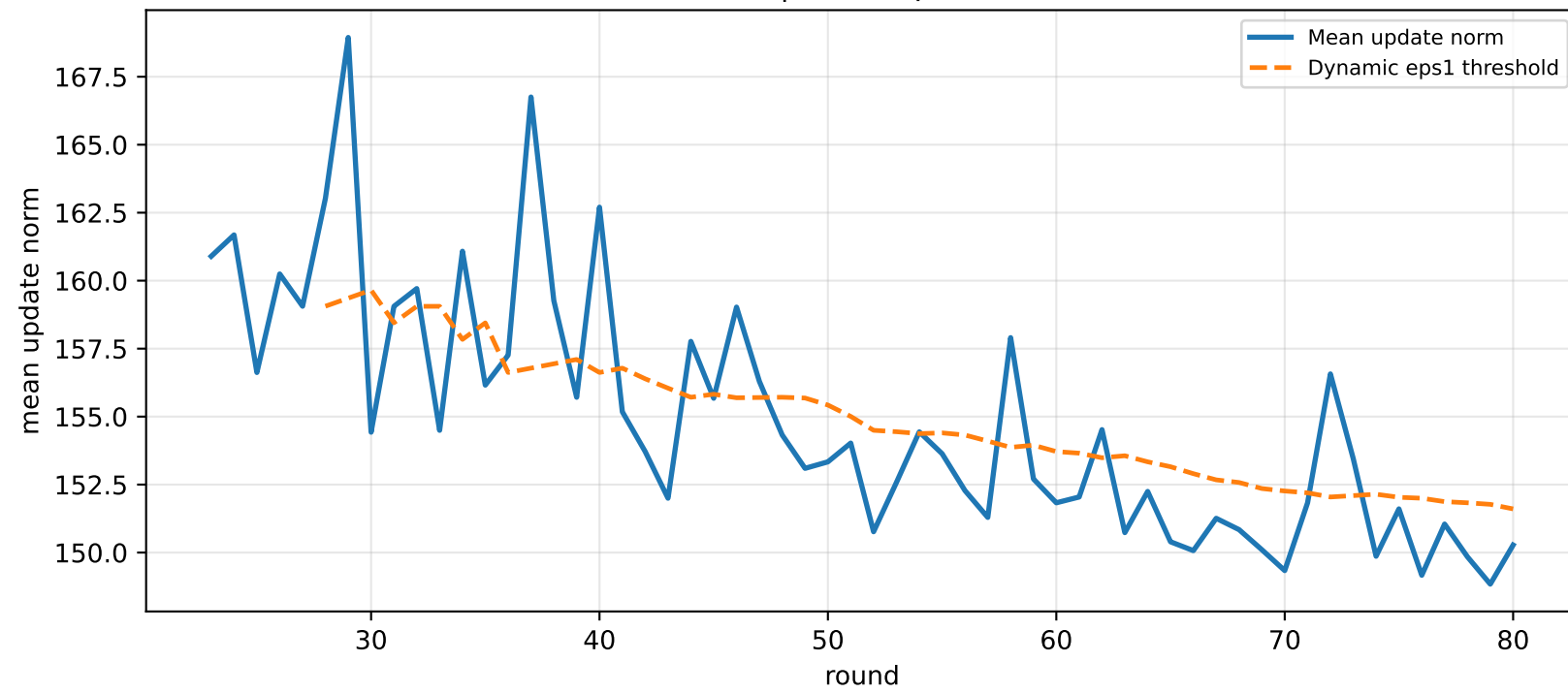
Cluster 0 | Mean update norm



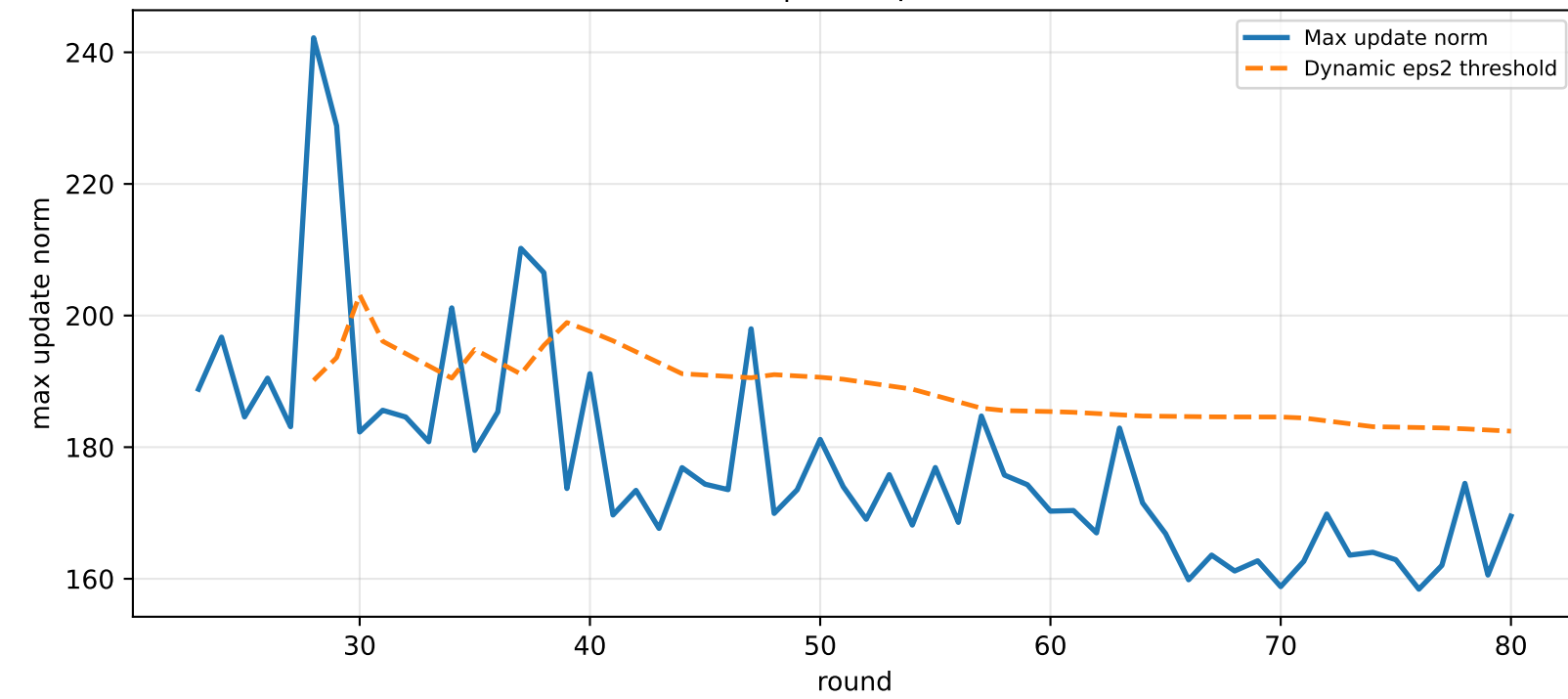
Cluster 0 | Max update norm



Cluster 1 | Mean update norm



Cluster 1 | Max update norm



Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

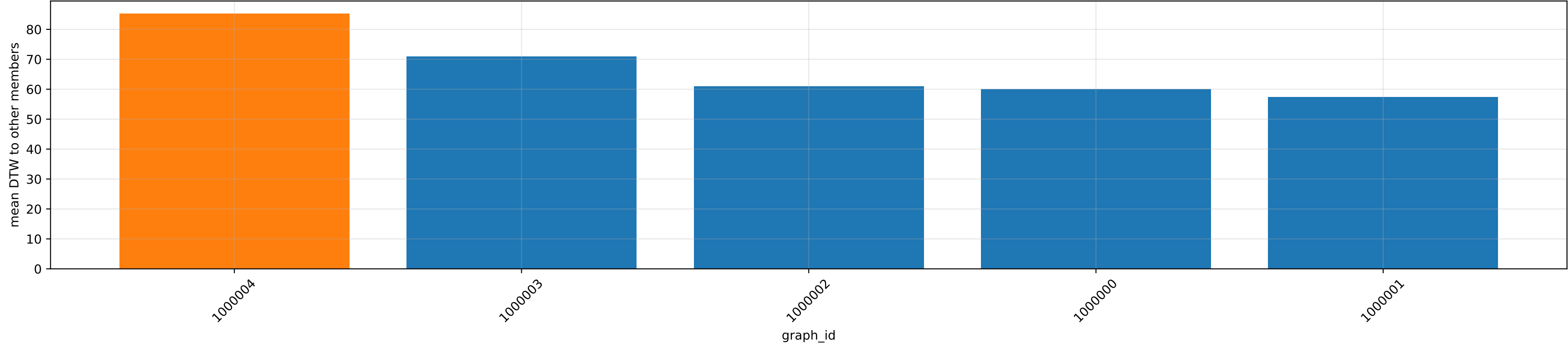
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	16	0	1 vs 4	1000004	1000004	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
85.203	nan	54.716

Per-member mean DTW



Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

First applied split pairwise DTW metadata | seed=2

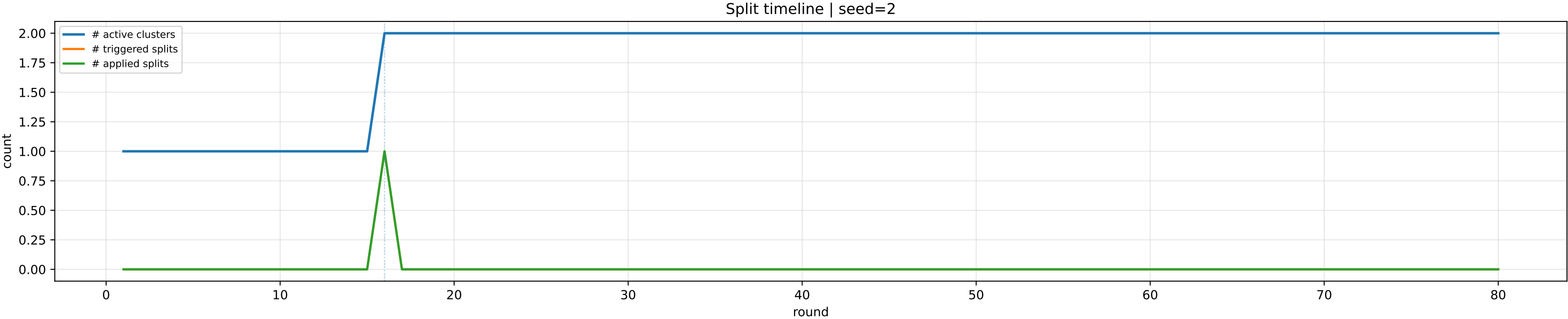
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	16	0	5	1000004.0	0.048	85.203	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 102.5 at (1000003, 1000004)

graph_id	1000000	1000001	1000002	1000003	1000004
1000000	0.0	52.0	54.7	41.8	91.7
1000001	52.0	0.0	40.1	71.9	65.6
1000002	54.7	40.1	0.0	67.8	81.1
1000003	41.8	71.9	67.8	0.0	102.5
1000004	91.7	65.6	81.1	102.5	0.0

Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002



Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	16	16	1, 2	1:1, 2:4

Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002



Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002

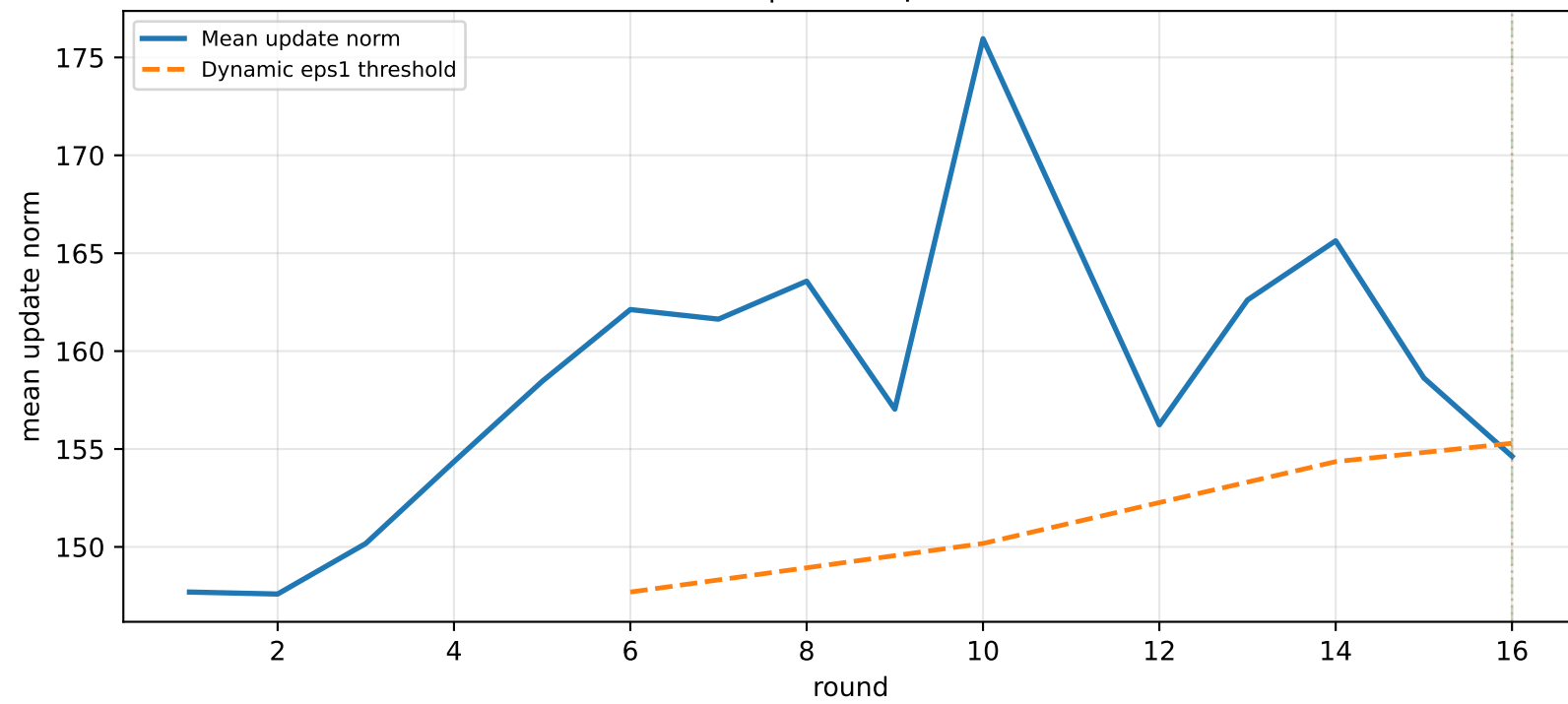
Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000004	4	masked_cycle6:1
2	4	1000000, 1000001, 1000002, 1000003	0, 1, 2, 3	masked_cycle2:1 , masked_cycle3:1 , masked_cycle4:1 , masked_cycle5:1

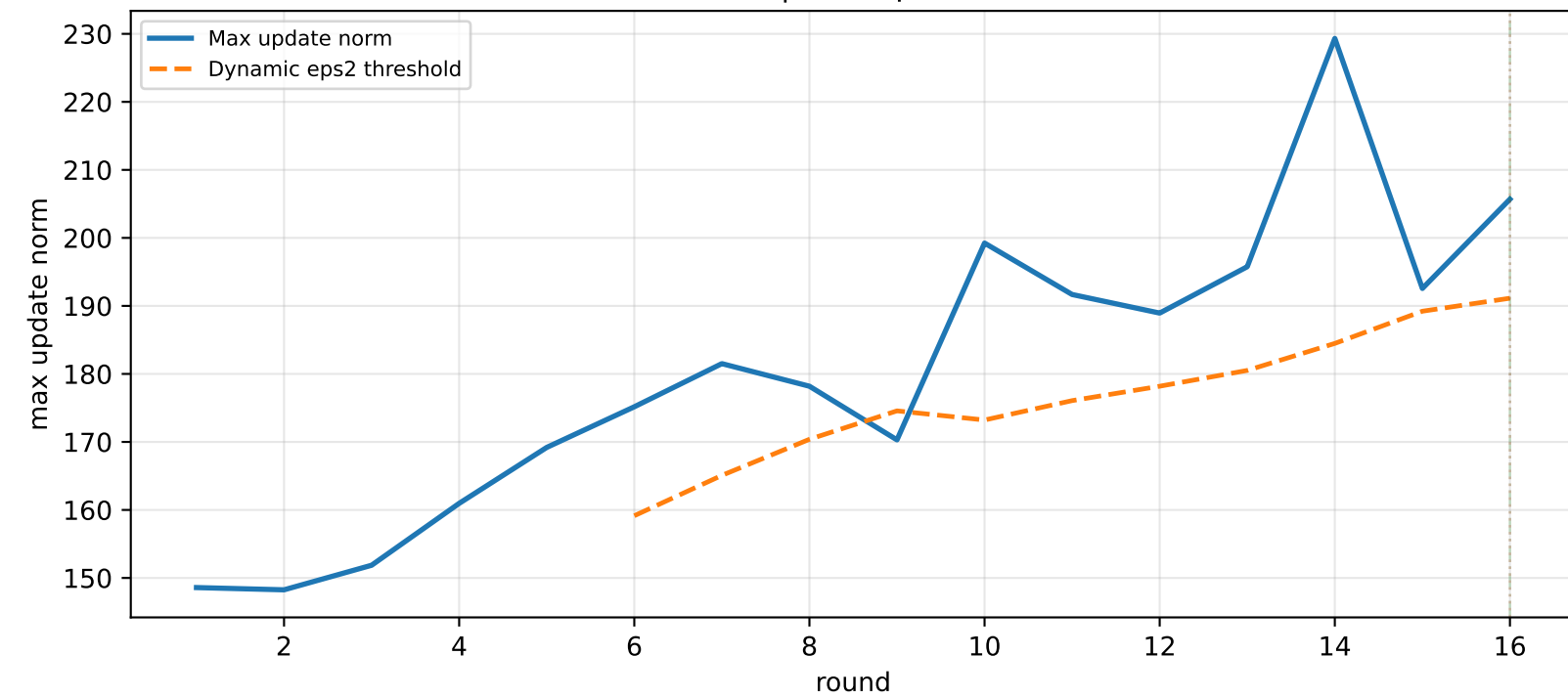
Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 |
family_centroid_jsd_mean=0.002

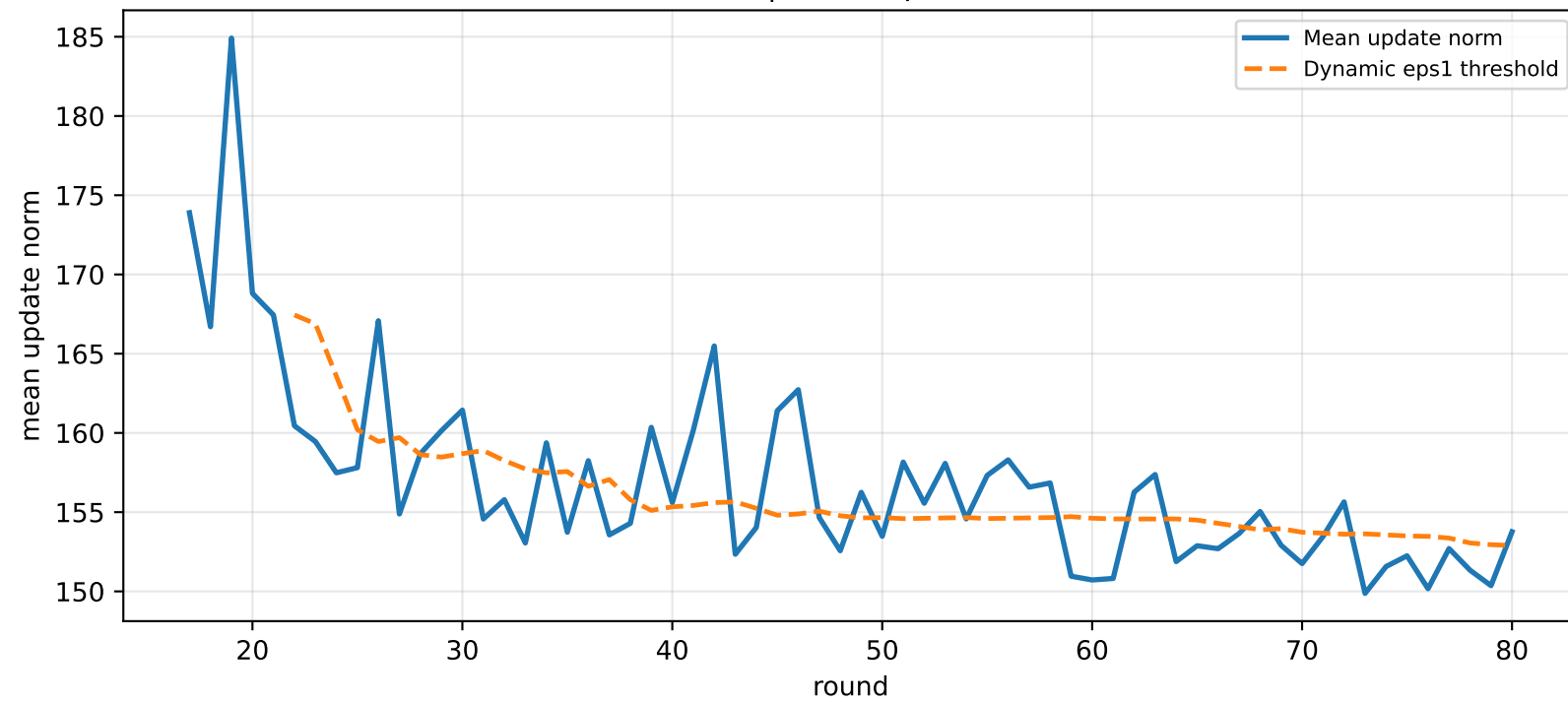
Cluster 0 | Mean update norm



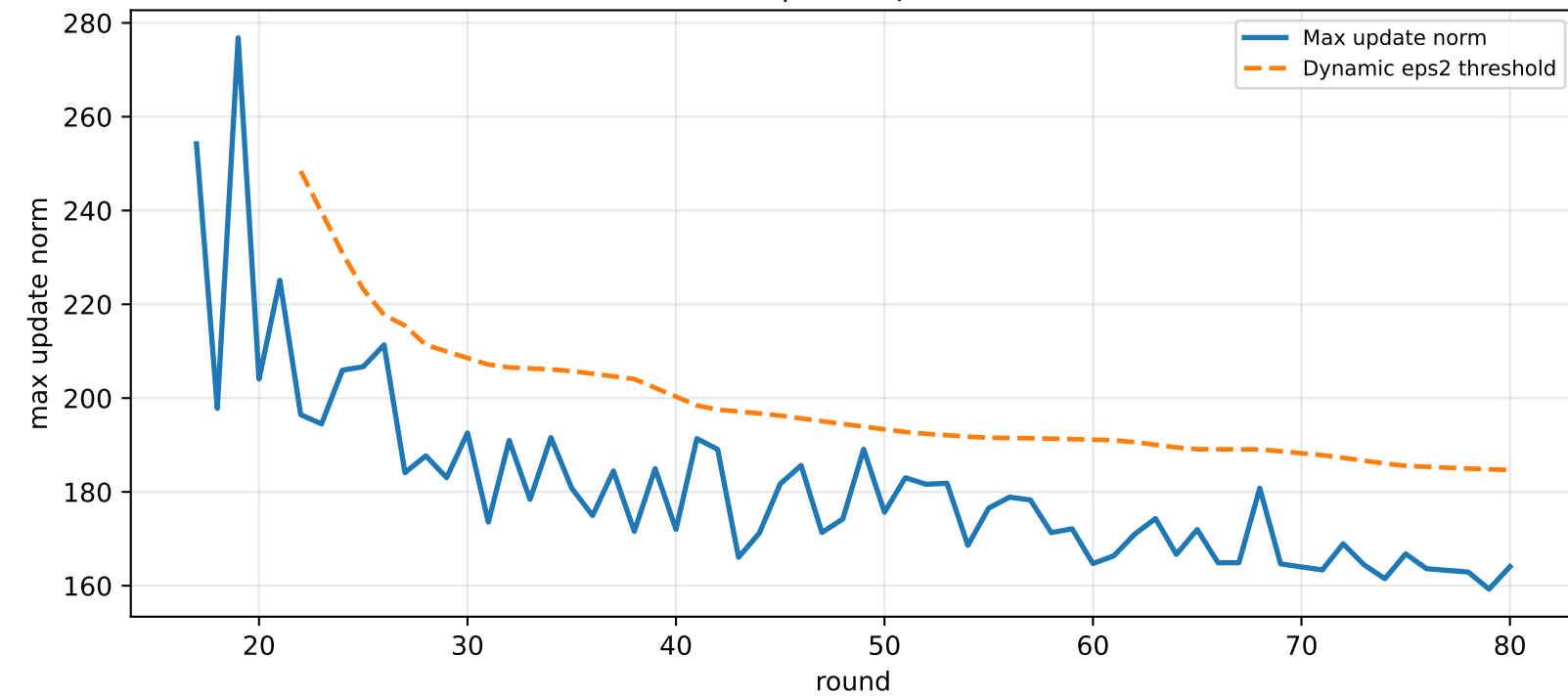
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



Cluster 2 | Max update norm



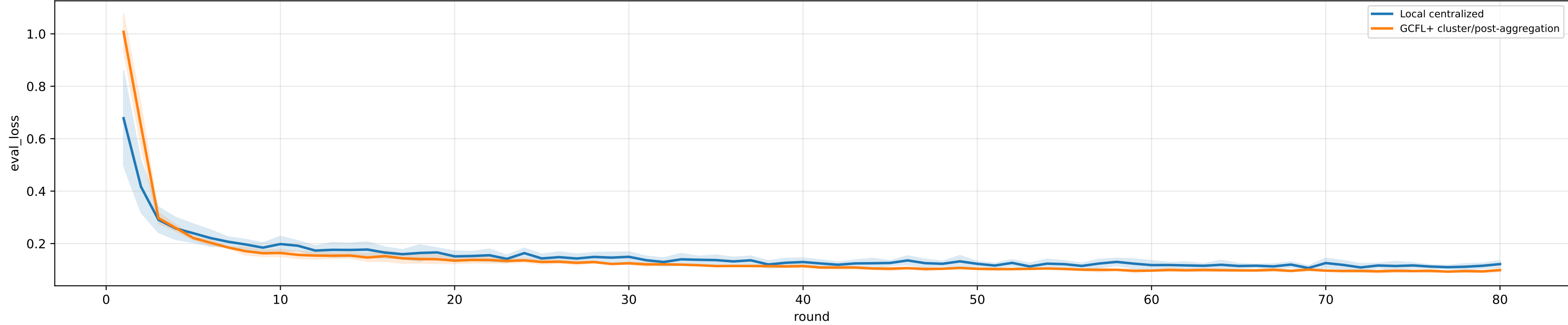
Controlled masking level: Low | GCFL+ diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

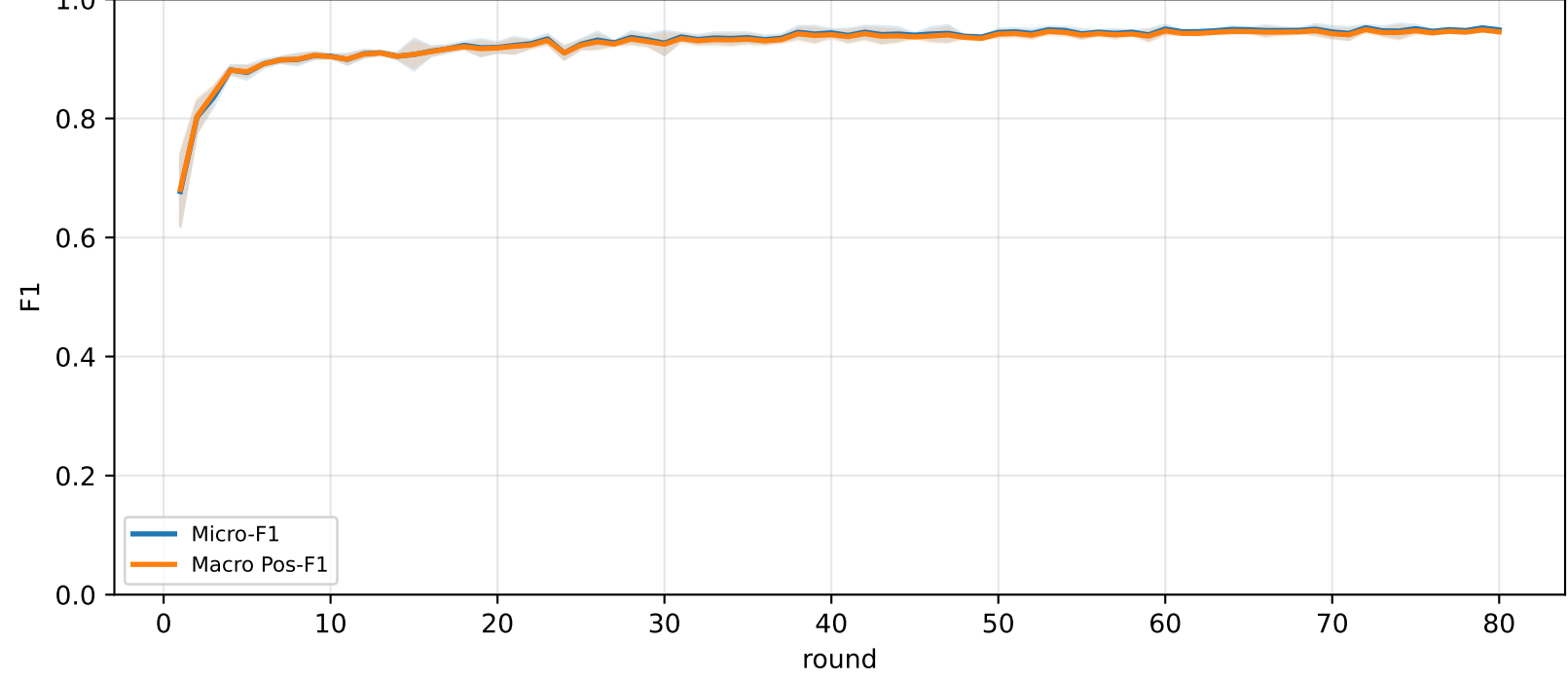
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.5% $\pm$ 0.2%	95.4% $\pm$ 0.2%	99.3% $\pm$ 0.0%	98.7% $\pm$ 0.2%	97.6% $\pm$ 0.3%	94.6% $\pm$ 0.0%	86.6% $\pm$ 0.6%
GCFL+	95.7% $\pm$ 0.1%	95.6% $\pm$ 0.1%	99.4% $\pm$ 0.1%	98.9% $\pm$ 0.1%	97.9% $\pm$ 0.3%	94.3% $\pm$ 0.7%	87.7% $\pm$ 0.8%

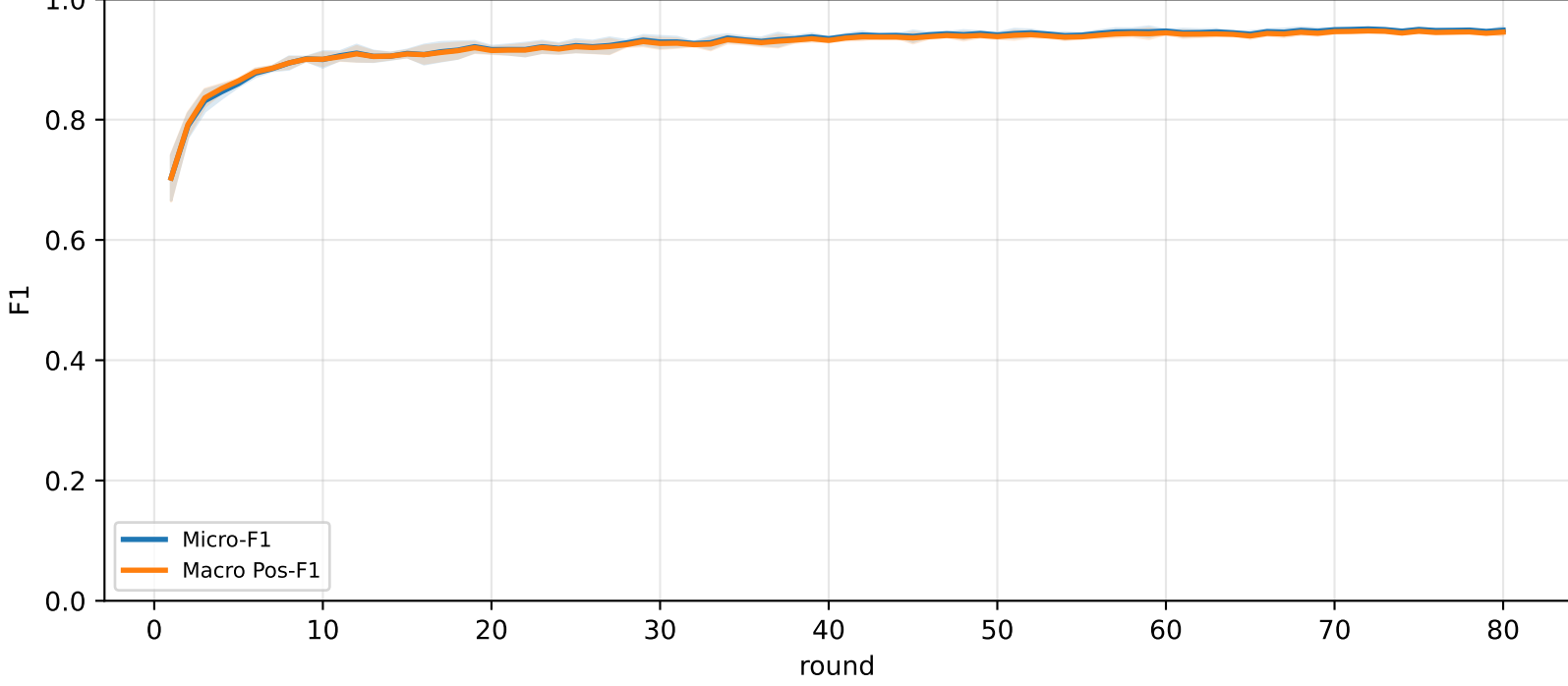
Validation loss comparison



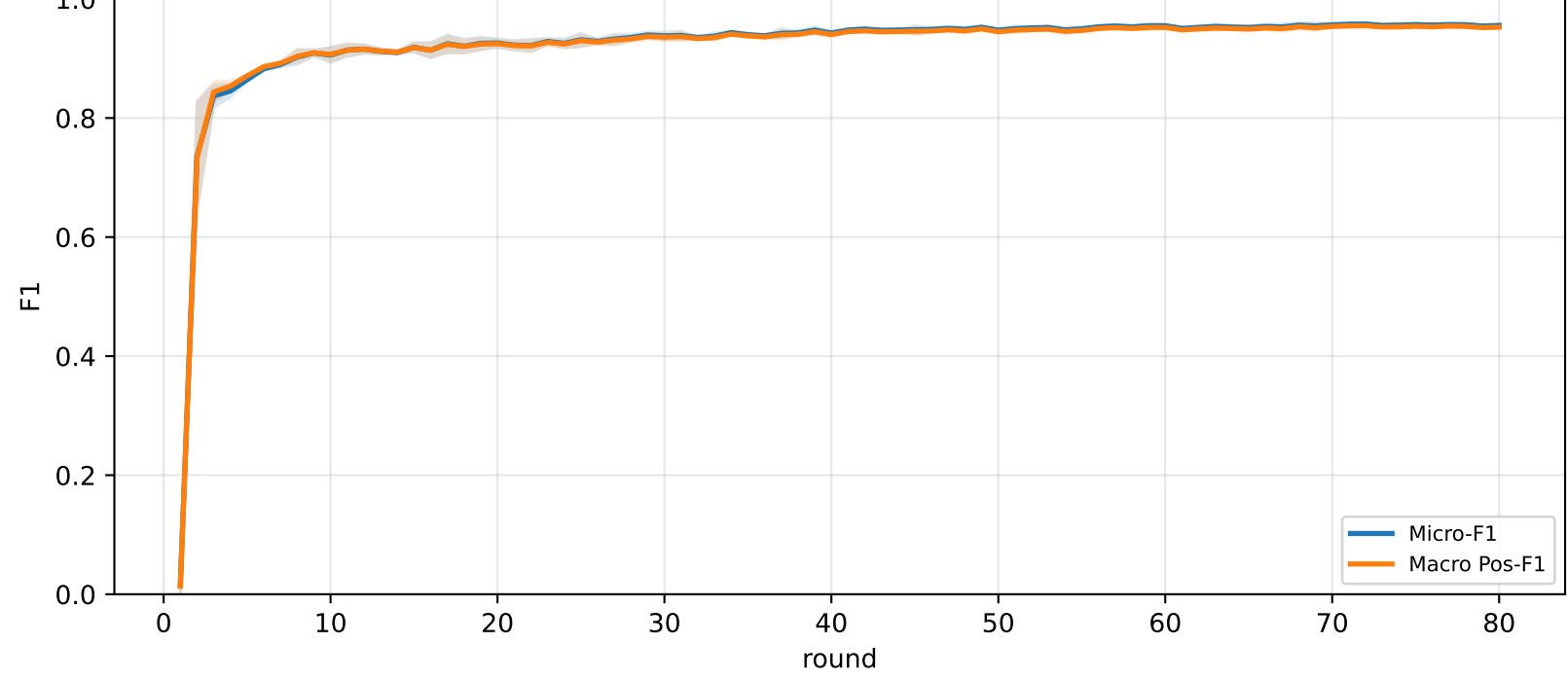
Pooled local centralized micro / macro F1



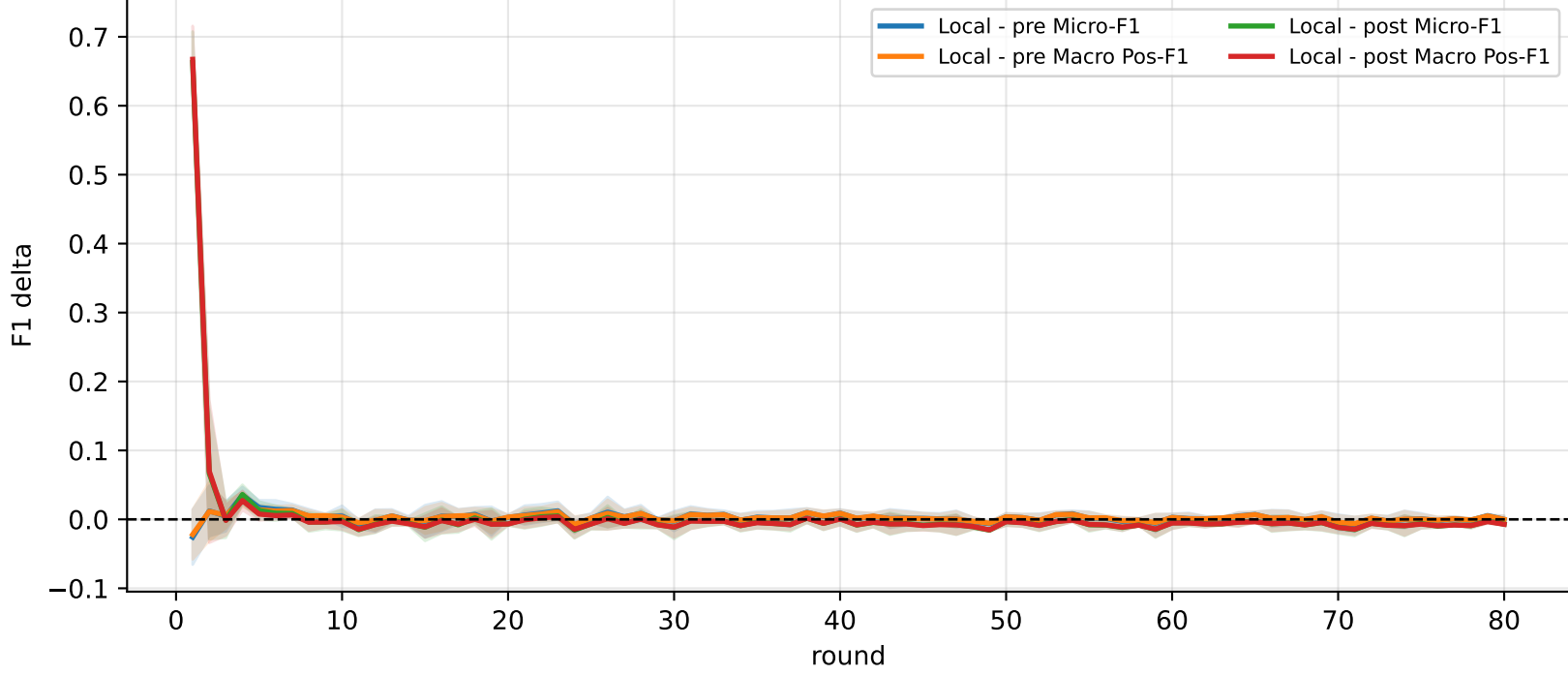
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

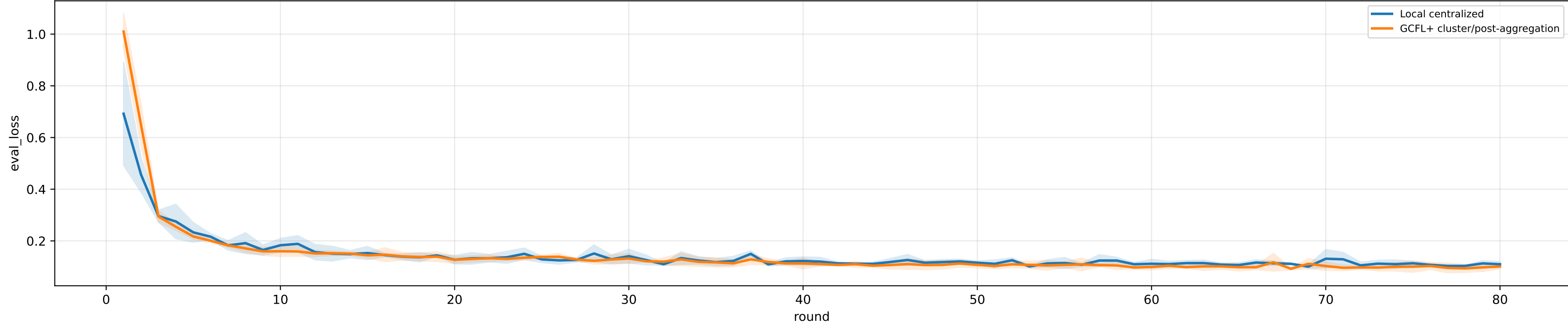


family=masked_cycle2 | n_clients=1 | cycle2=visible 22.1%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1%
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

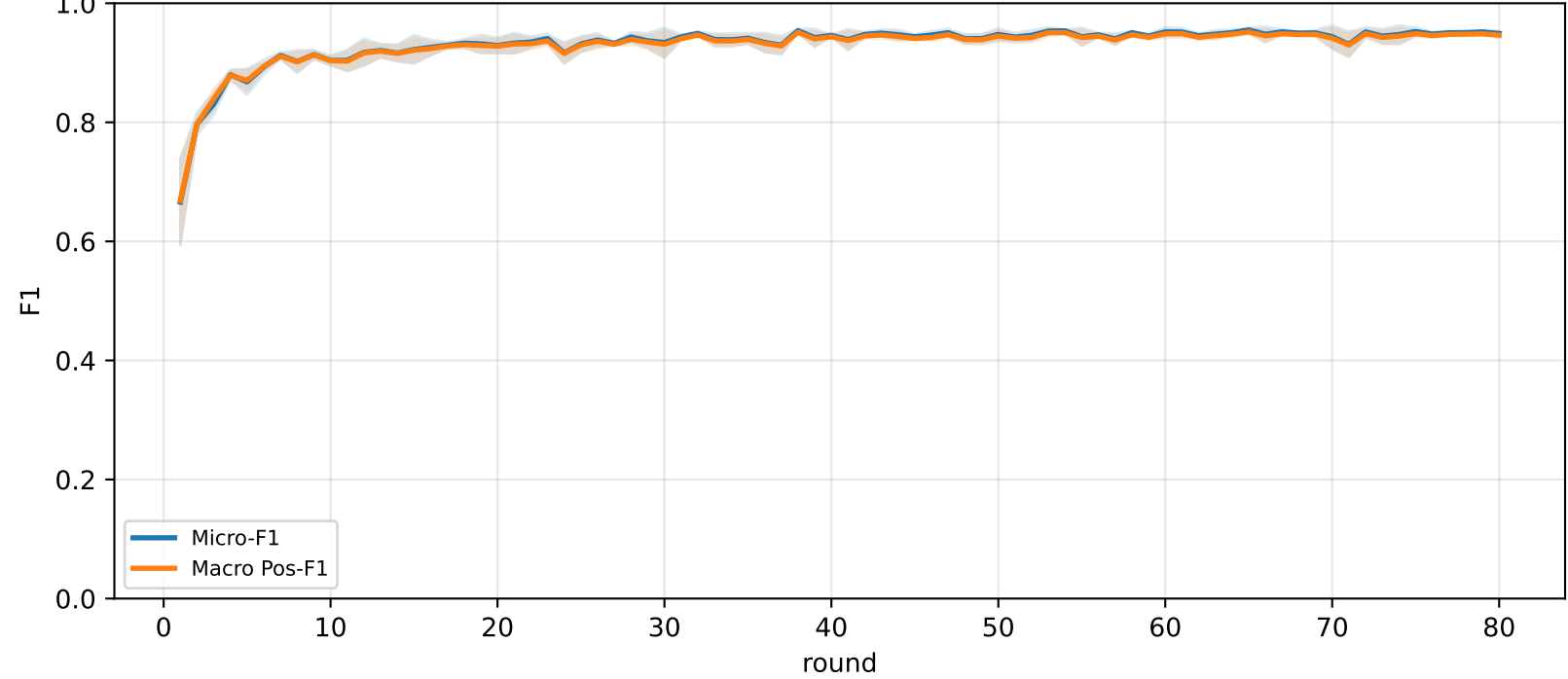
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.7% ± 0.3%	95.6% ± 0.3%	99.3% ± 0.0%	98.8% ± 0.1%	97.9% ± 0.2%	94.8% ± 0.6%	87.3% ± 0.8%
GCFL+	95.8% ± 0.2%	95.7% ± 0.3%	99.6% ± 0.3%	99.0% ± 0.2%	97.8% ± 0.4%	94.3% ± 0.9%	87.8% ± 0.5%

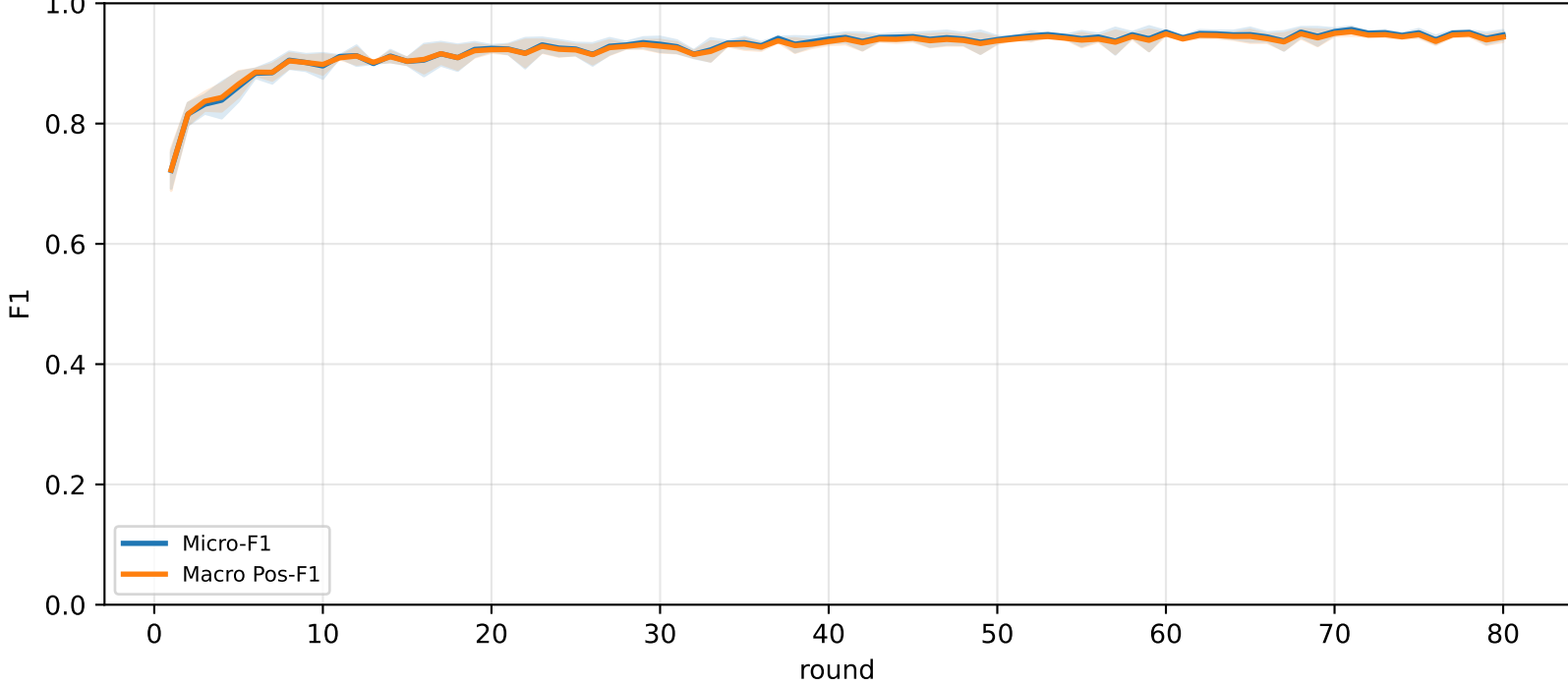
Validation loss comparison



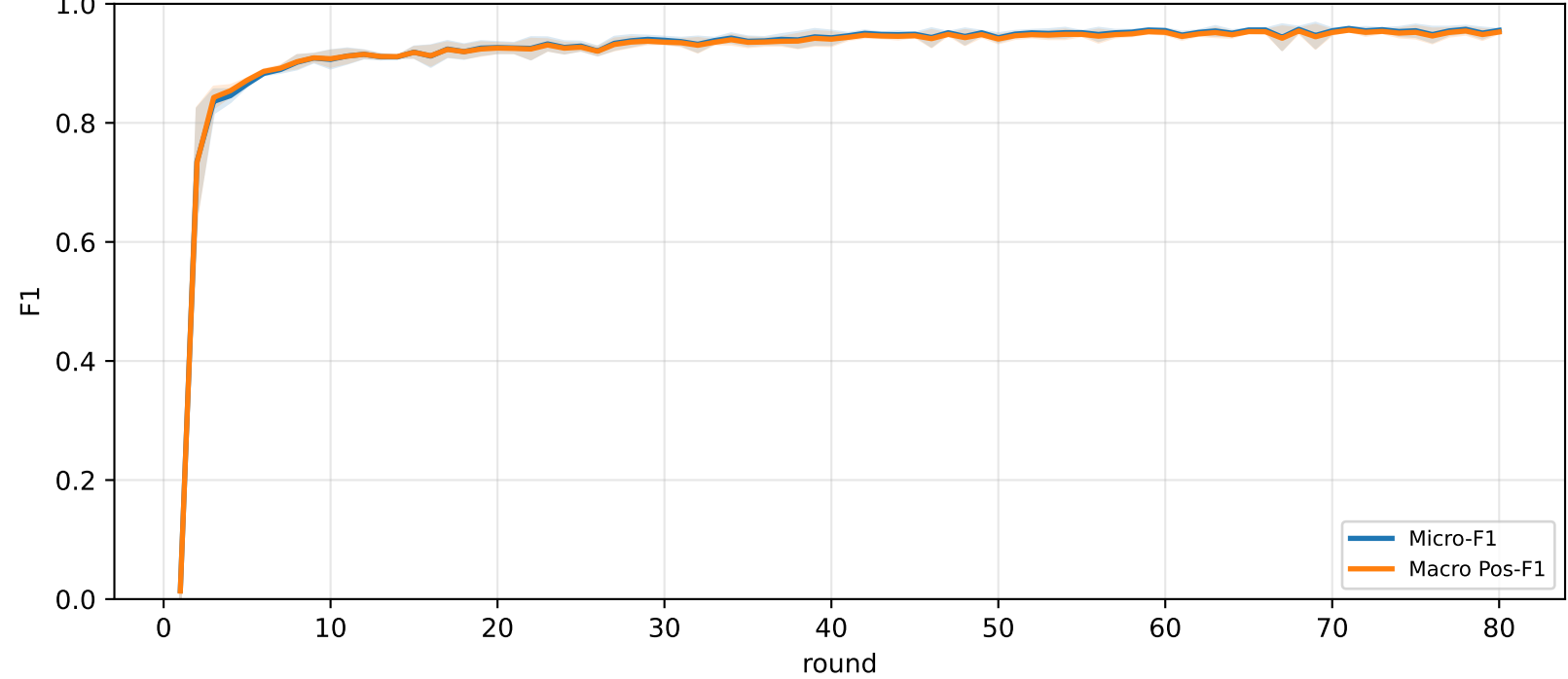
Pooled local centralized micro / macro F1



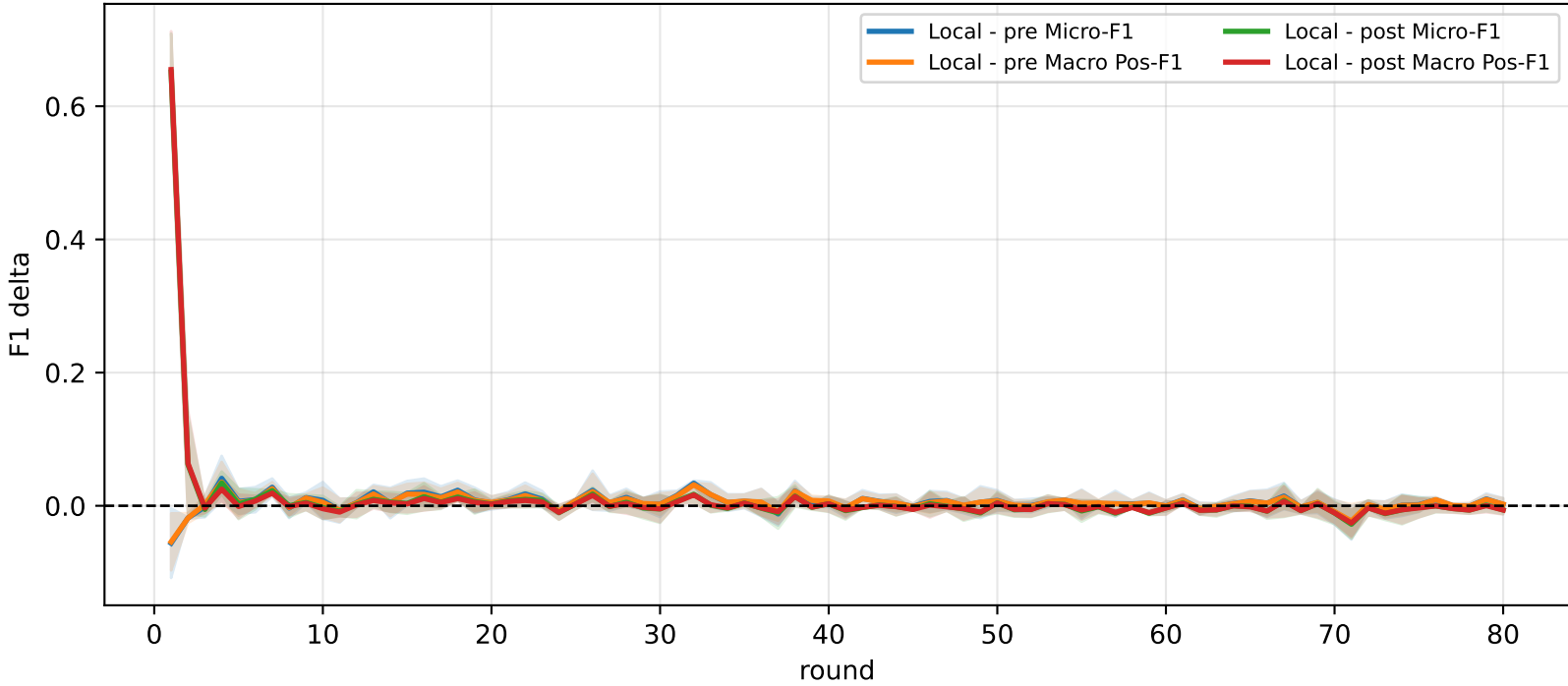
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

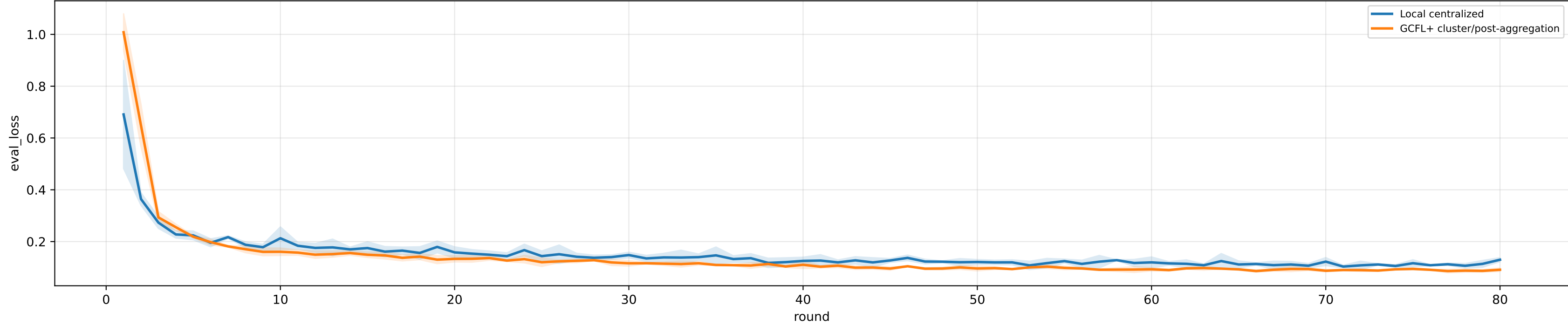


family=masked_cycle3 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 23.3%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1%
masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

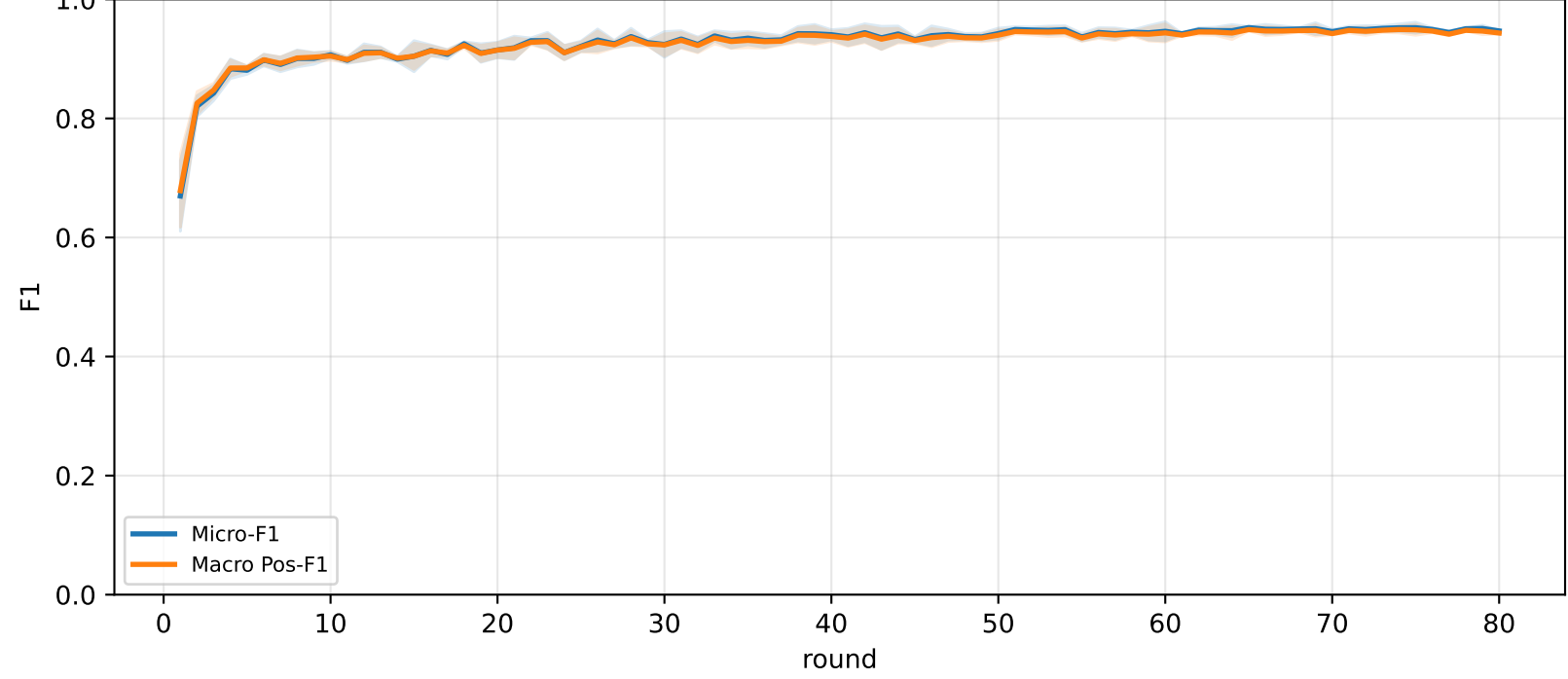
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.4% ± 0.2%	95.3% ± 0.2%	99.3% ± 0.1%	98.7% ± 0.3%	97.2% ± 0.6%	94.0% ± 0.3%	87.3% ± 0.6%
GCFL+	95.9% ± 0.2%	95.8% ± 0.2%	99.5% ± 0.2%	98.9% ± 0.2%	97.7% ± 0.4%	94.7% ± 1.1%	88.0% ± 0.6%

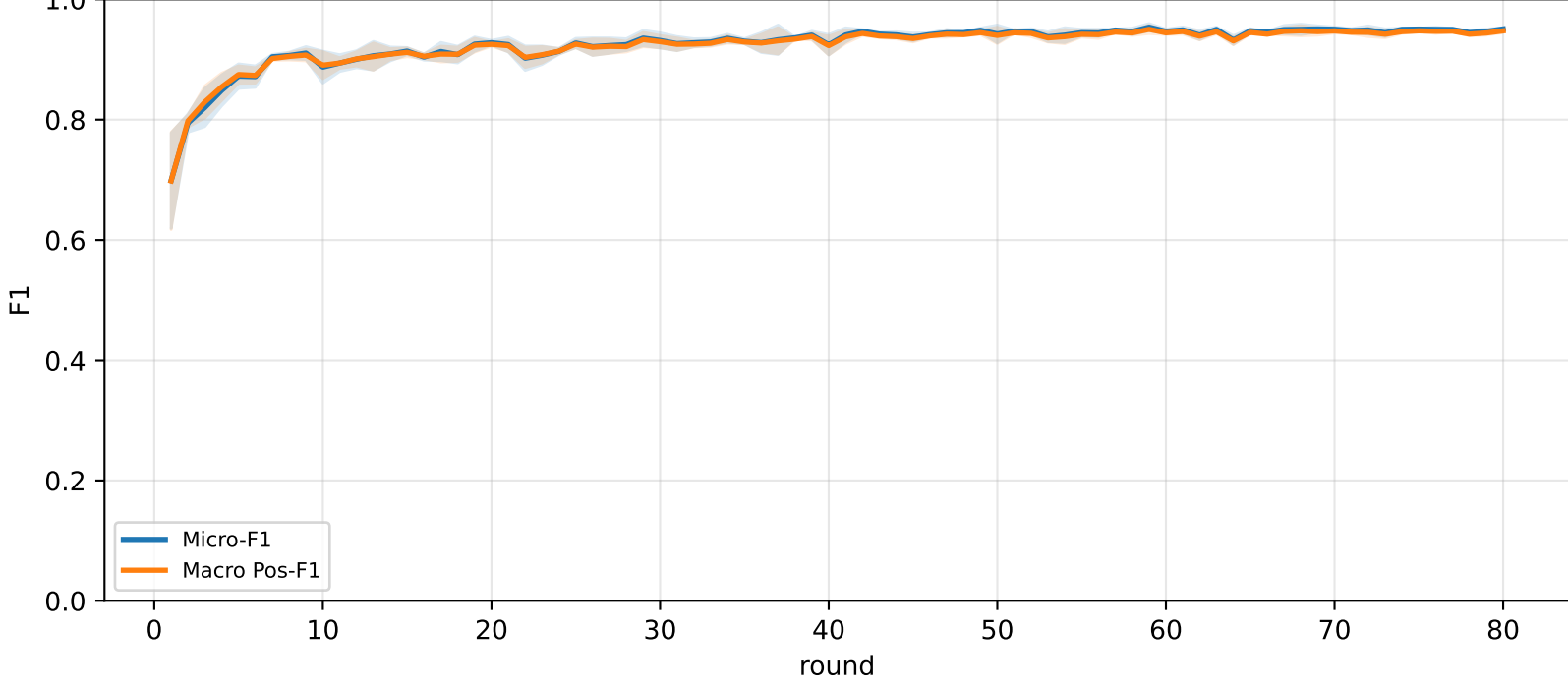
Validation loss comparison



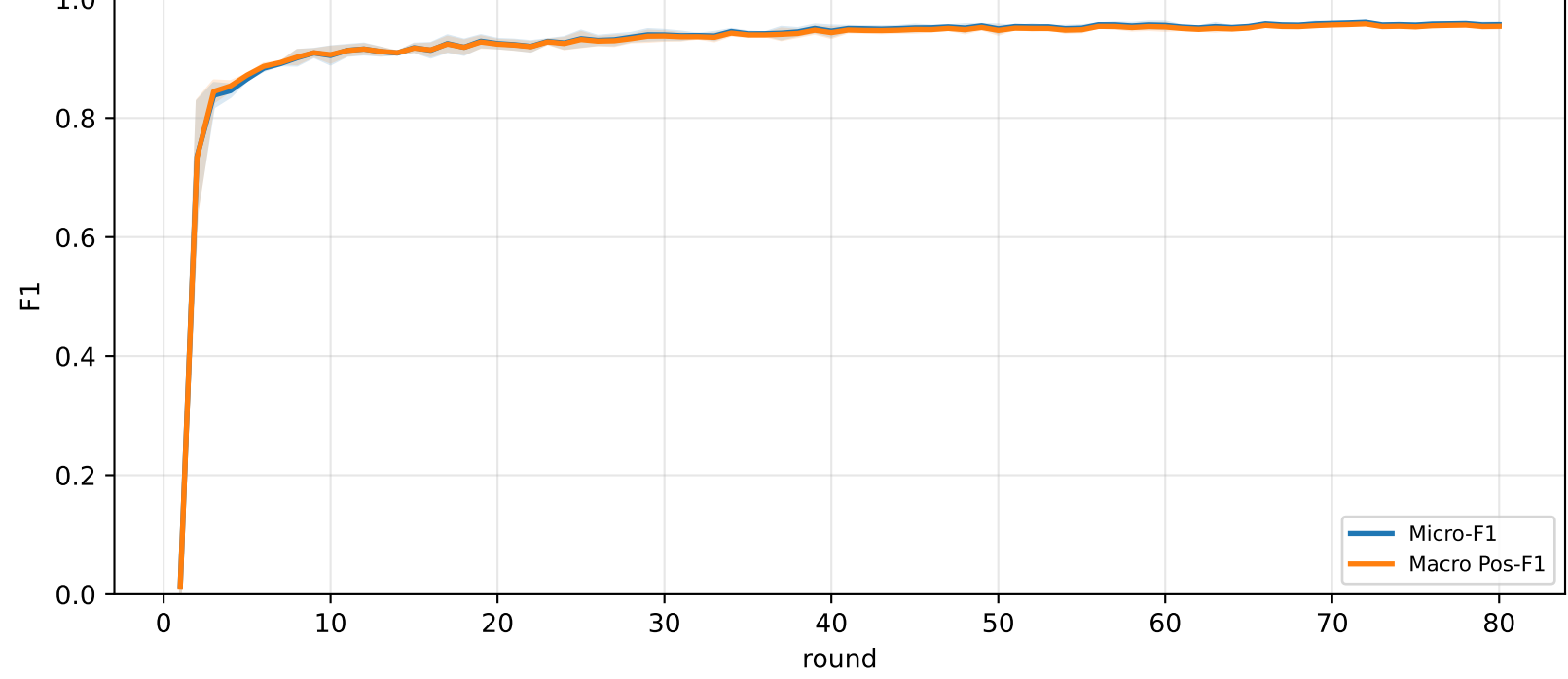
Pooled local centralized micro / macro F1



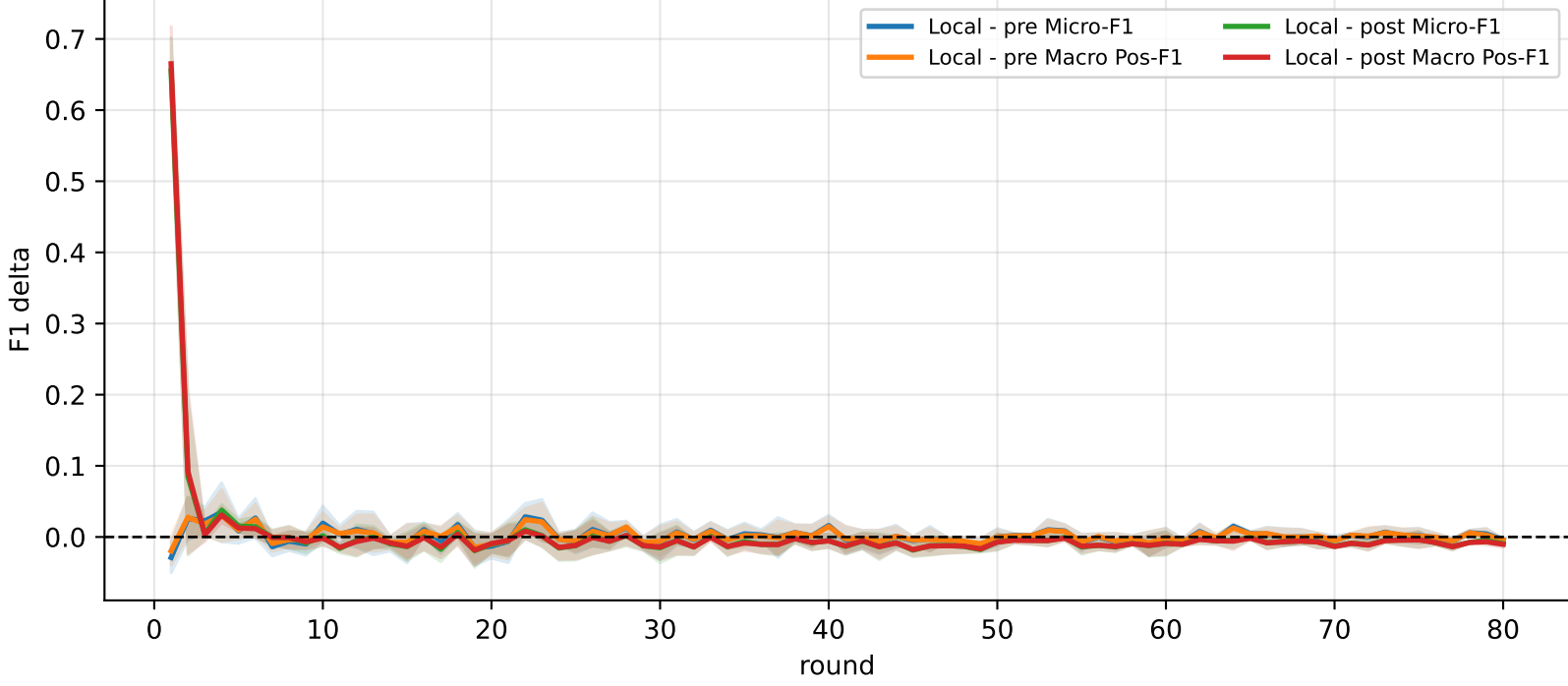
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



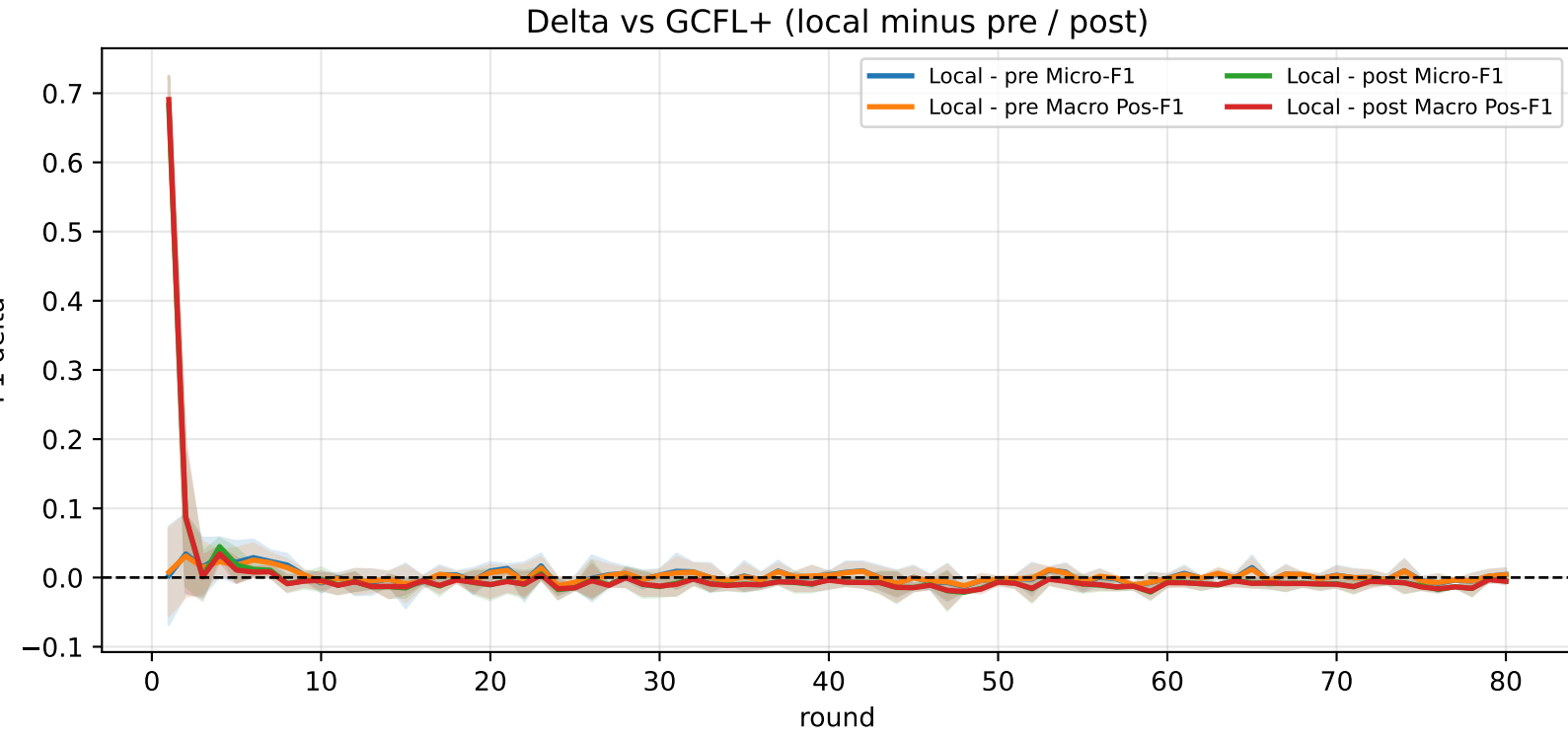
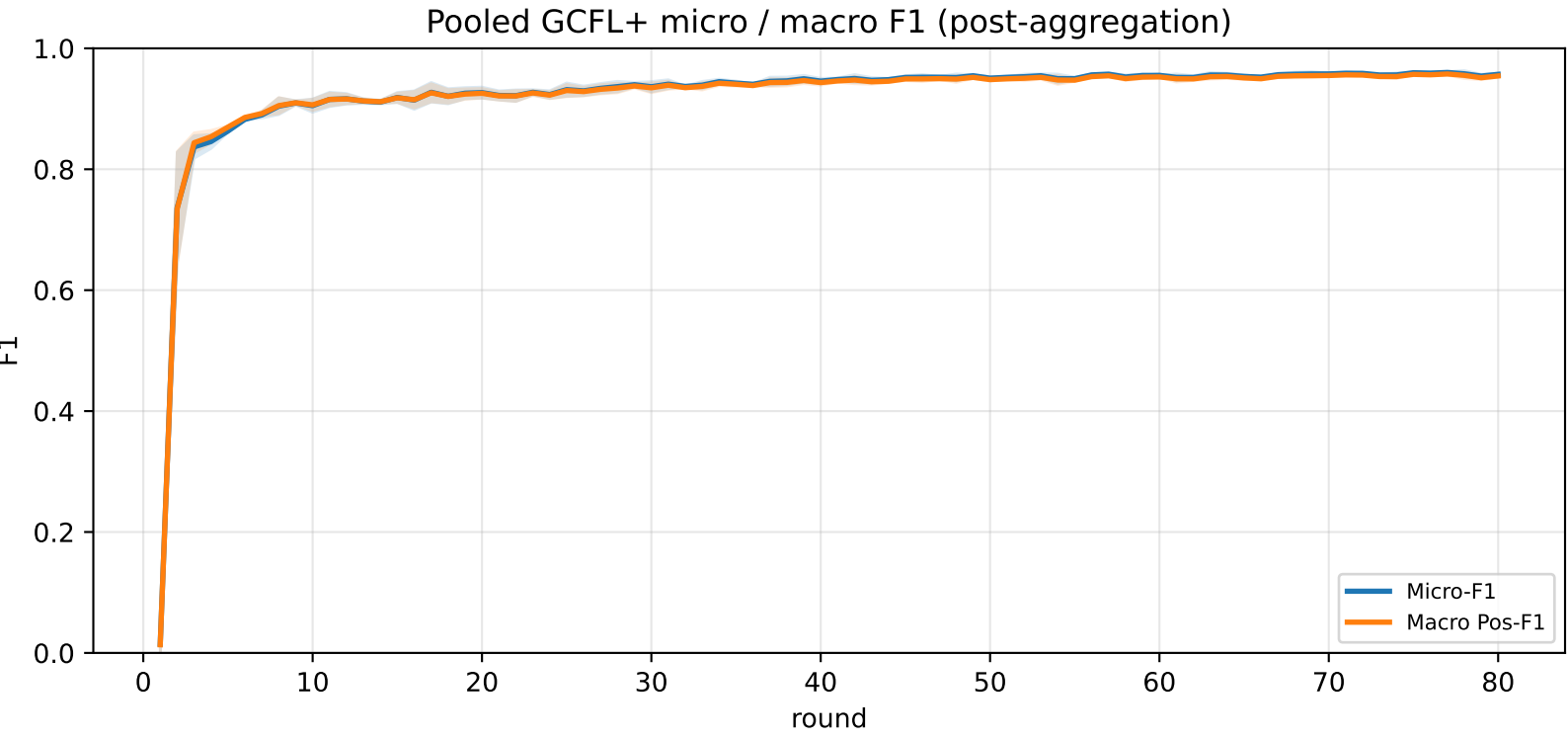
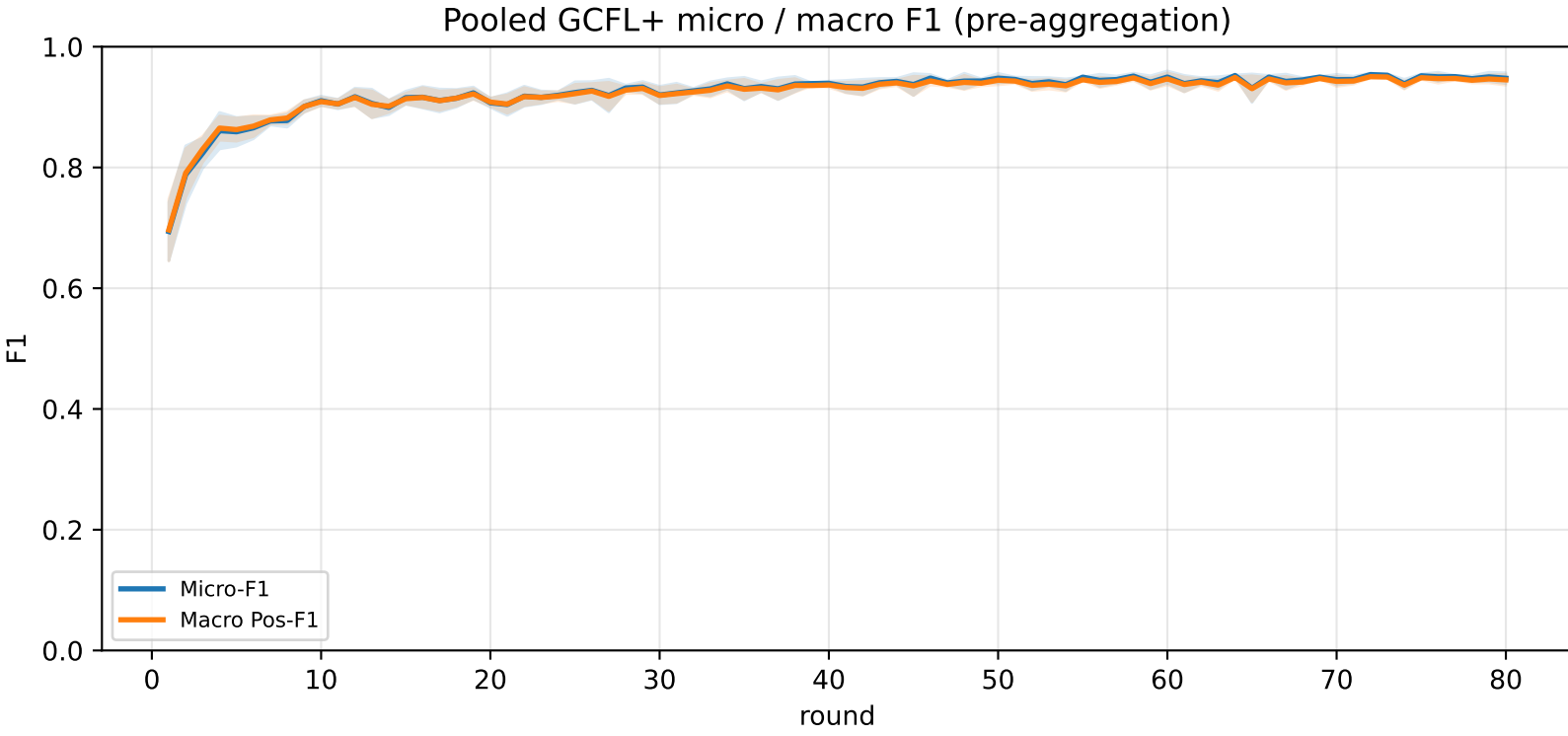
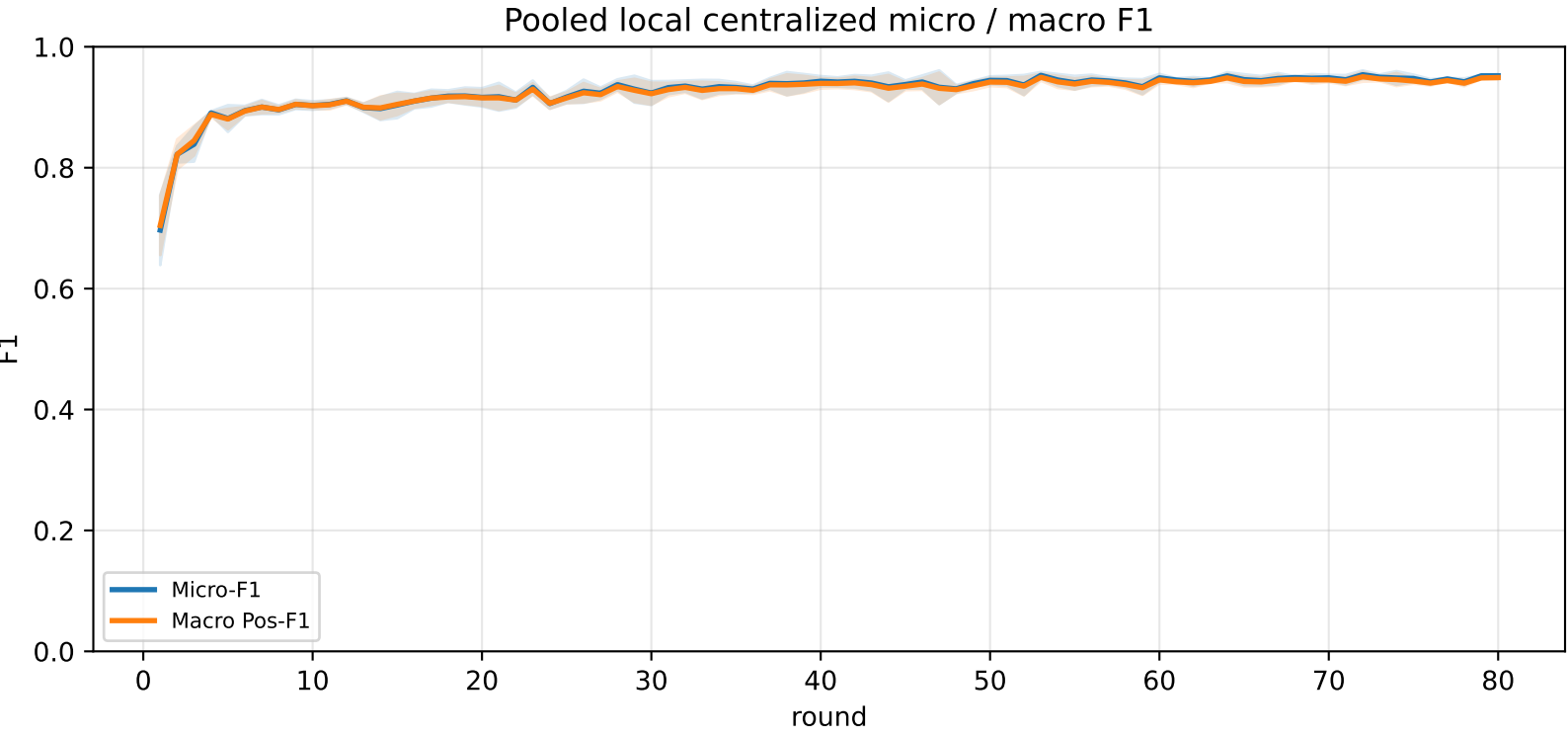
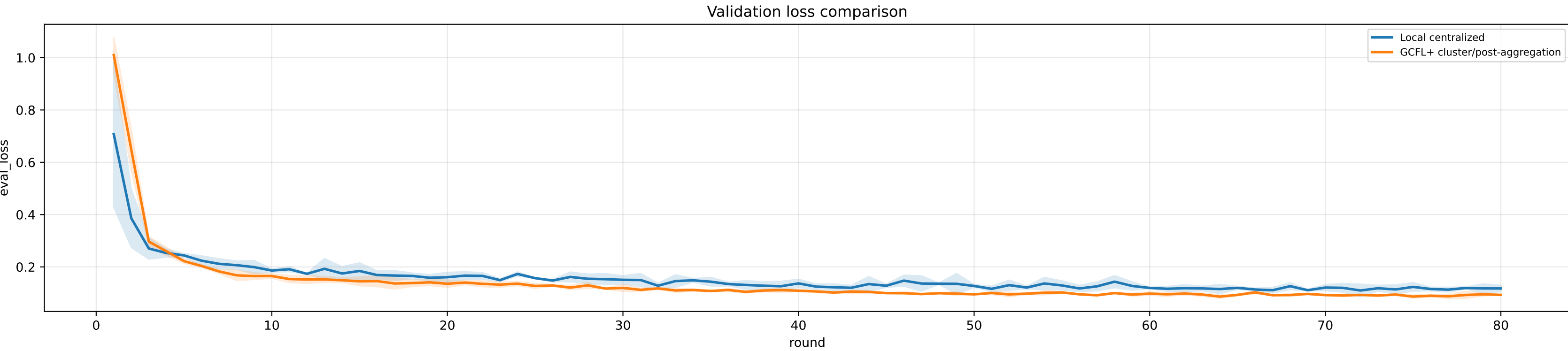
Delta vs GCFL+ (local minus pre / post)



Controlled masking level: Low | GCFL+ diagnostics | family=masked_cycle4 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle4 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 21.0%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1% | masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.5% ± 0.3%	95.4% ± 0.3%	99.4% ± 0.1%	98.6% ± 0.3%	97.3% ± 0.6%	94.7% ± 0.2%	86.9% ± 1.2%
GCFL+	95.7% ± 0.1%	95.6% ± 0.0%	99.3% ± 0.2%	98.8% ± 0.1%	98.0% ± 0.1%	94.3% ± 0.8%	87.7% ± 0.7%

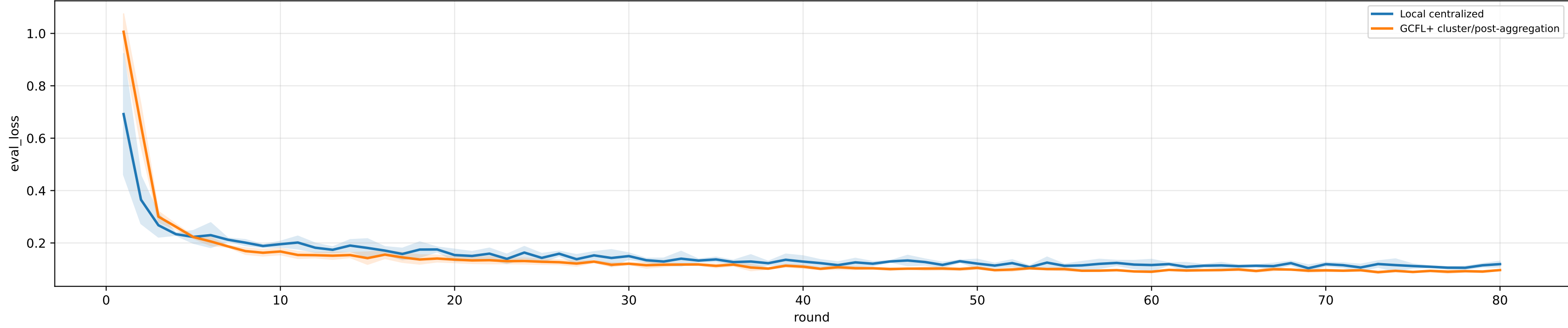


family=masked_cycle5 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 21.0%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1% | masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

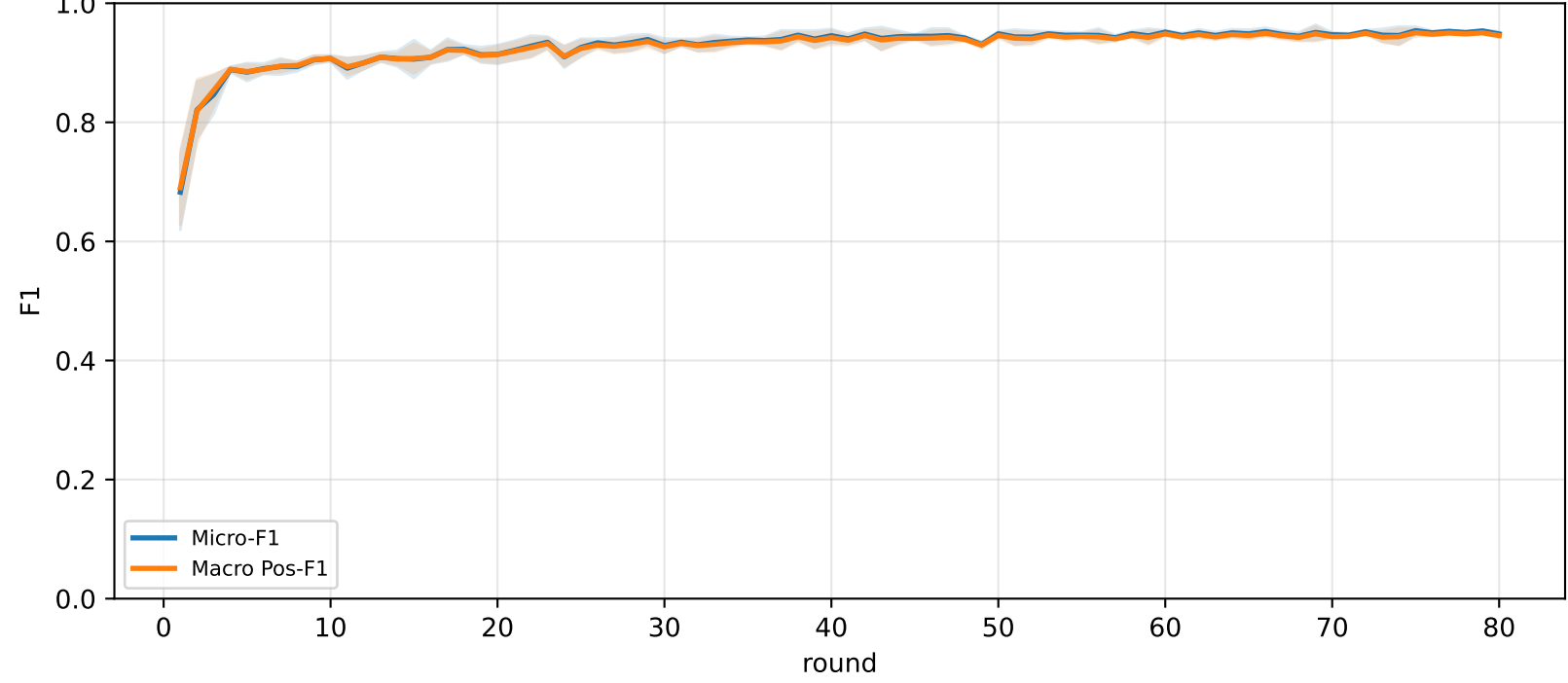
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.4% ± 0.2%	95.3% ± 0.2%	99.2% ± 0.2%	98.6% ± 0.1%	97.8% ± 0.1%	94.5% ± 0.2%	86.3% ± 1.0%
GCFL+	95.8% ± 0.2%	95.7% ± 0.2%	99.2% ± 0.1%	99.0% ± 0.2%	97.7% ± 0.5%	94.3% ± 1.0%	88.4% ± 0.3%

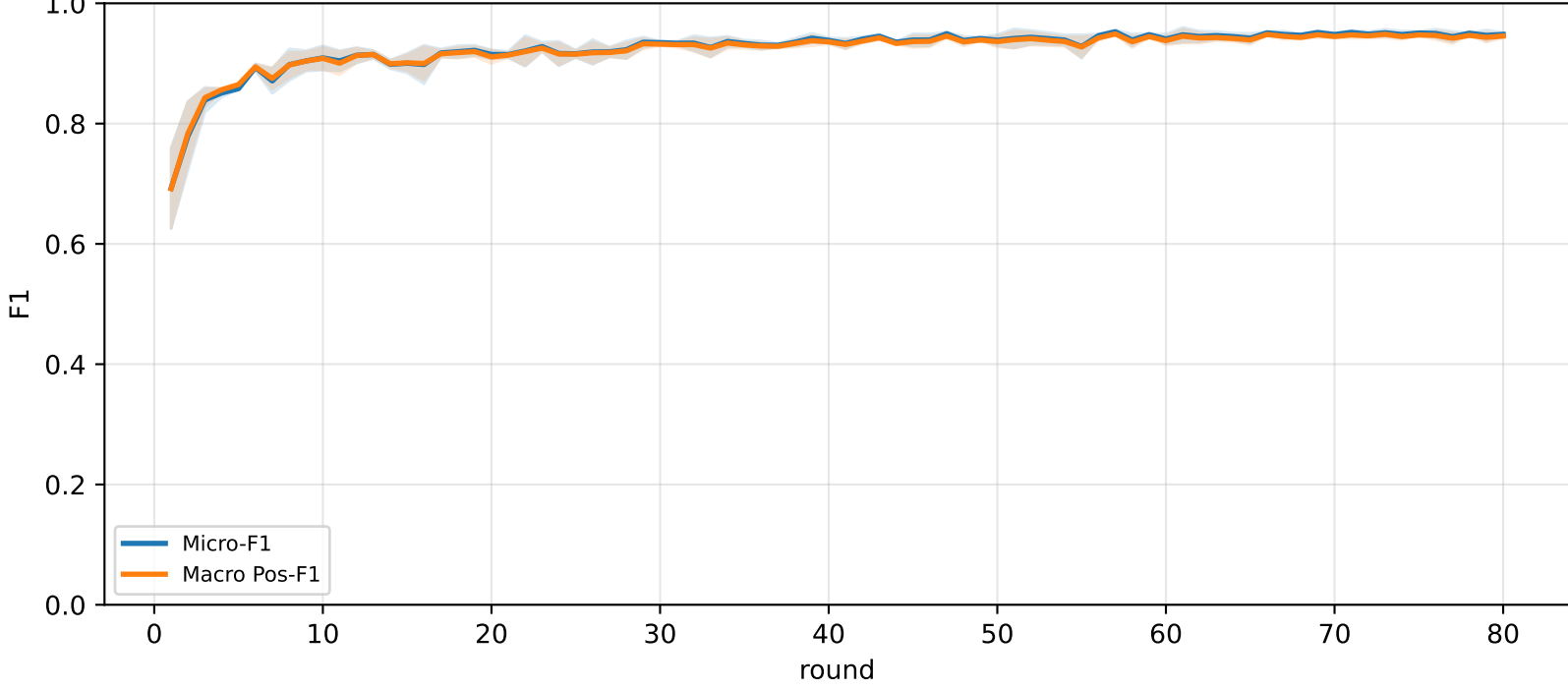
Validation loss comparison



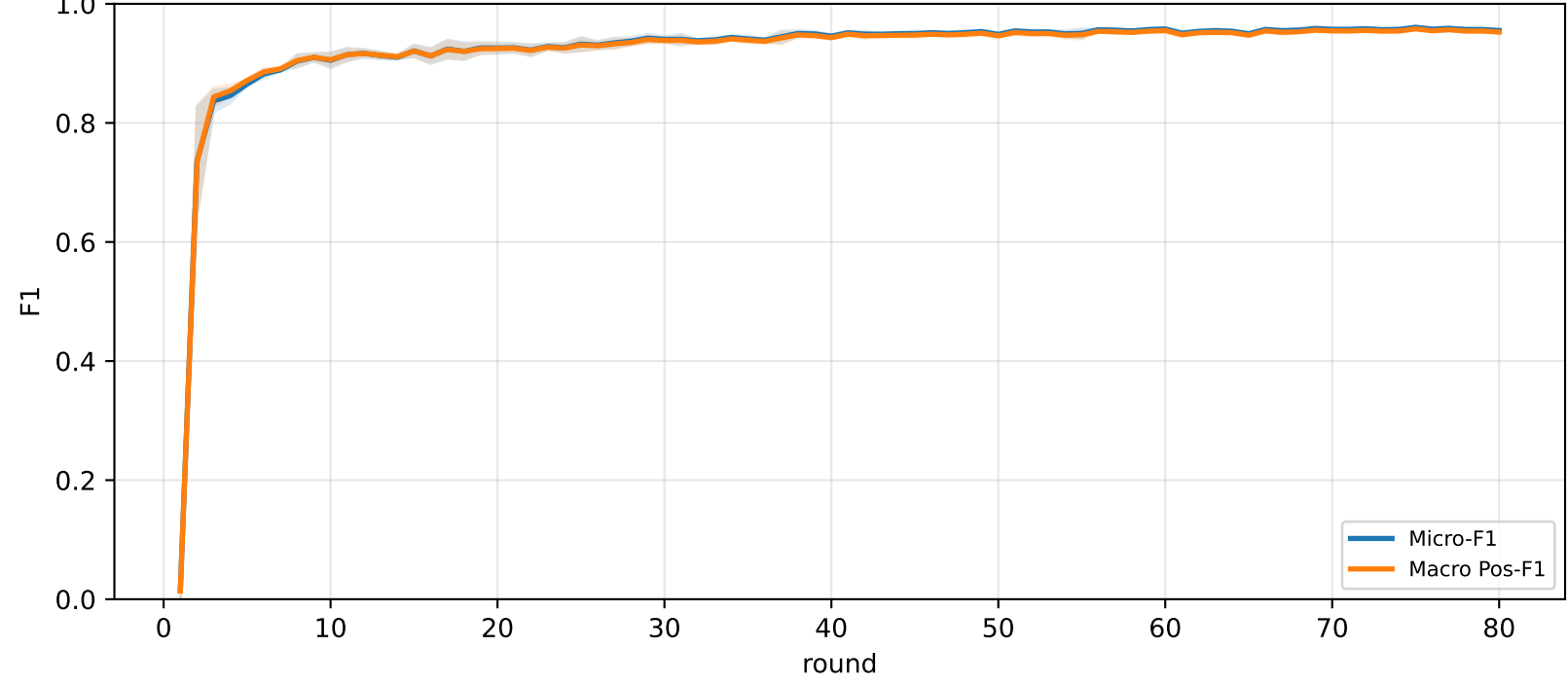
Pooled local centralized micro / macro F1



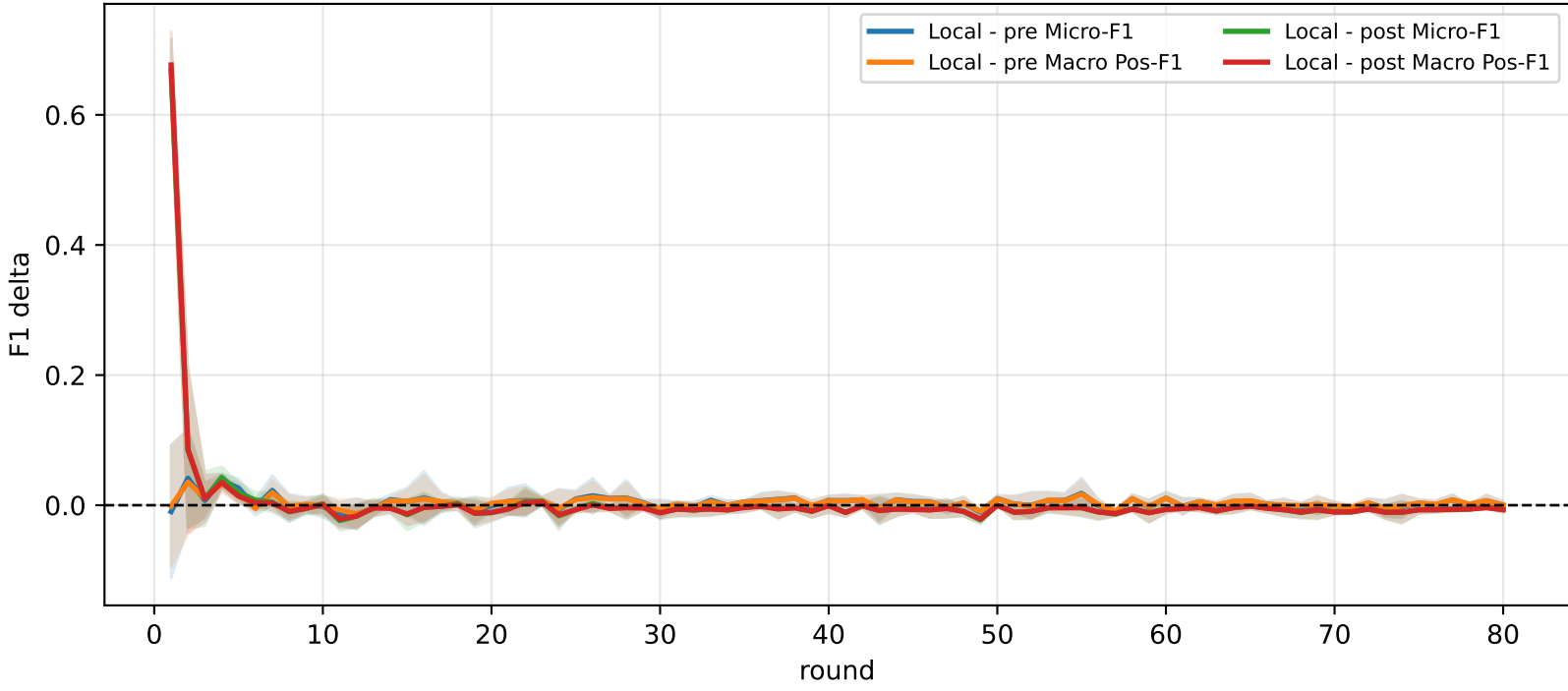
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



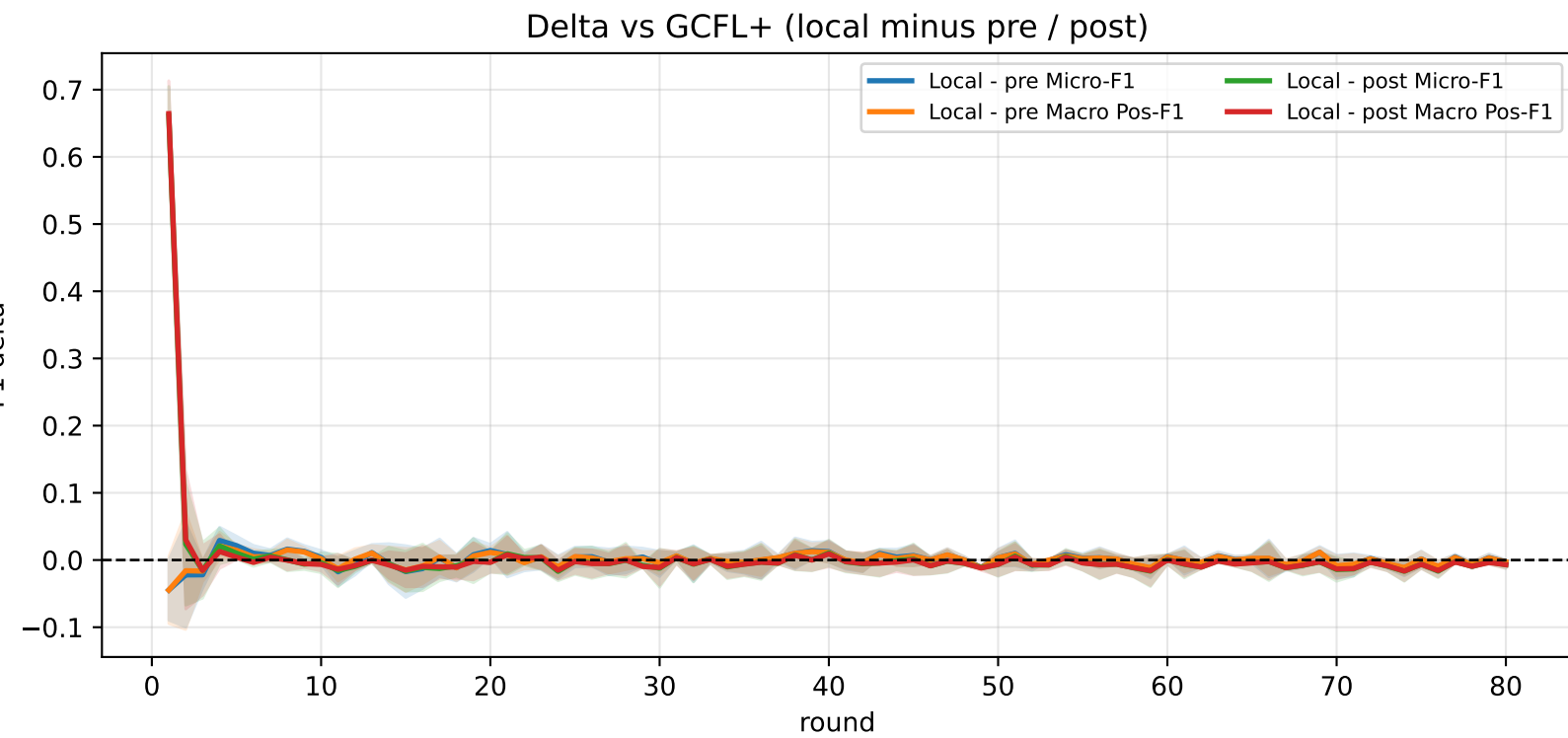
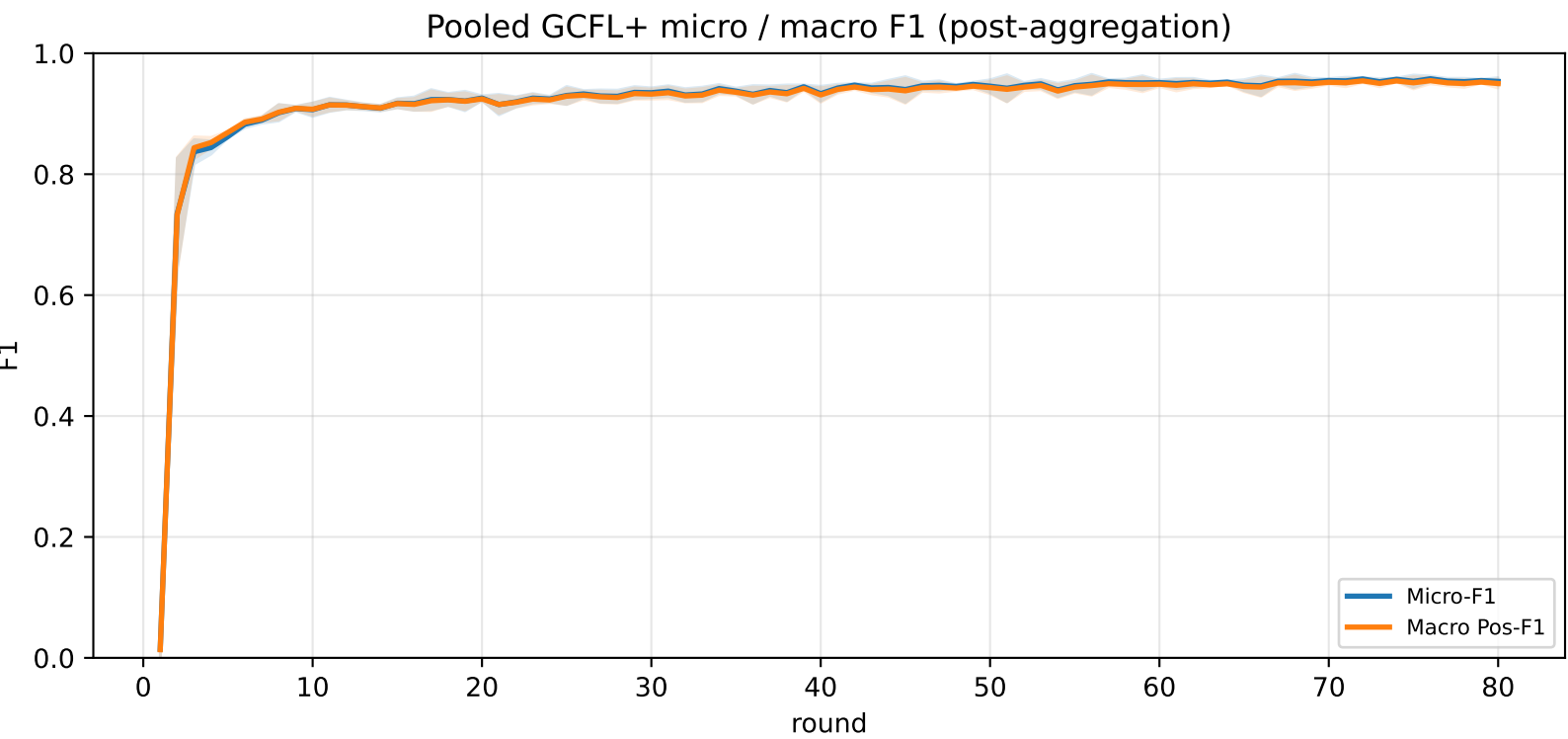
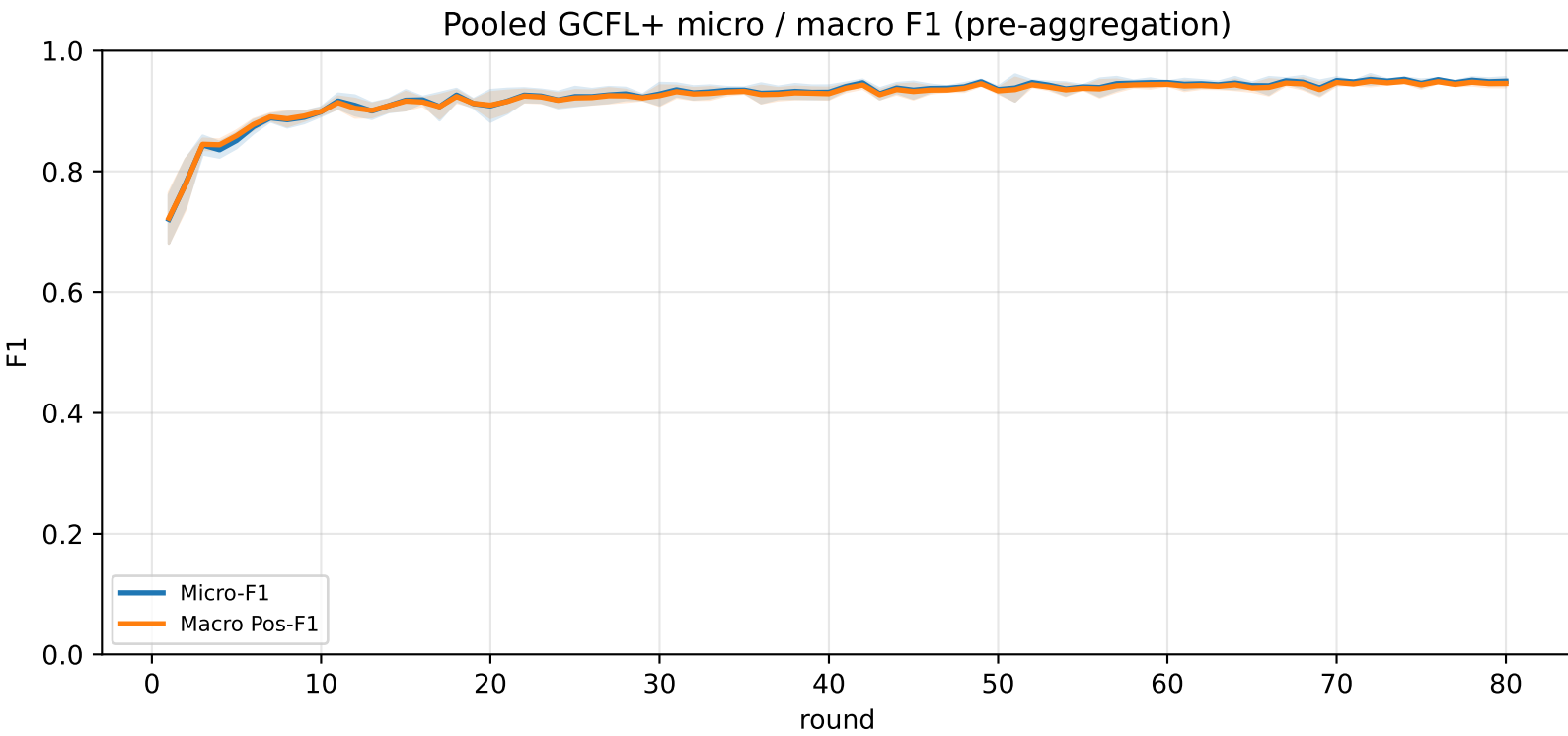
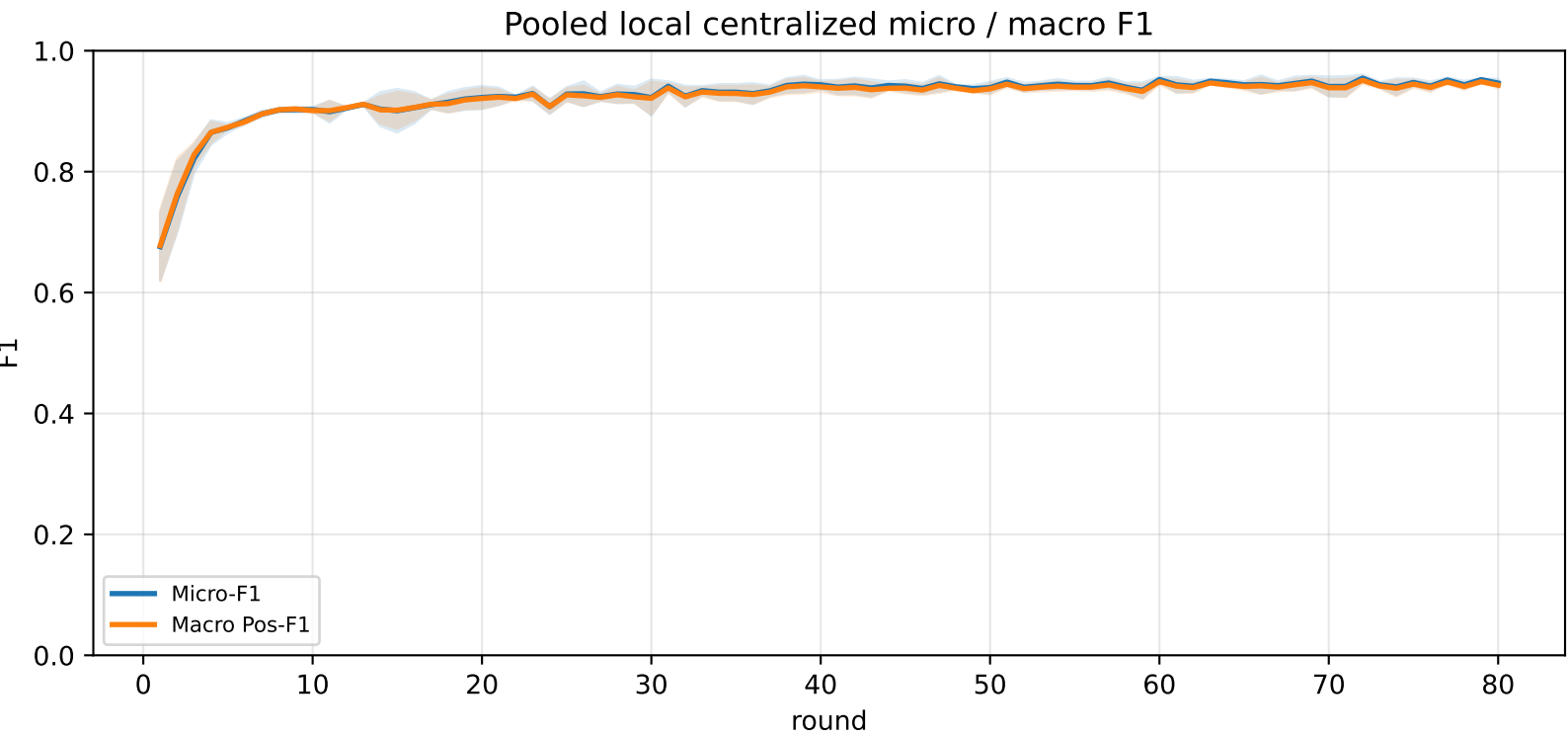
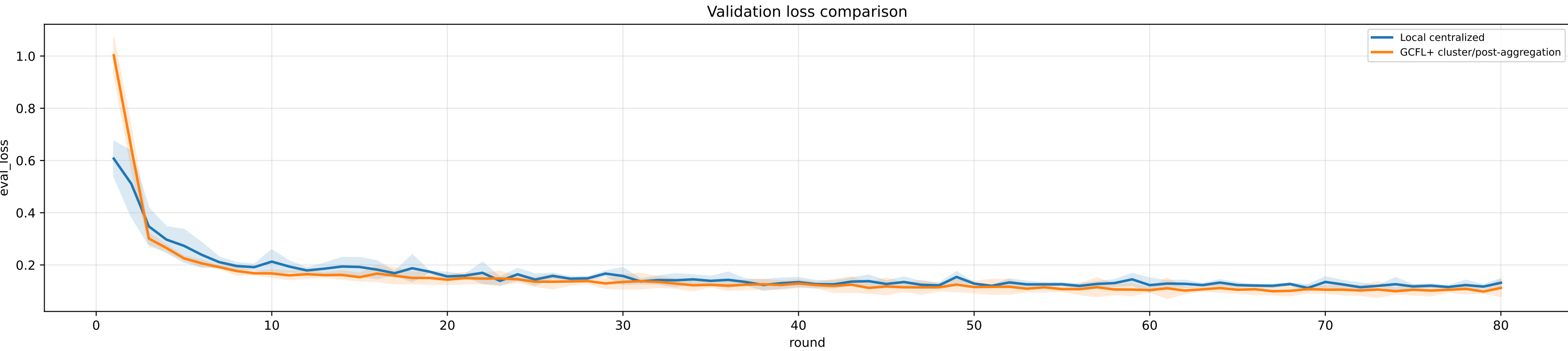
Delta vs GCFL+ (local minus pre / post)



Controlled masking level: Low | GCFL+ diagnostics | family=masked_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle6 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 20.9%±0.0% / true 26.1% | masking=low | mask_fraction=0.20 | task_jsd_mean=0.002 | task_jsd_median=0.002 | task_jsd_max=0.002 | family_centroid_jsd_mean=0.002

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.4% ± 0.3%	95.2% ± 0.3%	99.1% ± 0.2%	98.7% ± 0.4%	97.8% ± 0.4%	94.9% ± 0.4%	85.6% ± 0.5%
GCFL+	95.5% ± 0.4%	95.4% ± 0.4%	99.3% ± 0.1%	98.9% ± 0.3%	98.2% ± 0.5%	93.8% ± 0.6%	86.5% ± 2.0%



Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

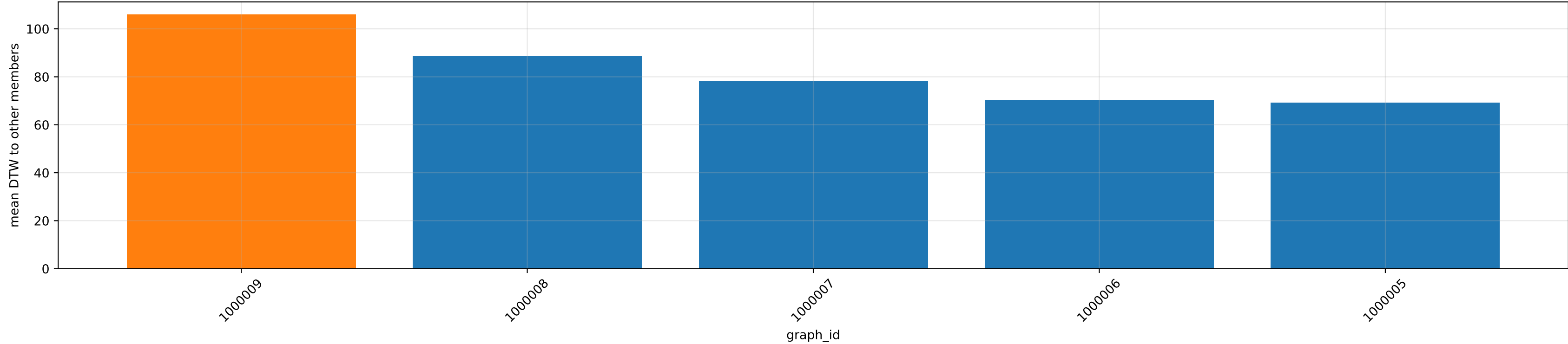
First applied split summary | seed=0

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
0	23	0	1 vs 4	1000009	1000009	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
105.927	nan	66.759

Per-member mean DTW



Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019

First applied split pairwise DTW metadata | seed=0

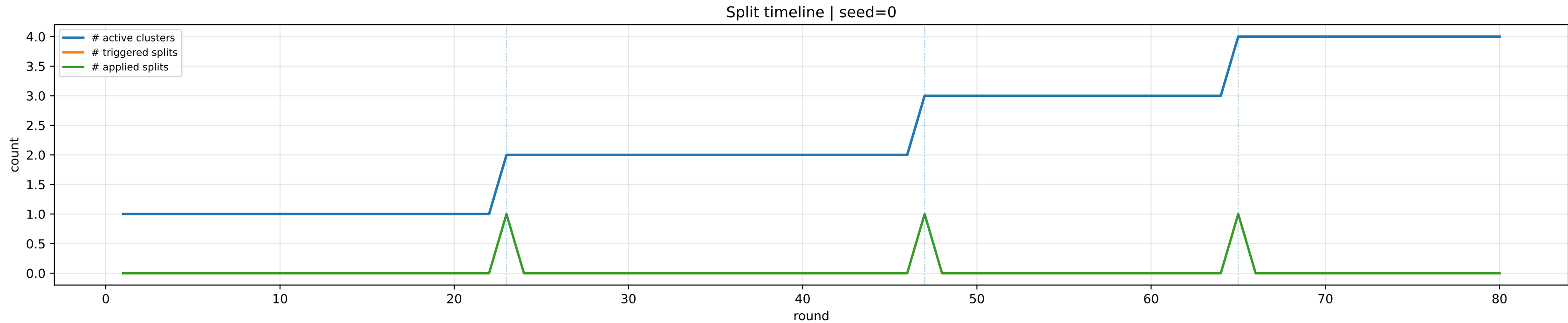
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
0	23	0	5	1000009.0	0.038	105.927	1

Pairwise DTW table | first applied split | seed=0
Global max DTW = 116.8 at (1000006, 1000009)

graph_id	1000005	1000006	1000007	1000008	1000009
1000005	0.0	42.4	56.5	93.8	84.2
1000006	42.4	0.0	62.7	60.0	116.8
1000007	56.5	62.7	0.0	85.3	107.9
1000008	93.8	60.0	85.3	0.0	114.9
1000009	84.2	116.8	107.9	114.9	0.0

Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5
clients | seed=0 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

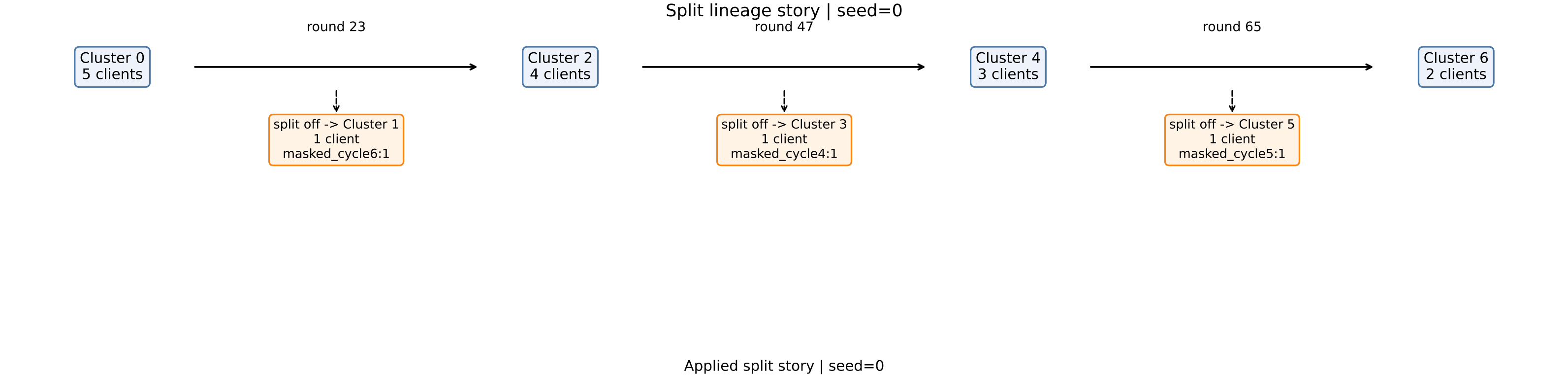


Seed summary | seed=0

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0	23, 47, 65	23, 47, 65	1, 3, 5, 6	1:1, 3:1, 5:1, 6:2

Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019



Applied split story seed=0							
round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
23	0	5	158.212 < 158.664 , 227.589 > 205.297	C1 (1 client)	masked_cycle6:1	C2 (4 clients)	masked_cycle2:1 , masked_cycle3:1 , masked_cycle4:1 , masked_cycle5:1
47	2	4	154.668 < 160.758 , 209.254 > 208.049	C3 (1 client)	masked_cycle4:1	C4 (3 clients)	masked_cycle2:1 , masked_cycle3:1 , masked_cycle5:1
65	4	3	157.533 < 157.943 , 192.360 > 183.192	C5 (1 client)	masked_cycle5:1	C6 (2 clients)	masked_cycle2:1 , masked_cycle3:1

Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

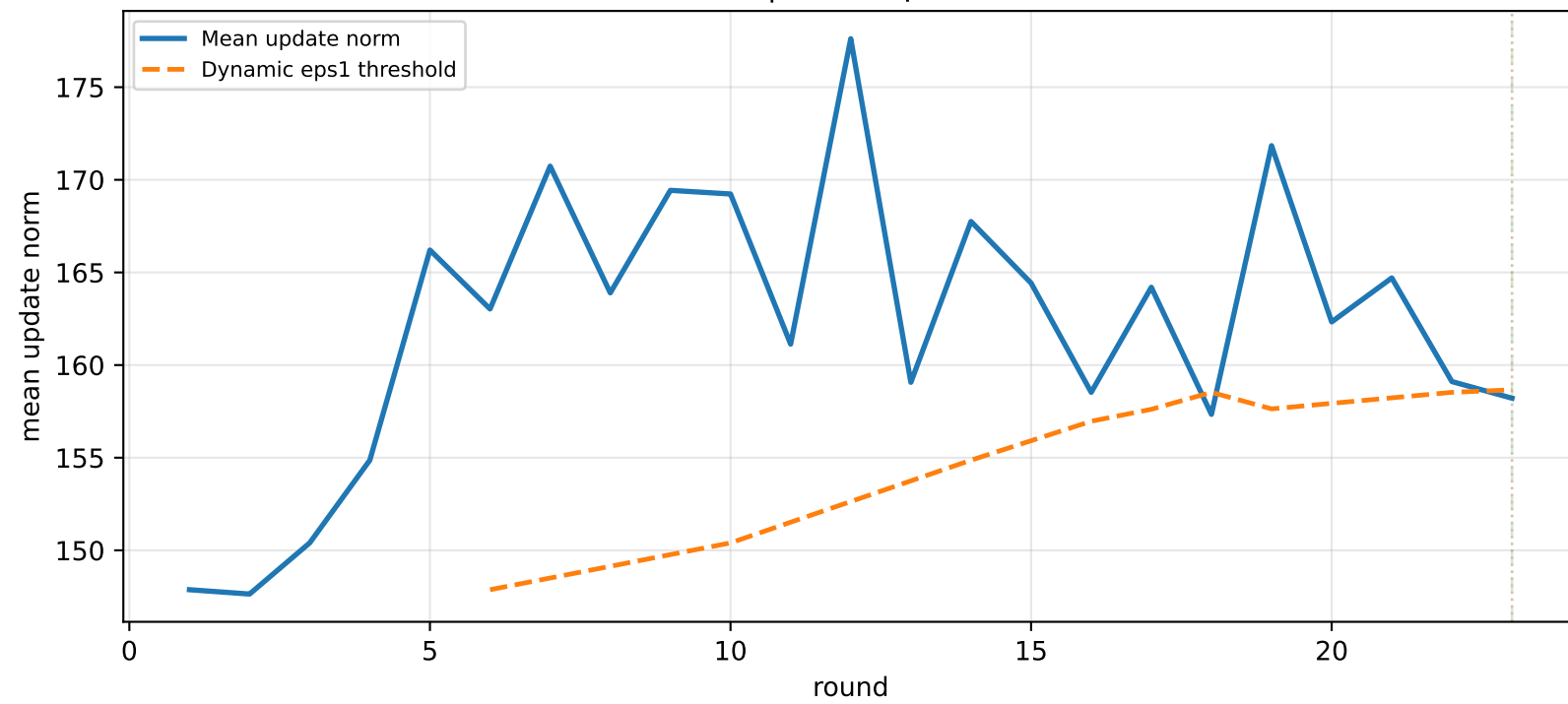
Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000009	4	masked_cycle6:1
3	1	1000007	2	masked_cycle4:1
5	1	1000008	3	masked_cycle5:1
6	2	1000005, 1000006	0, 1	masked_cycle2:1 , masked_cycle3:1

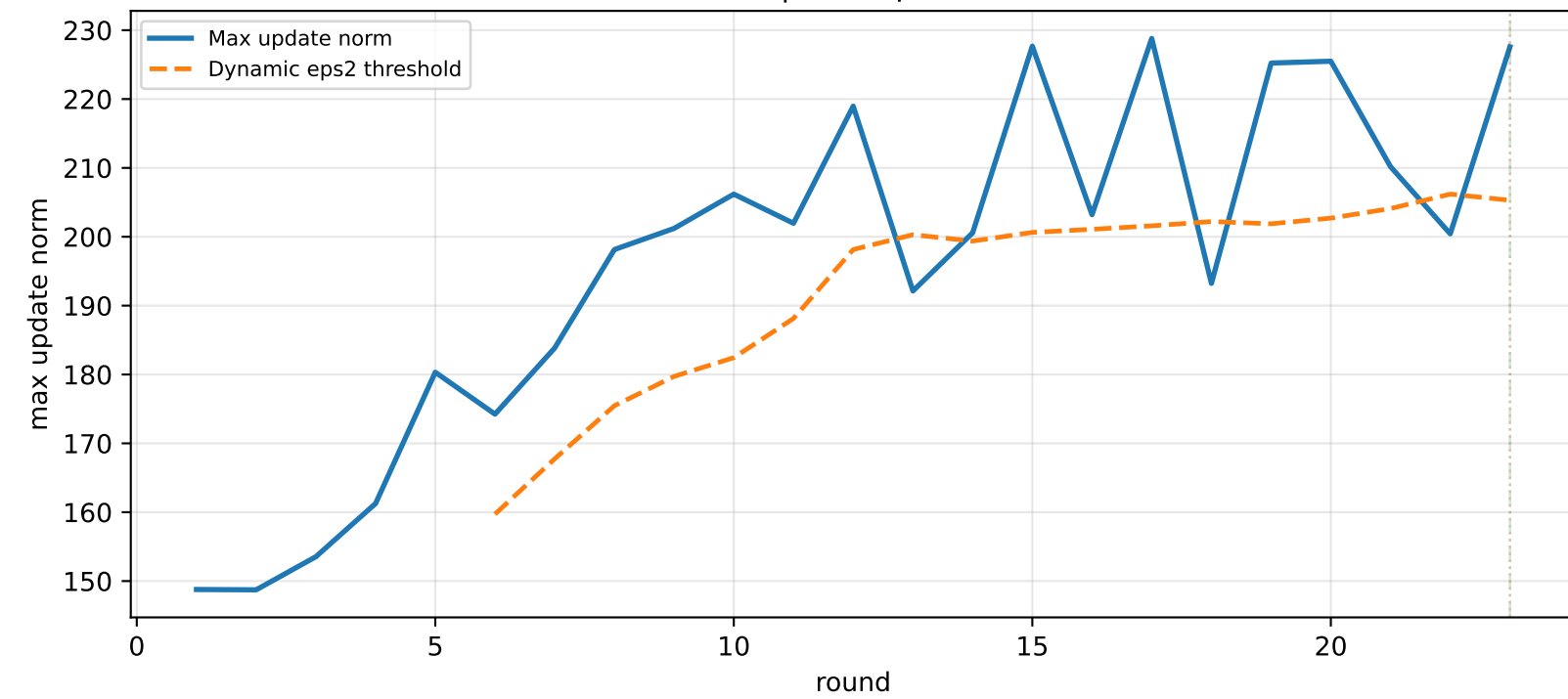
Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

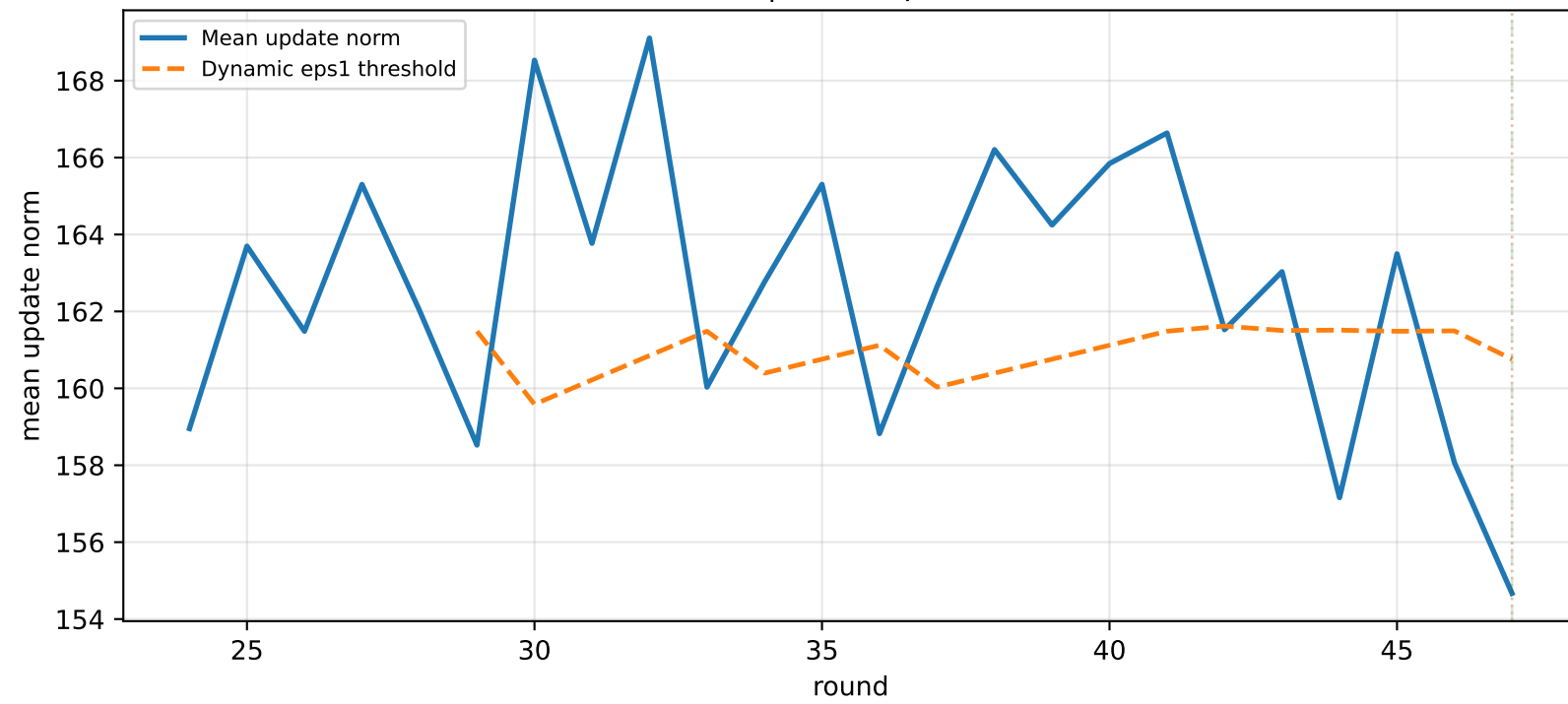
Cluster 0 | Mean update norm



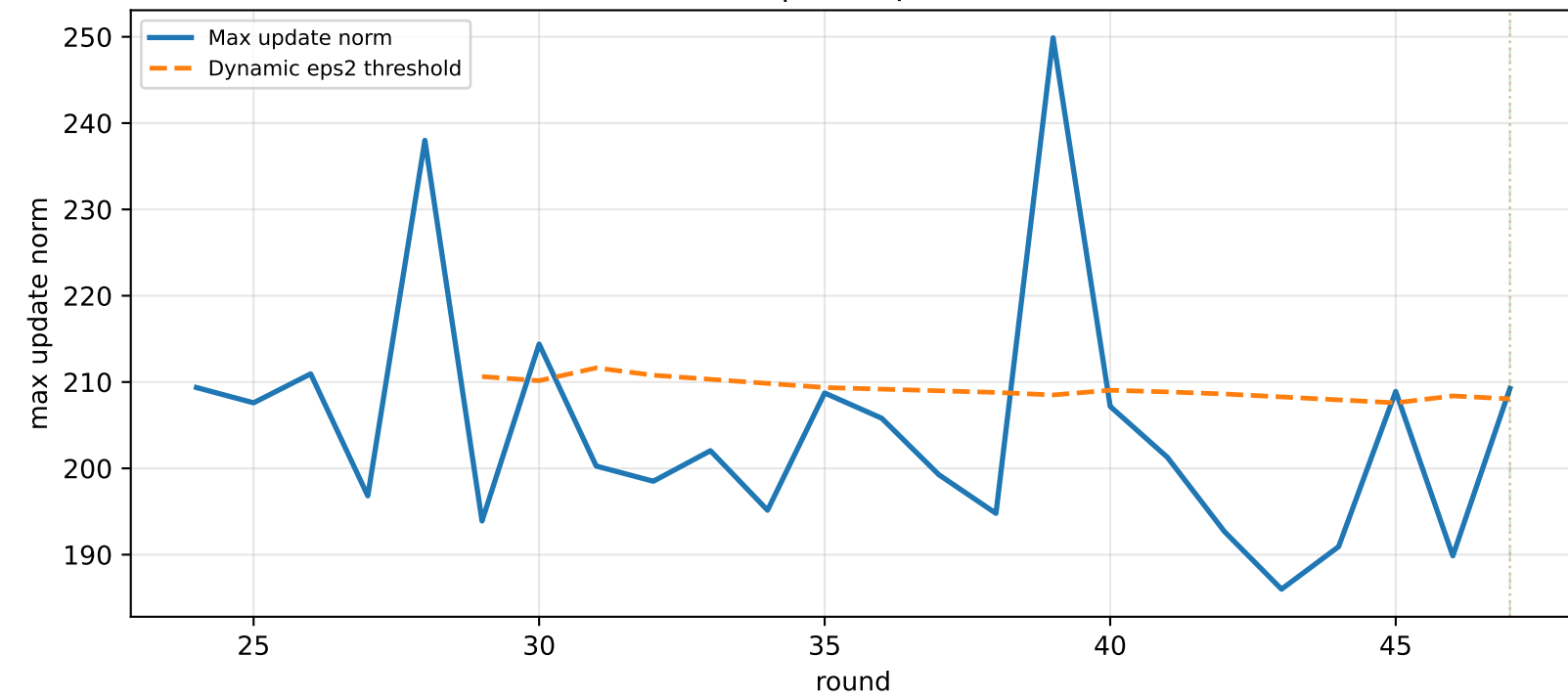
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



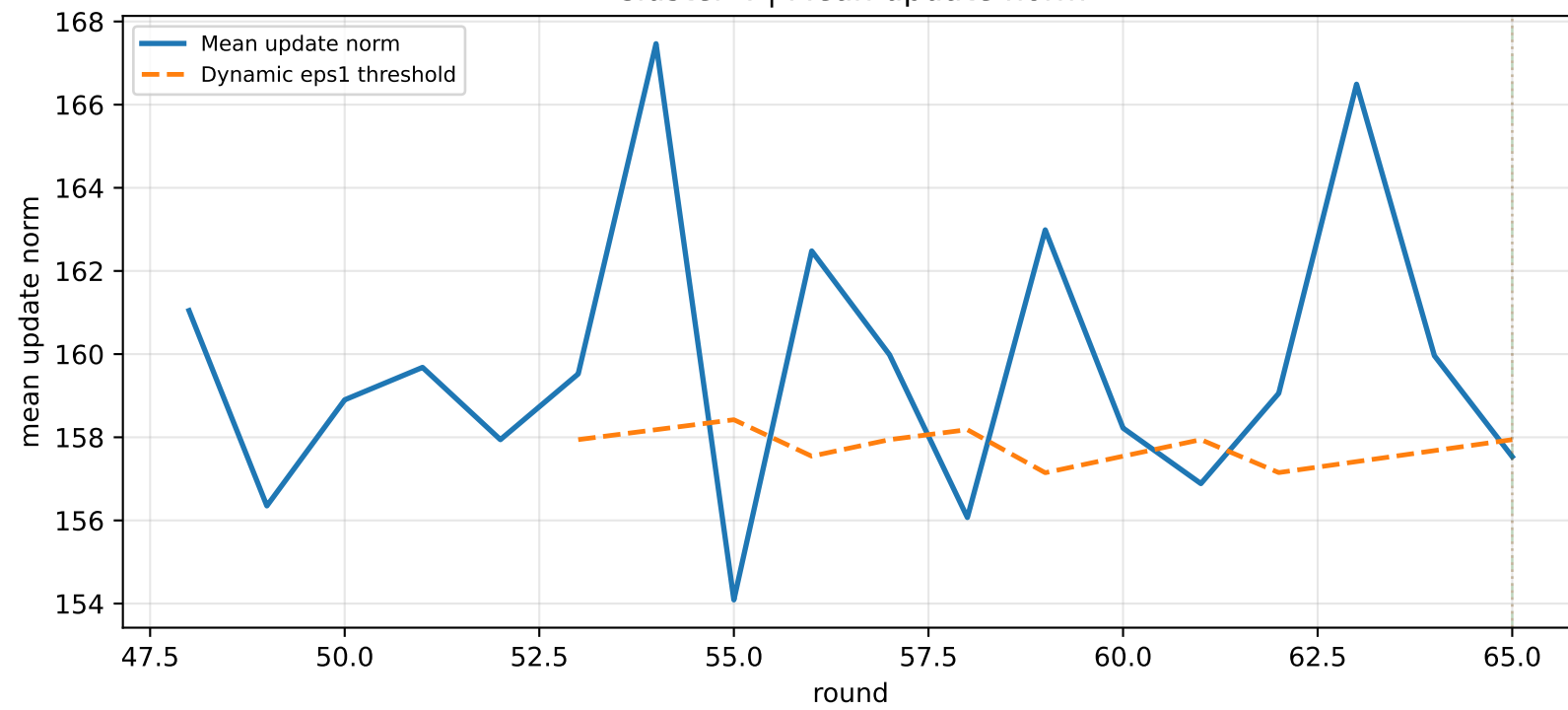
Cluster 2 | Max update norm



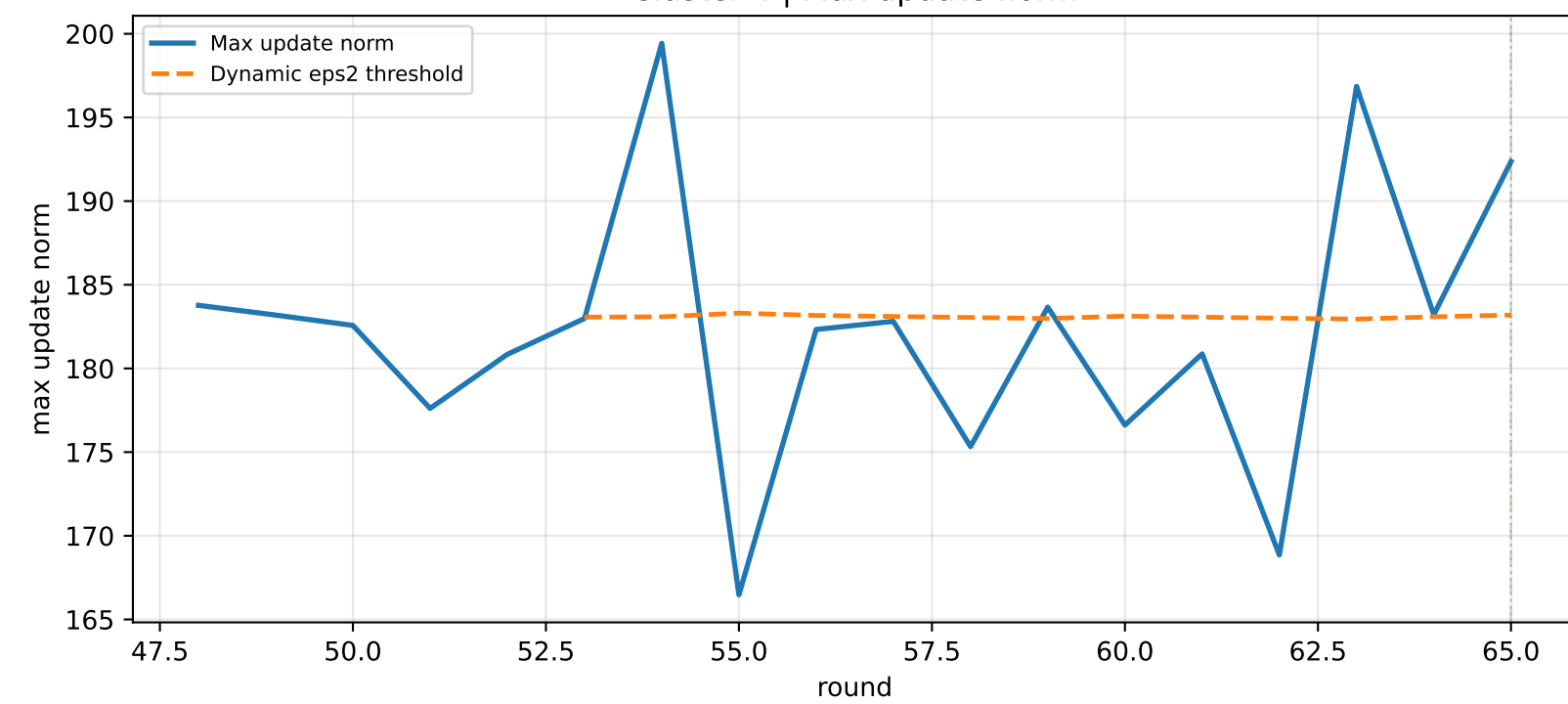
Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

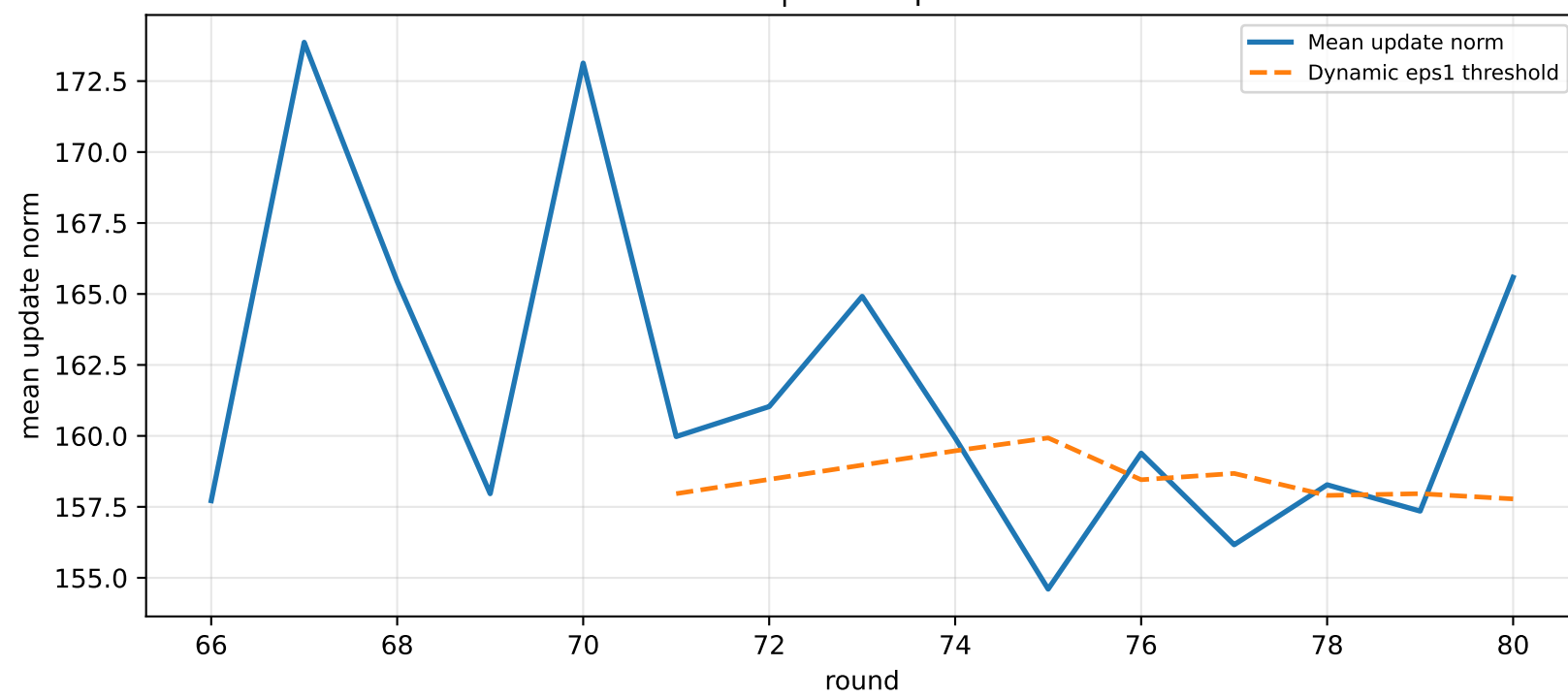
Cluster 4 | Mean update norm



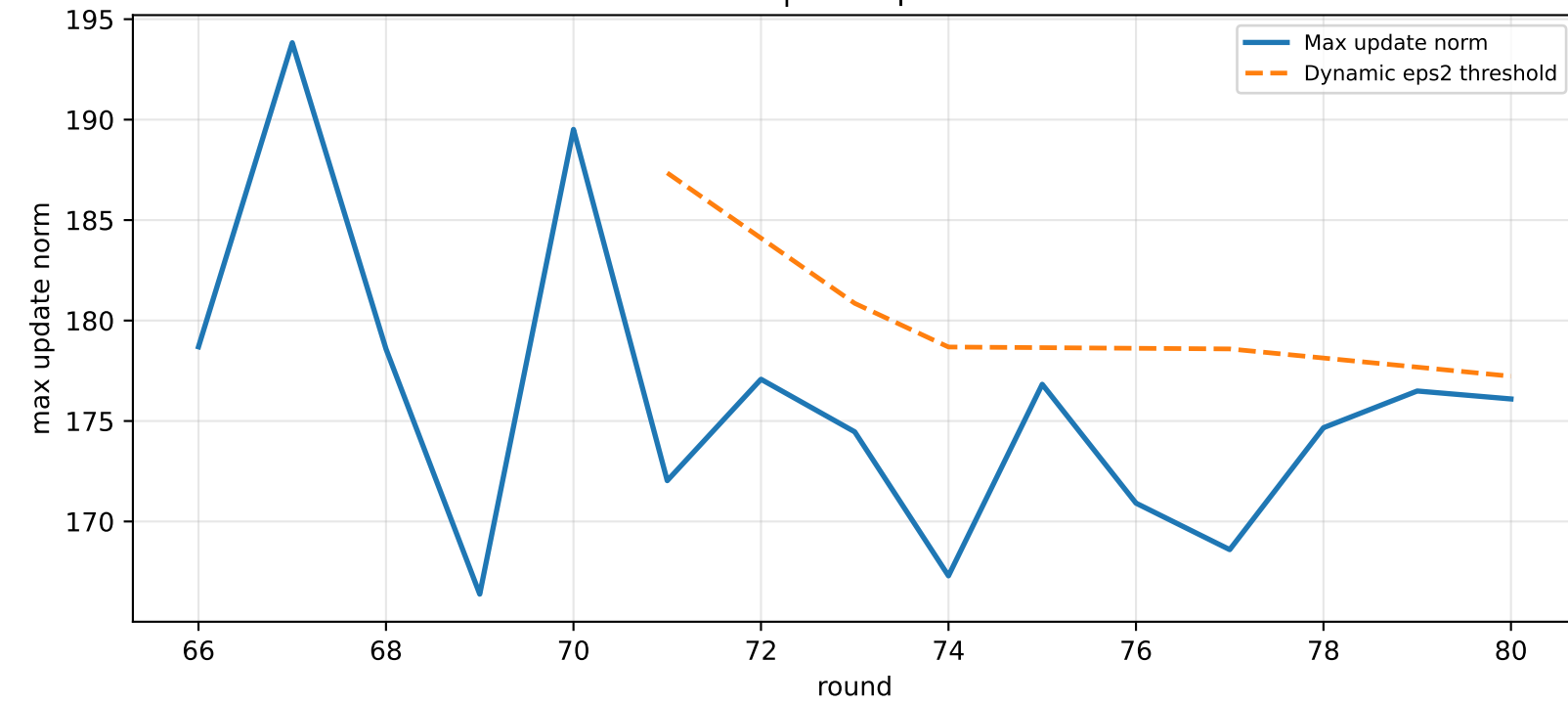
Cluster 4 | Max update norm



Cluster 6 | Mean update norm

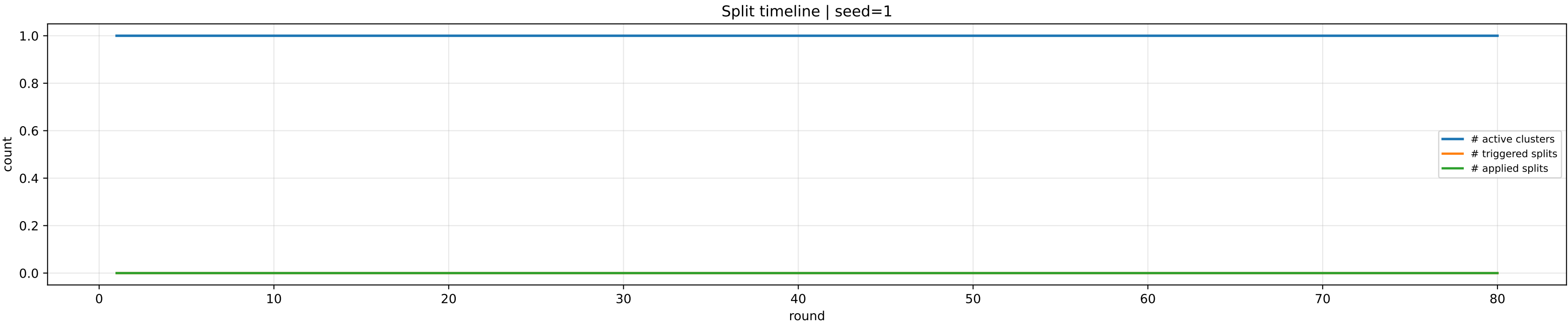


Cluster 6 | Max update norm



Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019



Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0			0	0:5

Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | final active clusters

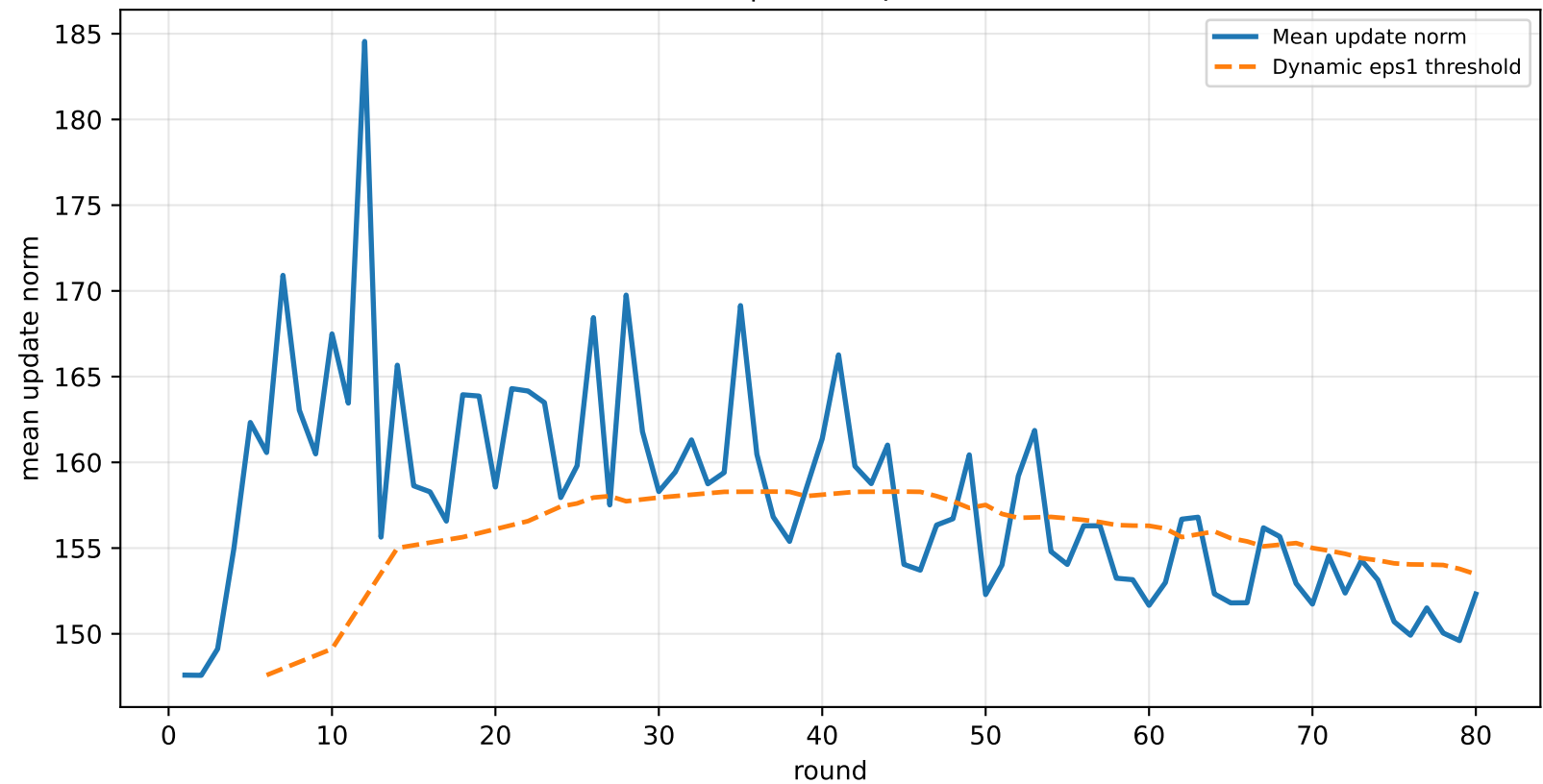
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

Final active clusters | seed=1

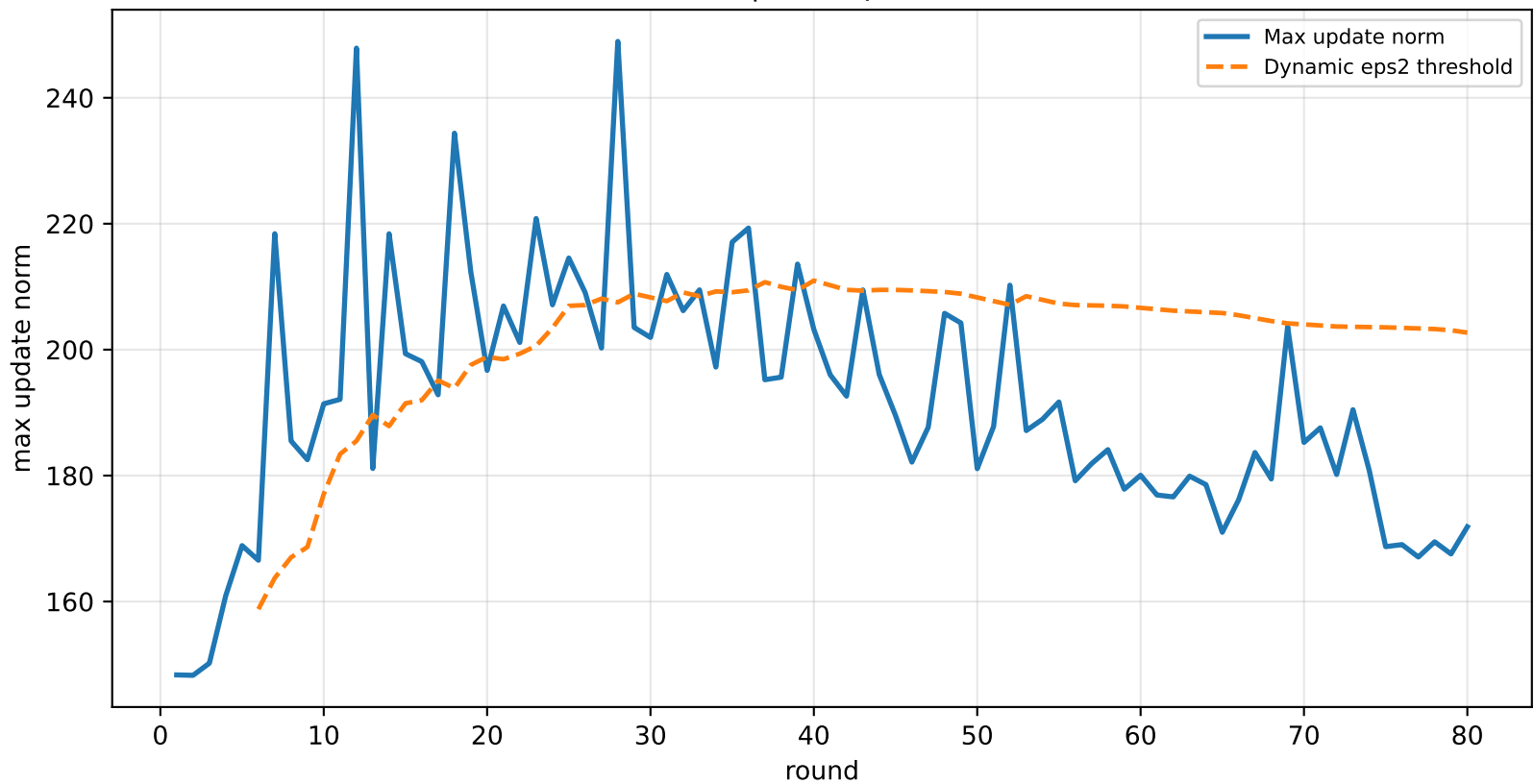
cluster_id	cluster_size	members	member_indices	family_mix
0	5	1000005, 1000006, 1000007, 1000008, 1000009	0, 1, 2, 3, 4	masked_cycle2:1 , masked_cycle3:1 , masked_cycle4:1 , masked_cycle5:1 , masked_cycle6:1

Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5
clients | seed=1 | norm / threshold panels (0-0)
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

Cluster 0 | Mean update norm



Cluster 0 | Max update norm



Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

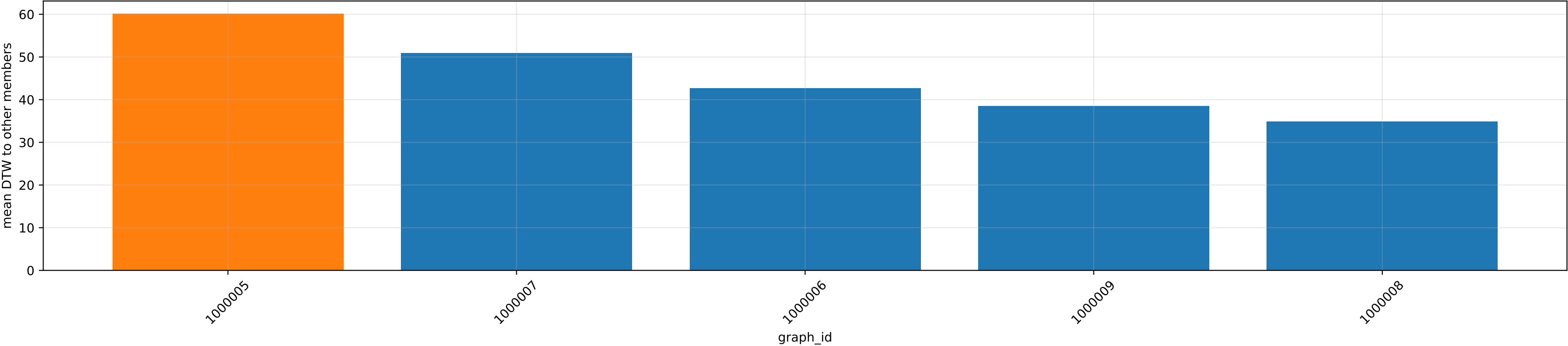
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	20	0	4 vs 1	1000005	1000005	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
60.105	35.605	nan

Per-member mean DTW



Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019

First applied split pairwise DTW metadata | seed=2

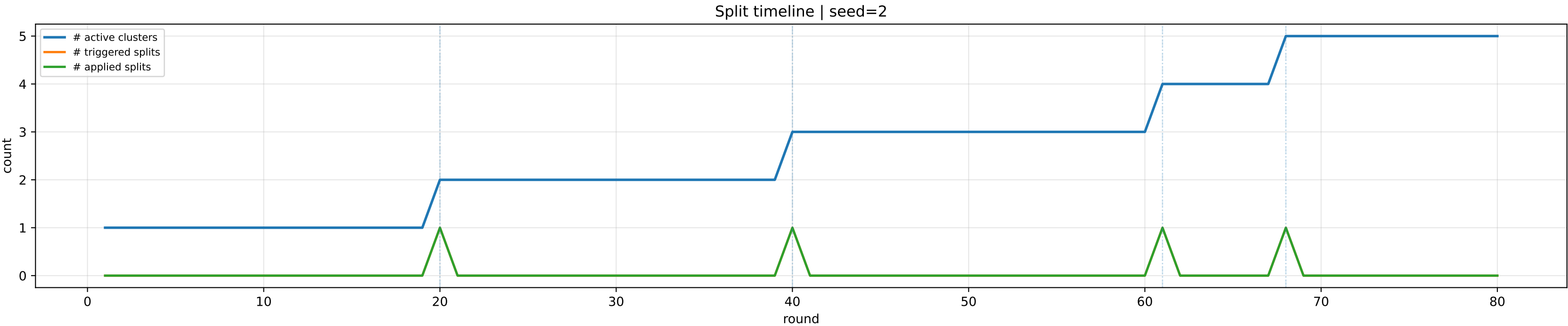
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	20	0	5	1000005.0	0.070	60.105	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 82.3 at (1000005, 1000007)

graph_id	1000005	1000006	1000007	1000008	1000009
1000005	0.0	64.9	82.3	53.8	39.5
1000006	64.9	0.0	30.2	31.3	44.1
1000007	82.3	30.2	0.0	37.5	53.6
1000008	53.8	31.3	37.5	0.0	16.9
1000009	39.5	44.1	53.6	16.9	0.0

Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019

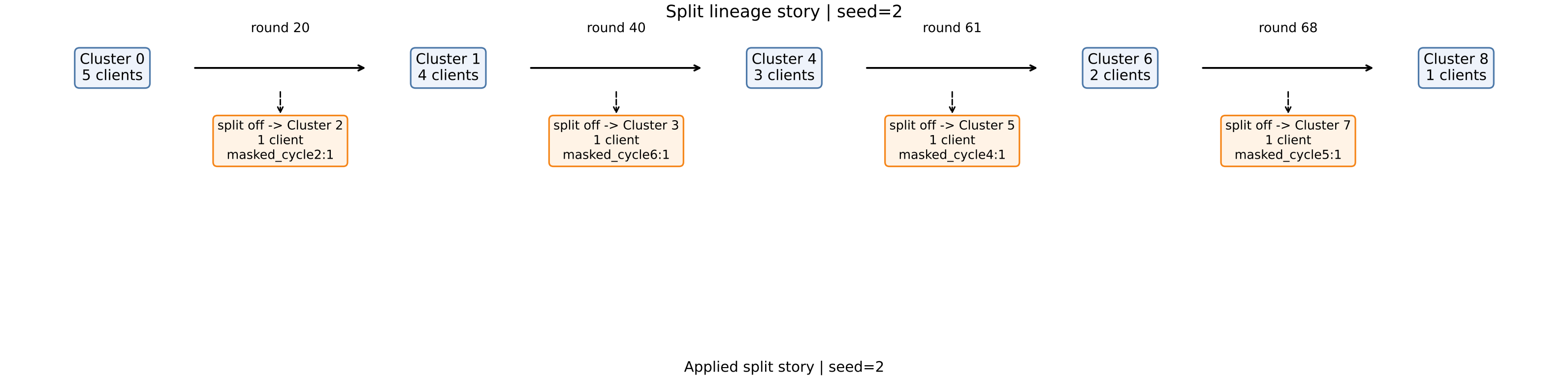


Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	20, 40, 61, 68	20, 40, 61, 68	2, 3, 5, 7, 8	2:1, 3:1, 5:1, 7:1, 8:1

Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019



Applied split story seed=2							
round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
20	0	5	154.483 < 154.777 , 194.740 > 192.336	C2 (1 client)	masked_cycle2:1	C1 (4 clients)	masked_cycle3:1 , masked_cycle4:1 , masked_cycle5:1 , masked_cycle6:1
40	1	4	155.562 < 157.269 , 194.421 > 193.954	C3 (1 client)	masked_cycle6:1	C4 (3 clients)	masked_cycle3:1 , masked_cycle4:1 , masked_cycle5:1
61	4	3	157.036 < 157.338 , 189.574 > 186.716	C5 (1 client)	masked_cycle4:1	C6 (2 clients)	masked_cycle3:1 , masked_cycle5:1
68	6	2	161.826 < 162.020 , 177.952 > 176.739	C7 (1 client)	masked_cycle5:1	C8 (1 clients)	masked_cycle3:1

Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | final active clusters

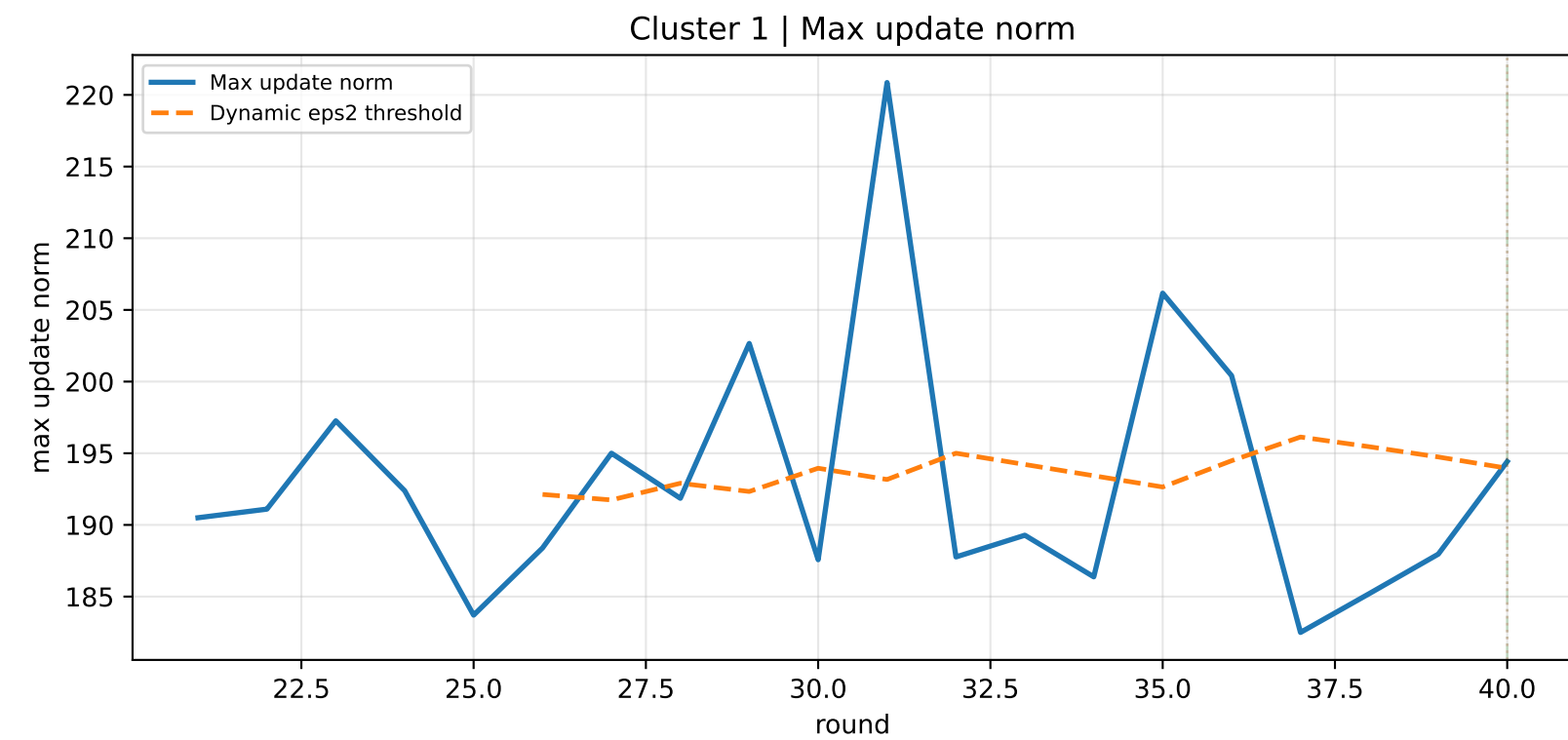
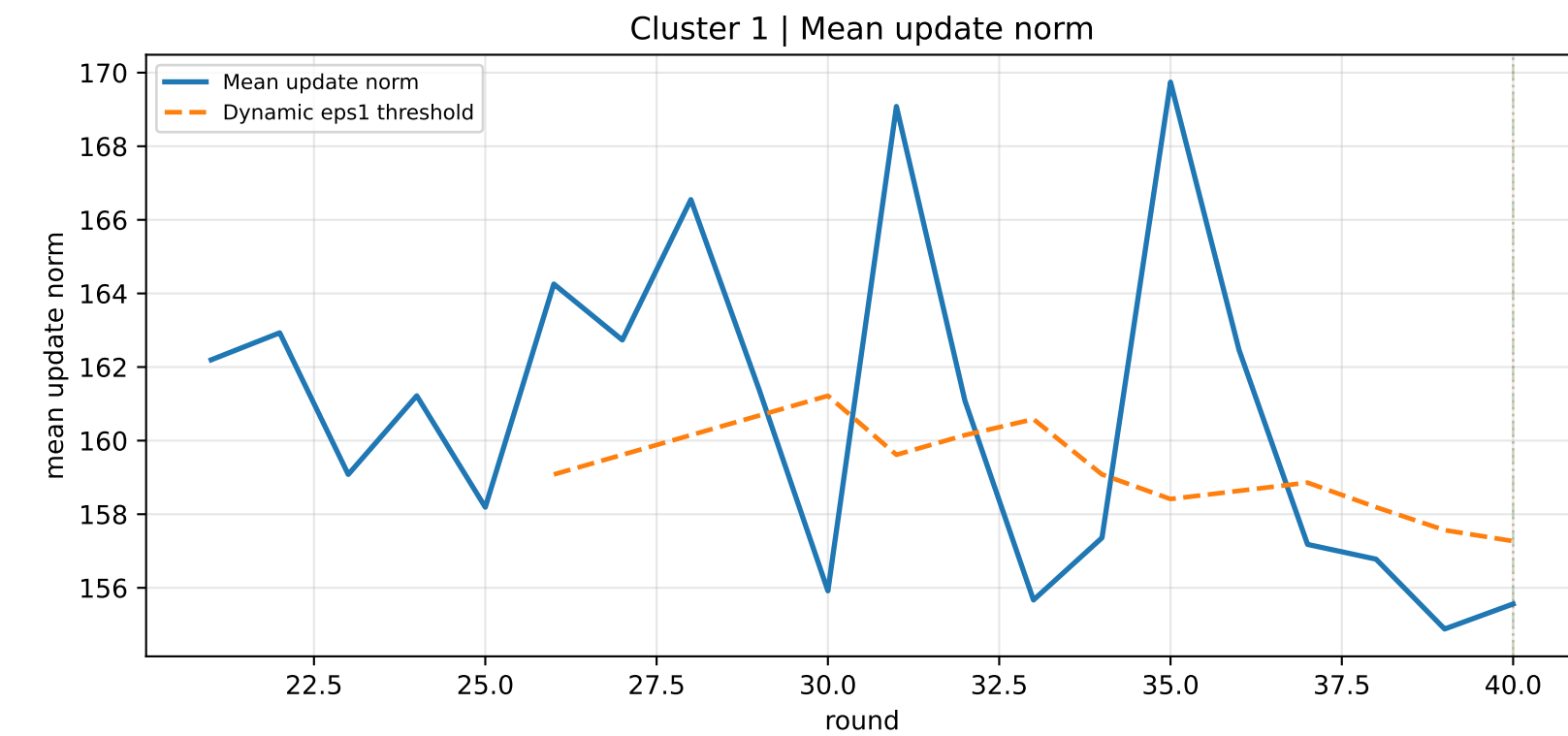
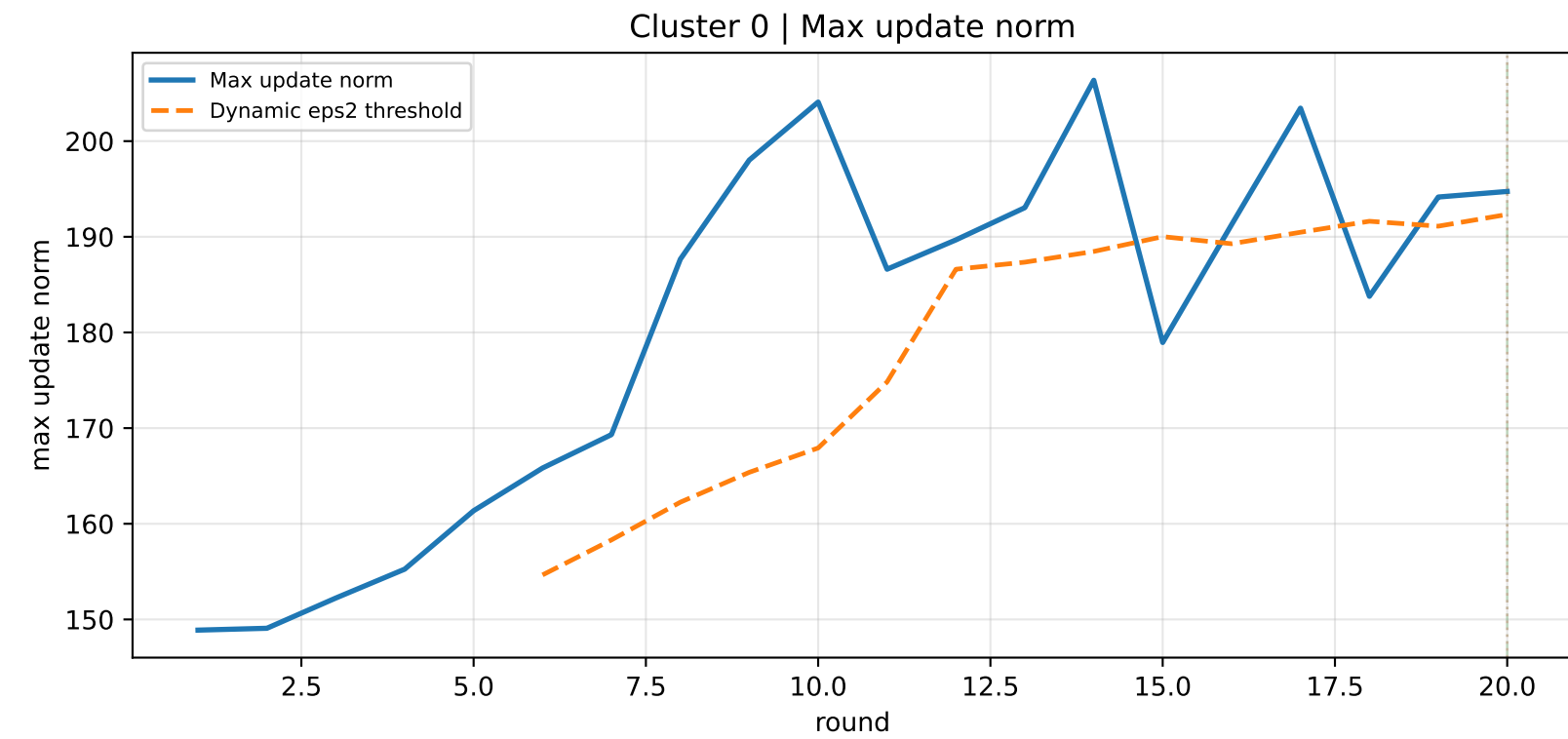
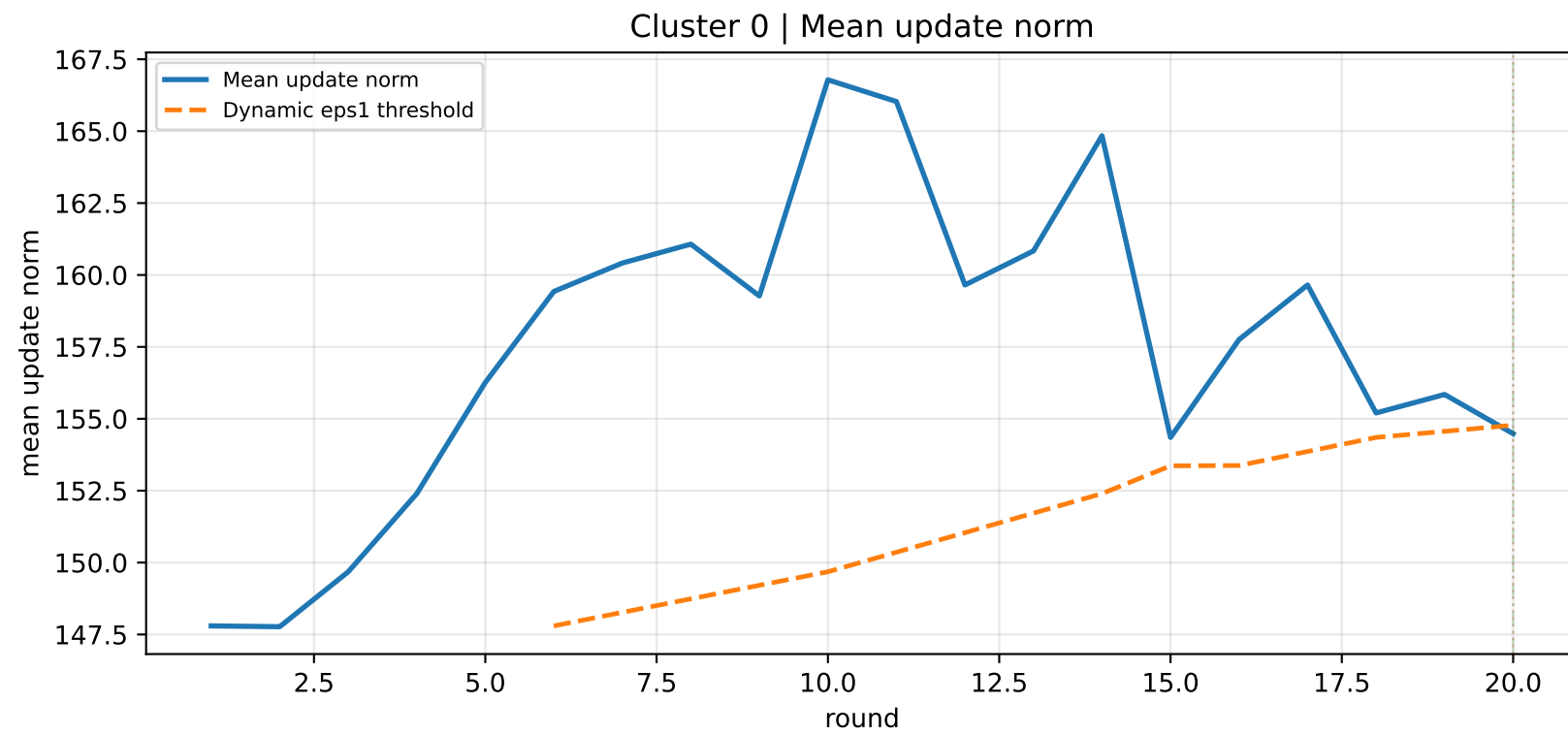
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
2	1	1000005	0	masked_cycle2:1
3	1	1000009	4	masked_cycle6:1
5	1	1000007	2	masked_cycle4:1
7	1	1000008	3	masked_cycle5:1
8	1	1000006	1	masked_cycle3:1

Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | norm / threshold panels (0-1)

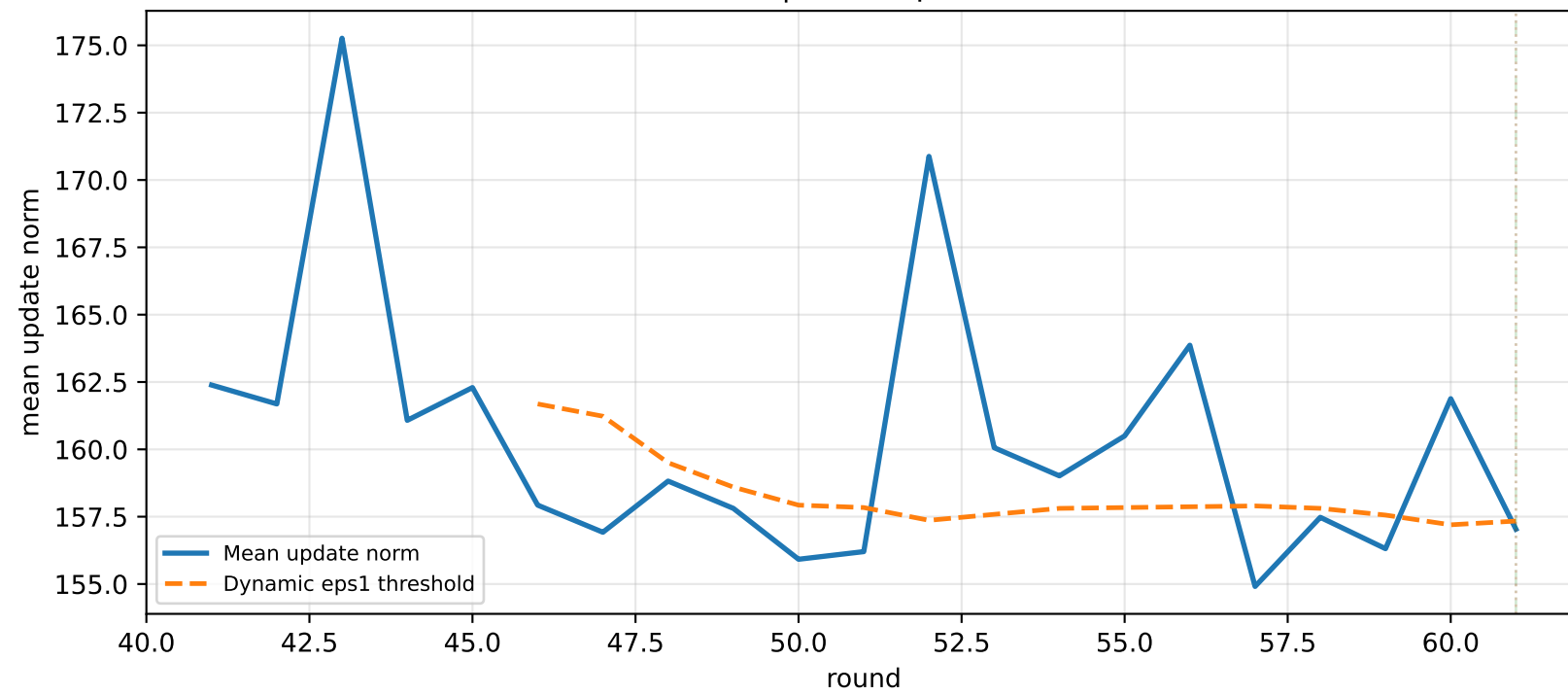
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019



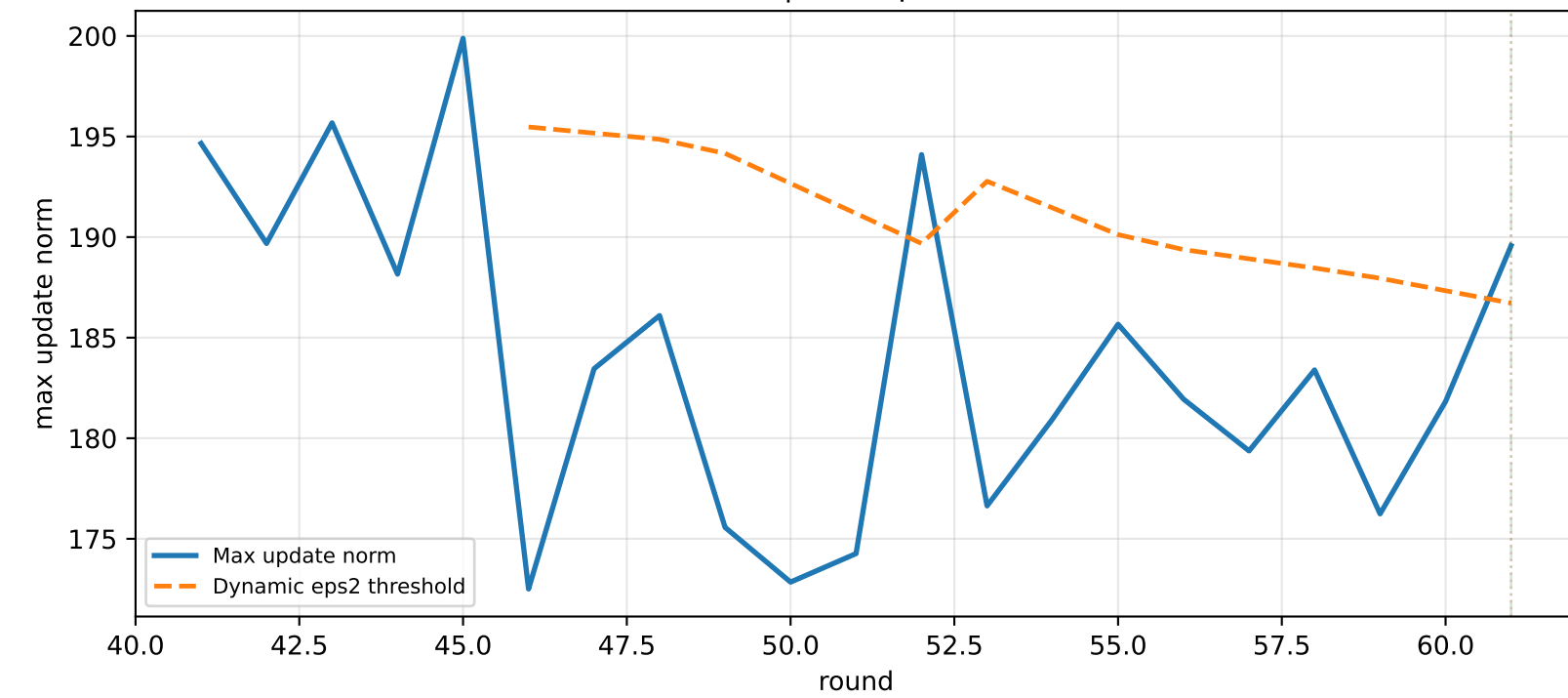
Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 |
family_centroid_jsd_mean=0.019

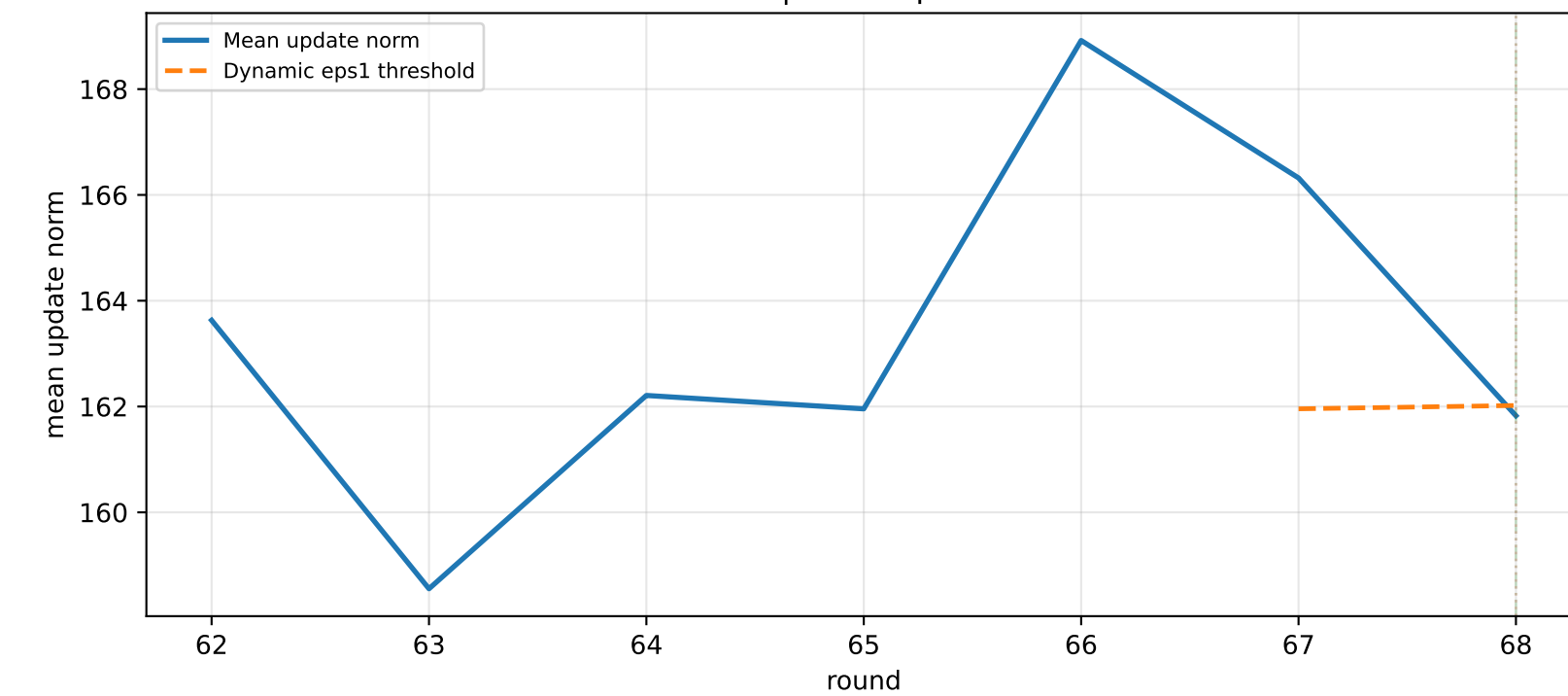
Cluster 4 | Mean update norm



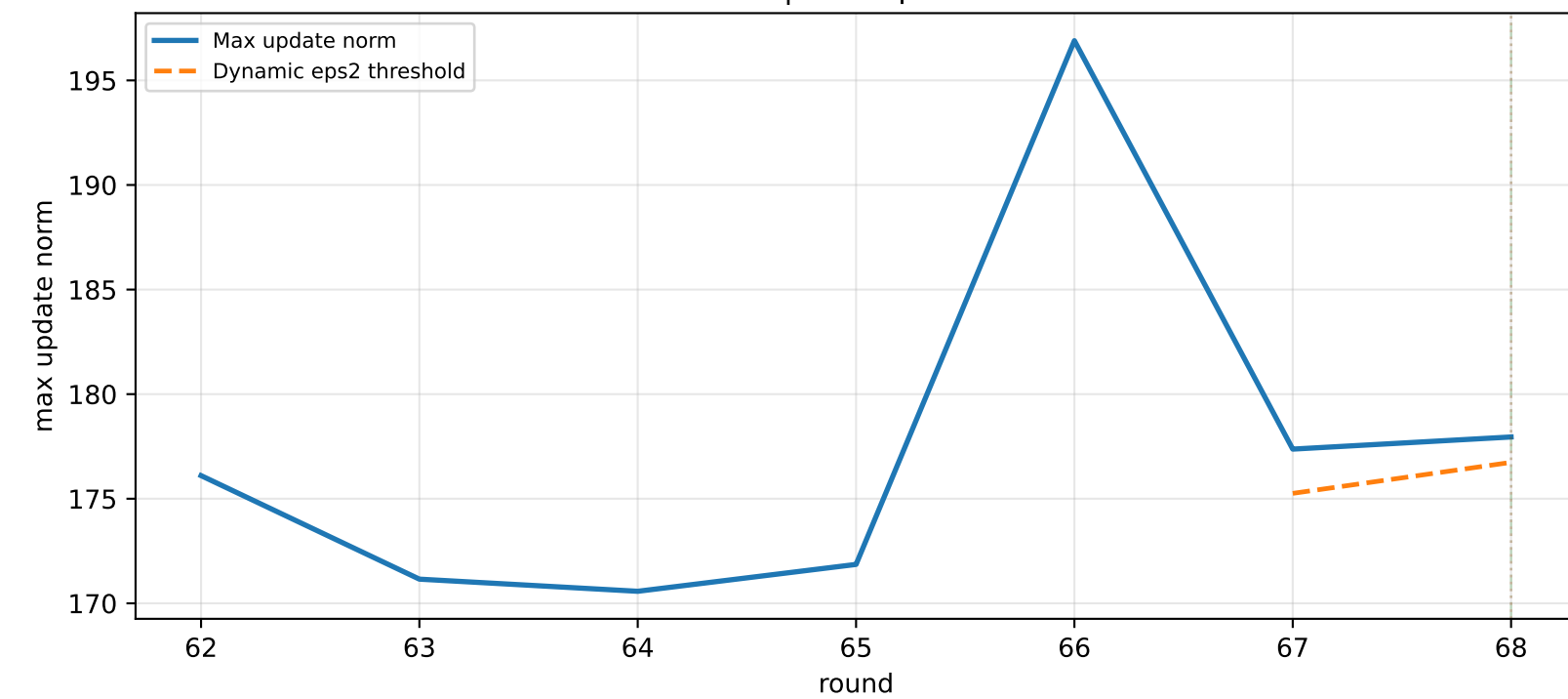
Cluster 4 | Max update norm



Cluster 6 | Mean update norm



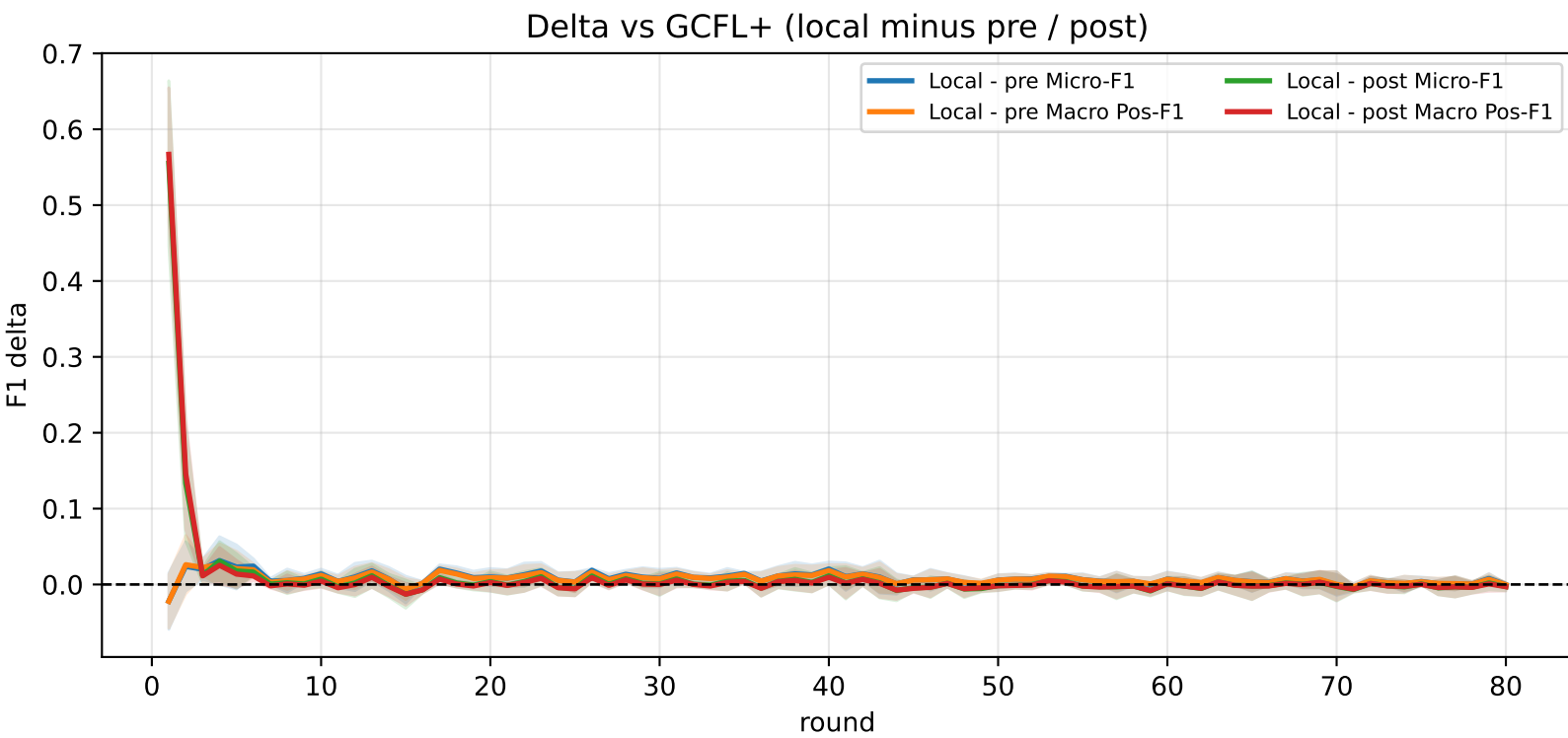
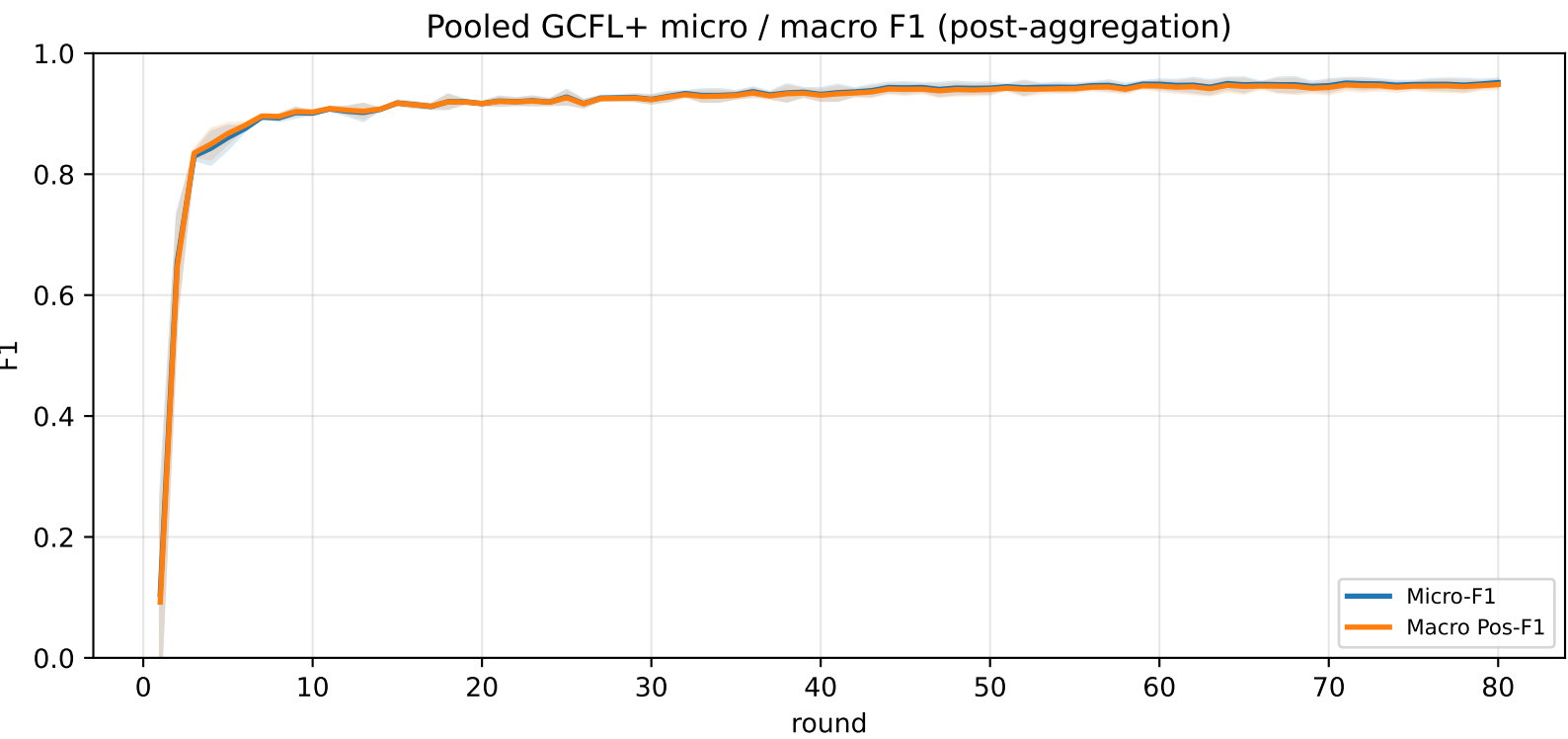
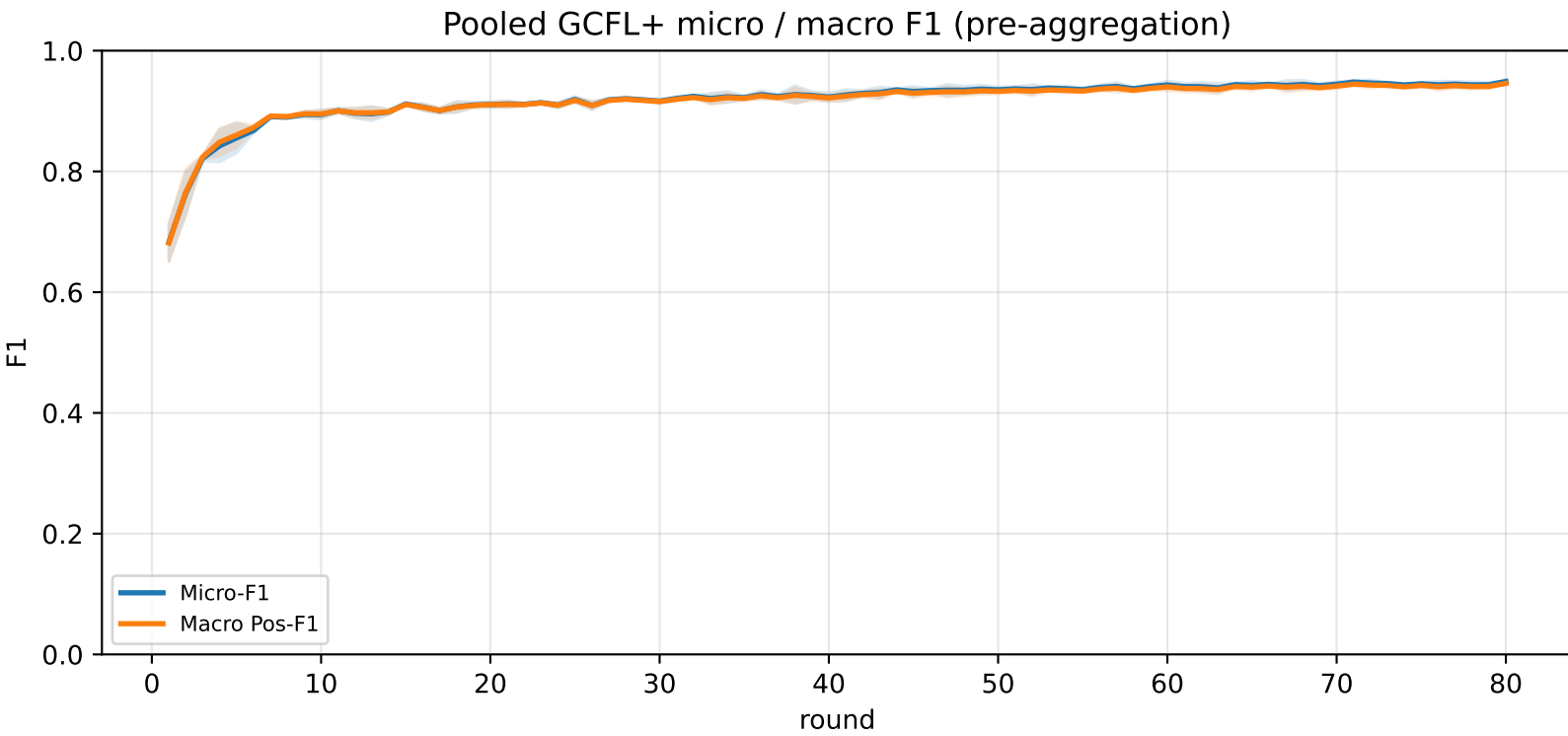
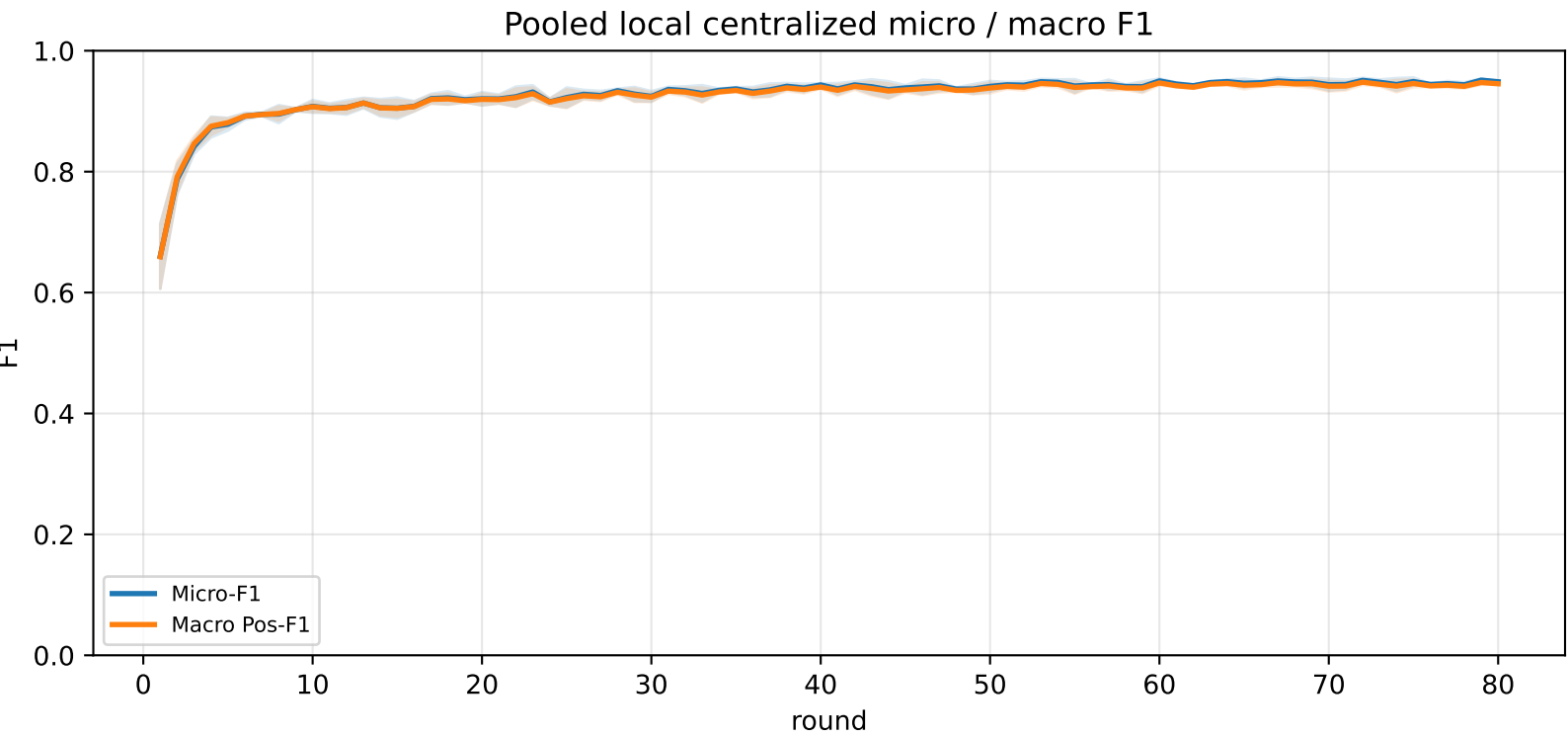
Cluster 6 | Max update norm



Controlled masking level: Mid | GCFL+ diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019

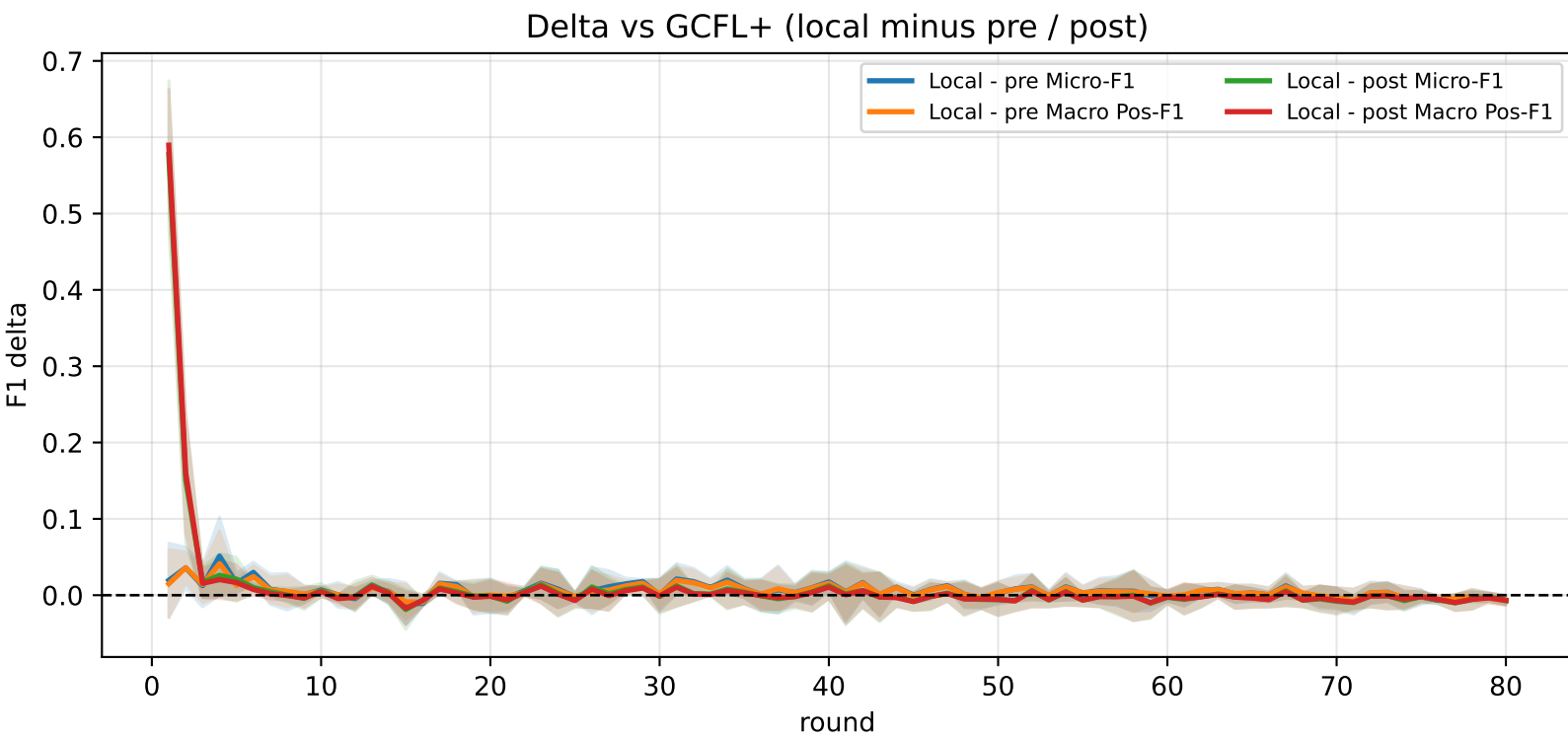
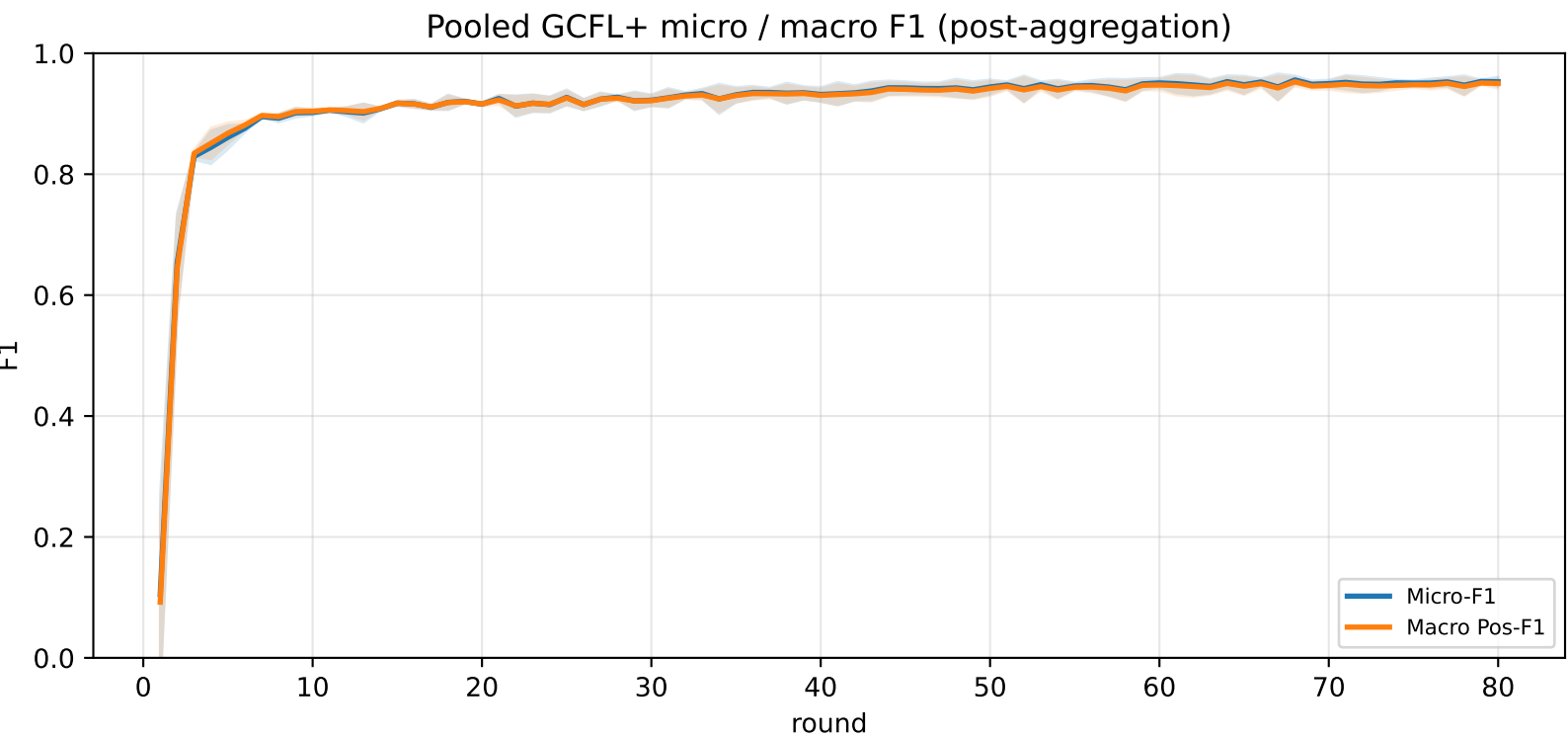
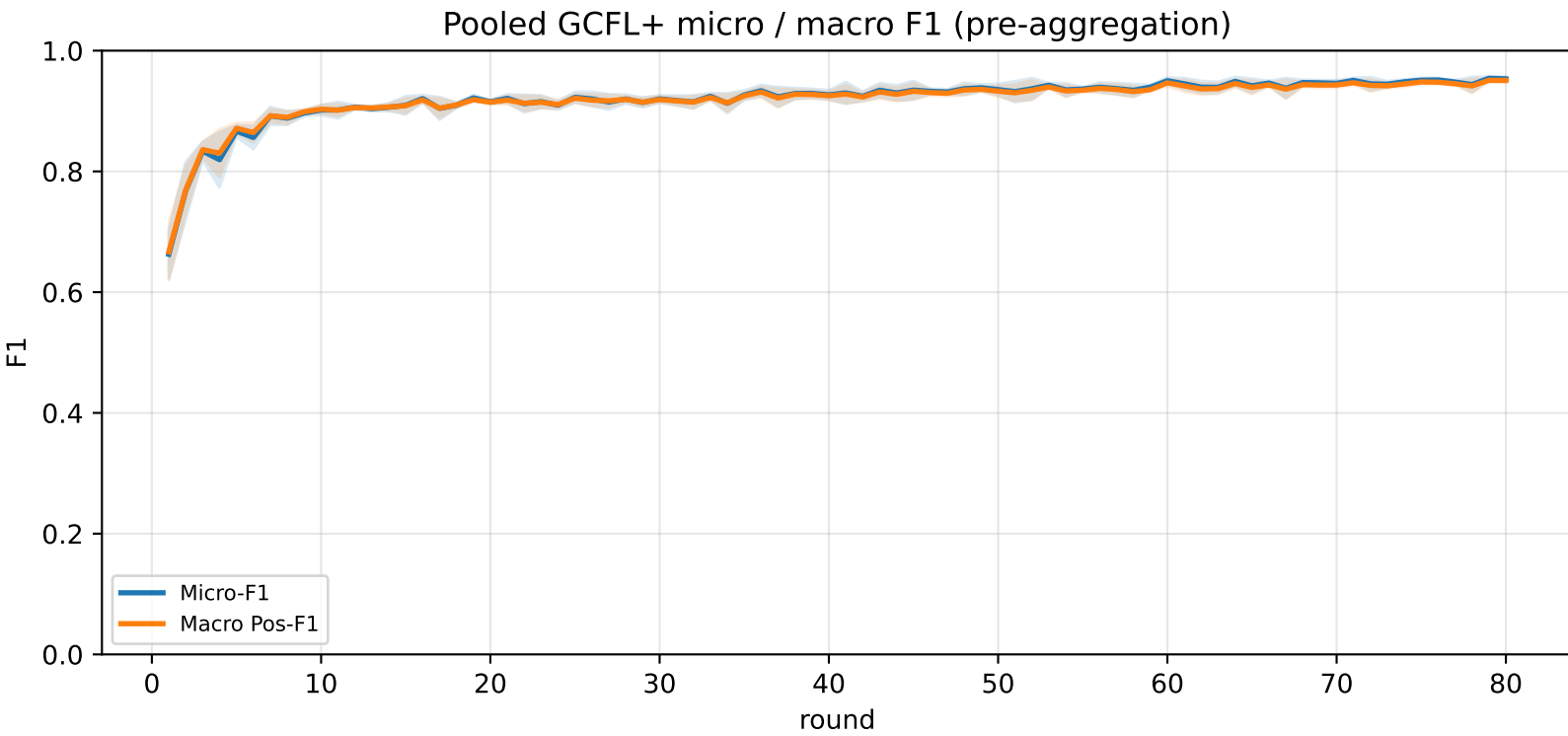
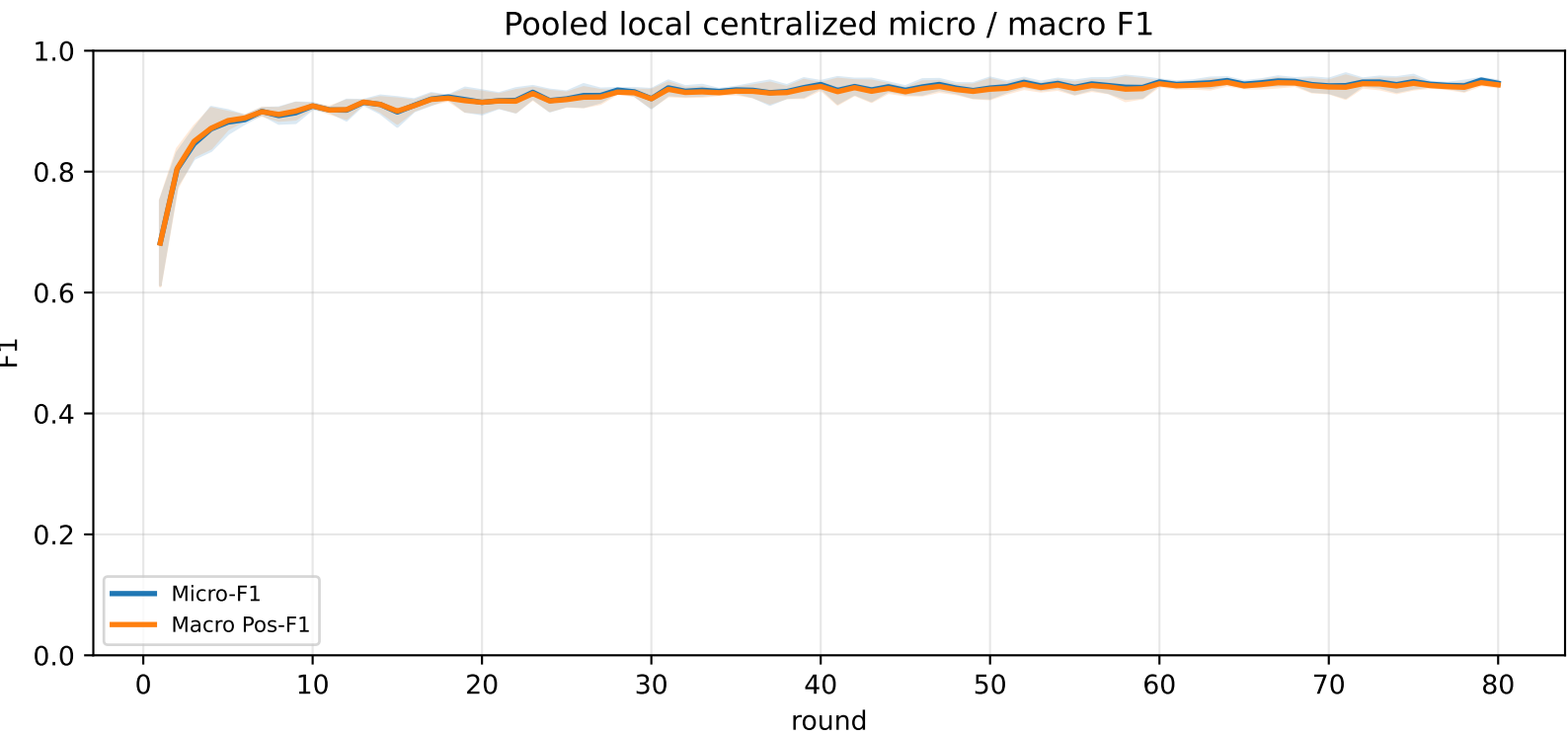
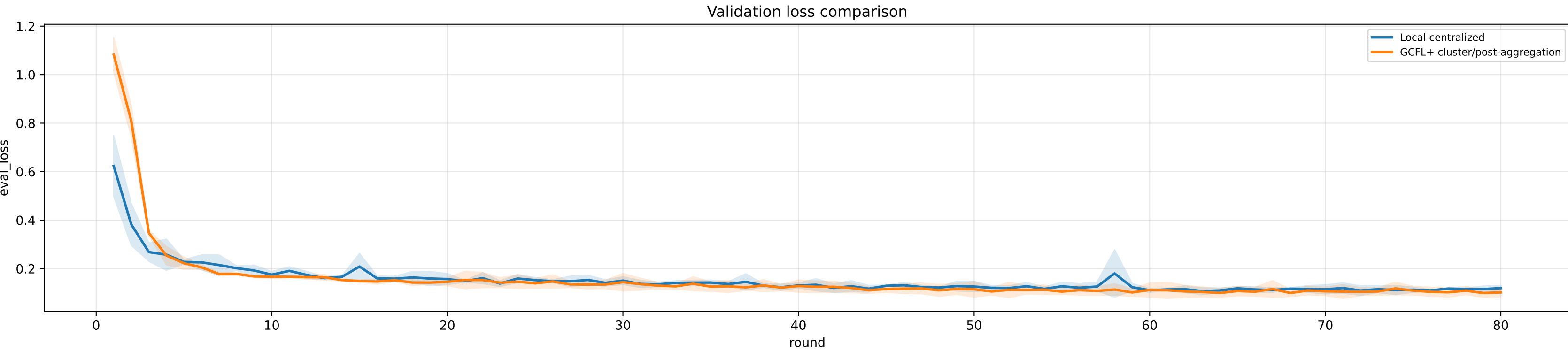
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.2% ± 0.0%	95.1% ± 0.1%	99.3% ± 0.0%	98.7% ± 0.2%	97.4% ± 0.3%	94.5% ± 0.4%	85.8% ± 0.5%
GCFL+	95.1% ± 0.6%	95.0% ± 0.6%	99.2% ± 0.1%	98.8% ± 0.2%	97.4% ± 0.3%	94.3% ± 0.8%	85.2% ± 1.9%



Controlled masking level: Mid | GCFL+ diagnostics | family=masked_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle2 | n_clients=1 | cycle2=visible 13.8%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1%
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019

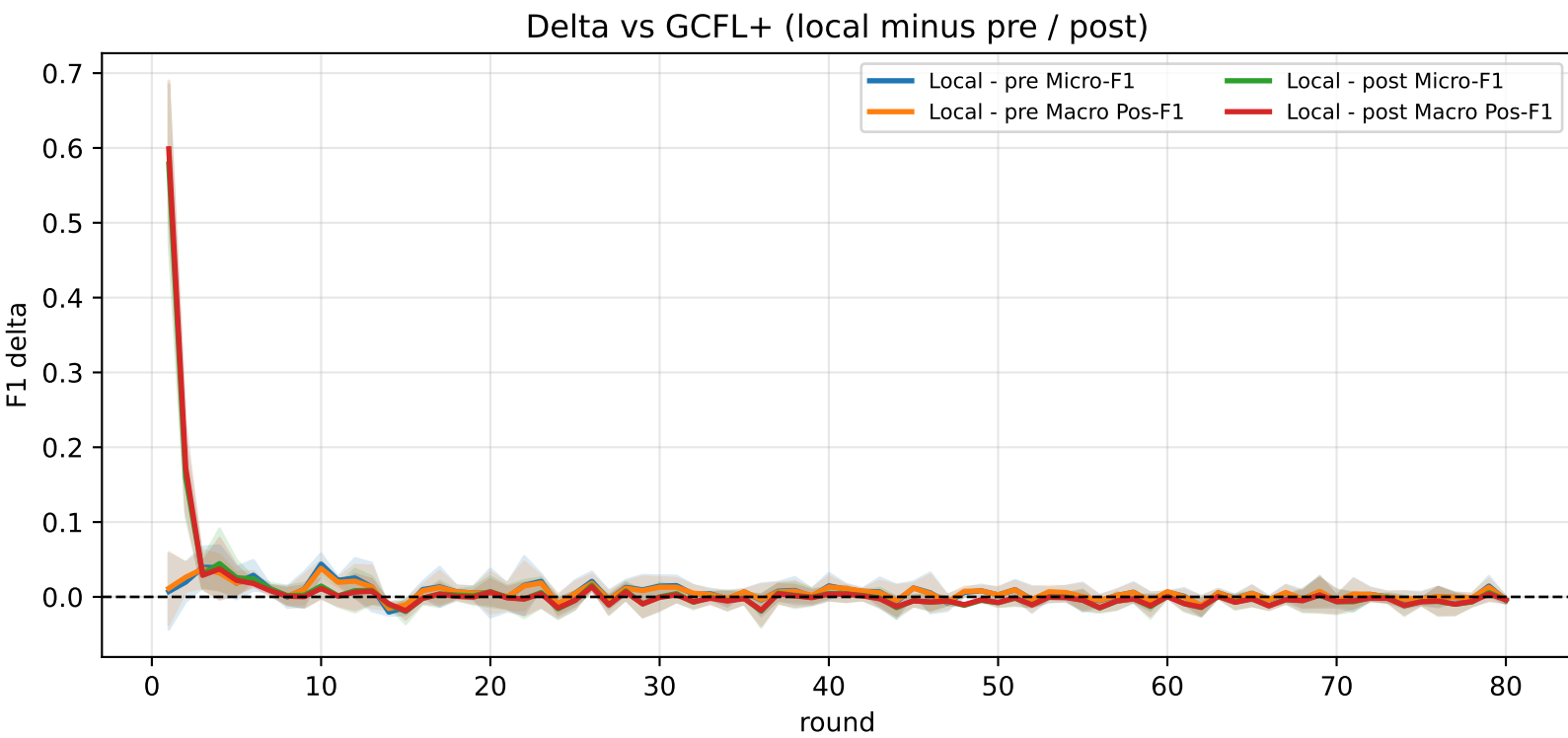
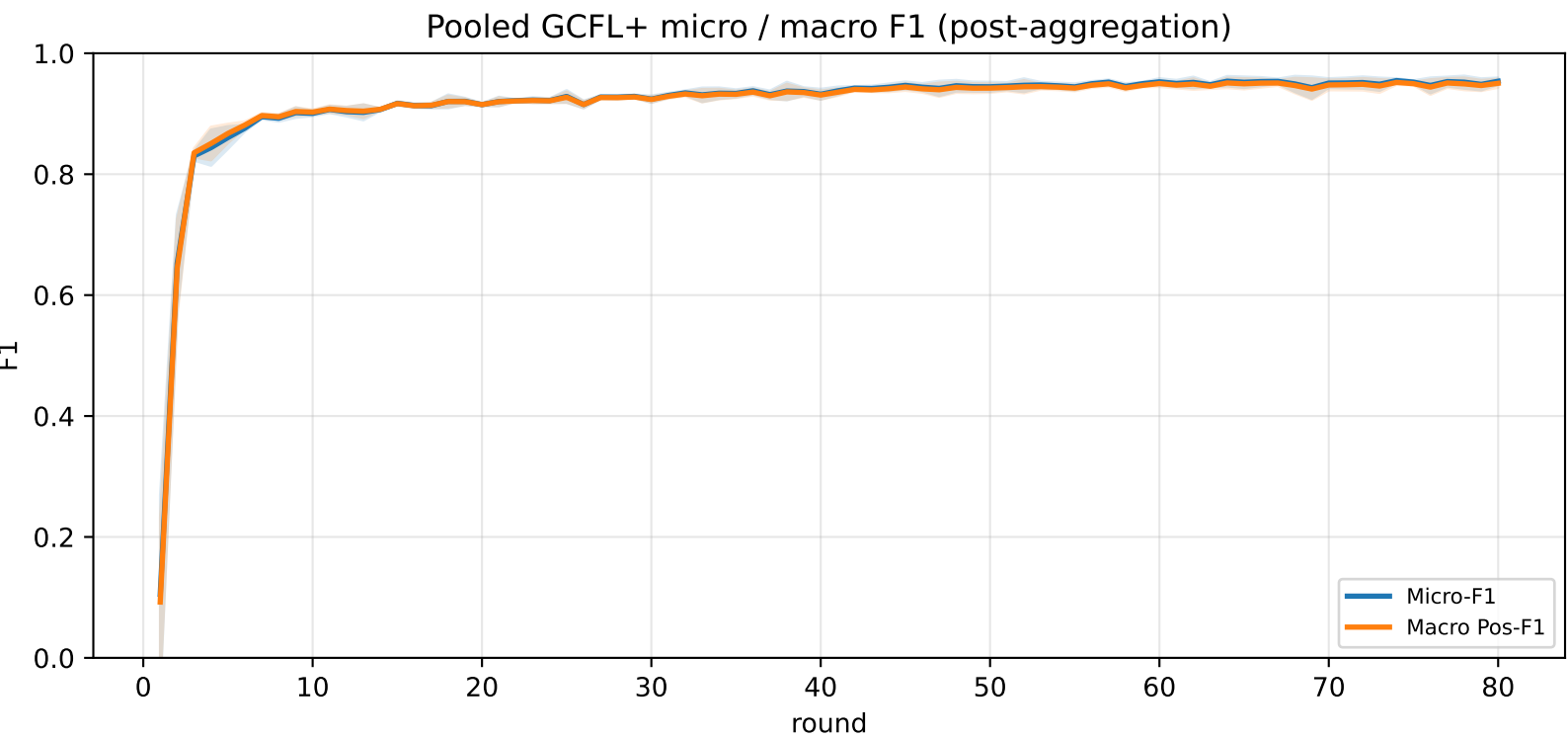
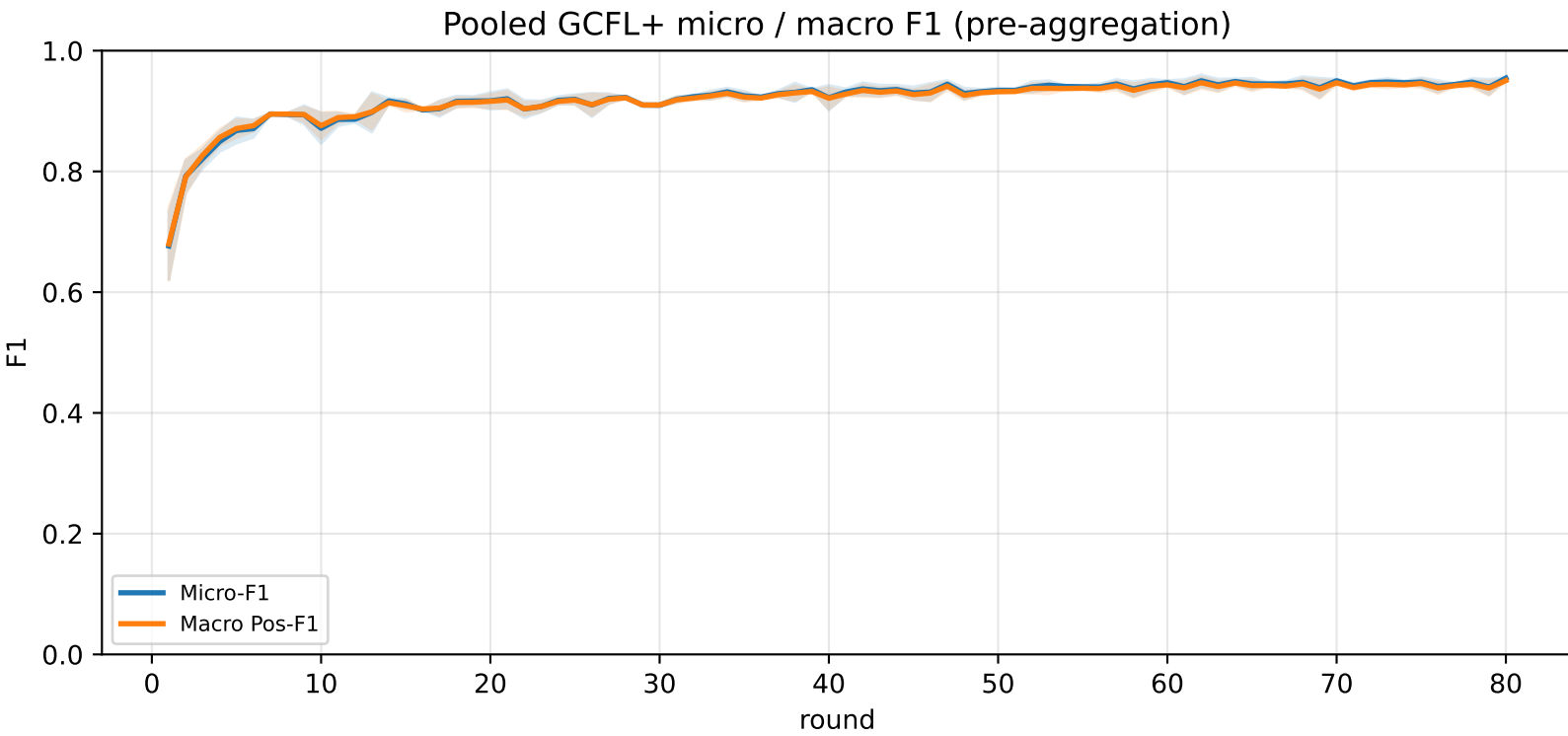
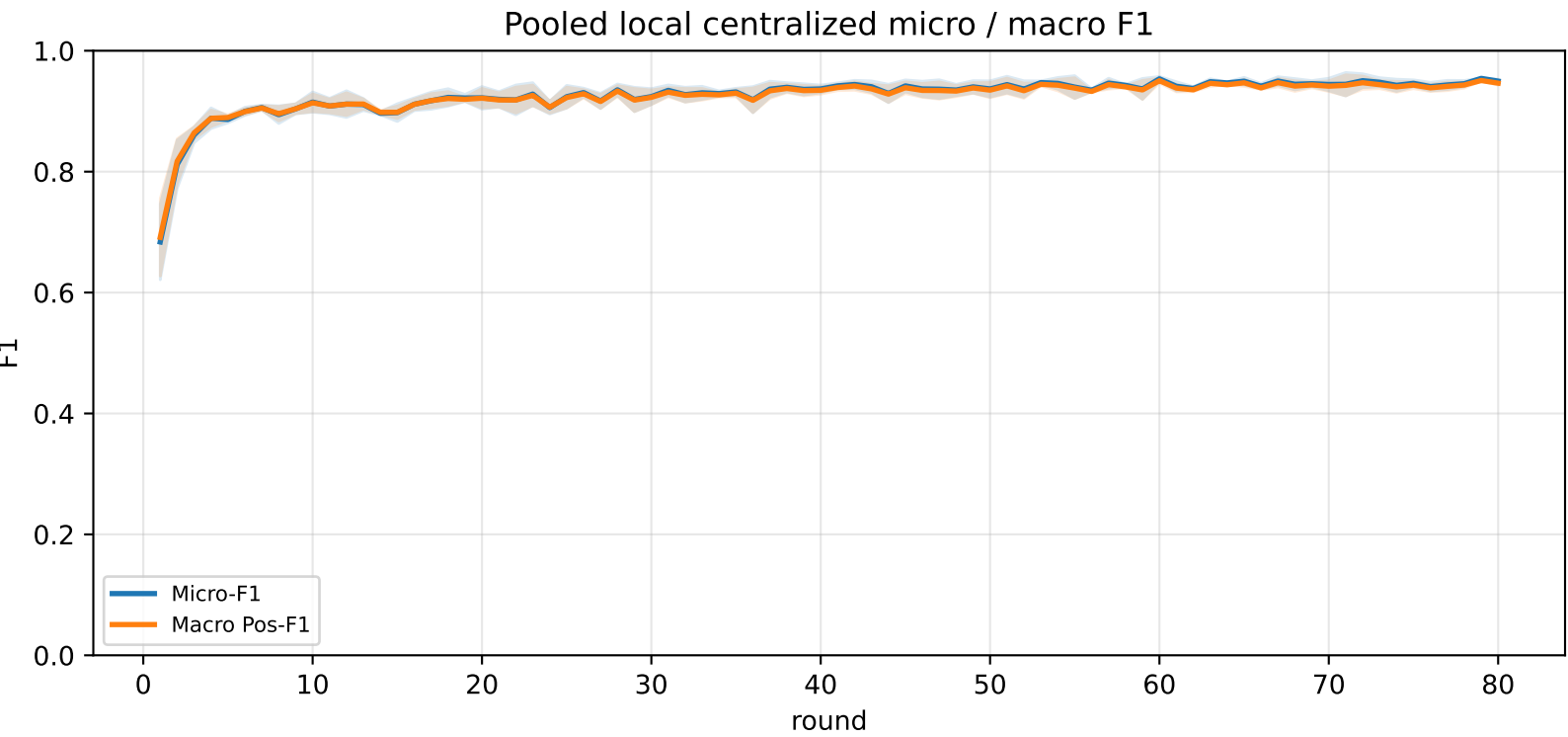
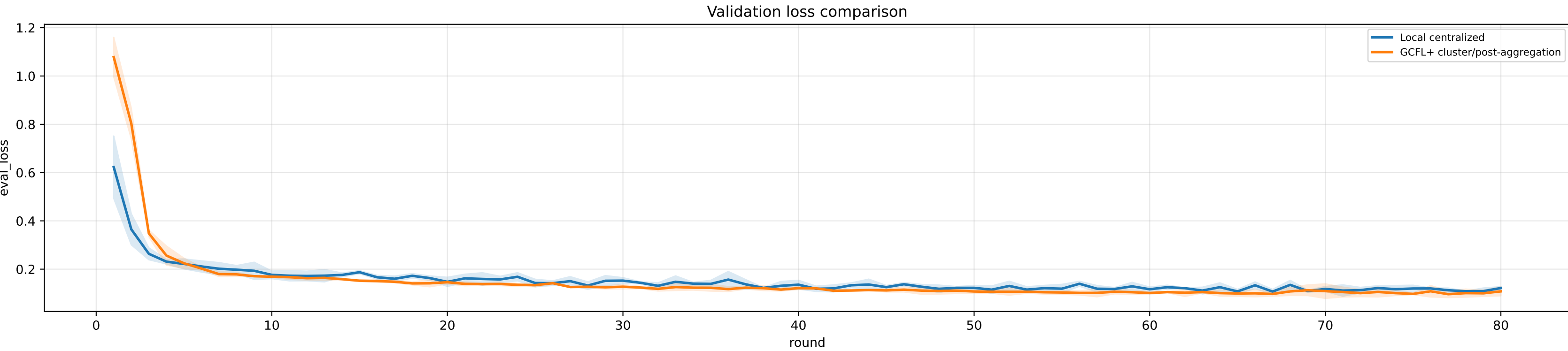
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.6% ± 0.1%	95.4% ± 0.0%	99.3% ± 0.2%	98.7% ± 0.0%	97.9% ± 0.3%	95.0% ± 0.0%	86.3% ± 0.5%
GCFL+	94.8% ± 1.1%	94.8% ± 1.0%	99.0% ± 0.2%	98.5% ± 0.3%	97.3% ± 0.8%	94.1% ± 1.0%	85.0% ± 2.8%



Controlled masking level: Mid | GCFL+ diagnostics | family=masked_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle3 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 14.6%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1% | masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.5% ± 0.3%	95.4% ± 0.3%	99.3% ± 0.1%	98.9% ± 0.2%	97.7% ± 0.4%	94.4% ± 0.7%	86.8% ± 1.0%
GCFL+	95.4% ± 0.2%	95.3% ± 0.2%	99.4% ± 0.2%	99.0% ± 0.2%	97.6% ± 0.1%	94.7% ± 0.3%	86.0% ± 1.3%

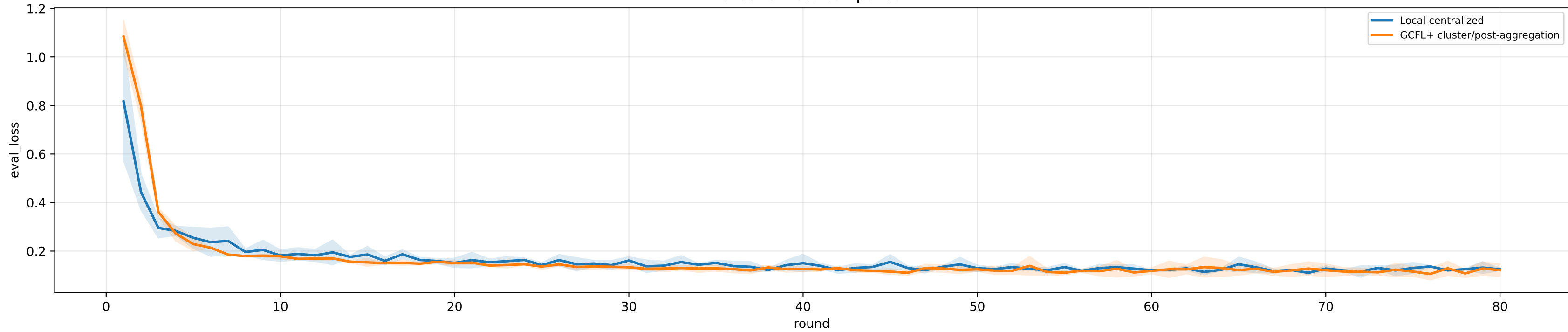


family=masked_cycle4 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 13.1%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1%
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019

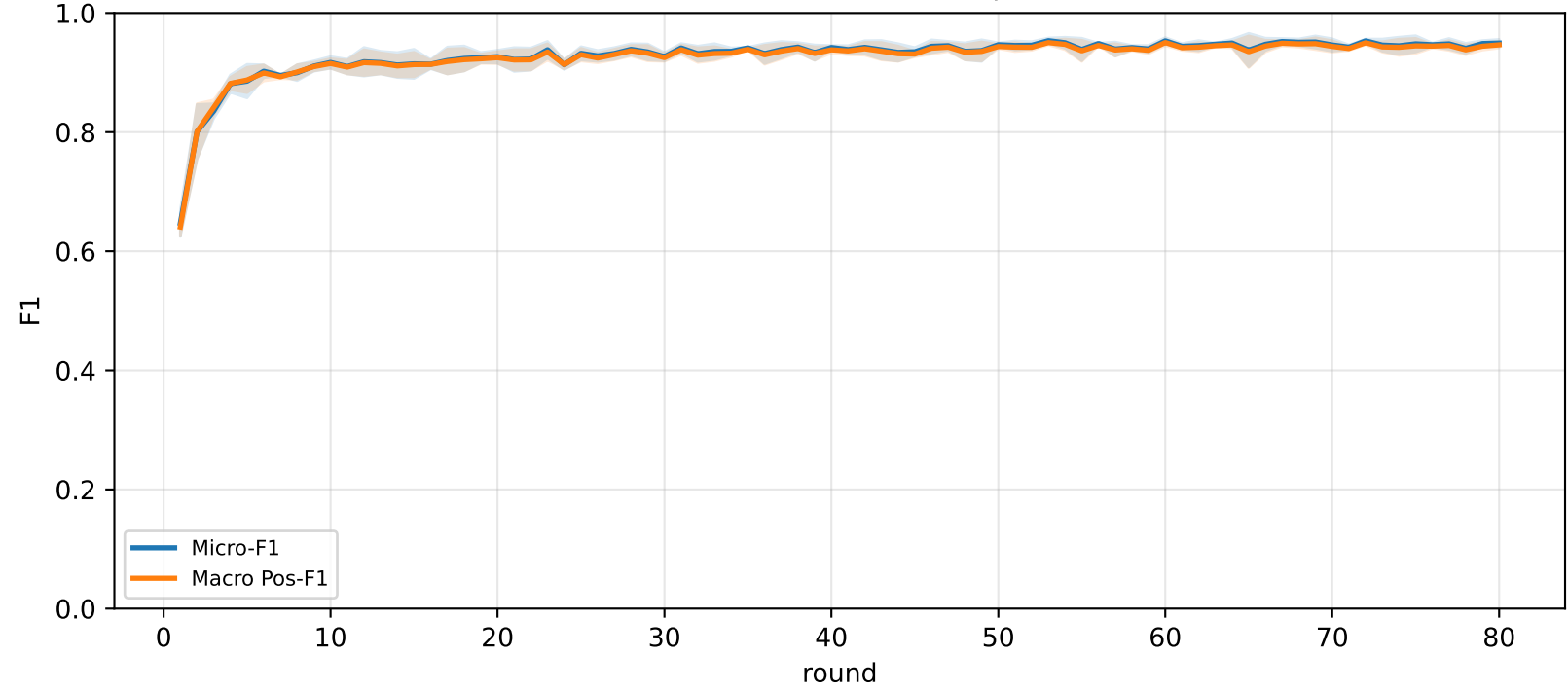
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.3% ± 0.3%	95.2% ± 0.3%	99.3% ± 0.1%	98.7% ± 0.2%	96.6% ± 1.3%	94.8% ± 0.3%	86.6% ± 0.4%
GCFL+	95.2% ± 0.5%	95.1% ± 0.5%	99.3% ± 0.1%	98.5% ± 0.4%	97.5% ± 0.3%	94.6% ± 0.6%	85.6% ± 1.3%

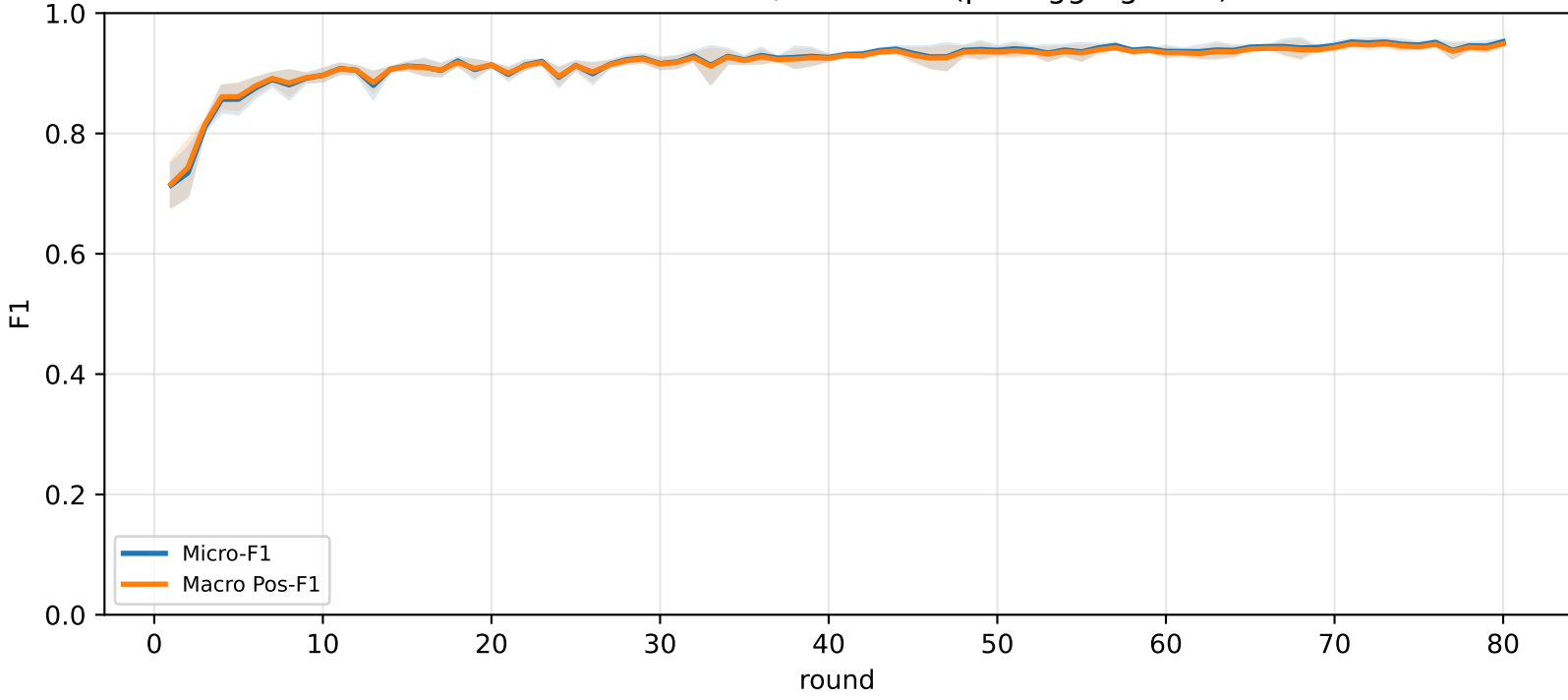
Validation loss comparison



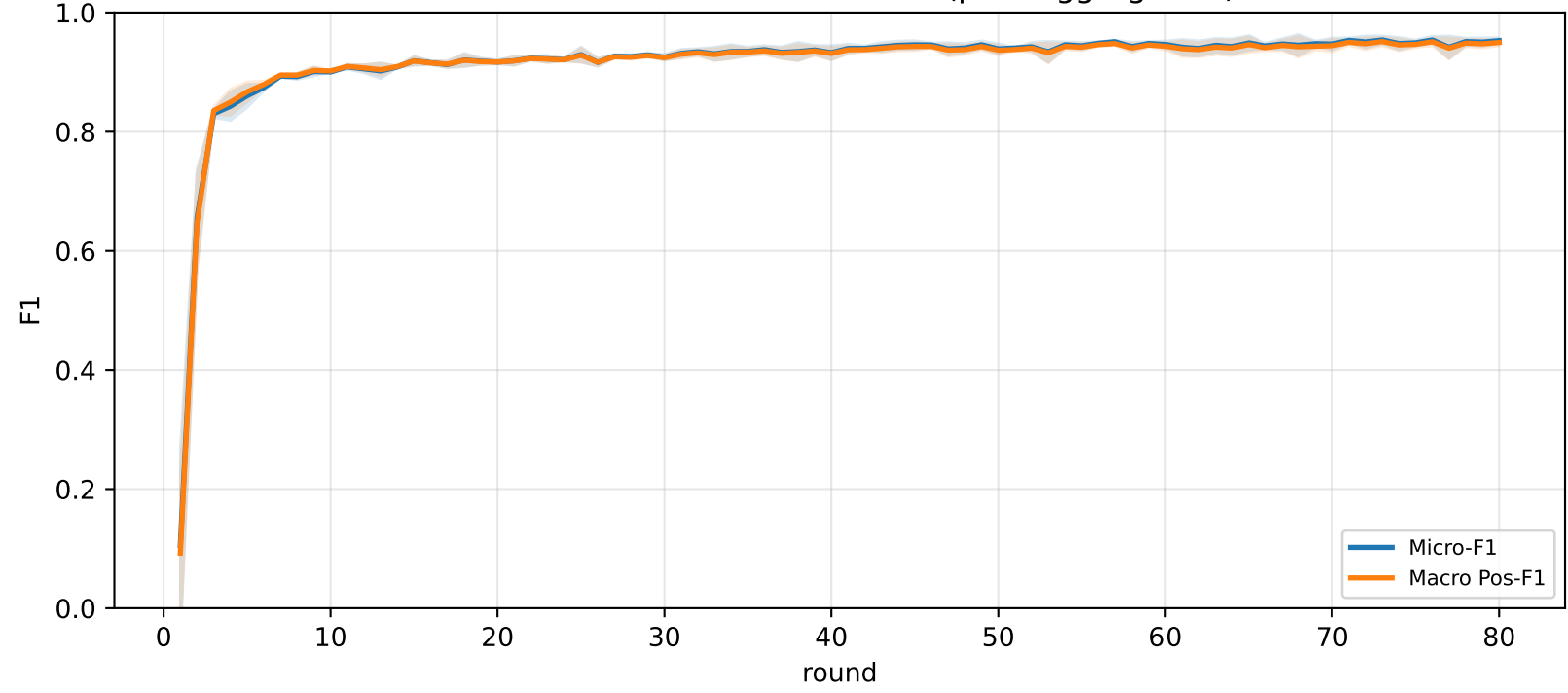
Pooled local centralized micro / macro F1



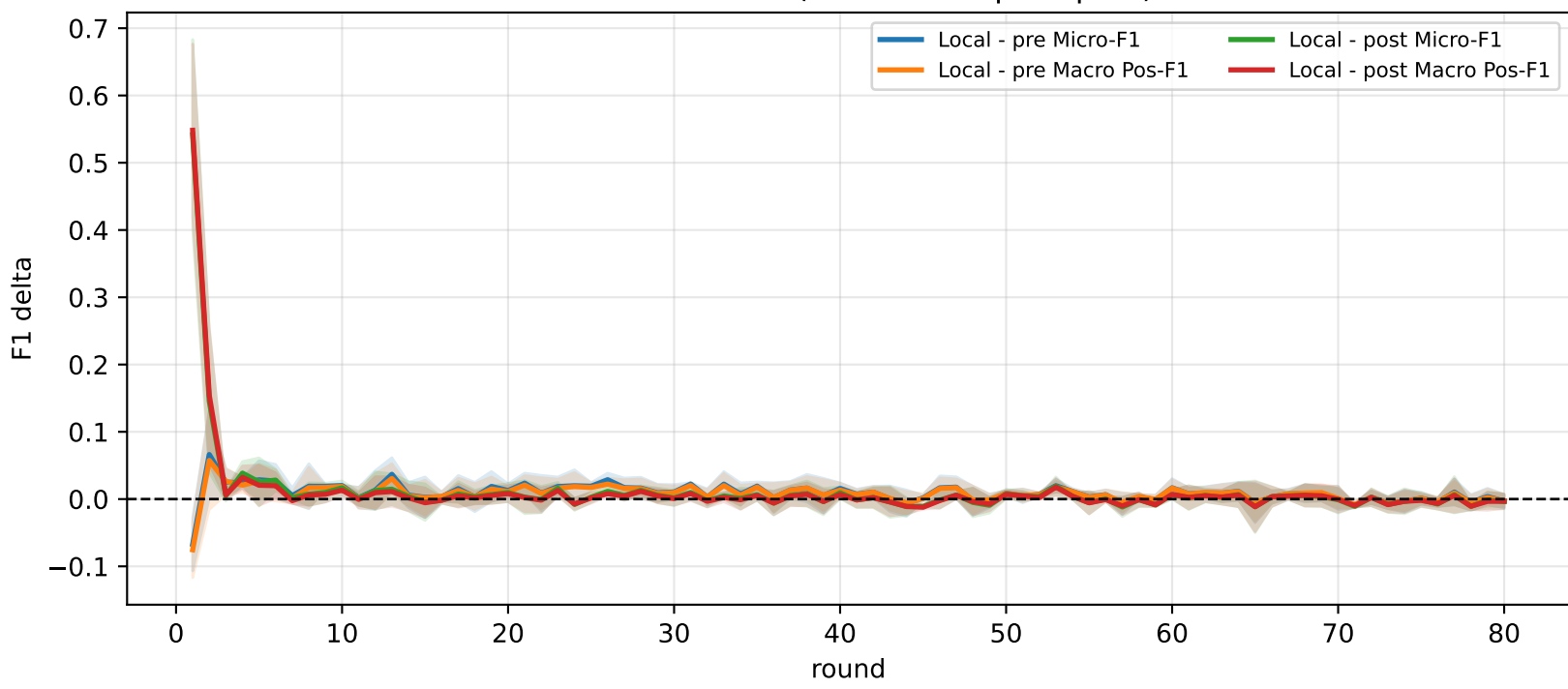
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)



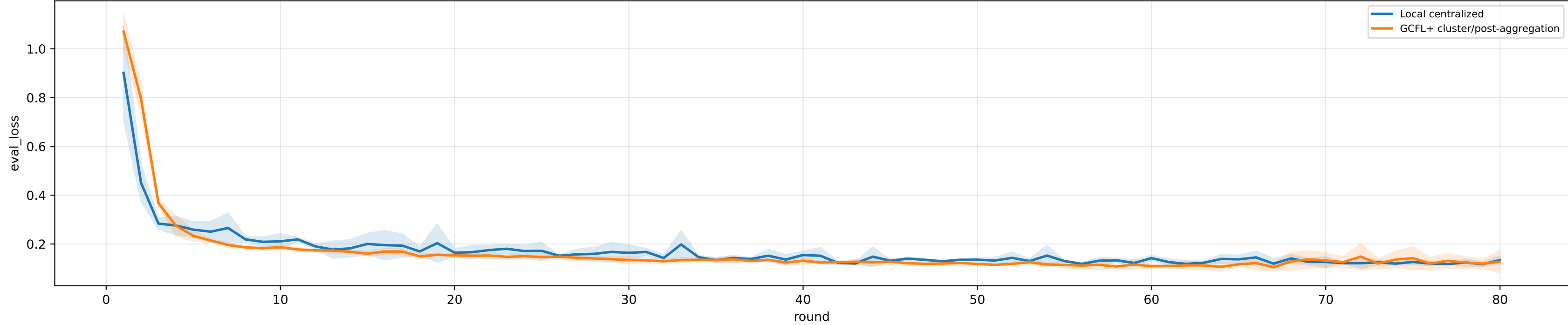
Controlled masking level: Mid | GCFL+ diagnostics | family=masked_cycle5 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle5 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 13.1%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1%
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019

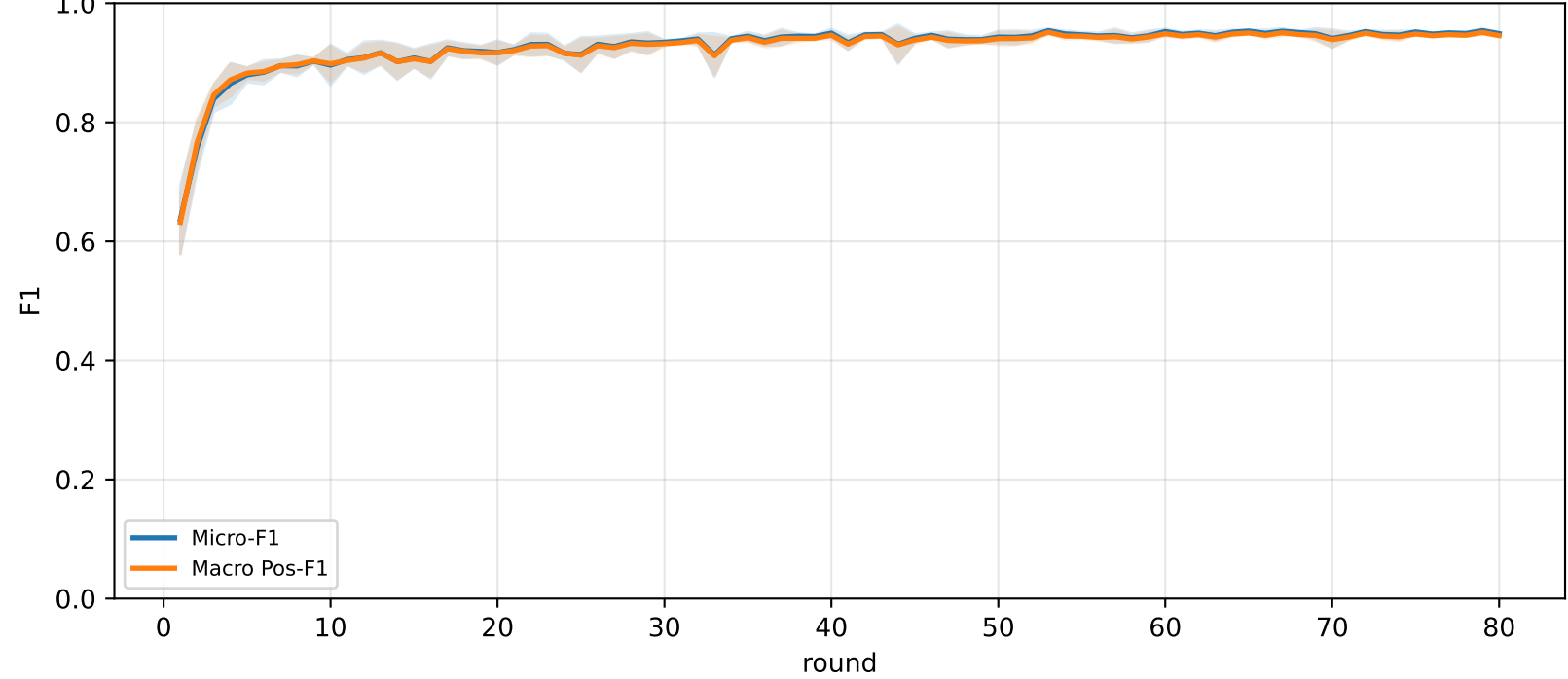
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.2% ± 0.1%	95.1% ± 0.2%	99.3% ± 0.0%	98.6% ± 0.5%	97.3% ± 0.1%	94.2% ± 0.4%	85.9% ± 0.1%
GCFL+	95.2% ± 0.5%	95.1% ± 0.6%	99.4% ± 0.2%	99.0% ± 0.1%	97.5% ± 0.4%	94.4% ± 1.2%	85.1% ± 1.0%

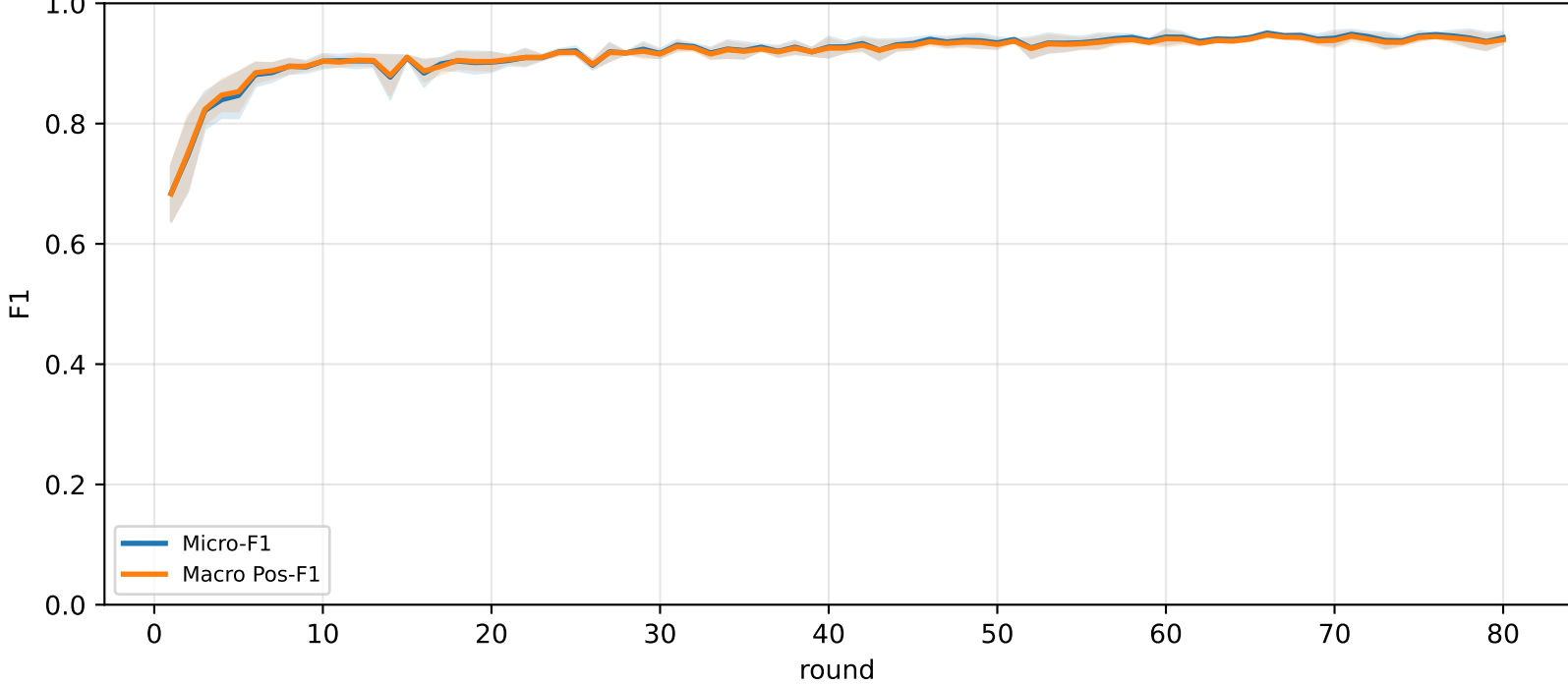
Validation loss comparison



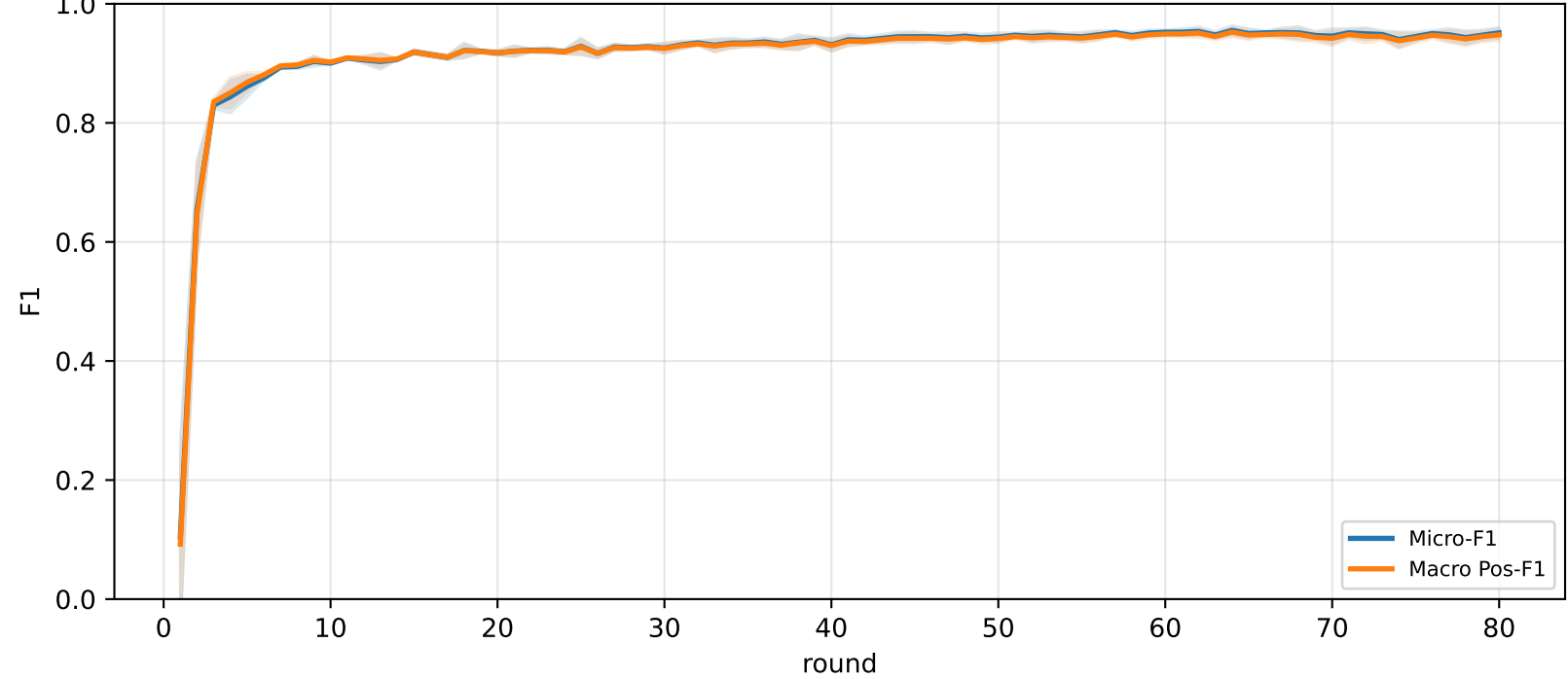
Pooled local centralized micro / macro F1



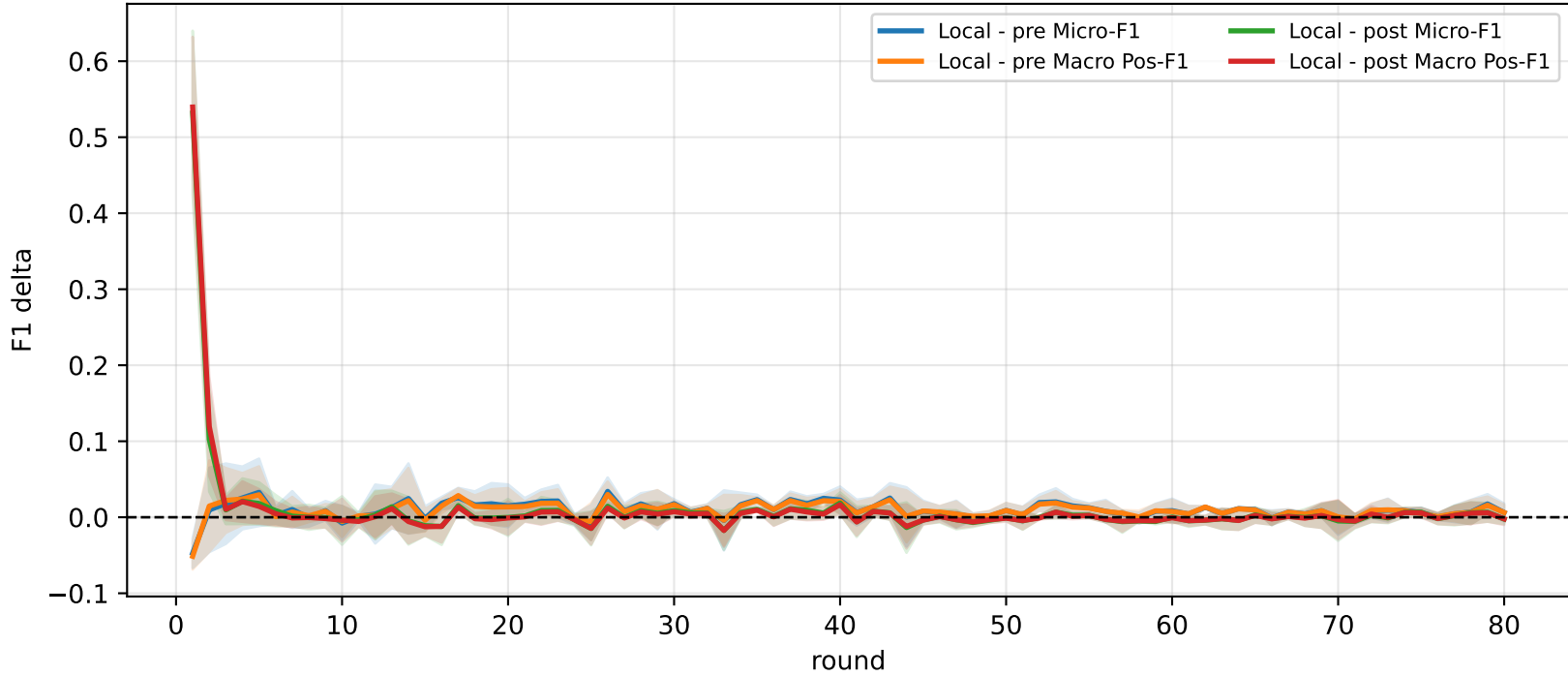
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



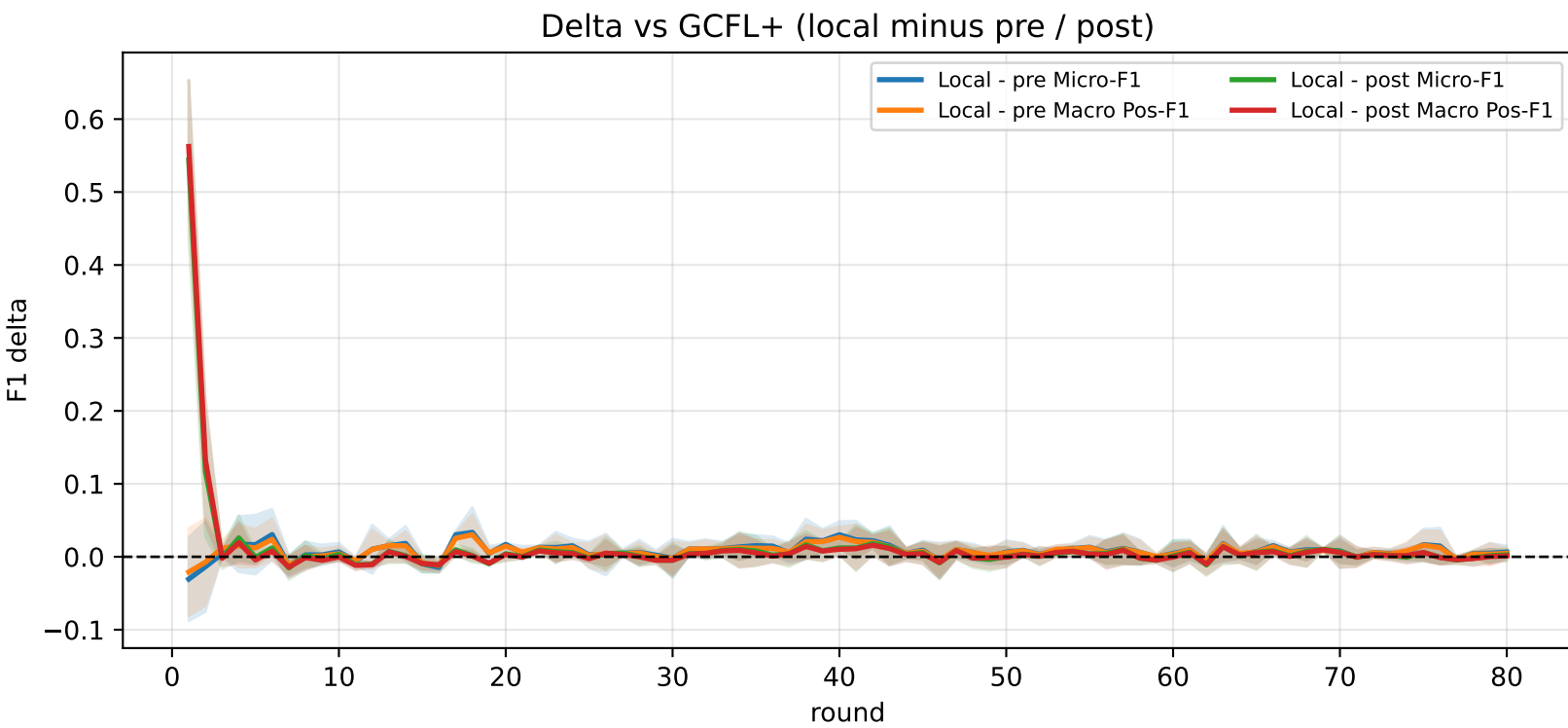
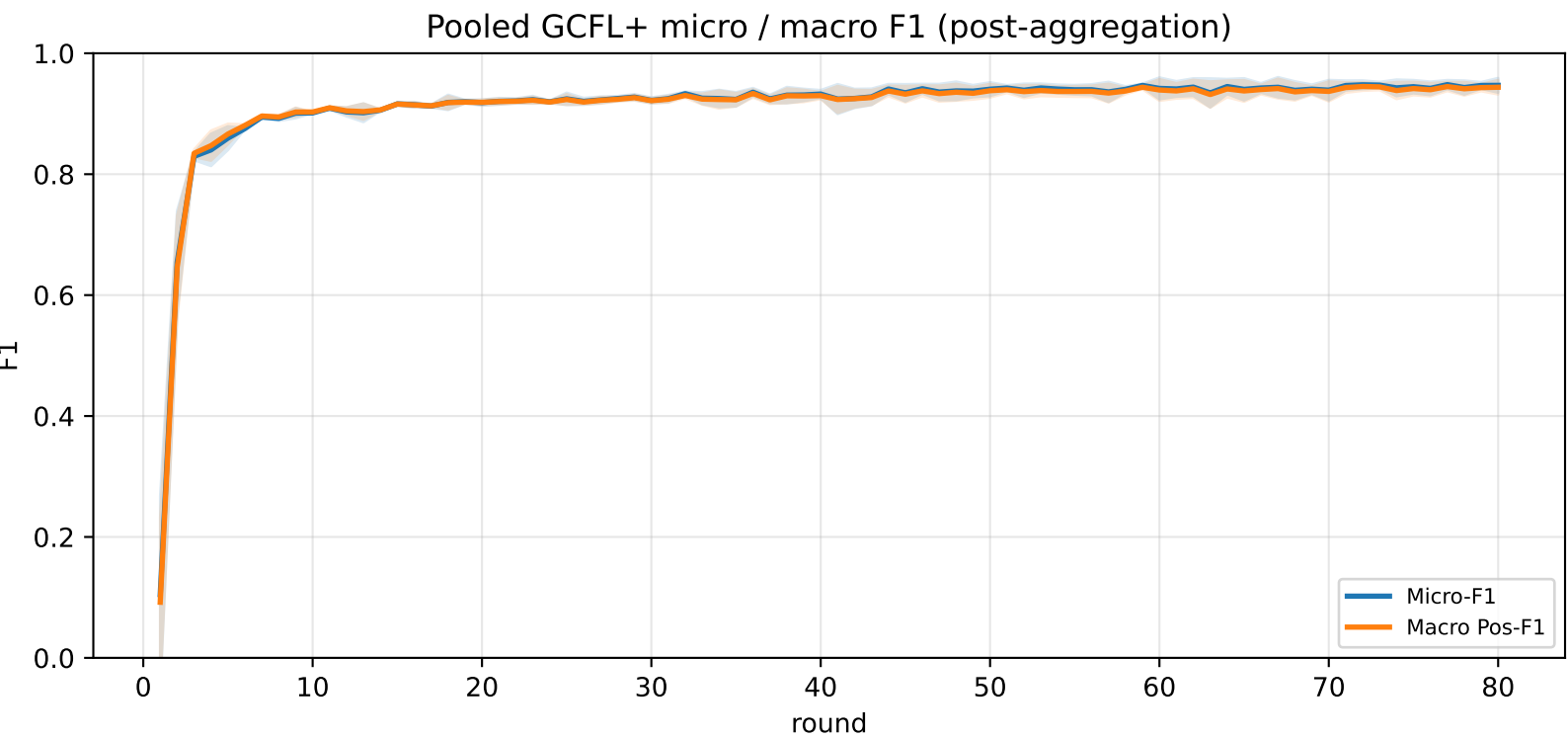
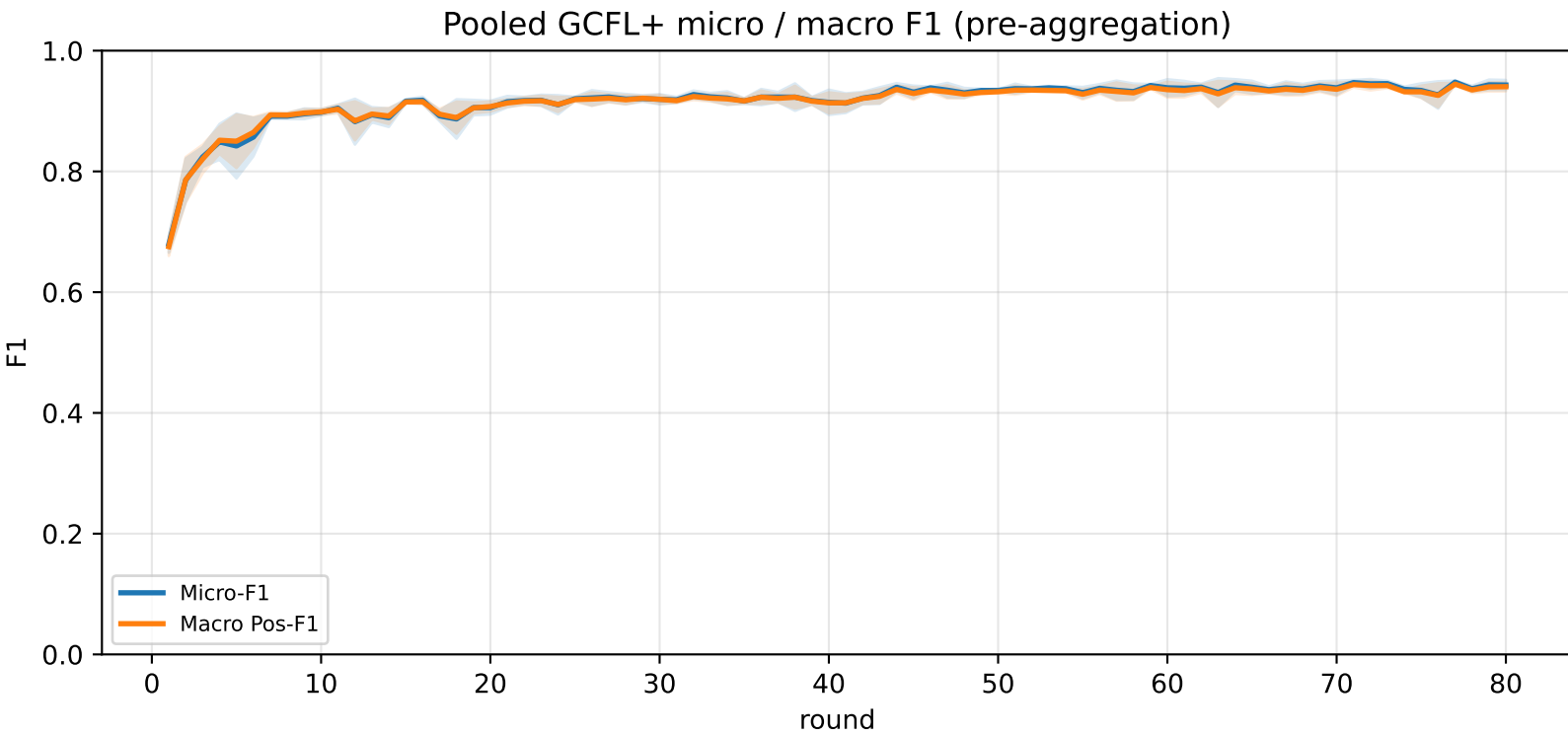
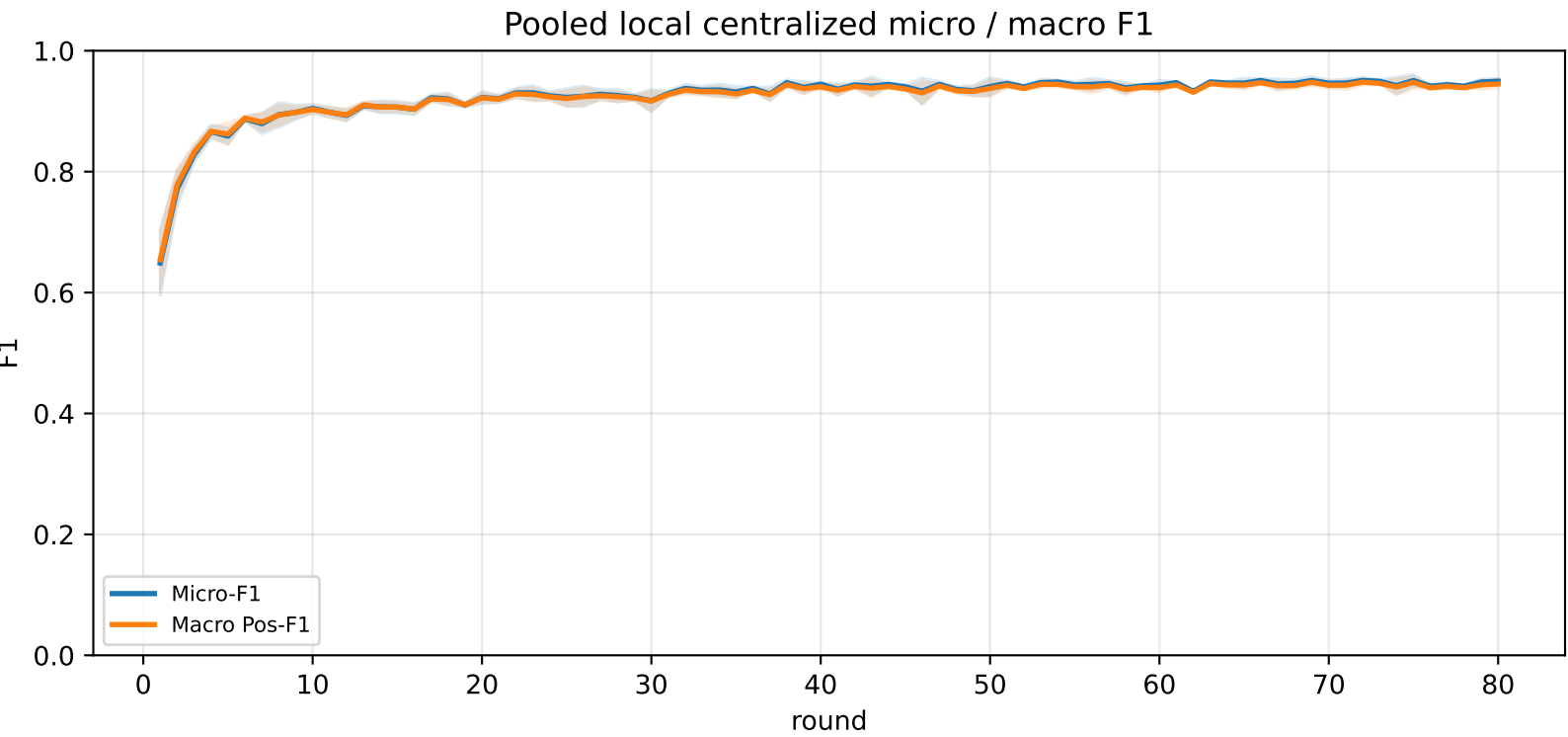
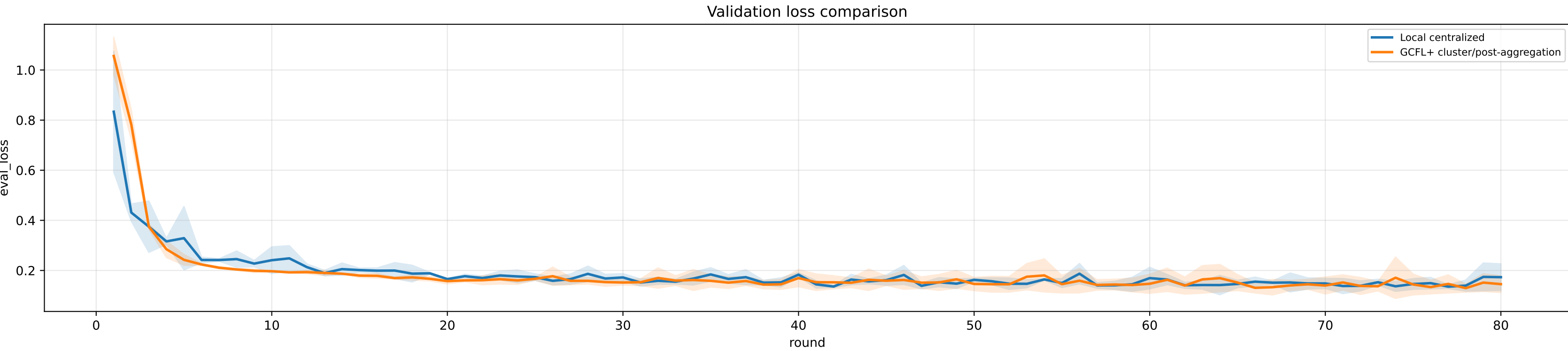
Delta vs GCFL+ (local minus pre / post)



Controlled masking level: Mid | GCFL+ diagnostics | family=masked_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle6 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 13.1%±0.0% / true 26.1%
masking=mid | mask_fraction=0.50 | task_jsd_mean=0.019 | task_jsd_median=0.019 | task_jsd_max=0.020 | family_centroid_jsd_mean=0.019

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.6% ± 0.4%	94.6% ± 0.4%	99.5% ± 0.1%	98.5% ± 0.4%	97.3% ± 0.2%	94.3% ± 1.0%	83.4% ± 2.5%
GCFL+	94.7% ± 1.0%	94.6% ± 1.1%	98.8% ± 0.6%	98.8% ± 0.4%	97.2% ± 0.5%	93.5% ± 1.7%	84.6% ± 3.1%



Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=0 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

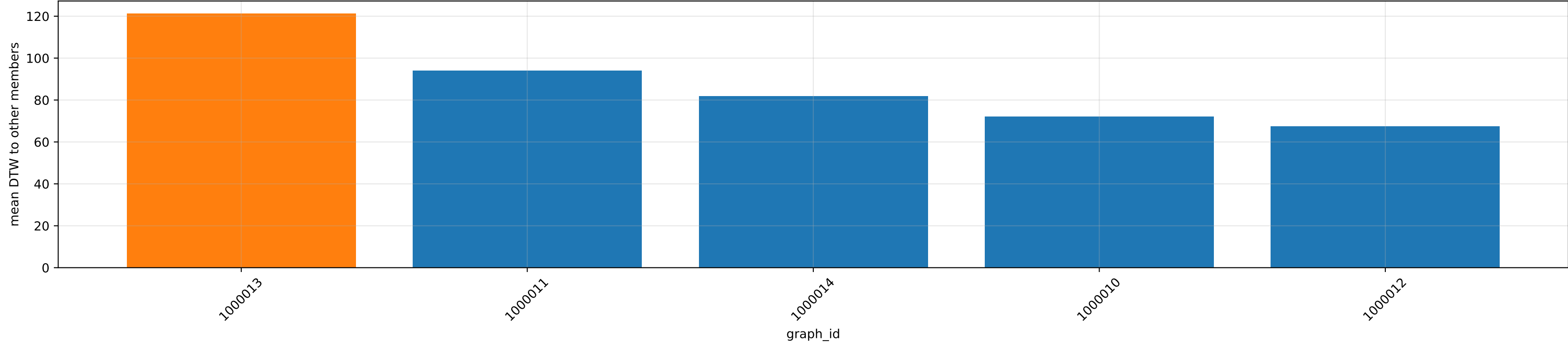
First applied split summary | seed=0

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
0	46	0	1 vs 4	1000013	1000013	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
121.189	nan	64.763

Per-member mean DTW



Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=0 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

First applied split pairwise DTW metadata | seed=0

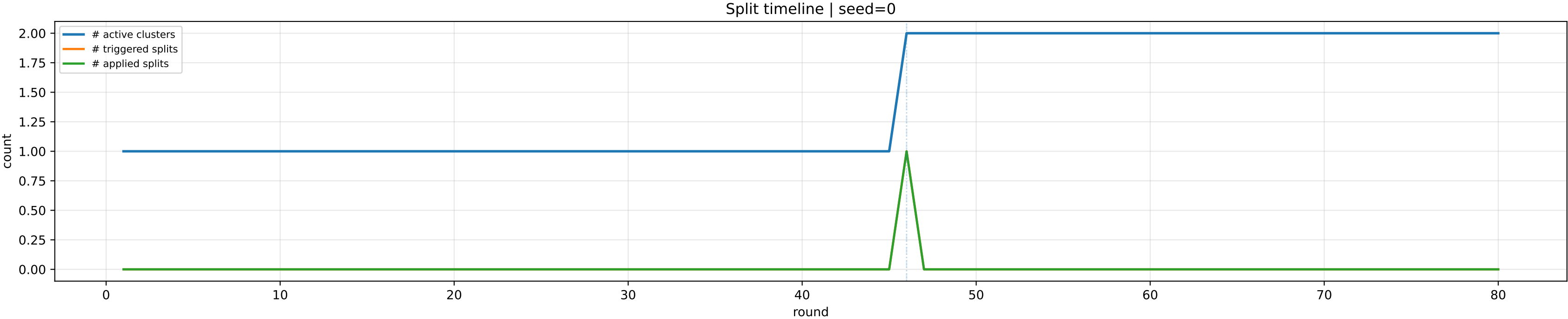
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
0	46	0	5	1000013.0	0.033	121.189	1

Pairwise DTW table | first applied split | seed=0
Global max DTW = 138.5 at (1000011, 1000013)

graph_id	1000010	1000011	1000012	1000013	1000014
1000010	0.0	84.8	47.9	108.2	47.6
1000011	84.8	0.0	48.2	138.5	104.5
1000012	47.9	48.2	0.0	118.2	55.6
1000013	108.2	138.5	118.2	0.0	119.8
1000014	47.6	104.5	55.6	119.8	0.0

Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=0 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069



Seed summary | seed=0

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0	46	46	1, 2	1:1, 2:4

Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=0 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069



Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

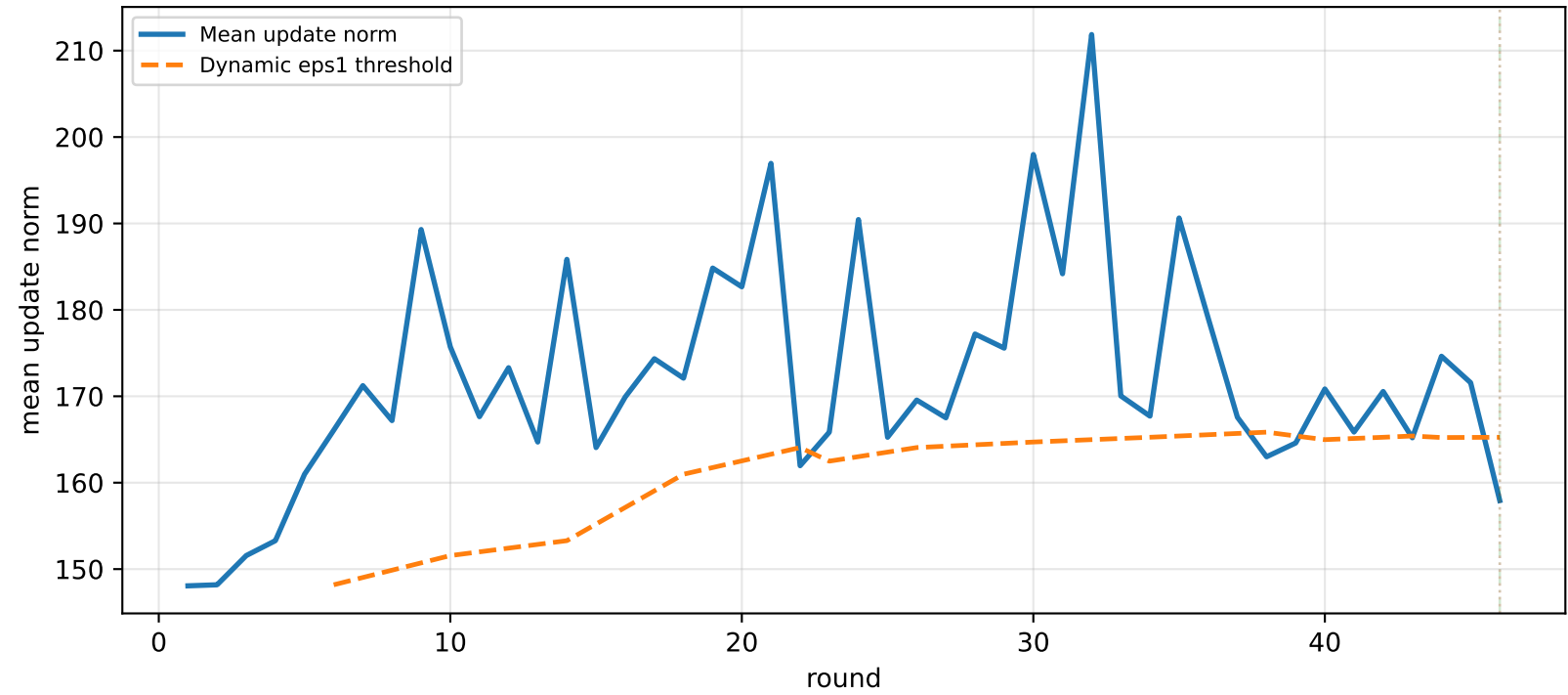
Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000013	3	masked_cycle5:1
2	4	1000010, 1000011, 1000012, 1000014	0, 1, 2, 4	masked_cycle2:1 , masked_cycle3:1 , masked_cycle4:1 , masked_cycle6:1

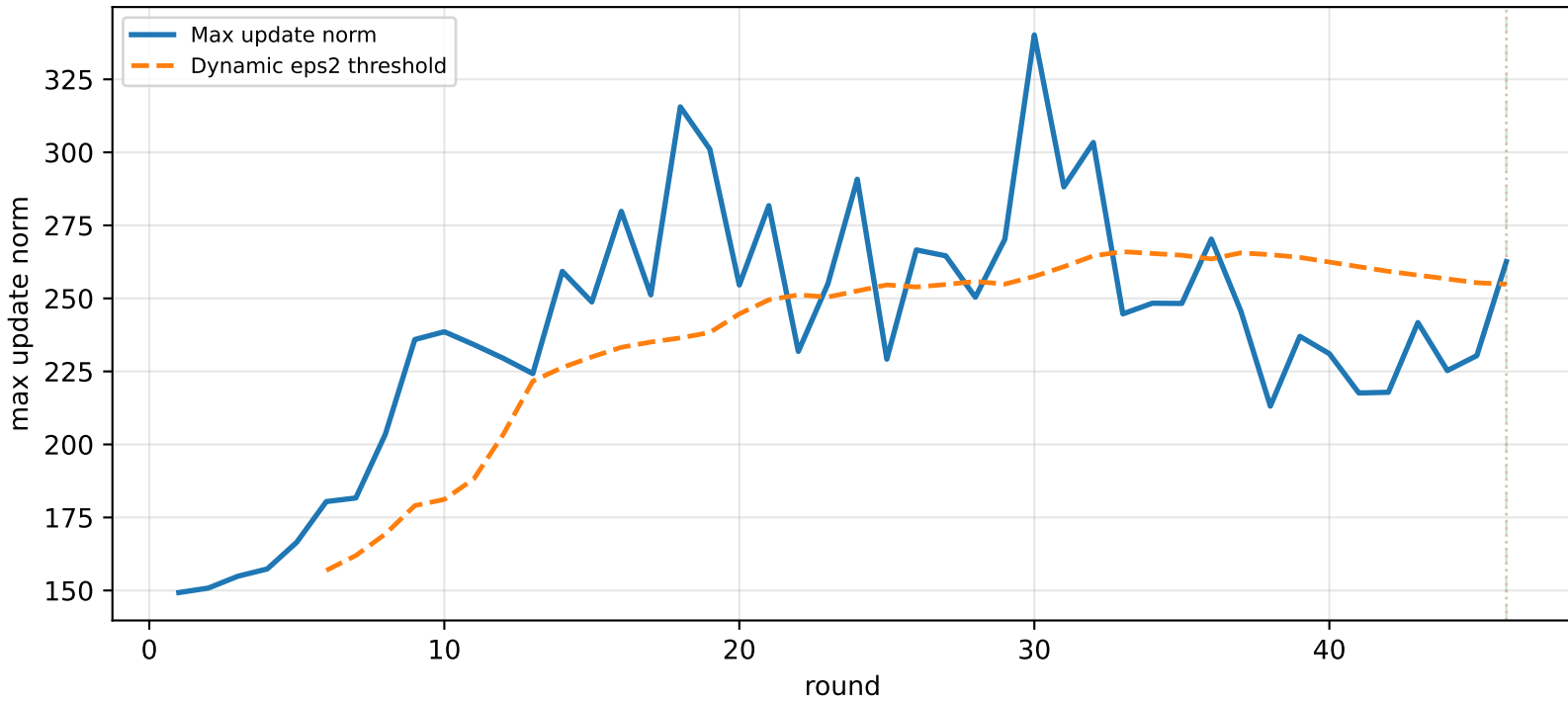
Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=0 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

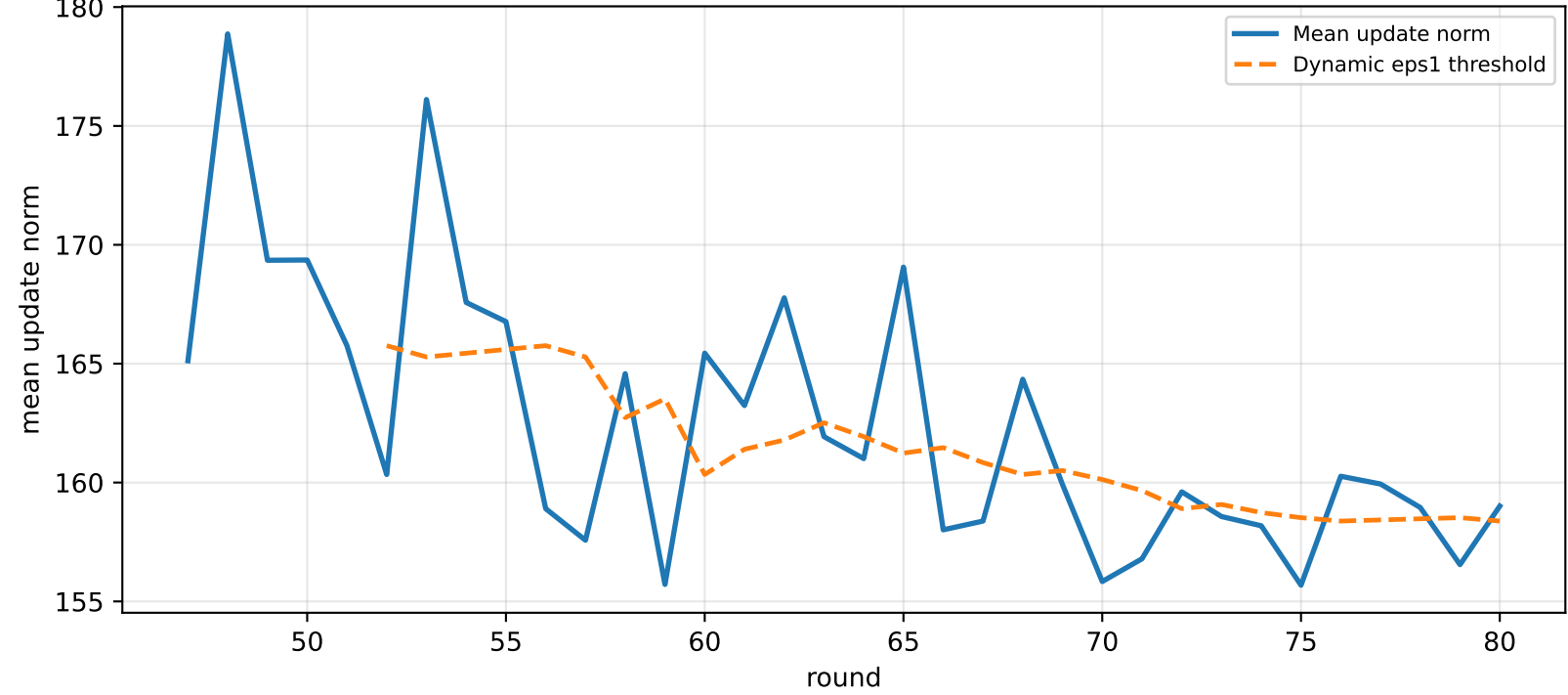
Cluster 0 | Mean update norm



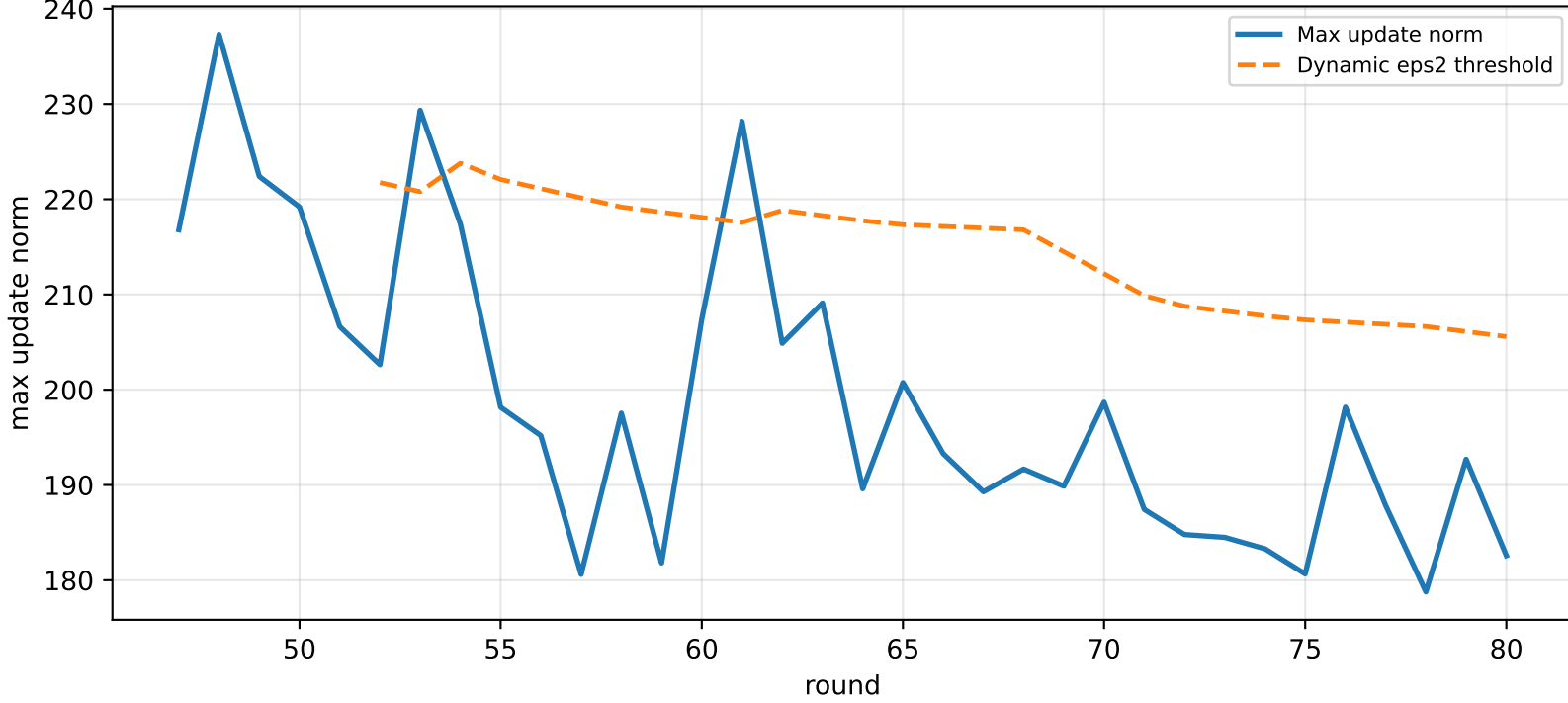
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



Cluster 2 | Max update norm



Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=1 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

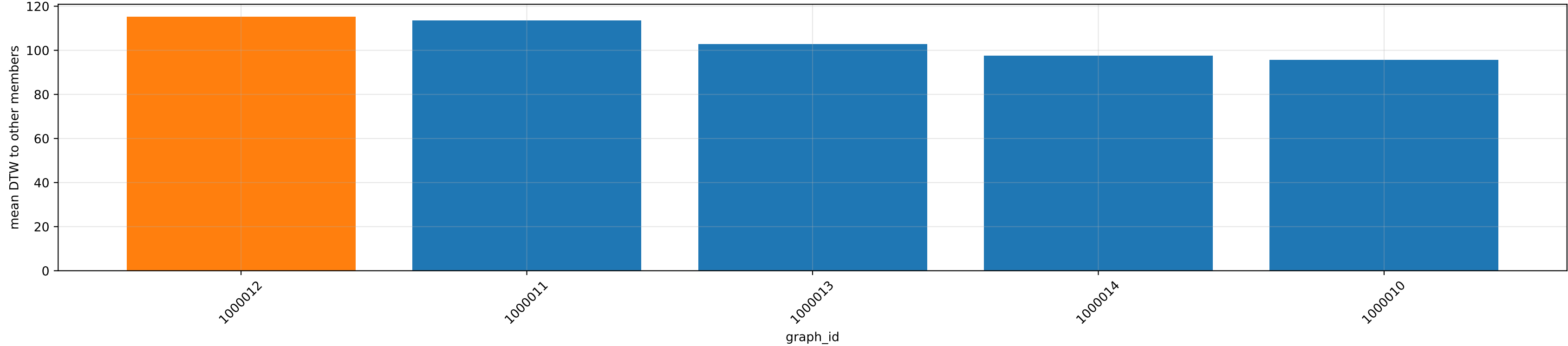
First applied split summary | seed=1

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
1	51	0	1 vs 4	1000012	1000012	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
115.137	nan	98.151

Per-member mean DTW



Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=1 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

First applied split pairwise DTW metadata | seed=1

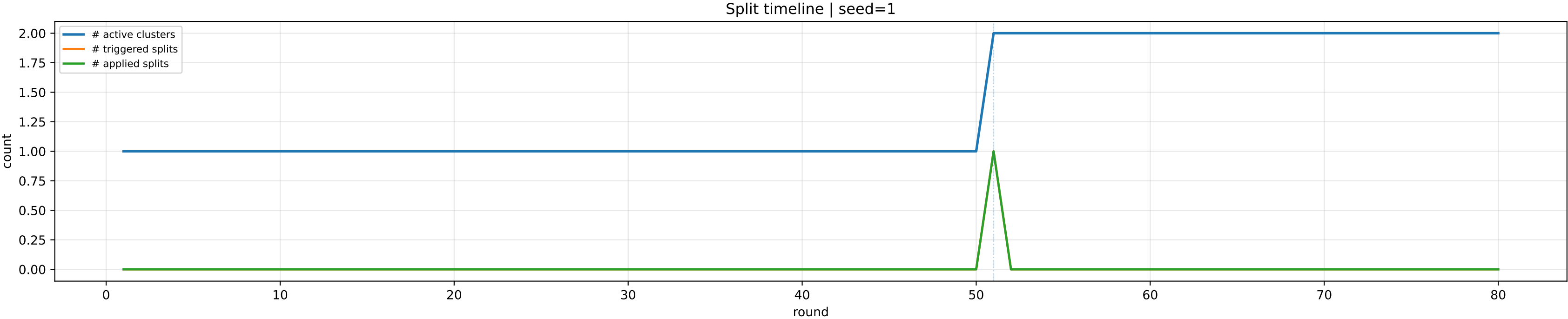
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
1	51	0	5	1000012.0	0.035	115.137	1

Pairwise DTW table | first applied split | seed=1
Global max DTW = 139.6 at (1000011, 1000012)

graph_id	1000010	1000011	1000012	1000013	1000014
1000010	0.0	92.5	101.7	100.2	88.4
1000011	92.5	0.0	139.6	103.8	118.4
1000012	101.7	139.6	0.0	121.3	97.9
1000013	100.2	103.8	121.3	0.0	85.6
1000014	88.4	118.4	97.9	85.6	0.0

Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=1 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069



Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0	51	51	1, 2	1:1, 2:4

Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=1 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069



Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=1 | final active clusters

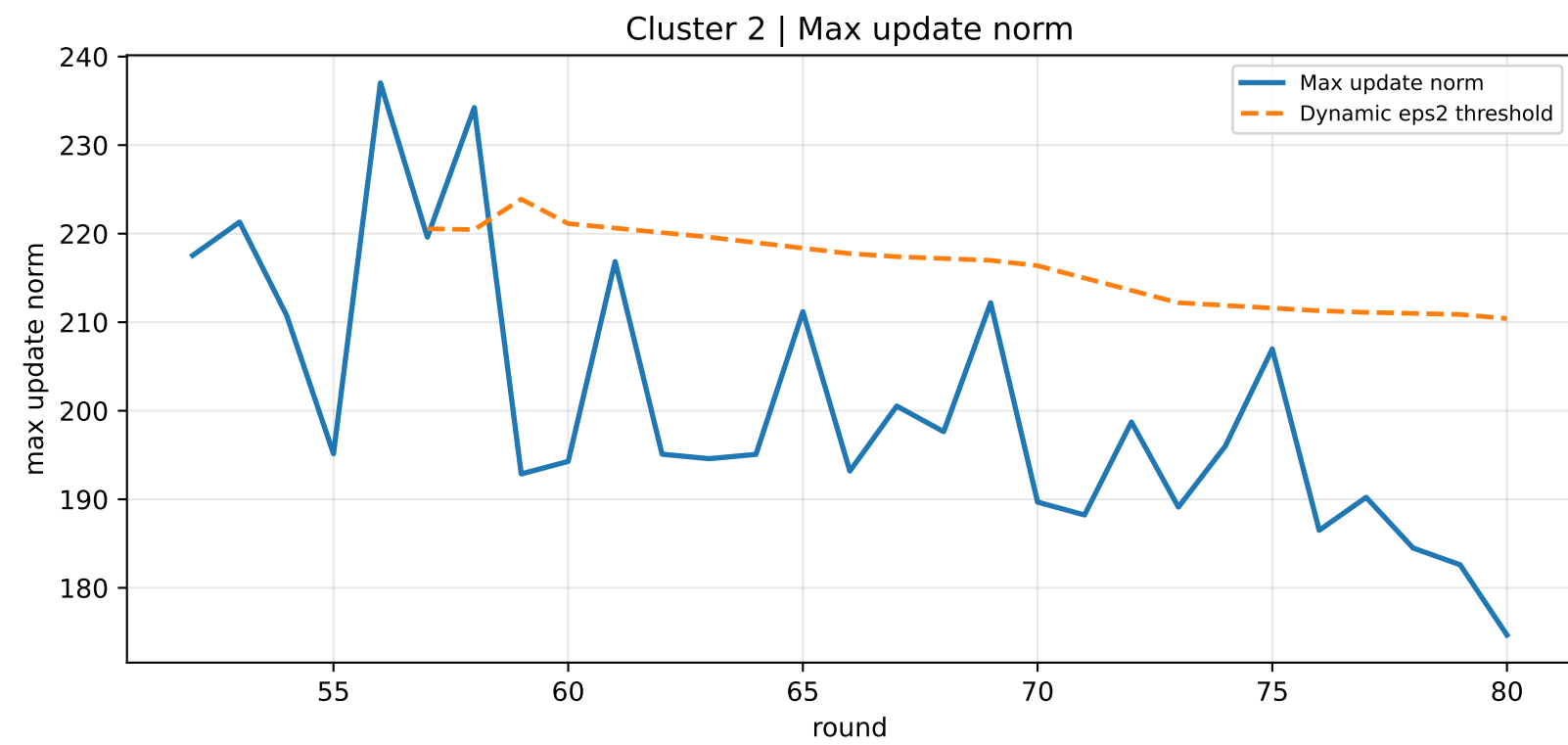
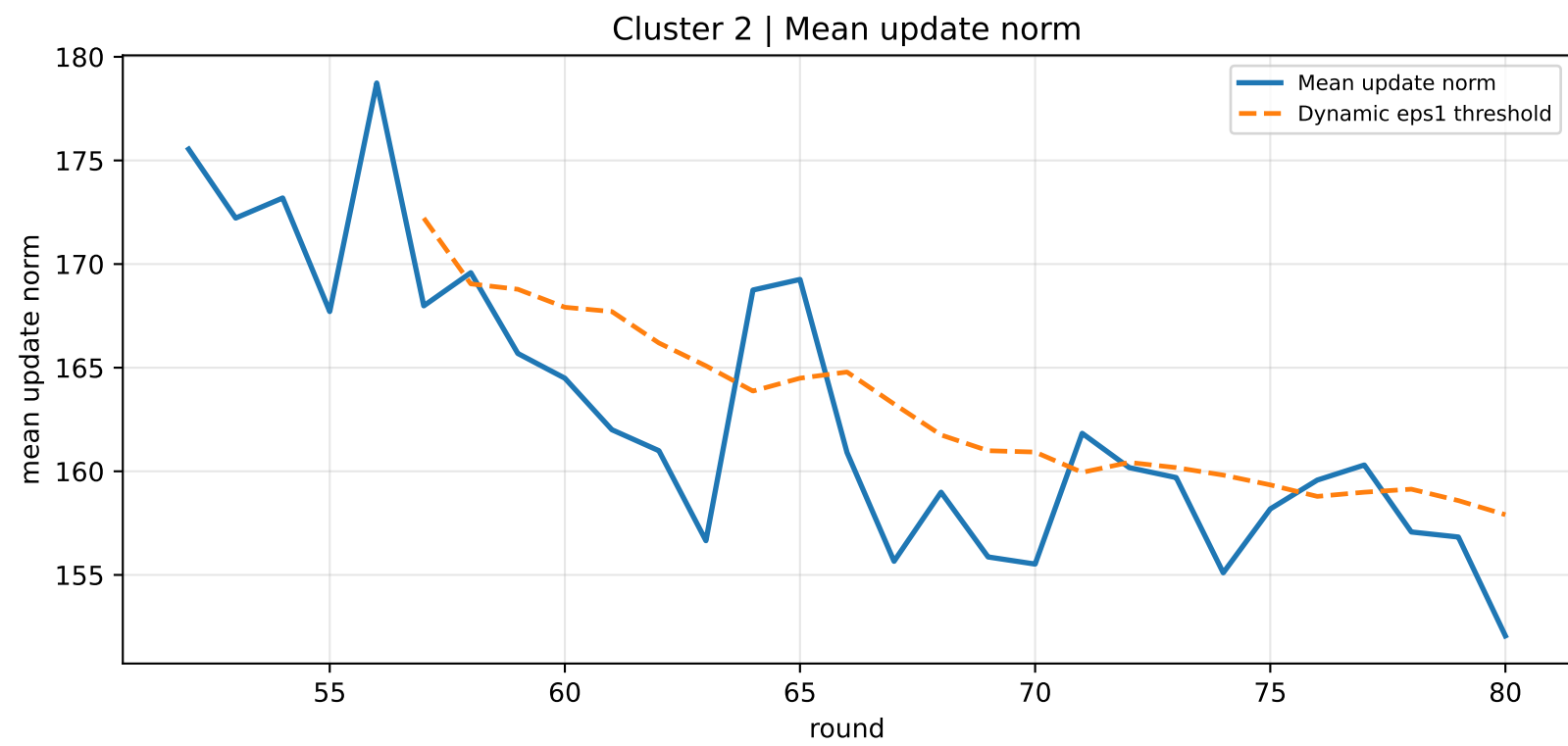
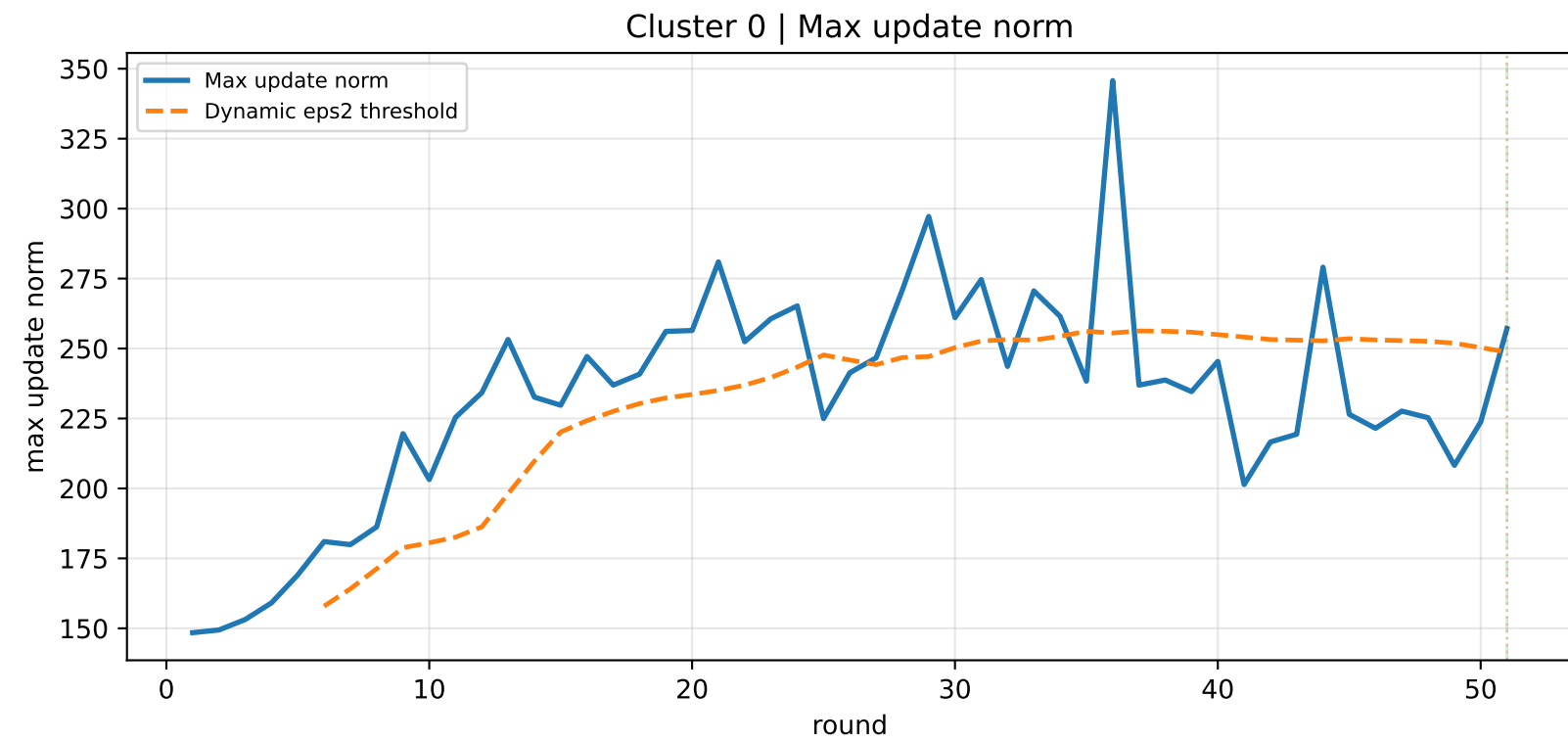
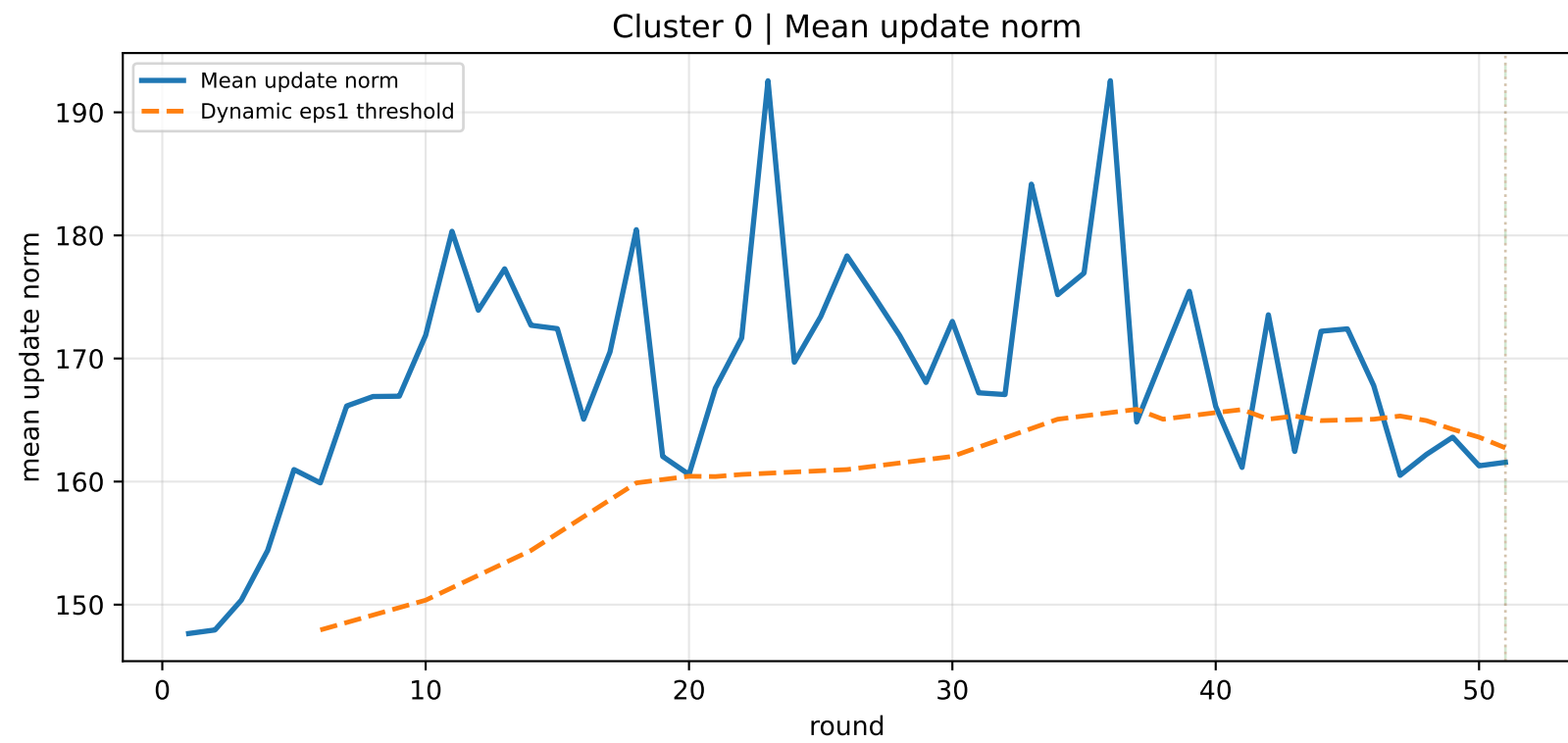
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

Final active clusters | seed=1

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000012	2	masked_cycle4:1
2	4	1000010, 1000011, 1000013, 1000014	0, 1, 3, 4	masked_cycle2:1 , masked_cycle3:1 , masked_cycle5:1 , masked_cycle6:1

**Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=1 | norm / threshold panels (0-2)**

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069



Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

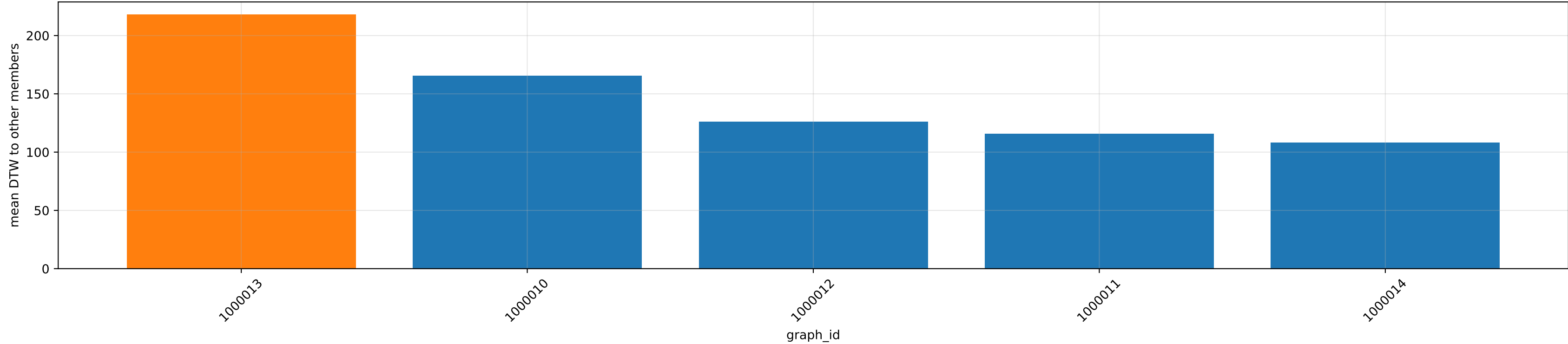
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	43	0	1 vs 4	1000013	1000013	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
217.957	nan	99.197

Per-member mean DTW



Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over 5 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

First applied split pairwise DTW metadata | seed=2

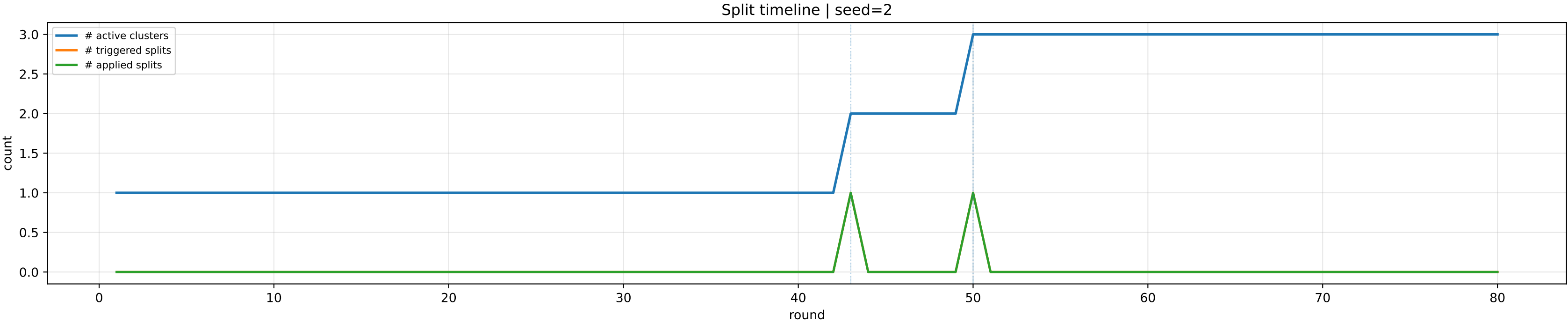
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	43	0	5	1000013.0	0.019	217.957	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 294.3 at (1000010, 1000013)

graph_id	1000010	1000011	1000012	1000013	1000014
1000010	0.0	135.0	127.1	294.3	105.3
1000011	135.0	0.0	97.2	176.9	54.3
1000012	127.1	97.2	0.0	203.5	76.2
1000013	294.3	176.9	203.5	0.0	197.2
1000014	105.3	54.3	76.2	197.2	0.0

Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=2 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

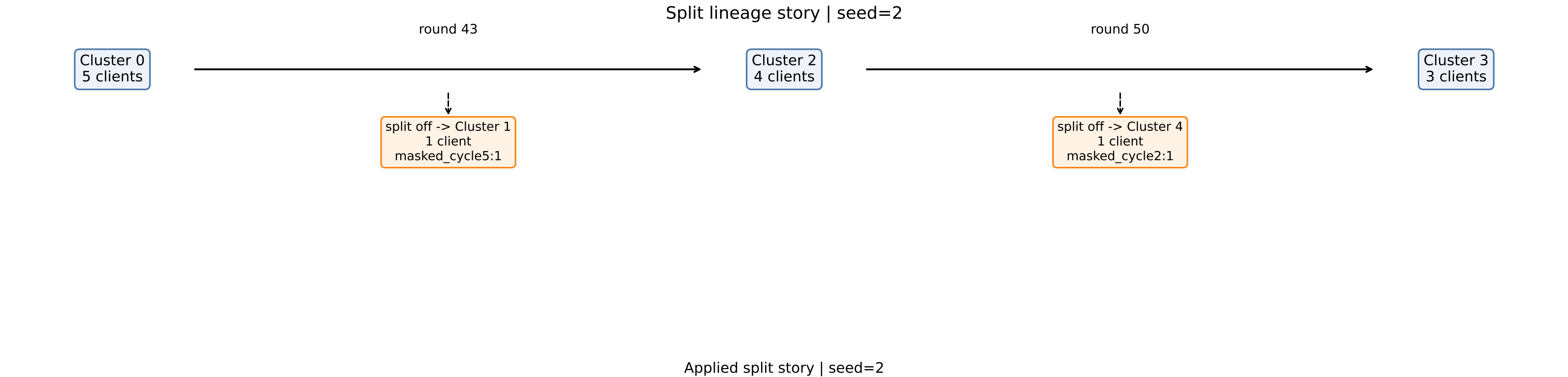


Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	43, 50	43, 50	1, 3, 4	1:1, 3:3, 4:1

Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=2 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069



Applied split story seed=2							
round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
43	0	5	164.255 < 165.064 , 273.964 > 265.838	C1 (1 client)	masked_cycle5:1	C2 (4 clients)	masked_cycle2:1 , masked_cycle3:1 , masked_cycle4:1 , masked_cycle6:1
50	2	4	164.298 < 165.042 , 303.754 > 240.125	C4 (1 client)	masked_cycle2:1	C3 (3 clients)	masked_cycle3:1 , masked_cycle4:1 , masked_cycle6:1

Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=2 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

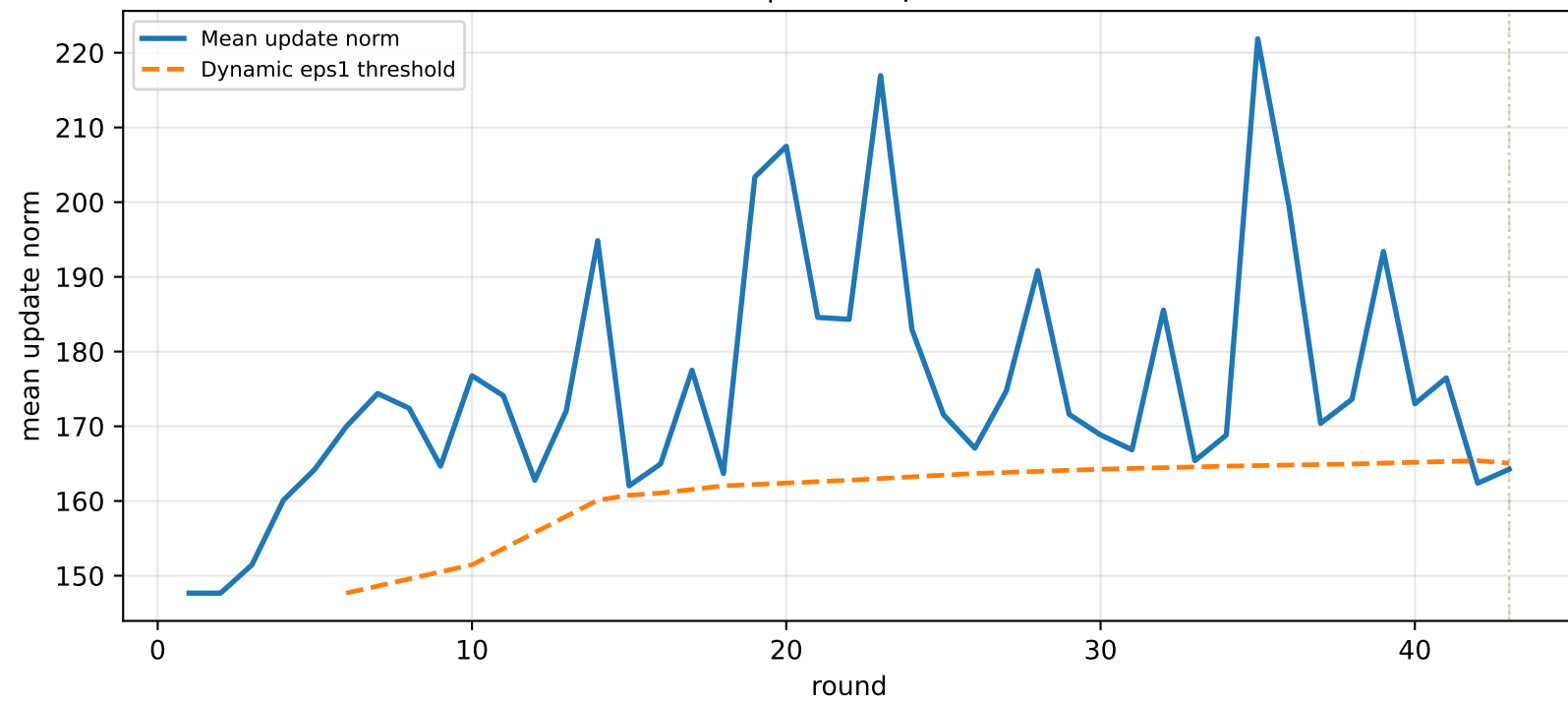
Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
1	1	1000013	3	masked_cycle5:1
3	3	1000011, 1000012, 1000014	1, 2, 4	masked_cycle3:1 , masked_cycle4:1 , masked_cycle6:1
4	1	1000010	0	masked_cycle2:1

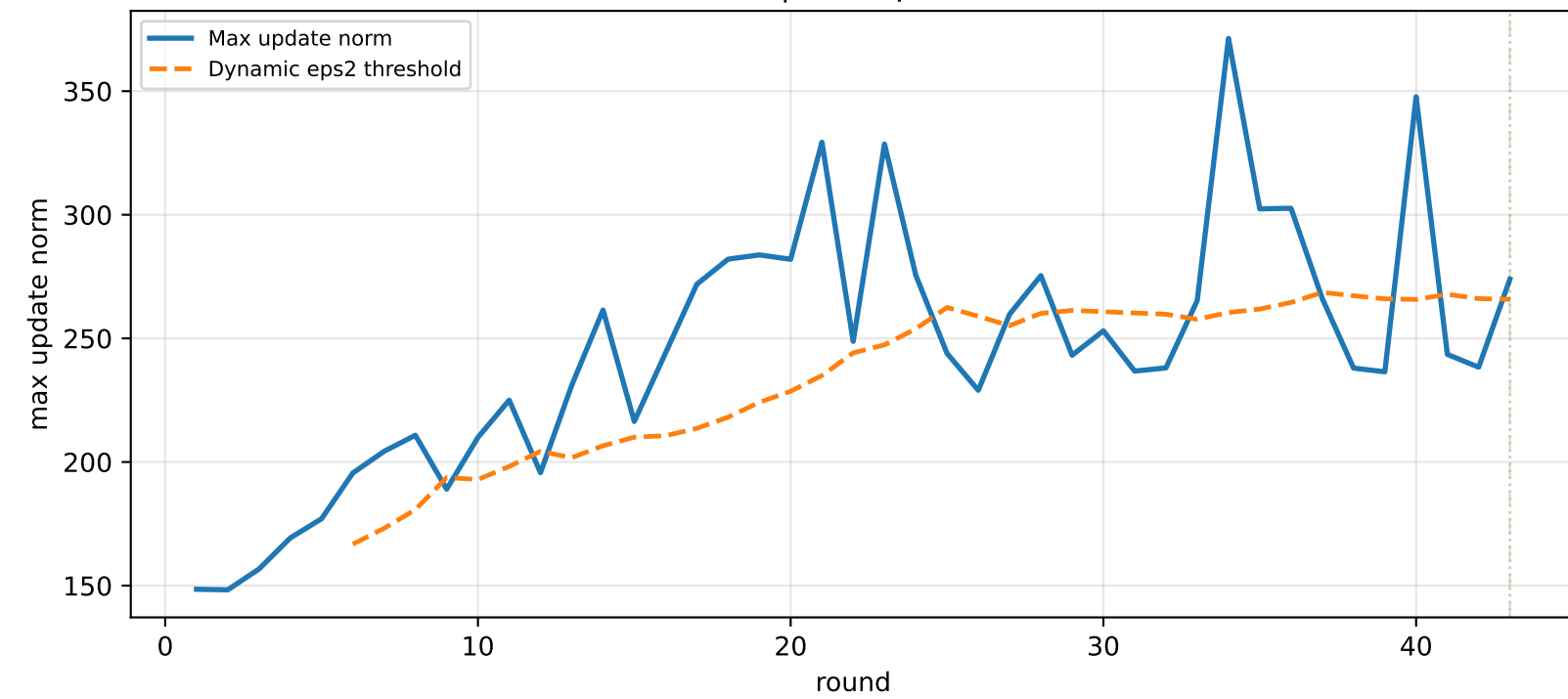
**Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=2 | norm / threshold panels (0-2)**

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

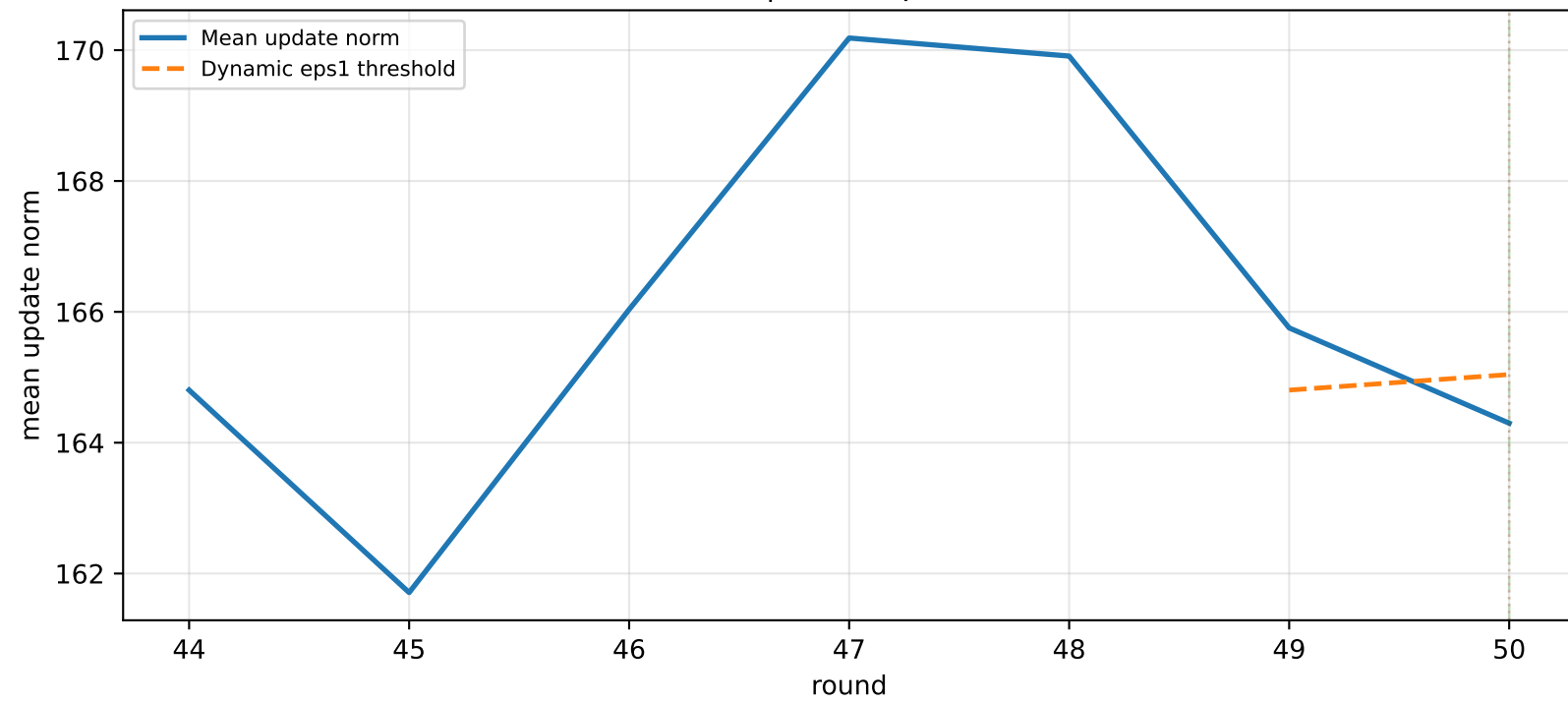
Cluster 0 | Mean update norm



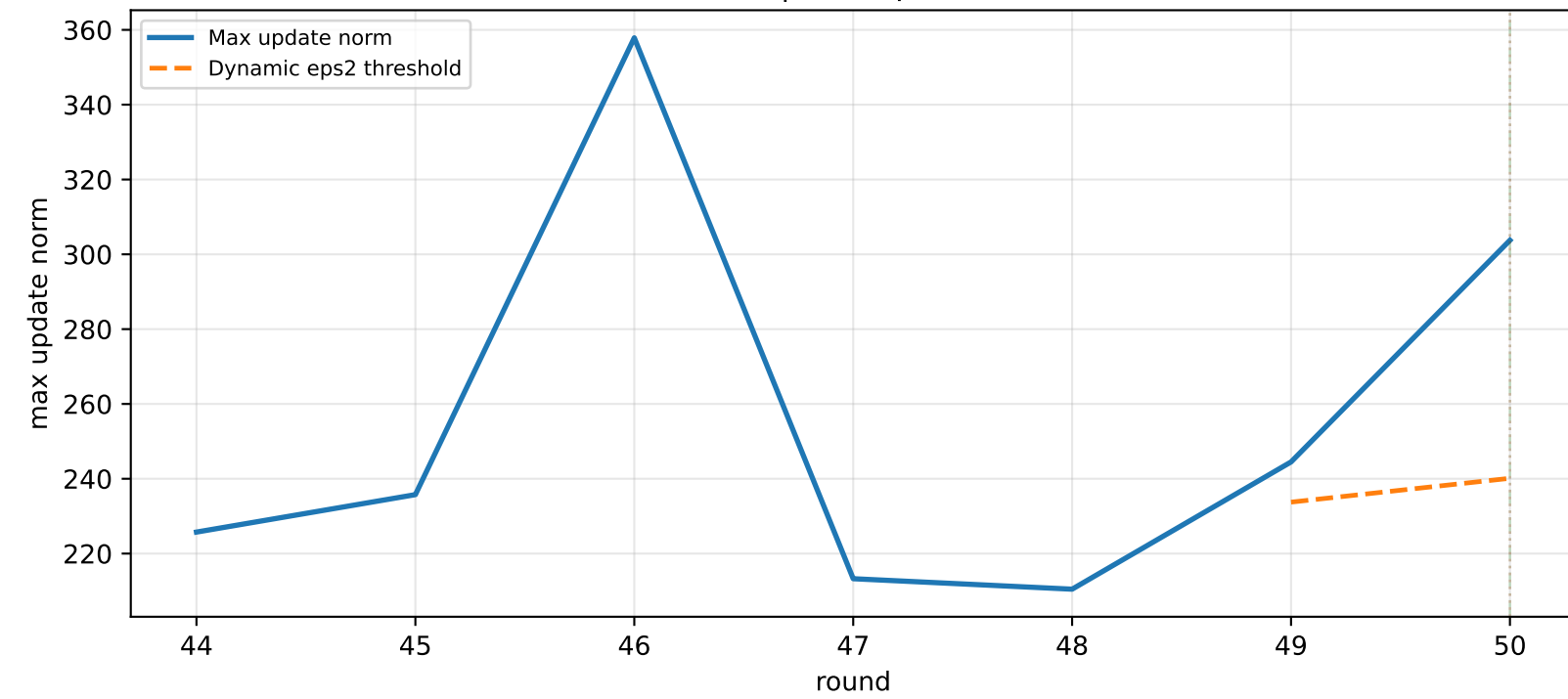
Cluster 0 | Max update norm



Cluster 2 | Mean update norm

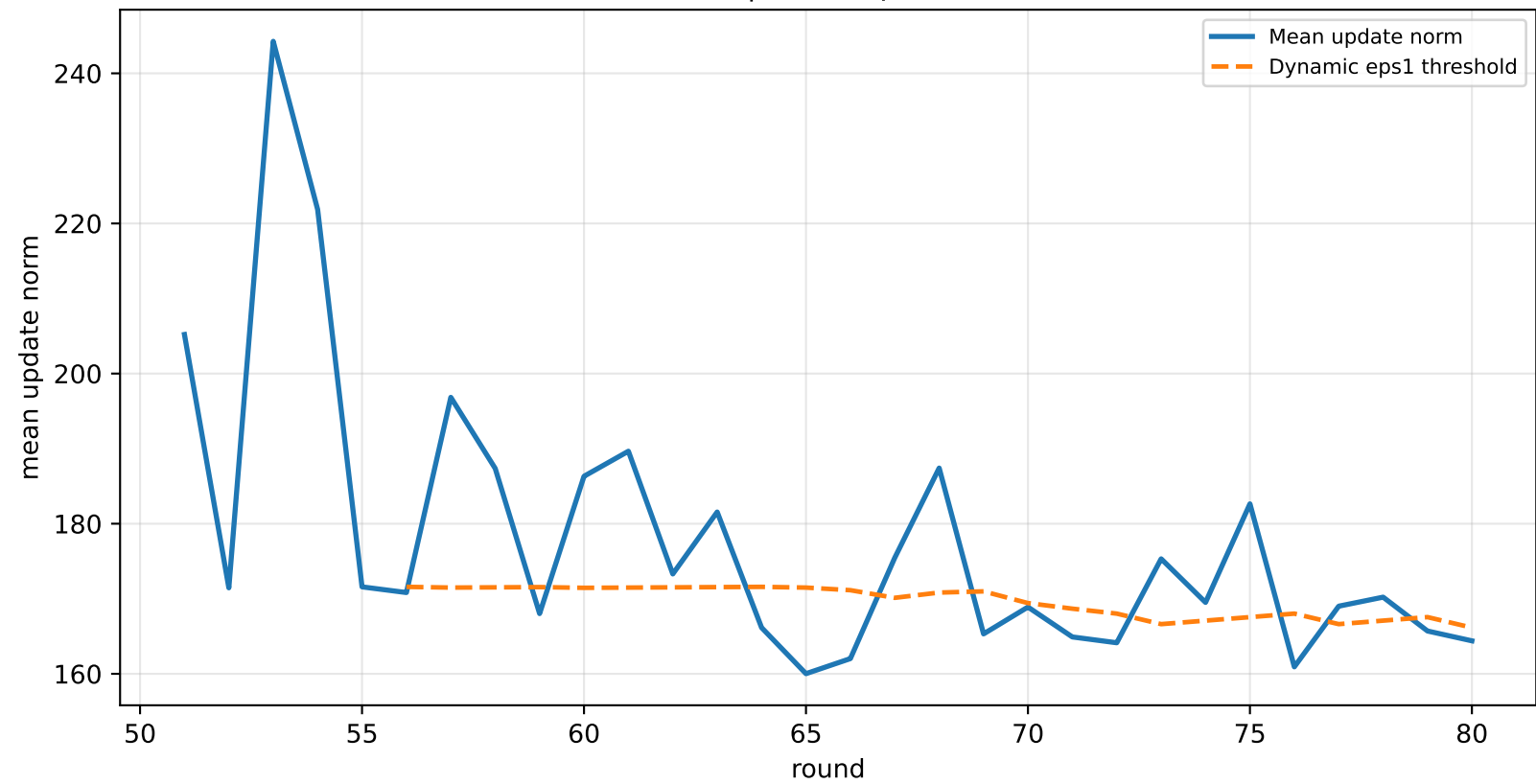


Cluster 2 | Max update norm

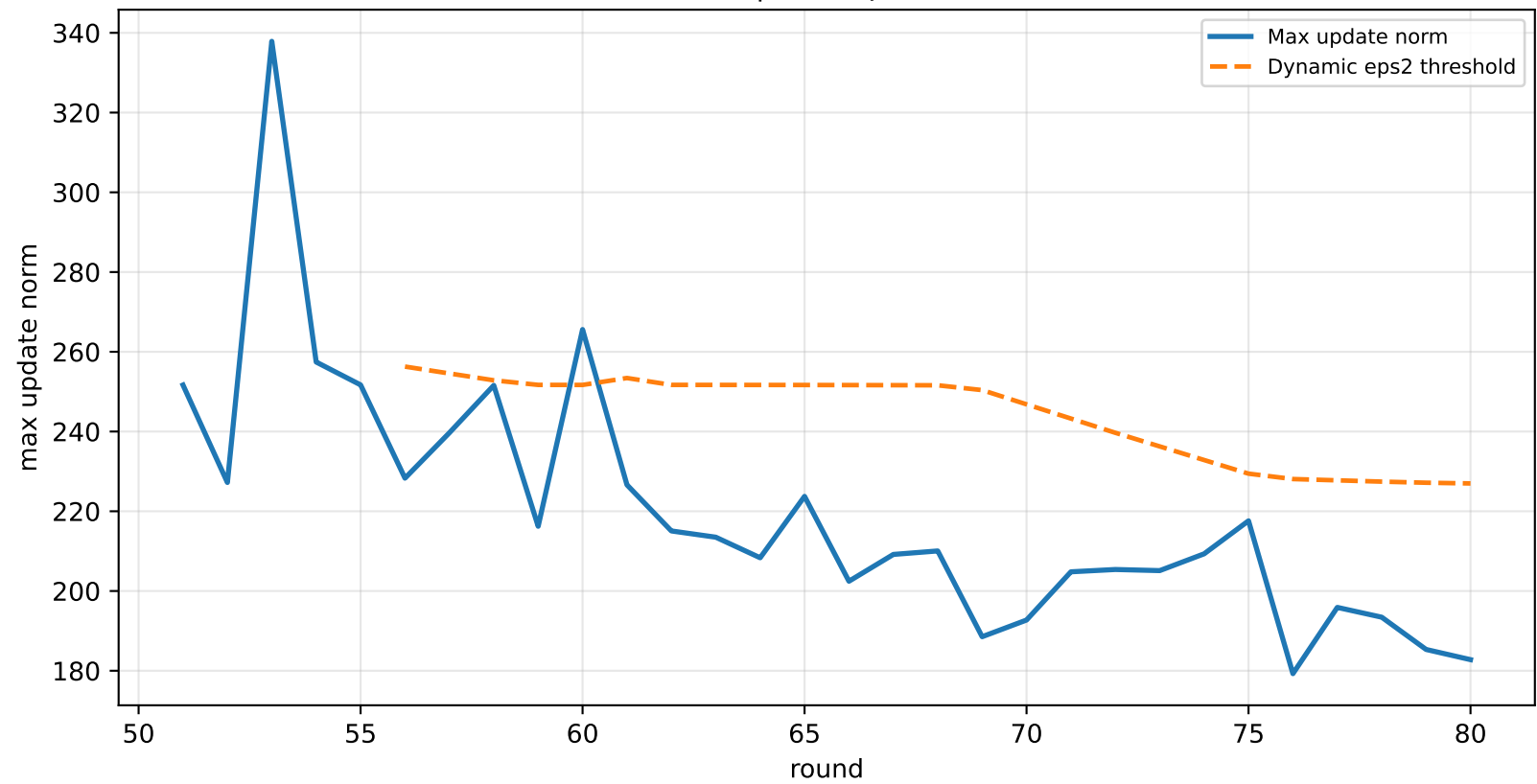


Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | seed=2 | norm / threshold panels (3-3)
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

Cluster 3 | Mean update norm



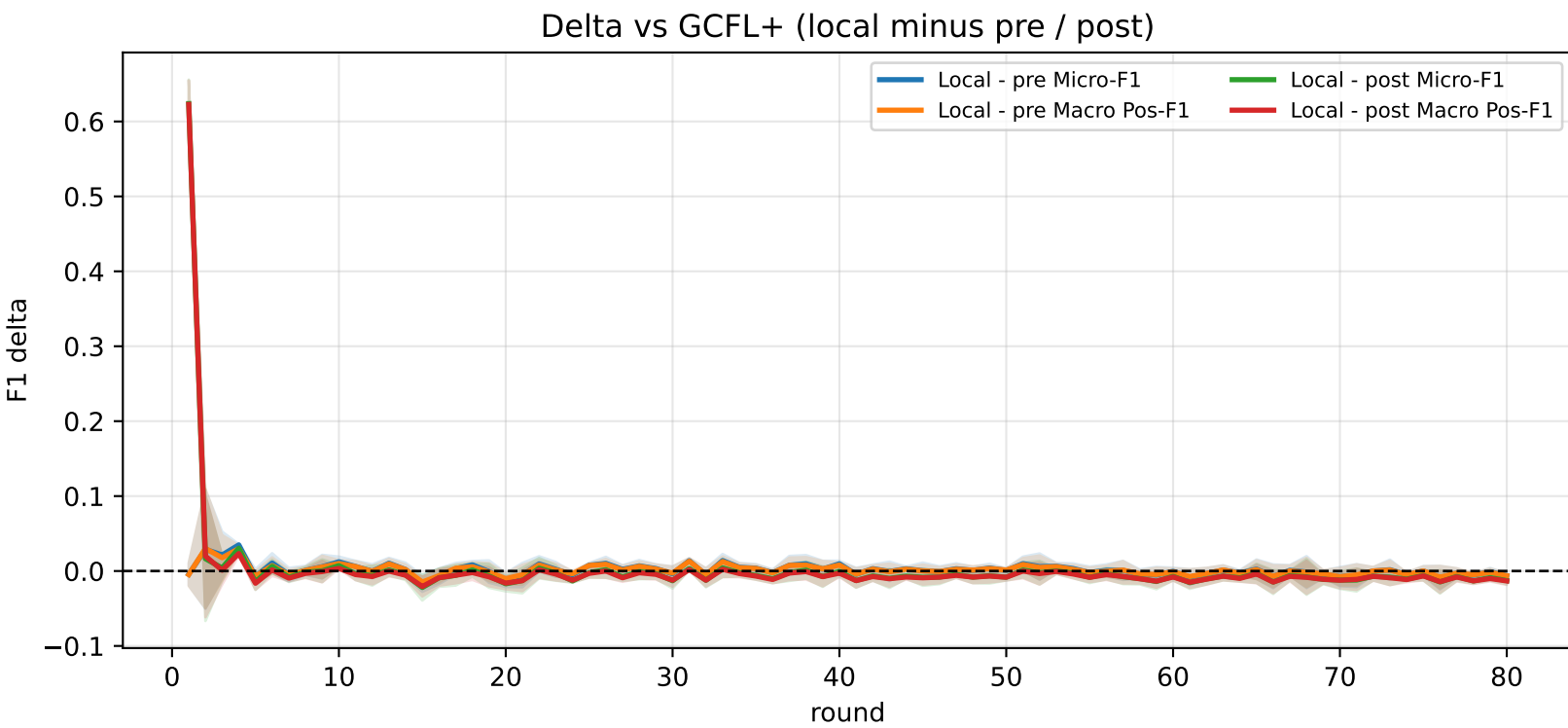
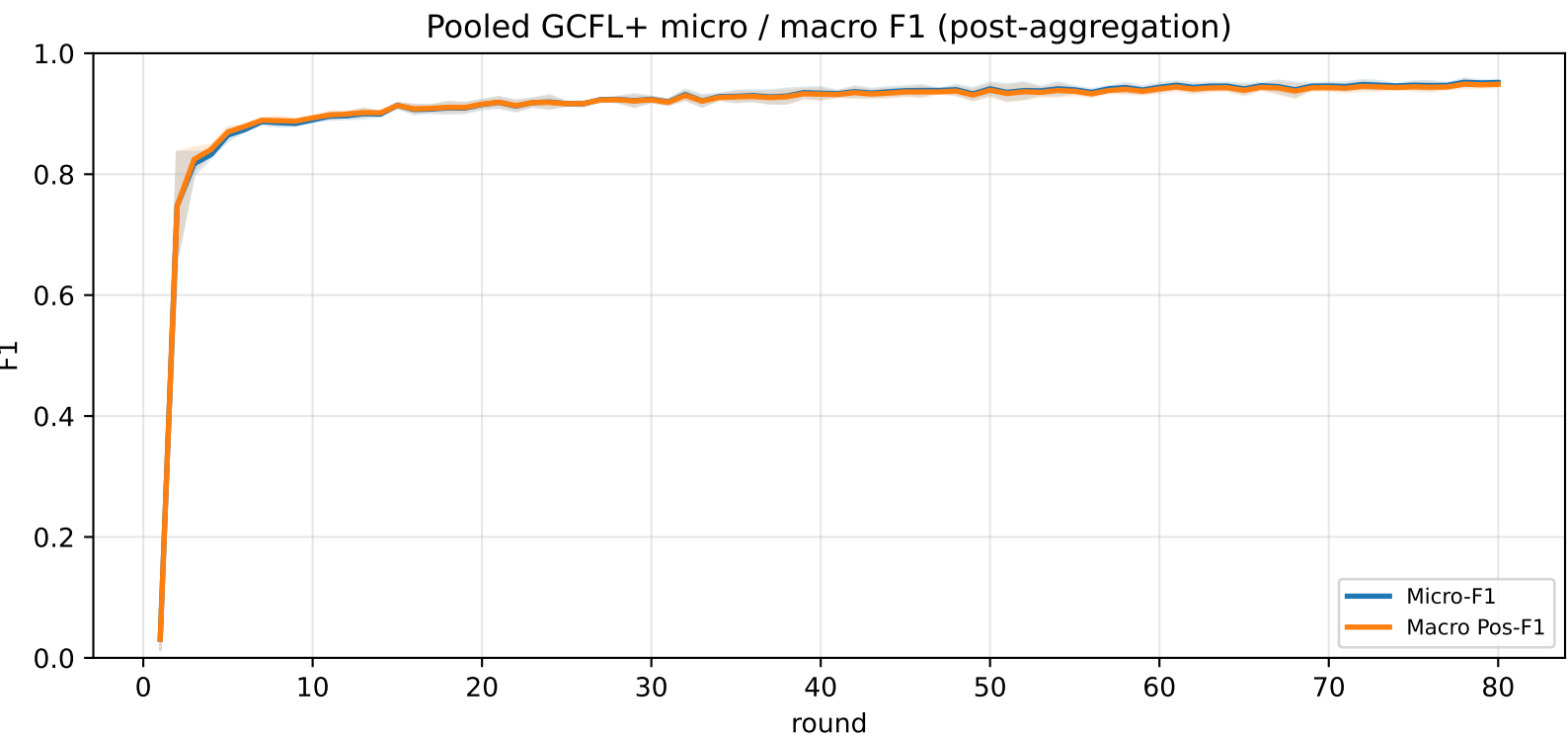
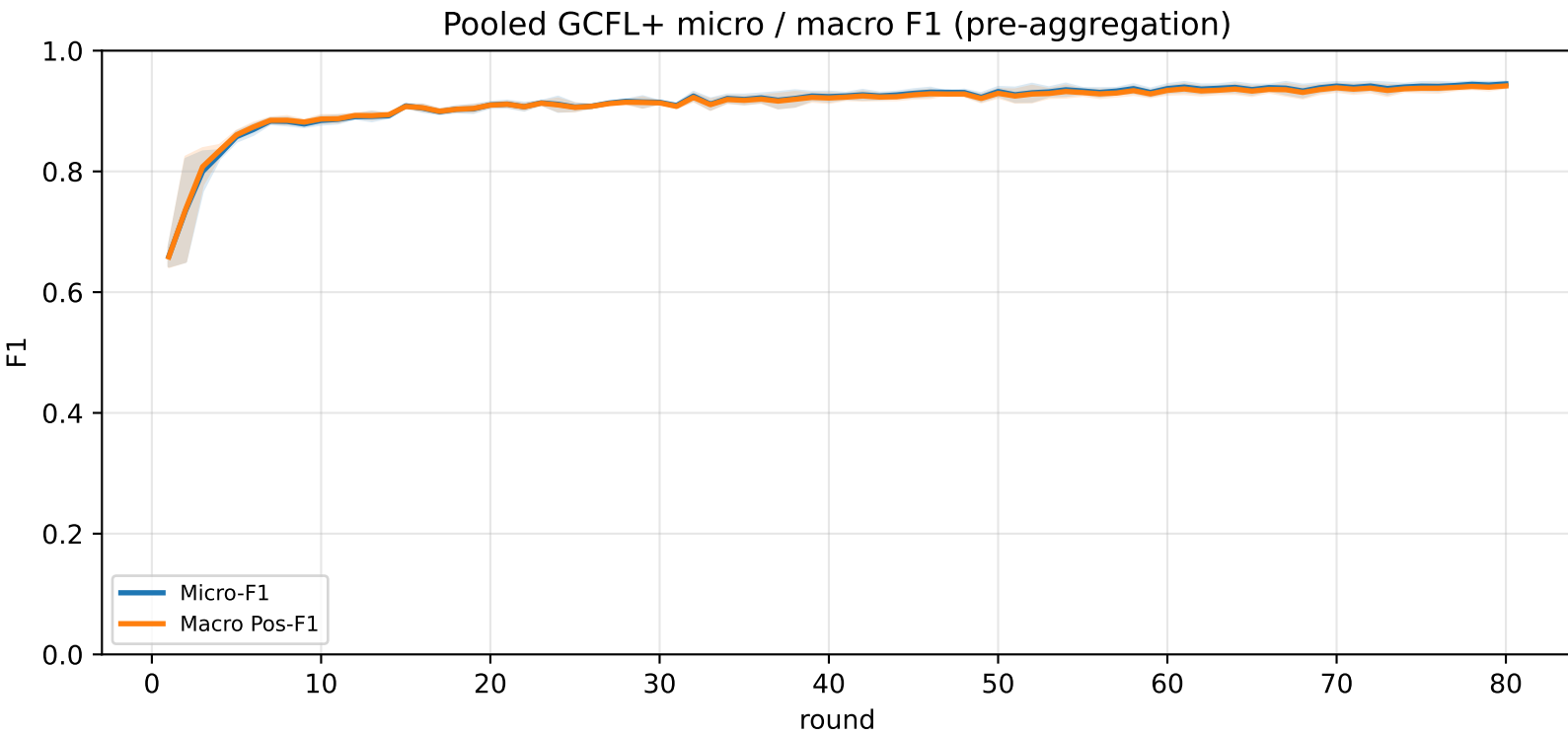
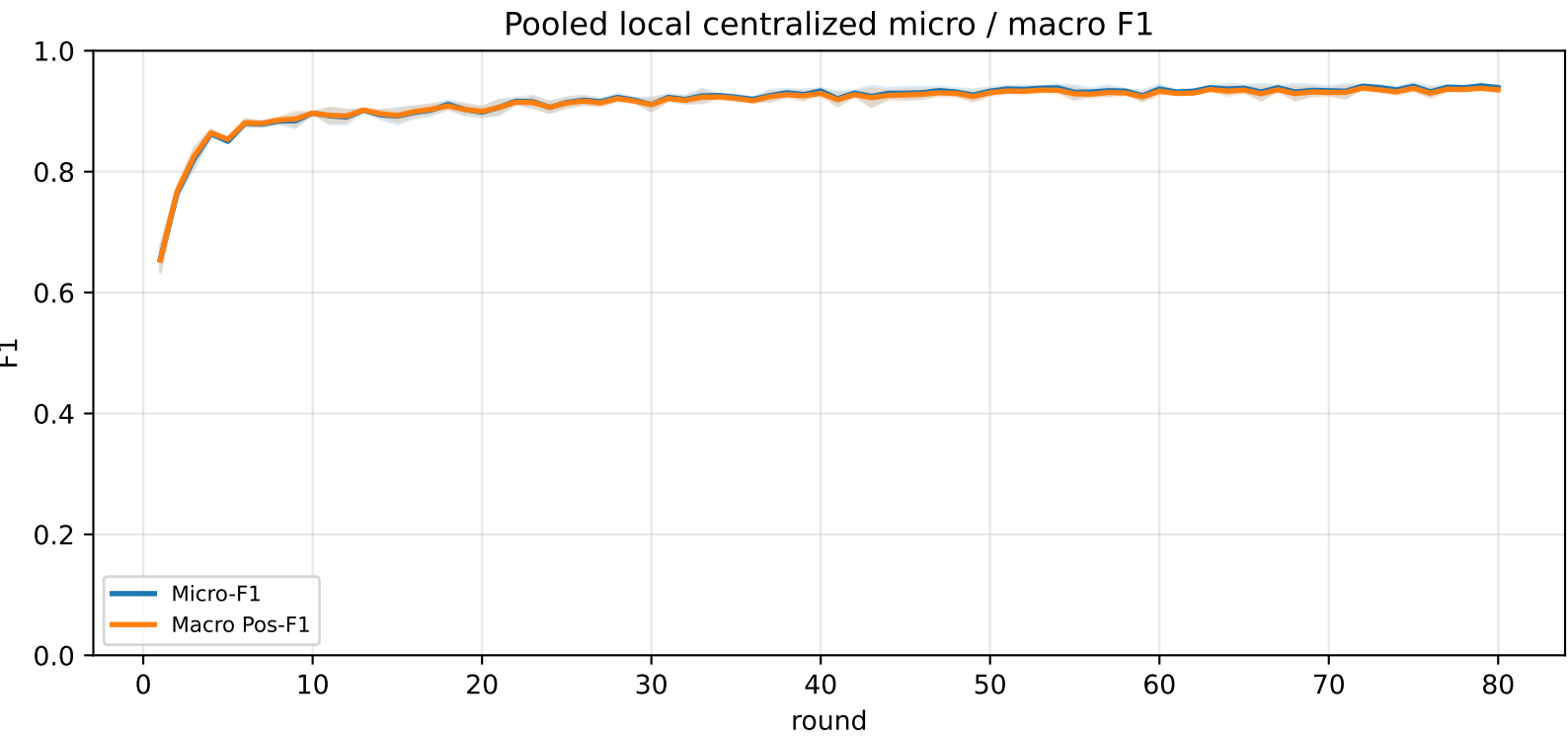
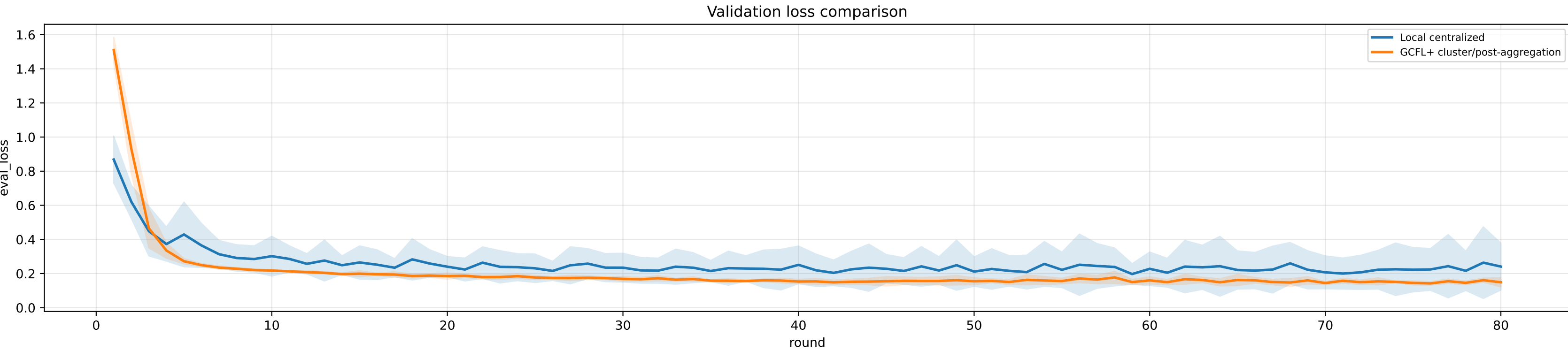
Cluster 3 | Max update norm



Controlled masking level: High | GCFL+ diagnostics | aggregated / pooled over
5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

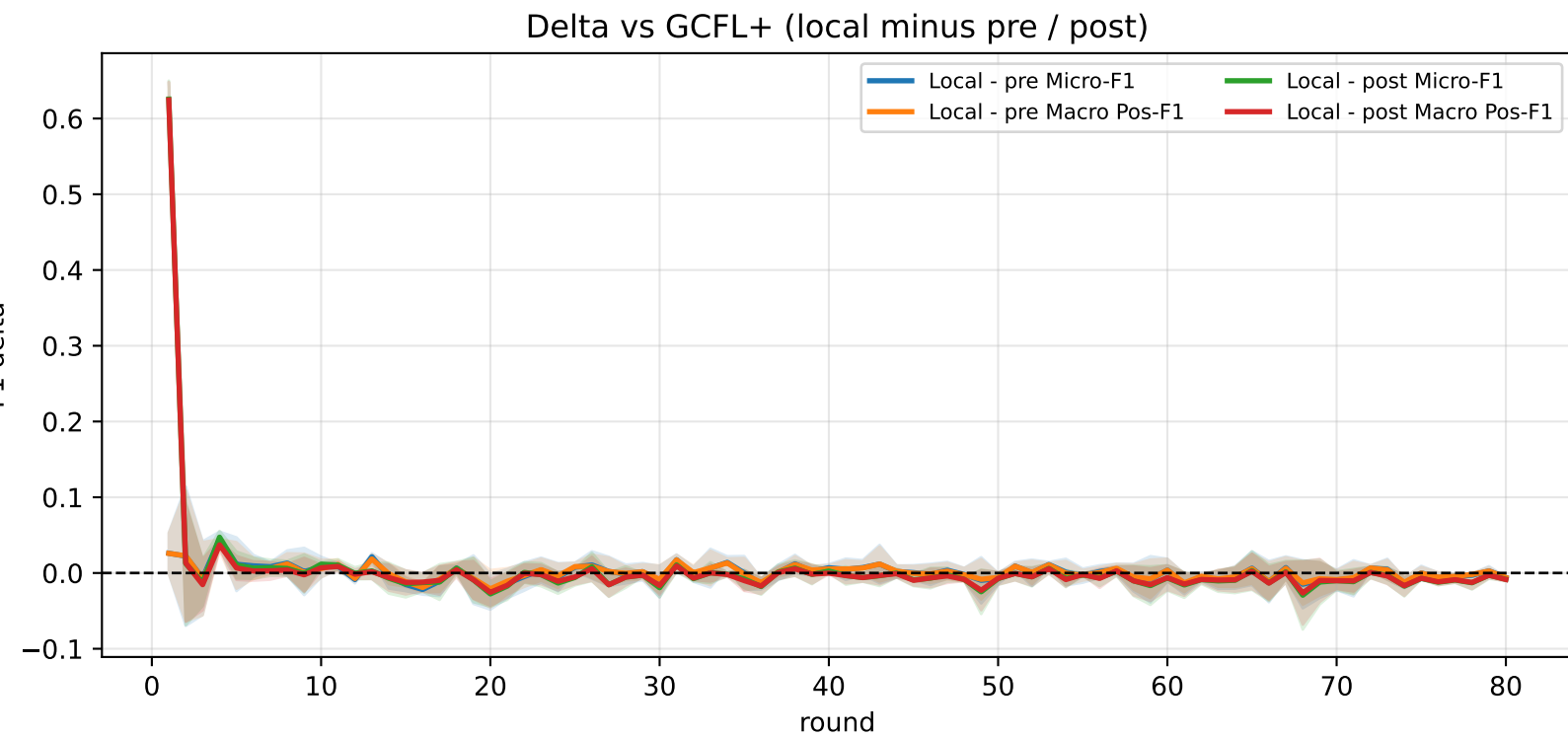
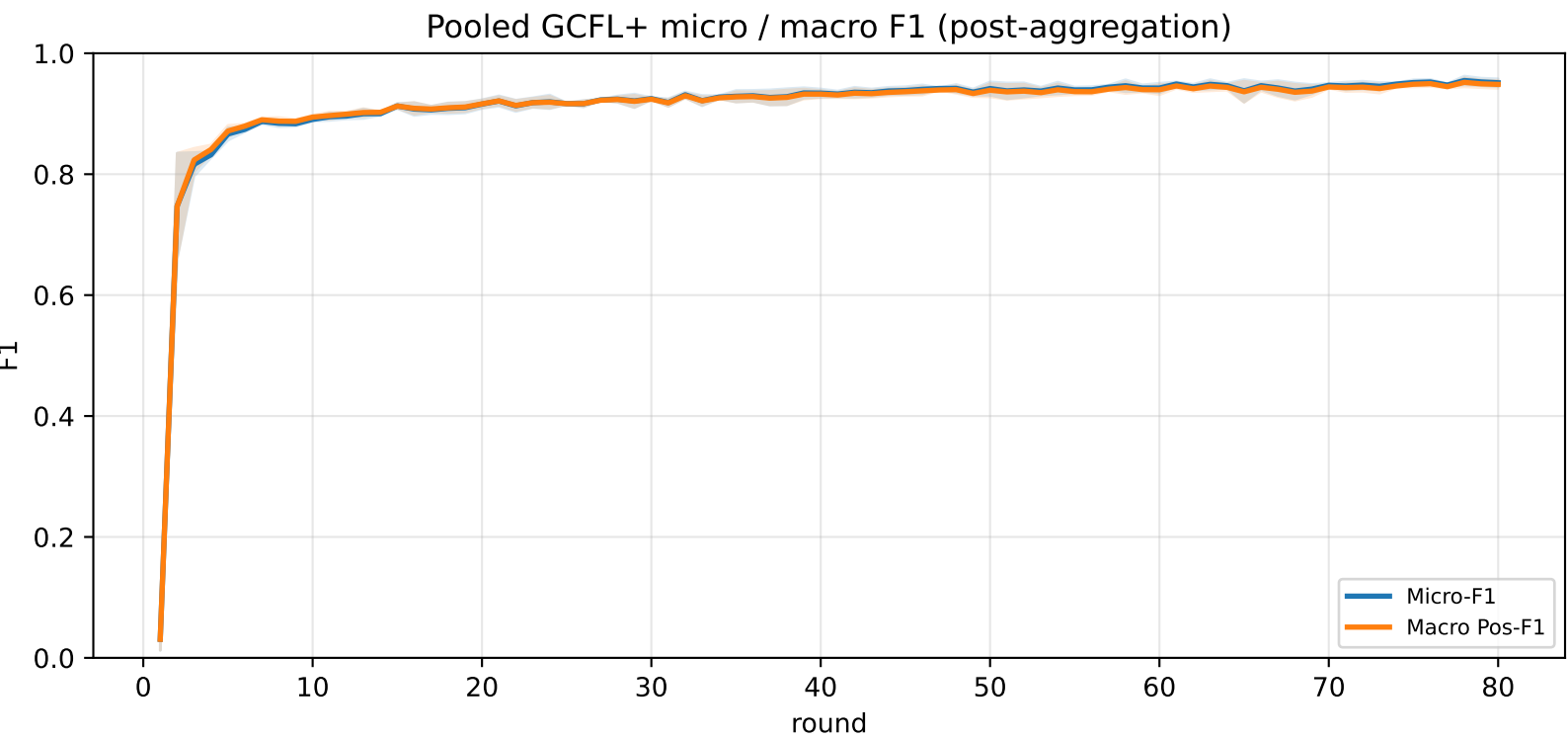
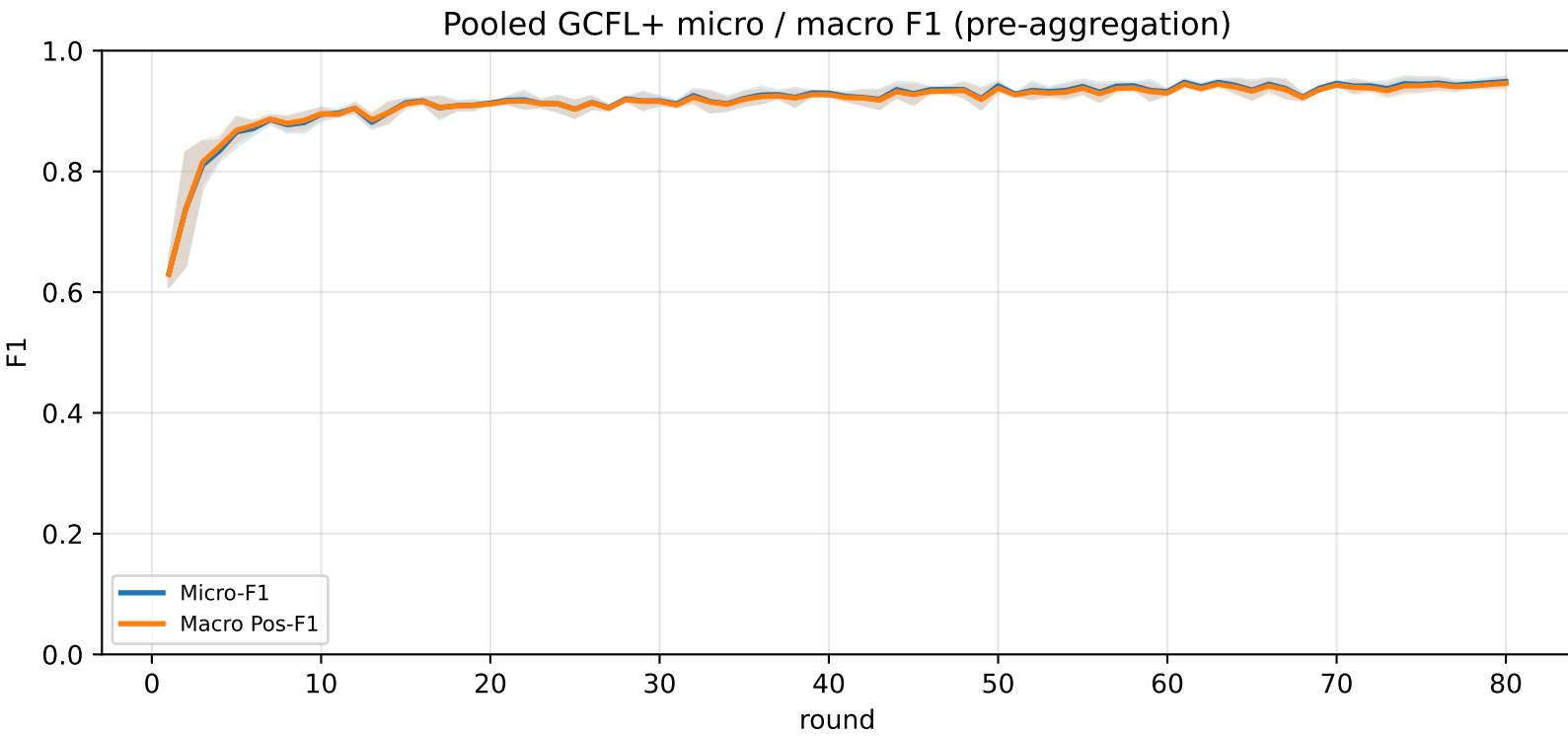
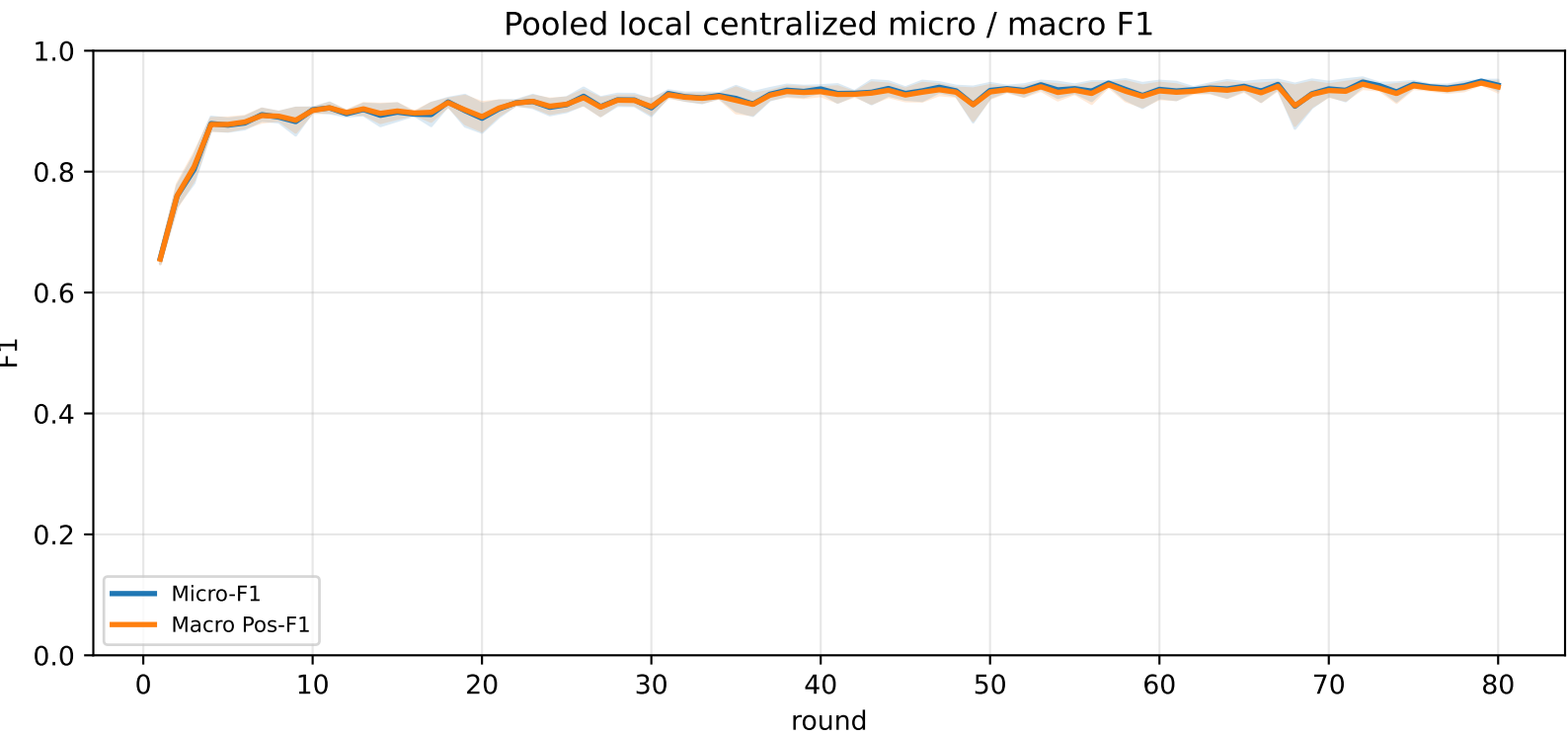
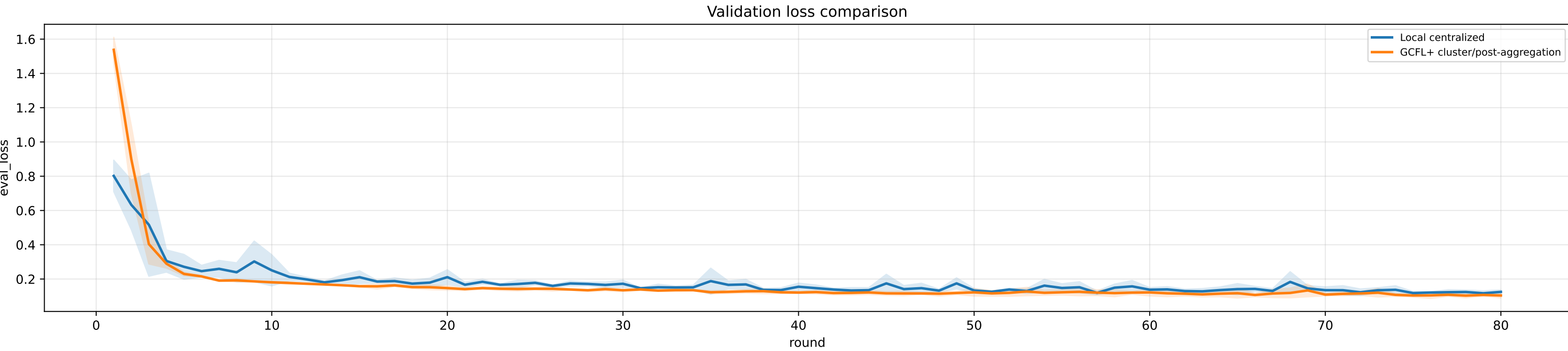
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.6% ± 0.1%	93.6% ± 0.1%	98.8% ± 0.5%	98.2% ± 0.5%	96.8% ± 0.2%	92.1% ± 0.3%	81.7% ± 0.6%
GCFL+	94.6% ± 0.4%	94.6% ± 0.3%	99.3% ± 0.1%	98.7% ± 0.1%	97.2% ± 0.4%	93.2% ± 0.5%	84.4% ± 1.5%



Controlled masking level: High | GCFL+ diagnostics | family=masked_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle2 | n_clients=1 | cycle2=visible 5.5%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1%
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

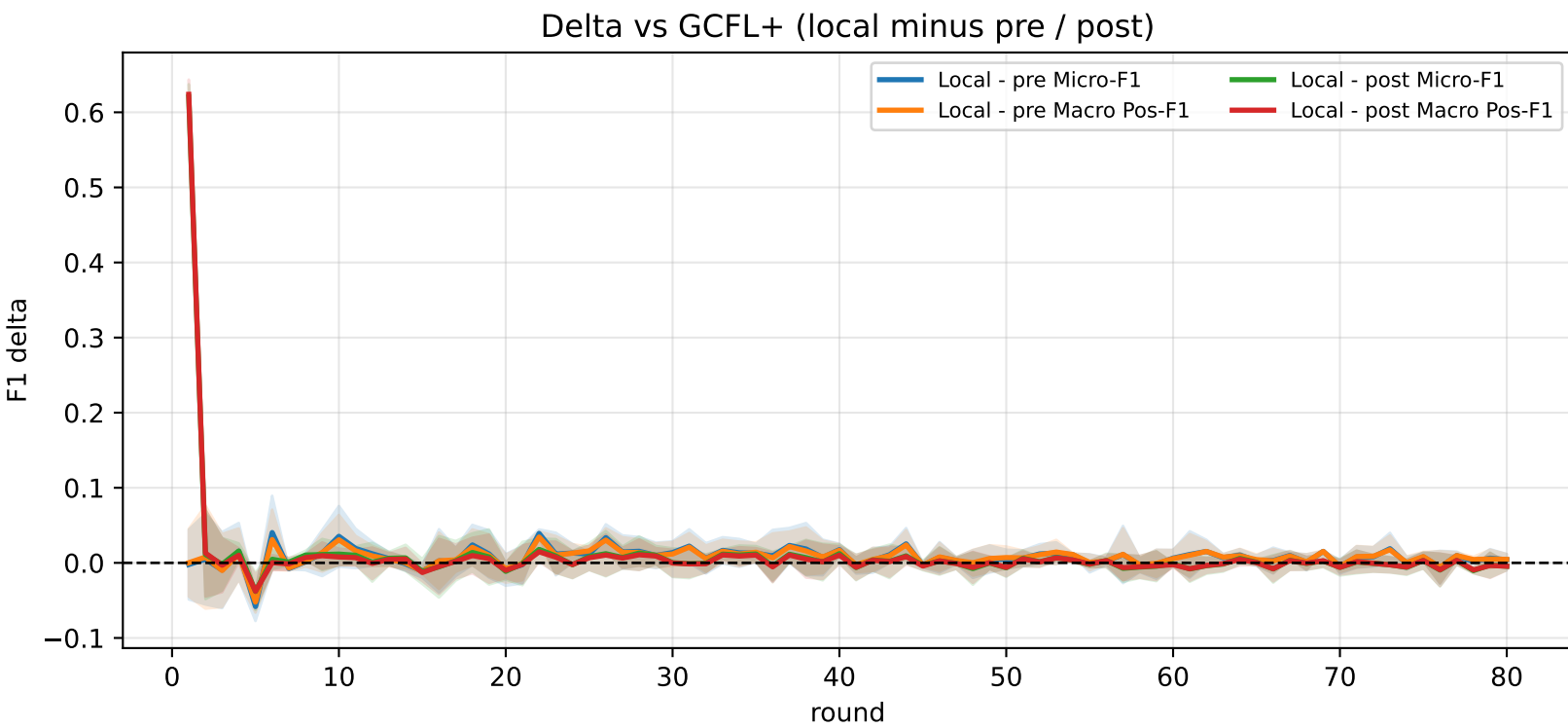
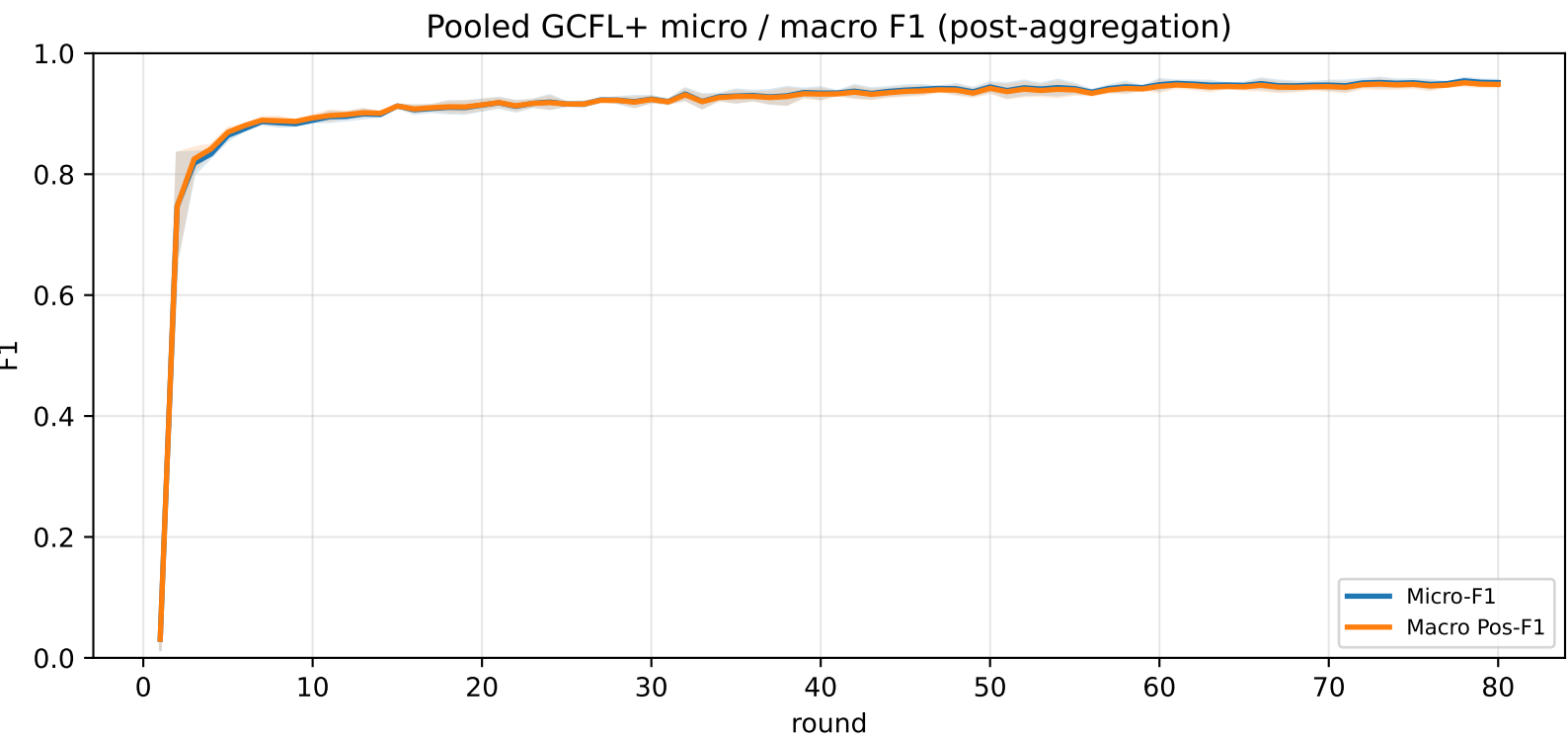
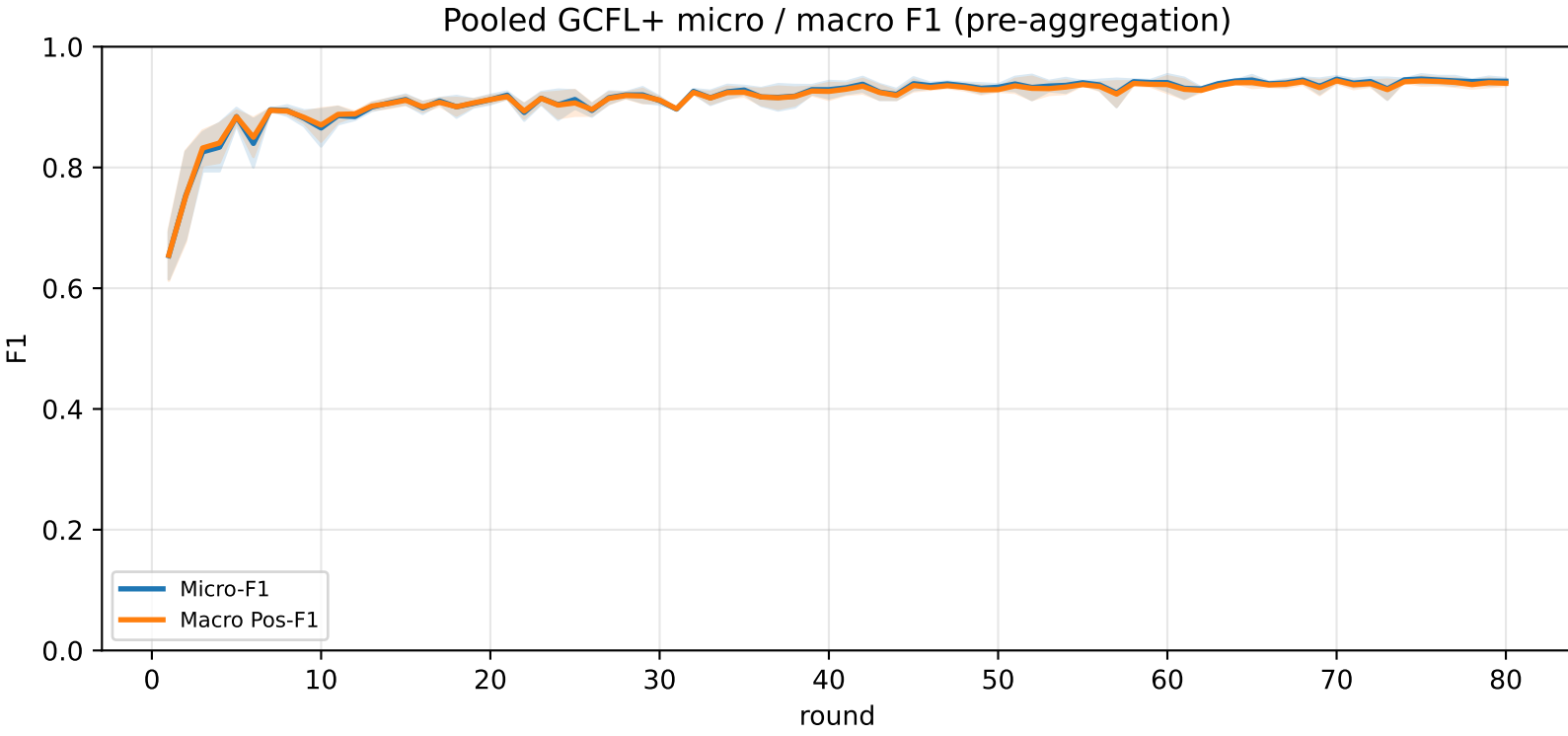
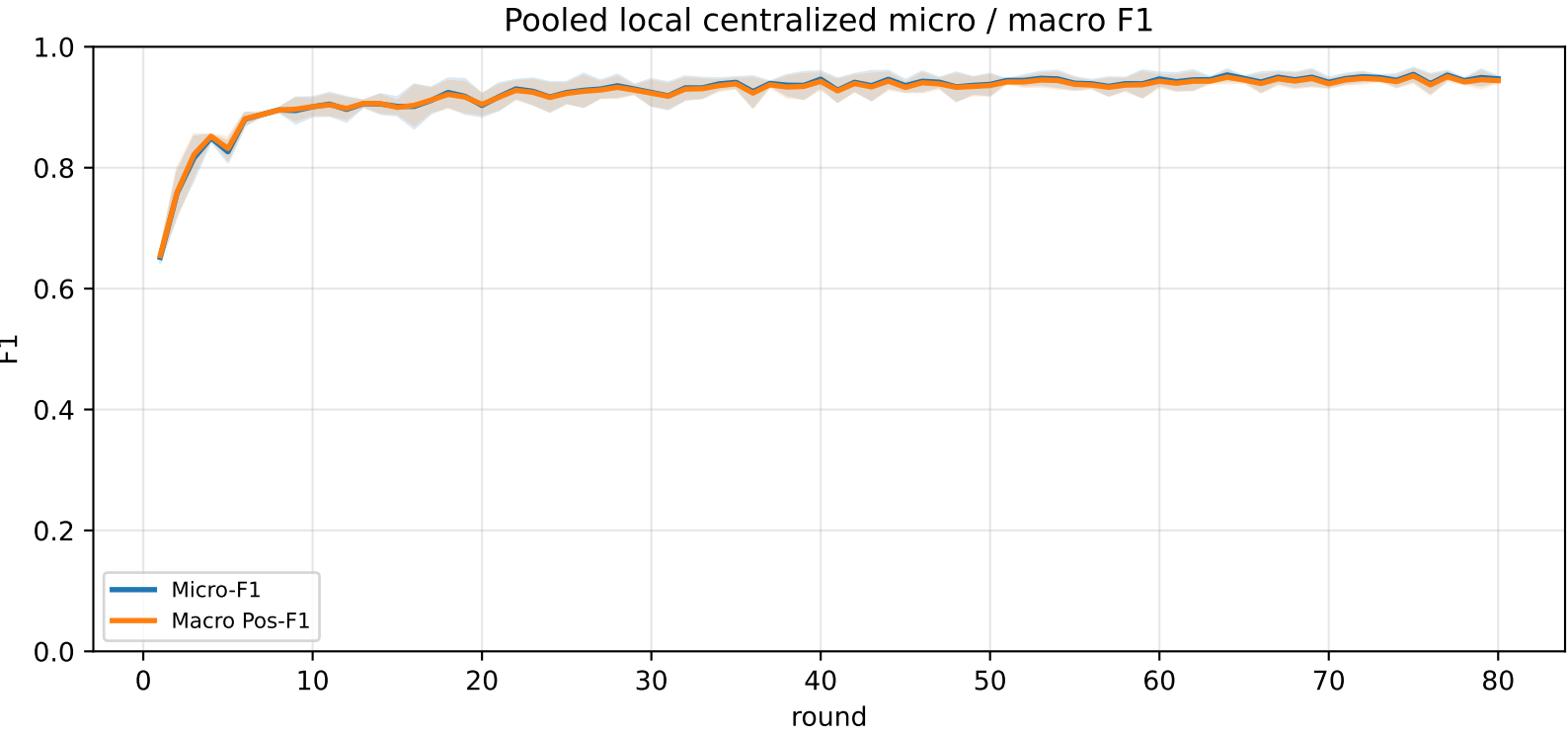
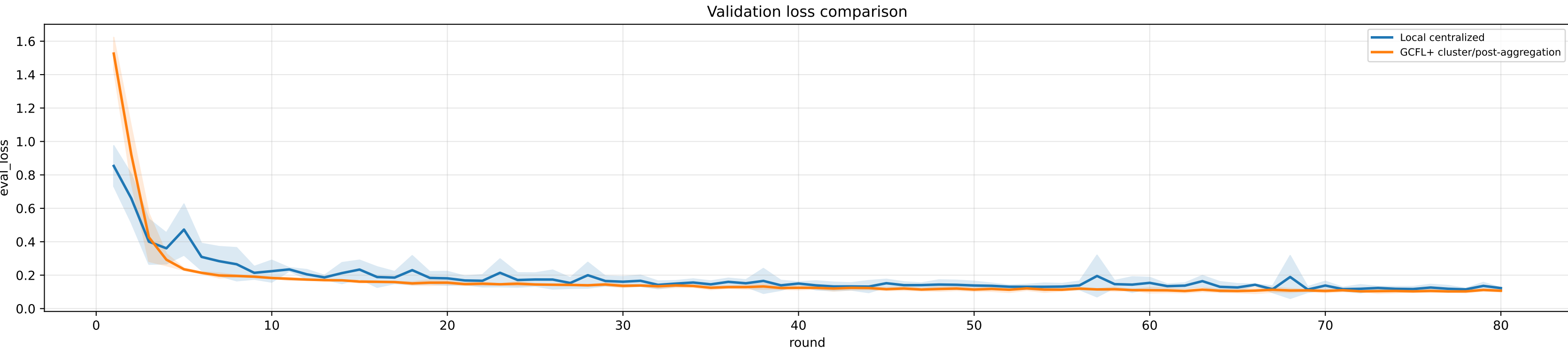
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.1%	94.8% ± 0.1%	99.3% ± 0.0%	98.8% ± 0.2%	97.8% ± 0.2%	93.9% ± 0.4%	84.5% ± 1.0%
GCFL+	94.6% ± 0.1%	94.5% ± 0.2%	99.3% ± 0.2%	98.8% ± 0.1%	97.3% ± 0.3%	92.9% ± 1.3%	84.5% ± 0.9%



Controlled masking level: High | GCFL+ diagnostics | family=masked_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

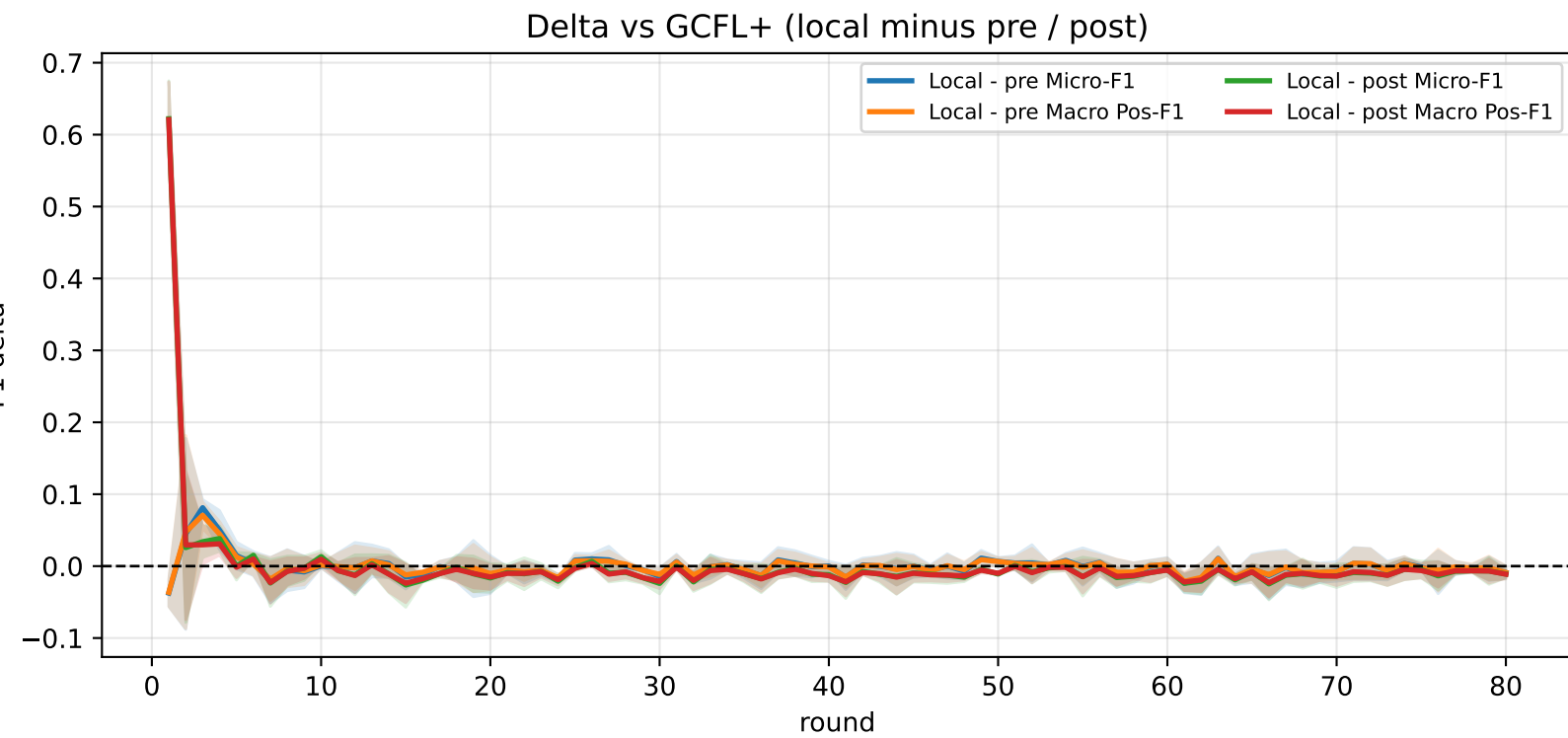
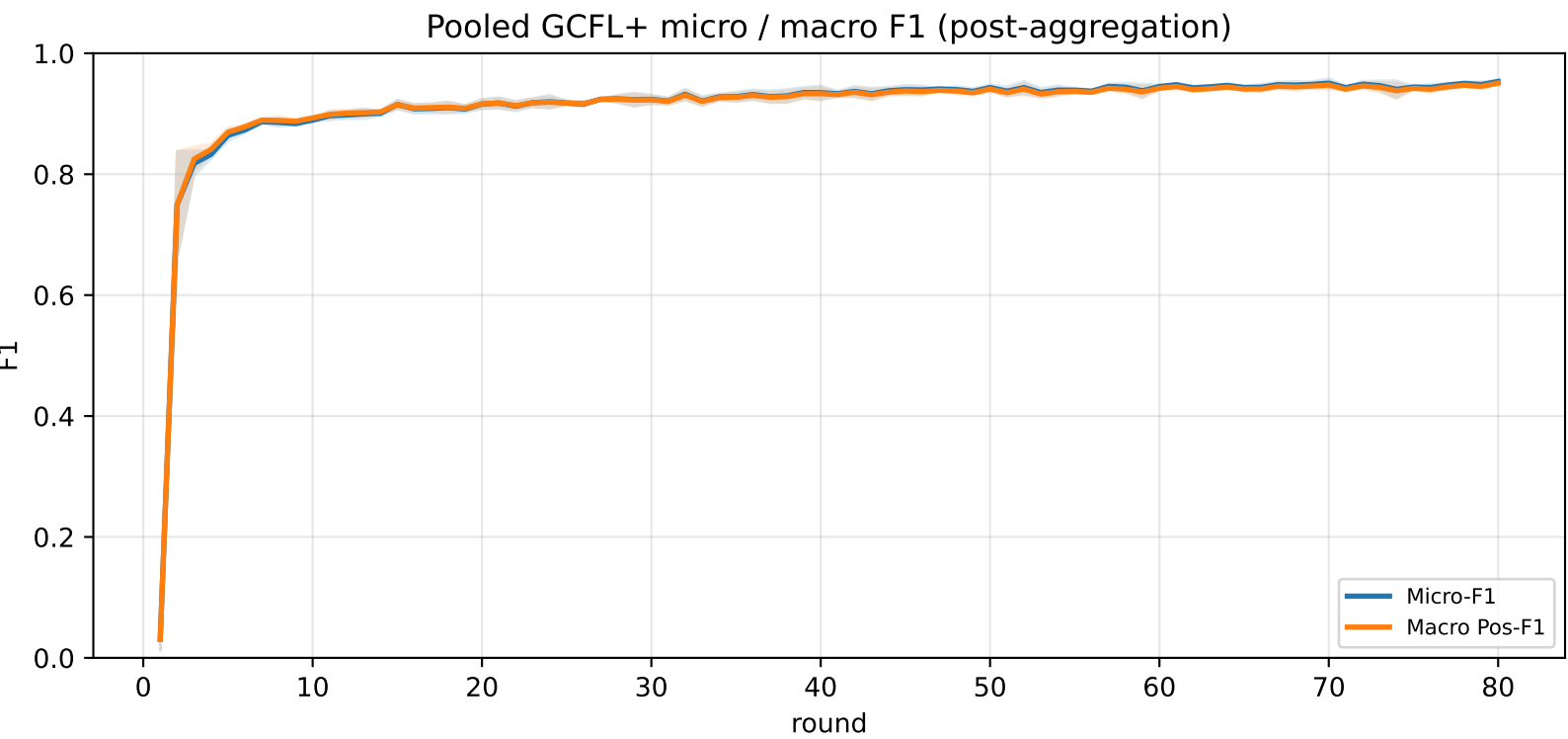
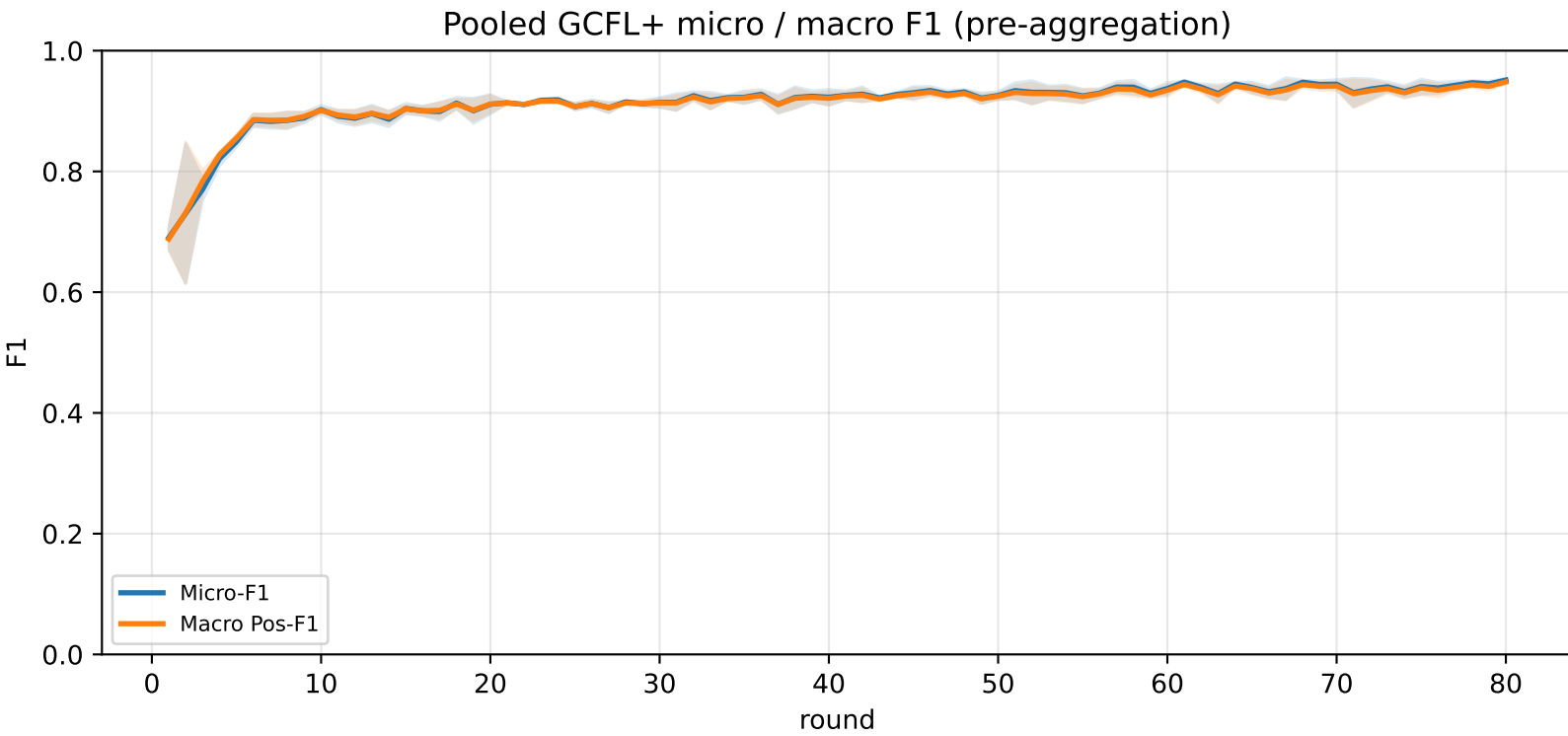
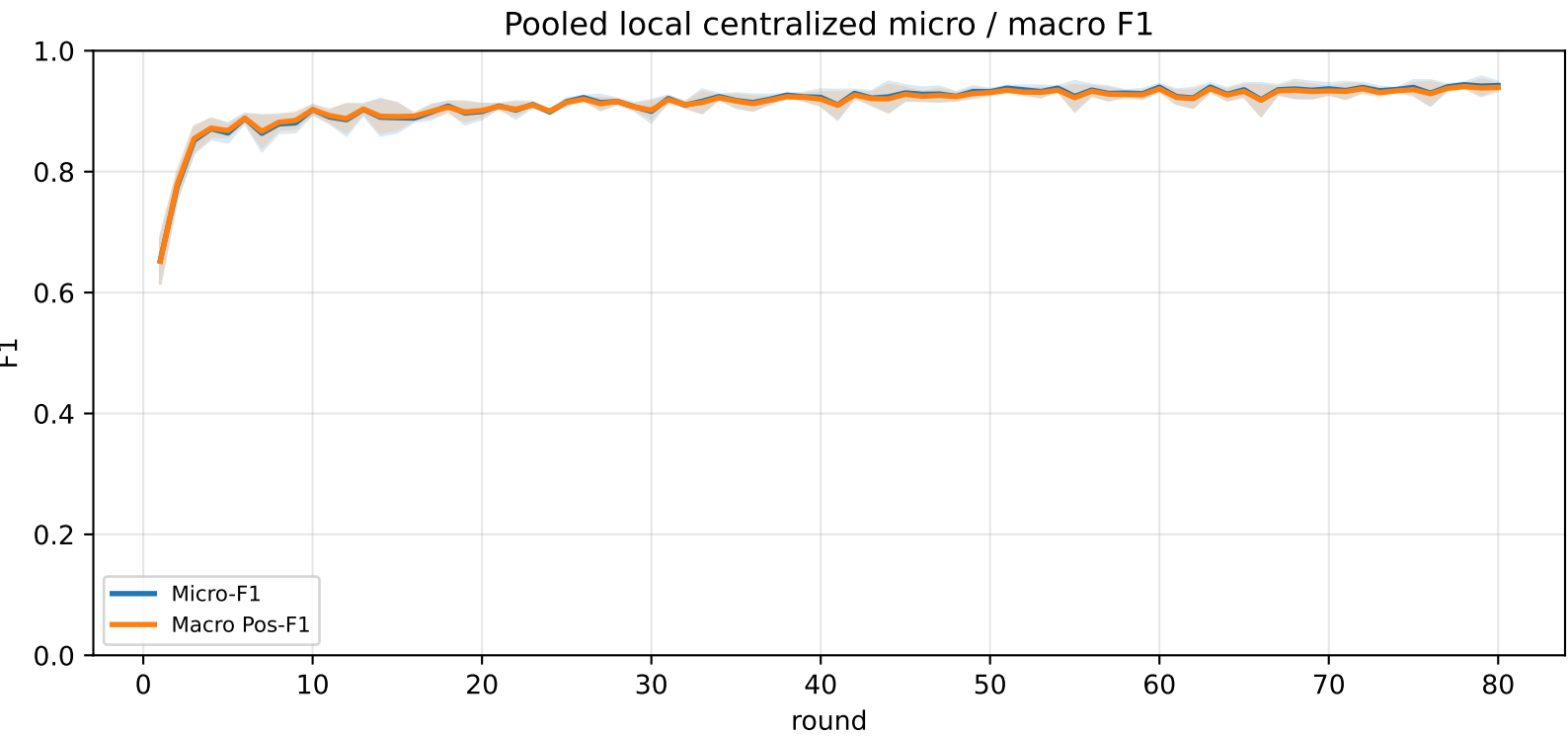
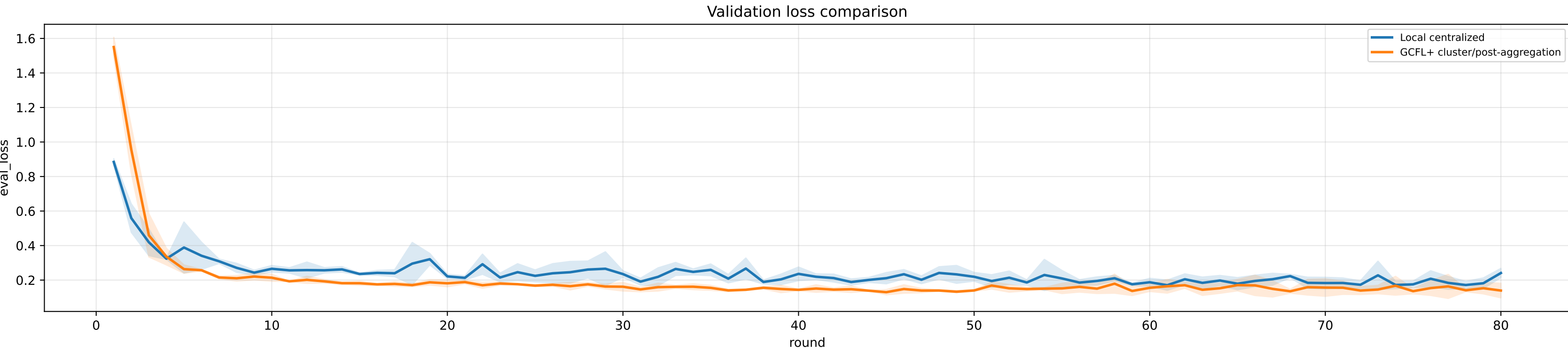
family=masked_cycle3 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 5.8%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1%
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.1% ± 0.8%	95.0% ± 0.8%	99.3% ± 0.1%	98.2% ± 0.9%	97.5% ± 0.5%	94.3% ± 0.4%	85.9% ± 2.5%
GCFL+	94.8% ± 0.3%	94.7% ± 0.3%	99.2% ± 0.1%	98.6% ± 0.4%	97.4% ± 0.1%	93.9% ± 0.4%	84.6% ± 1.7%



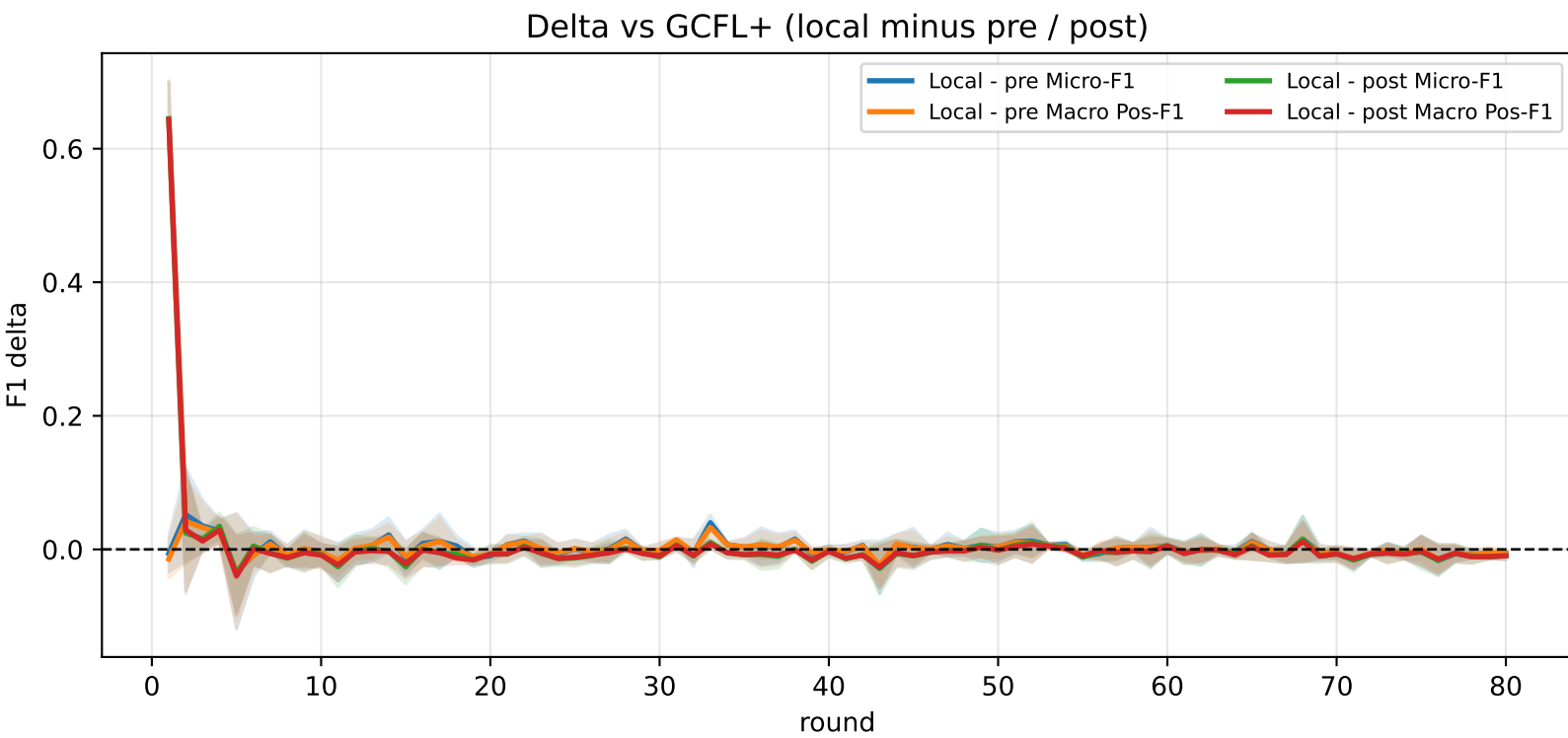
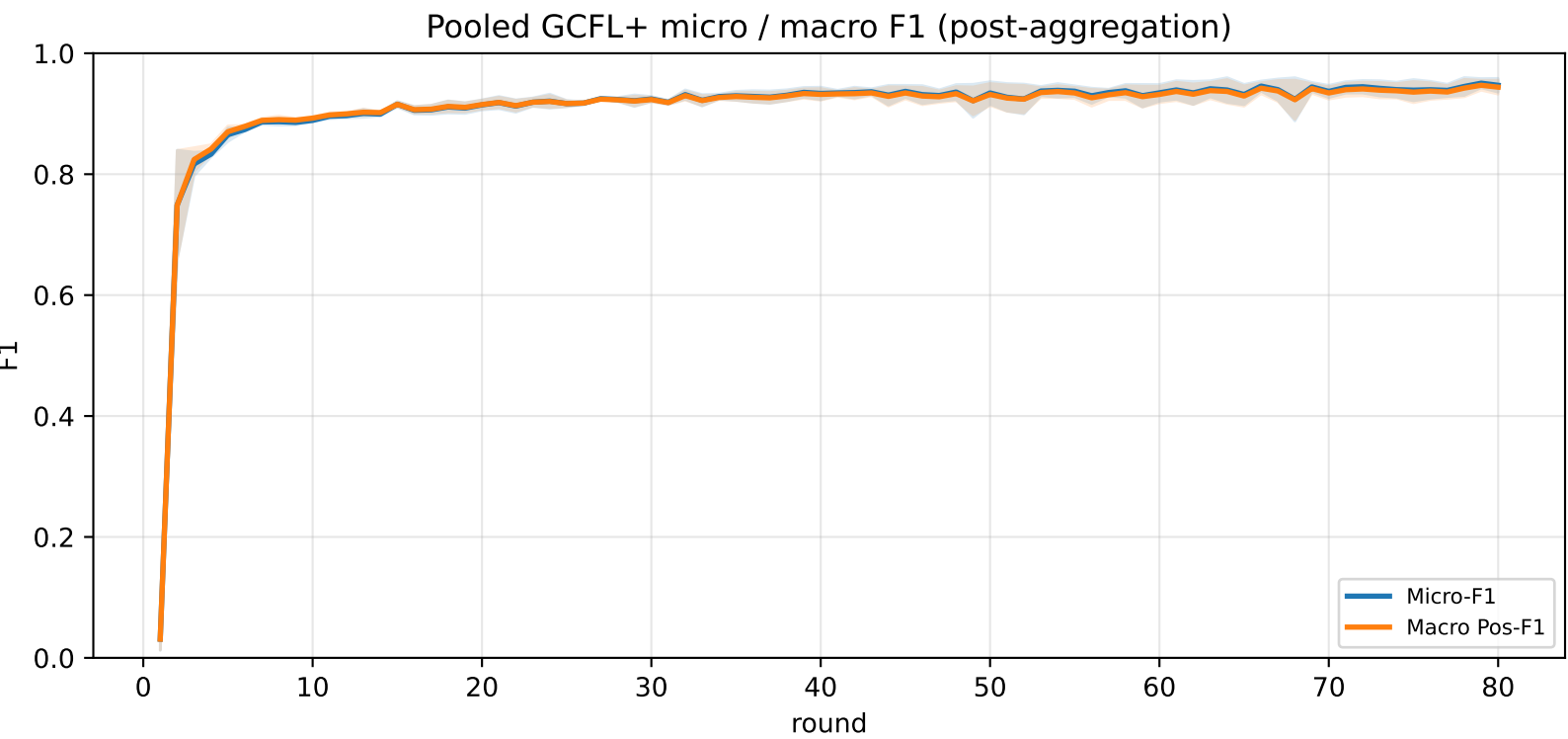
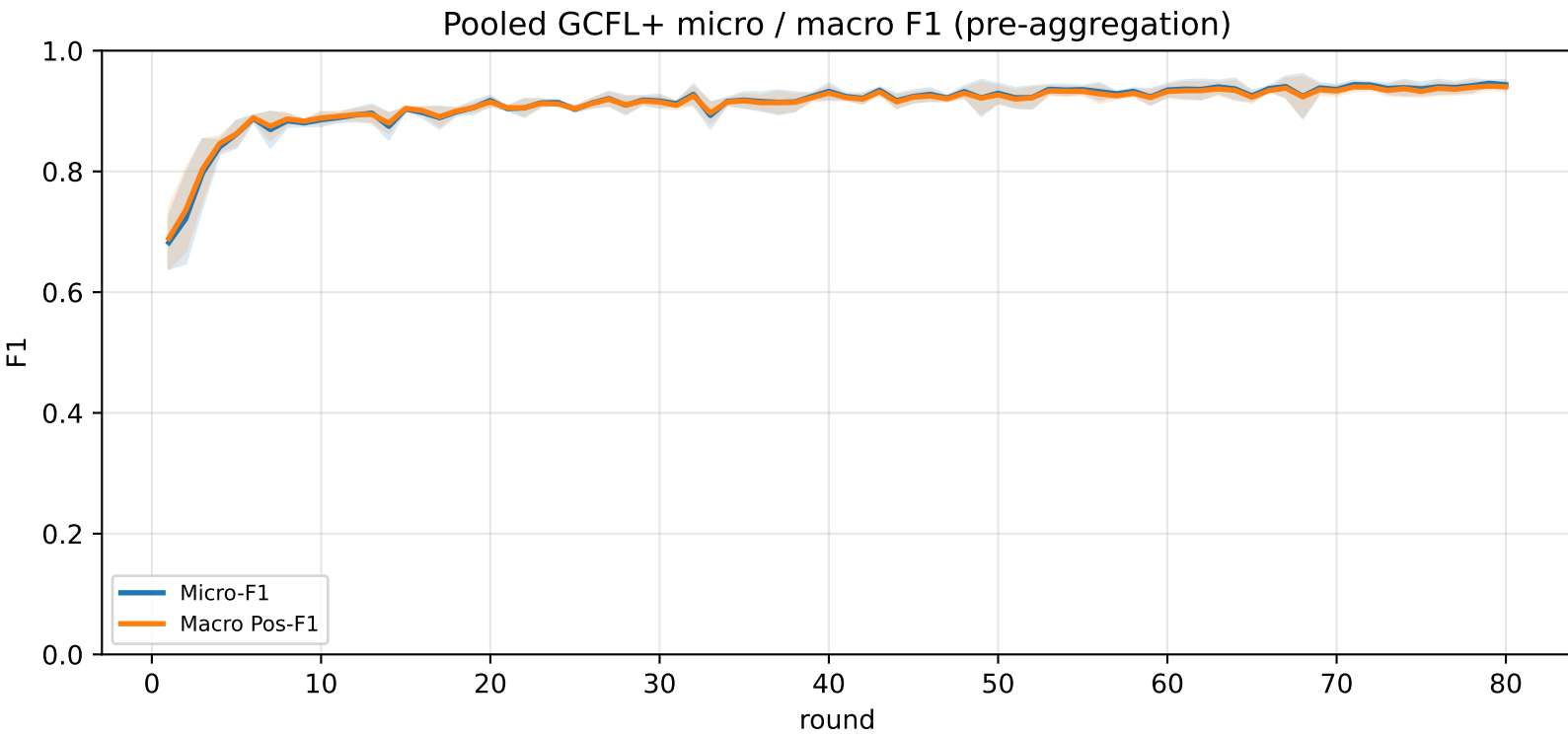
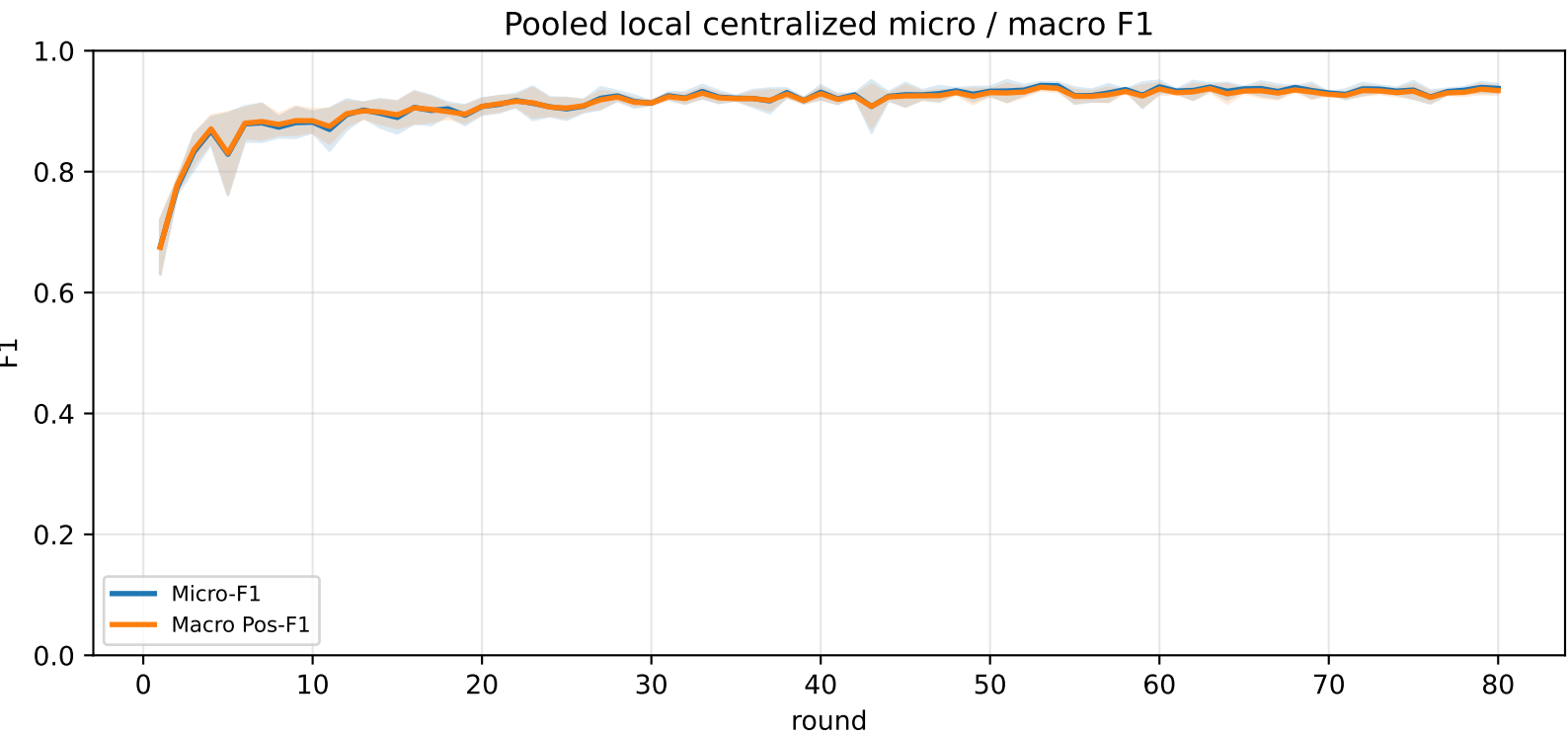
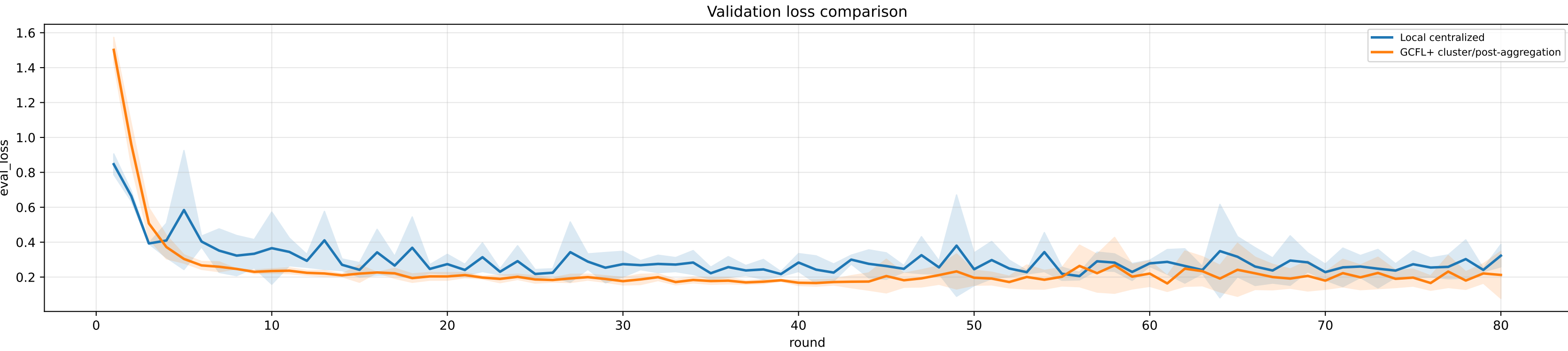
family=masked_cycle4 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 5.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1% | masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.4% ± 0.2%	93.5% ± 0.2%	99.2% ± 0.0%	98.2% ± 0.5%	96.5% ± 0.8%	92.3% ± 0.9%	81.1% ± 1.2%
GCFL+	95.1% ± 0.2%	95.0% ± 0.1%	99.3% ± 0.1%	98.7% ± 0.3%	97.4% ± 0.9%	93.7% ± 0.4%	86.0% ± 1.6%



family=masked_cycle5 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 5.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1% | masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.9% ± 0.5%	92.9% ± 0.4%	99.0% ± 0.2%	98.0% ± 0.4%	95.5% ± 0.4%	90.0% ± 1.4%	82.1% ± 1.9%
GCFL+	93.6% ± 1.1%	93.6% ± 1.0%	99.3% ± 0.3%	98.6% ± 0.3%	96.6% ± 0.9%	91.6% ± 1.7%	82.2% ± 2.4%



Controlled masking level: High | GCFL+ diagnostics | family=masked_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle6 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% | cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 5.2%±0.0% / true 26.1%
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.5% ± 0.6%	91.7% ± 0.4%	97.3% ± 2.5%	97.9% ± 0.5%	96.8% ± 0.5%	90.4% ± 2.2%	76.1% ± 2.5%
GCFL+	94.9% ± 0.4%	94.8% ± 0.4%	99.4% ± 0.2%	98.8% ± 0.3%	97.3% ± 0.5%	94.0% ± 0.5%	84.8% ± 1.7%

